ELE 717 · POSTGRADUATE DIPLOMA · NIGERIAN DEFENCE ACADEMY

Solid State Electronics

Exam Companion: the 50 PGD examination questions, answered

Full model answers with key terms, marking checklists, common mistakes and variant questions, plus a formula sheet, three practice papers and last-minute flashcards.

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ORIENTATION

Start Here

This companion is built around one thing: the **50 PGD examination questions** for ELE 717 Solid State Electronics. Every question has a full model answer written the way you should write it in the exam, with the terms it uses defined, the marking points listed, the usual mistakes pointed out, and a variant question to test yourself.

It is separate from the week-by-week course. Use the weekly course to learn a topic properly; use this companion to turn that understanding into marks.

How each question is laid out

Every one of the 50 questions follows the same pattern, so you always know where to look:

Part	What it gives you
Question type, topic, full notes	Whether it is a definition, explanation, comparison, derivation or calculation, and which week of the weekly course teaches it in full.
What the examiner wants	What the command word asks for and what a complete answer contains.
Key terms	Every technical word the answer uses, defined in one line, so the question stands on its own.
Model answer	A full-mark answer: numbered points, equations with every symbol explained, diagrams to draw, and for calculations the Given → Find → Formula → Solution → Answer → What this means layout.
Marking checklist	The points a marker looks for. Compare your own answer against it, point by point.
Common mistakes	The errors that lose marks on that question.
Remember it	One sentence to fix the idea in your memory.
Try a variant	A similar question the examiner could set instead, with a short answer.

How to use it: three passes

1. **Understand.** Read the model answer for a question, slowly, with the key terms. If something is still unclear, open the week named under "Full notes".
2. **Write.** Cover the model answer. Write your own answer on paper, under exam conditions (about 1½ minutes per mark, or 15–20 minutes for a long question). Then mark it against the **marking checklist**. Every missing point is a mark you would have lost.
3. **Recall.** In the last days, use the **Last-minute flashcards** page: one line per question. Then sit the **practice papers**, timed.

Tick each page as done with the button at the bottom; your progress is remembered in this browser.

Command words: what each one asks for

Command word	What to do	How much to write
Define / What is	Give the precise meaning in one or two sentences, with the symbol and unit if there is one. Add an example.	Short, exact
State	Write the law or equation, with its symbols defined. No proof needed.	Short
Explain	Say *why* or *how*: give the mechanism, step by step, with a diagram if it helps.	Medium to long
Describe	Say *what happens*, in order (stages, procedure).	Medium
Distinguish / Differentiate	Give the features of each, side by side. Finish with a table.	Medium
Compare	Like distinguish, but cover similarities as well as differences, under the headings the question names. Finish with a table.	Medium to long
Derive	Start from stated assumptions and known laws; number each step; end with the boxed result.	Long
Calculate / Determine	Show the formula, the substitution with units, and the answer with its unit.	Short

Exam technique that earns marks

- **Define before you use.** Start long answers with one-line definitions of the key terms. Markers give marks for them, and it organises your answer.
- **Every equation needs its symbols.** Write "where n = electron concentration (m^{-3}), …" under each equation. An equation with undefined symbols often scores half.
- **Draw.** Band diagrams, profiles and I-V curves earn marks quickly. Label the axes, the levels (E_c, E_v, E_F) and the key quantities (qV_{bi}, W, Φ_B).
- **Units, always: use SI.** Work in metres throughout (m^{-3}, $m^2/V{\cdot}s$, V/m, F/m); the answer then comes out in A/m^2, m, V. Energies stay in eV. If a question gives data in centimetre units (as Q19, Q29, Q34 and Q38 do), convert them in your **Given** line first, e.g. 5×10^{16} cm^{-3} = 5×10^{22} m^{-3}. The conversion table is on the Formula and constants sheet.
- **Derivations: assumptions first.** List the assumptions, number the steps, put a box round the result. Even if you get stuck, the assumptions and the start earn marks.
- **Comparisons: end with a table.** It is quick for the marker to check and shows you covered every heading.
- **Say what the number means.** One sentence after a calculation ("the field is low enough for v = μE to hold") shows understanding.

Numbers you must know (silicon at 300 K)

This companion uses **SI units** throughout (metres, not centimetres); energies are in electron-volts.

Quantity	Value
Electronic charge q	1.6×10^{-19} C
Thermal voltage kT/q	0.0259 V (kT = 0.0259 eV)
Band gap E_g	1.12 eV (Ge 0.66 eV, GaAs 1.42 eV)
Intrinsic concentration n_i	1×10^{16} m^{-3} (= 1×10^{10} cm^{-3})
Effective densities of states N_c, N_v	2.8×10^{25}, 1.04×10^{25} m^{-3}
Mobilities μ_n, μ_p	0.135, 0.048 $m^2/V\cdot s$
Diffusion coefficients D_n, D_p	3.5×10^{-3}, 1.24×10^{-3} m^2/s
Permittivity ε_s	$11.7\varepsilon_0 = 1.04\times10^{-10}$ F/m ($\varepsilon_0 = 8.854\times10^{-12}$ F/m)
Electron affinity χ	4.05 eV
Lattice constant a	0.543 nm
Saturation velocity v_{sat}	1×10^5 m/s

The **Formula and constants sheet** page has the full list.

Question map

All 50 questions, what kind of answer each needs, and where to find it. "Full notes" is the week of the weekly course that teaches the topic.

Q	Topic	Type	Page	Full notes
1	Semiconductor vs conductor and insulator	Define, explain	Questions 1–13	Week 1
2	Elemental and compound semiconductors	Distinguish	Questions 1–13	Week 1
3	Why silicon	Explain	Questions 1–13	Week 1
4	Si, Ge and GaAs	Compare	Questions 1–13	Week 1
5	Crystal lattice and structure	Define, explain	Questions 1–13	Week 1
6	Miller indices	Define, explain	Questions 1–13	Week 1
7	Miller indices of intercepts 2a, 3b, 1c	Calculation	Questions 1–13	Week 1
8	Covalent, ionic and metallic bonding	Distinguish	Questions 1–13	Week 1
9	MG-Si to EG-Si	Describe	Questions 1–13	Week 2
10	Czochralski growth	Explain	Questions 1–13	Week 2
11	Advantages and limitations of CZ	State	Questions 1–13	Week 2
12	VPE, LPE and MBE principles	Explain	Questions 1–13	Week 2
13	VPE, LPE and MBE compared	Compare	Questions 1–13	Week 2
14	What doping is and why	Define, explain	Questions 14–21	Week 3
15	n-type and p-type	Distinguish	Questions 14–21	Week 3
16	Donors and acceptors	Distinguish	Questions 14–21	Week 3
17	Mechanism of dopant diffusion	Describe	Questions 14–21	Week 3
18	Fick's first and second laws	State, explain	Questions 14–21	Week 3
19	Constant-source diffusion profile	Explain, calculation	Questions 14–21	Week 3
20	Ion implantation and its advantages	Explain	Questions 14–21	Week 3
21	Diffusion vs ion implantation	Compare	Questions 14–21	Week 3
22	Assumptions of the quantum model	Explain	Questions 22–34	Week 4
23	Schrödinger equation	State, explain	Questions 22–34	Week 4
24	Kronig–Penney model	Explain	Questions 22–34	Week 4
25	Valence band, conduction band, gap	Distinguish	Questions 22–34	Week 4
26	Direct and indirect band gaps	Distinguish	Questions 22–34	Week 4
27	Intrinsic semiconductor; derive n_i	Define, derive	Questions 22–34	Weeks 4–5
28	Intrinsic and extrinsic concentrations	Distinguish	Questions 22–34	Week 4
29	n and p when $n = p = n_i$	Calculation	Questions 22–34	Week 4
30	Fermi–Dirac function	Explain	Questions 22–34	Week 5
31	Fermi–Dirac at absolute zero	State, explain	Questions 22–34	Week 5
32	Density of states	Define, explain	Questions 22–34	Week 5
33	Fermi level and doping	Explain	Questions 22–34	Week 5
34	Majority carriers for $N_D = 5\times10^{22}$ m^{-3} (5×10^{16} cm^{-3})	Calculation	Questions 22–34	Weeks 4–5

Q	Topic	Type	Page	Full notes
35	Mobility and what affects it	Define, explain	Questions 35–46	Week 7
36	Scattering mechanisms	Explain	Questions 35–46	Week 7
37	Drift current density	Derive	Questions 35–46	Week 7
38	Electron drift current density	Calculation	Questions 35–46	Week 7
39	Diffusion current and Fick's law	Explain, state	Questions 35–46	Week 7
40	Einstein relation	State, explain	Questions 35–46	Week 7
41	Velocity saturation and hot carriers	Explain	Questions 35–46	Week 8
42	Generation and recombination	Define	Questions 35–46	Week 8
43	SRH recombination	Explain	Questions 35–46	Week 8
44	Minority-carrier lifetime	Define, explain	Questions 35–46	Week 8
45	Continuity equation	Explain	Questions 35–46	Week 9
46	Surface recombination velocity	Define, explain	Questions 35–46	Week 9
47	Depletion region and built-in potential	Explain	Questions 47–50	Week 10
48	Shockley diode equation	Derive	Questions 47–50	Week 10
49	Schottky diode vs ohmic contact	Distinguish	Questions 47–50	Week 11
50	I-V and C-V characterisation	Explain	Questions 47–50	Week 12

The five calculations (Q7, Q19, Q29, Q34, Q38) and the three derivations (Q27, Q37, Q48) are the most reliable marks in the paper: learn them until you can write them without looking.

Questions 1–13 — Material Physics and Semiconductor Growth

Module 1, part 1: what a semiconductor is, the main materials, crystal structure and Miller indices, bonding, and how silicon is purified and grown into crystals and epitaxial layers. Each question stands on its own: its key terms are defined with it, so you can start anywhere.

Symbols and abbreviations used on this page

Every symbol, constant and abbreviation on this page is listed here, even if another page already defined it, so you can study this page on its own.

Constants and units

Symbol	Meaning	Units / typical value
q	Magnitude of the electronic charge	1.6×10^{-19} C
k	Boltzmann constant	8.617×10^{-5} eV/K = 1.381×10^{-23} J/K
kT	Thermal energy	0.0259 eV (often 0.02585 eV) at 300 K
T	Absolute temperature	kelvin (K); room temperature = 300 K
h	Planck constant	6.626×10^{-34} J·s
eV	Electron-volt, a unit of energy	1 eV = 1.6×10^{-19} J
µm	Micrometre (micron)	1 µm = 10^{-6} m

Symbols

Symbol	Meaning	Units / typical value
E_c (E_C)	Energy of the bottom of the conduction band	eV
E_v (E_V)	Energy of the top of the valence band	eV
E_g	Band-gap energy, $E_c - E_v$	eV; Si 1.12, Ge 0.66, GaAs 1.42 at 300 K
f(E)	Fermi–Dirac function: probability that a state at energy E is occupied	dimensionless, 0 to 1
n	Electron concentration (free electrons per m^3)	m^{-3}
p	Hole concentration (holes per m^3)	m^{-3}
n_i	Intrinsic carrier concentration ($n = p = n_i$ in pure material)	m^{-3}; Si ≈ 1×10^{16} at 300 K
k (wave vector)	Wave vector, $2\pi/\lambda$; measures the electron's momentum, $p = \hbar k$	m^{-1}
λ	Wavelength	μm or nm; λ(μm) = 1.24/E(eV)
a	Lattice constant: edge length of the cubic unit cell	nm; Si 0.543, Ge and GaAs 0.565
(h k l), {h k l}	Miller indices of a plane; family of equivalent planes	integers
[u v w], ⟨u v w⟩	A crystal direction; family of equivalent directions	integers
d_{hkl}	Spacing between neighbouring (h k l) planes, $a/\sqrt{(h^2 + k^2 + l^2)}$	nm
k_0	Segregation coefficient, C_s/C_l at the freezing interface	dimensionless; B 0.8, P 0.35
C_s	Surface concentration of a diffusion (or solid concentration when freezing)	m^{-3}
C_l, C_0	Concentration in the melt (liquid); C_0 at the start	m^{-3}
f (growth)	Fraction of the melt already solidified	0 to 1
C(x, t)	Dopant concentration at depth x and time t	m^{-3}
μ, μ_n, μ_p	Mobility: drift velocity per unit field (electrons, holes)	$m^2/(V\cdot s)$; Si μ_n 0.135, μ_p 0.048
σ	Conductivity, $q(n\mu_n + p\mu_p)$ (in Week 8, σ is also a capture cross-section)	S/m (siemens per metre)
ρ	Resistivity, $1/\sigma$	Ω·m
α, β, γ (cell)	Angles between the edges of a unit cell	degrees; 90° for cubic
v_{pull}	Pull rate of a Czochralski crystal	mm/min
r (atom)	Atomic radius (hard-sphere model), half the nearest-neighbour distance	nm

Abbreviations

Abbreviation	Meaning
Si	Silicon
Ge	Germanium
GaAs	Gallium arsenide
GaN	Gallium nitride
SiC	Silicon carbide
InP	Indium phosphide
GaP	Gallium phosphide
SiO_2	Silicon dioxide (silica), the oxide insulator
FCC	Face-centred cubic lattice
III–V	Compound of Group III and Group V elements (e.g. GaAs)
MG-Si	Metallurgical-grade silicon (≈ 98 % pure)
EG-Si	Electronic-grade silicon (99.9999999 % or better)
CZ	Czochralski crystal growth
FZ	Float-zone crystal growth
CVD	Chemical vapour deposition
VPE	Vapour-phase epitaxy
LPE	Liquid-phase epitaxy
MBE	Molecular-beam epitaxy
MOCVD	Metal-organic chemical vapour deposition
UHV	Ultra-high vacuum (≈ 10^{-10} Torr)
RHEED	Reflection high-energy electron diffraction (counts atomic layers during MBE)
MOS	Metal–oxide–semiconductor
MOSFET	Metal–oxide–semiconductor field-effect transistor
CMOS	Complementary MOS (n- and p-channel MOSFETs together)
MESFET	Metal–semiconductor field-effect transistor
HEMT	High-electron-mobility transistor
BJT	Bipolar junction transistor
HBT	Heterojunction bipolar transistor
IC	Integrated circuit
VLSI	Very-large-scale integration
LED	Light-emitting diode
RF	Radio frequency
IGBT	Insulated-gate bipolar transistor
MEMS	Micro-electro-mechanical systems

Abbreviation	Meaning
LCM	Lowest common multiple
KOH	Potassium hydroxide (an anisotropic silicon etchant)
VCSEL	Vertical-cavity surface-emitting laser

Questions on this page

Q	Question
1	Define a semiconductor and explain how its electrical conductivity differs from that of a conductor and an insulator.
2	Distinguish between elemental semiconductors and compound semiconductors, giving two examples of each.
3	Explain why silicon is widely used in semiconductor device fabrication.
4	Compare the properties and applications of silicon (Si), germanium (Ge), and gallium arsenide (GaAs).
5	What is a crystal lattice? Explain the significance of crystal structure in semiconductor materials.
6	Define Miller indices and explain how they are used to identify crystal planes.
7	Determine the Miller indices of a crystal plane having intercepts of 2a, 3b, and 1c.
8	Explain the difference between covalent, ionic, and metallic bonding and identify the dominant bonding mechanism in silicon.
9	Describe the major stages involved in converting metallurgical-grade silicon (MG-Si) into electronic-grade silicon (EG-Si).
10	Explain the principle and operation of the Czochralski (CZ) crystal growth method.
11	State four advantages and two limitations of the Czochralski method.
12	Explain the principles of vapour-phase epitaxy (VPE), liquid-phase epitaxy (LPE), and molecular-beam epitaxy (MBE).
13	Compare VPE, LPE, and MBE in terms of growth mechanism, temperature, precision, and applications.

Model answers

Q1. Define a semiconductor and explain how its electrical conductivity differs from that of a conductor and an insulator.

> **Question type:** Definition and explanation · **Topic:** classification of solids by conductivity · **Full notes:** Week 1 (Introduction to Semiconductor Materials)

What the examiner wants. First a precise definition of a semiconductor (conductivity between metals and insulators, and controllable). Then an explanation, using energy bands, of why the three classes of material conduct so differently, including how each responds to temperature. A labelled band sketch and a short comparison table earn the full marks.

Key terms.

- **Conductivity (σ):** how easily a material carries electric current, in S/m; its inverse is resistivity $\rho = 1/\sigma$ in Ω·m.
- **Energy band:** a wide range of closely spaced allowed electron energies formed when many atoms join into a solid.
- **Valence band:** the highest band that is full at 0 K; it holds the electrons that form the bonds.

- **Conduction band:** the band above the valence band; electrons in it are free to move and carry current.
- **Band gap (E_g):** the forbidden energy range between the top of the valence band (E_v) and the bottom of the conduction band (E_c), $E_g = E_c - E_v$, in electron-volts (eV).
- **Hole:** the empty place left in a bond when an electron is freed; it moves through the crystal like a positive charge.
- **Doping:** adding a small, controlled amount of a chosen impurity to change the number of carriers.
- **kT:** the typical thermal energy of a particle; 0.0259 eV at 300 K (k = Boltzmann's constant, T = absolute temperature).

Model answer.

Definition. A semiconductor is a material whose electrical conductivity lies between that of a conductor and an insulator (resistivity roughly 10^{-5} to 10^{6} Ω·m) and can be changed over many orders of magnitude by temperature, light, electric field and, above all, by doping. Examples: silicon (Si), germanium (Ge), gallium arsenide (GaAs).

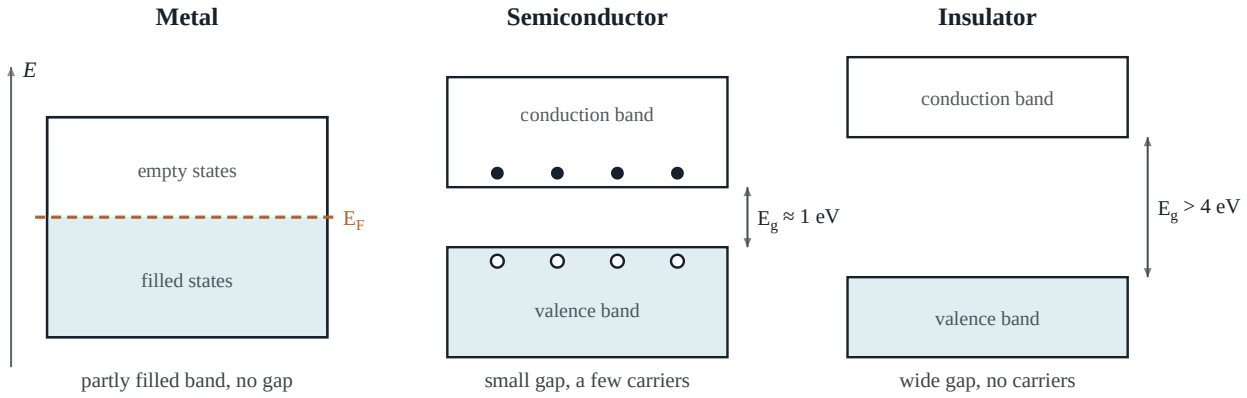
How conductivity depends on carriers. Current is carried by mobile charges, so

$$\sigma = q(n\mu_n + p\mu_p)$$

Where:

- σ = conductivity (S/m)
- q = electronic charge, 1.6×10^{-19} C
- n, p = concentrations of free electrons and holes (m^{-3})
- μ_n, μ_p = electron and hole mobilities, i.e. drift speed per unit field ($m^2/V\cdot s$)

The three classes differ mainly in **how many free carriers they have**, and that is decided by their energy-band structure.



Energy bands of a metal, a semiconductor and an insulator at room temperature: no gap, a small gap, and a wide gap.

Sketch in the exam: three band diagrams side by side, label "conduction band", "valence band", "E_g", and show a few electrons above a small gap for the semiconductor.

1. Conductor (metal). The valence and conduction bands overlap, or the top band is only partly filled, so there is no gap. About 10^{28} free electrons per m^3 are always present. Conductivity is very high (σ ≈ 10^{7}–10^{8} S/m; copper ρ = 1.7×10^{-8} Ω·m) and is hardly changed by impurities. When temperature rises the number of electrons stays the same but lattice vibrations scatter them more, so conductivity **falls** (positive temperature coefficient of resistance).

2. Insulator. The valence band is full and the conduction band empty, separated by a wide gap (E_g > about 4 eV; SiO_2 ≈ 9 eV). Room-temperature thermal energy (kT ≈ 0.026 eV) is far too small to lift electrons across, so there are

practically no free carriers. Resistivity is very high (10^8–10^{16} Ω·m) and changes little with temperature under normal conditions.

3. Semiconductor. The arrangement is the same as in an insulator (full valence band, empty conduction band at 0 K) but the gap is **small** (Si 1.12 eV, Ge 0.66 eV, GaAs 1.42 eV). At room temperature a small fraction of bonds is broken thermally: each freed electron enters the conduction band and leaves a hole in the valence band, so **both electrons and holes** conduct. Pure silicon has only $n = p = n_i = 1\times10^{16}$ m^{-3} at 300 K, so its conductivity is low; but:

- heating breaks more bonds (n_i rises roughly as $e^{-E_g/2kT}$), so conductivity **rises** with temperature (negative temperature coefficient of resistance);
- doping with about one phosphorus atom per million Si atoms (5×10^{22} m^{-3}) raises the conductivity more than a million times.

Comparison.

Property	Conductor	Semiconductor	Insulator
Band gap	None (bands overlap)	Small, ≈ 0.5–3.5 eV	Large, > 4 eV
Free carriers	≈ 10^{28} m^{-3}, fixed	Few, controllable; electrons **and** holes	Practically none
Resistivity (Ω·m)	≈ 10^{-8}	≈ 10^{-5} to 10^{6}	10^{8} to 10^{16}
Effect of heating	σ falls	σ rises	Very small
Effect of impurities	Small	Very large	Small
Examples	Cu, Al, Ag	Si, Ge, GaAs	Glass, SiO_2, rubber

Conclusion. Conductors have no gap and a fixed, huge number of carriers; insulators have a wide gap and essentially none; semiconductors have a small gap, two kinds of carrier, and a carrier density that we can engineer, which is what makes devices possible.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definition: conductivity between conductor and insulator **and** controllable (doping, temperature, light)
- ✓ at least two examples (Si, Ge, GaAs)
- ✓ labelled band diagram for all three classes
- ✓ conductor: overlapping bands, ≈ 10^{28} m^{-3} electrons, σ falls with temperature
- ✓ insulator: wide gap (> 4 eV, e.g. SiO_2 9 eV), no carriers
- ✓ semiconductor: small gap (Si 1.12 eV), electron–hole pairs created thermally, σ rises with temperature
- ✓ effect of doping on a semiconductor (orders of magnitude)
- ✓ typical resistivity ranges or a comparison table

Common mistakes.

- Defining a semiconductor only as "conductivity between a conductor and an insulator". The key word is *controllable*; without it you lose the main definition mark.
- Saying a semiconductor conducts "less when hot, like a metal". It is the opposite: heating creates carriers, so σ rises.
- Forgetting holes. Semiconductors have two carrier types; metals have only electrons.
- Drawing the semiconductor band diagram with overlapping bands, or the metal with a gap.

Remember it. "No gap, small gap, big gap": metal, semiconductor, insulator, and only the small gap can be tuned.

Try a variant. Explain why the resistance of a silicon sample falls when it is heated, while that of a copper wire rises. *Answer:* In copper the number of free electrons ($\approx 10^{28}$ m^{-3}) is fixed; heating only increases lattice vibrations, which scatter the electrons and lower their mobility, so resistance rises. In silicon heating breaks covalent bonds, creating electron–hole pairs; n_i grows roughly as $e^{-E_g/2kT}$, so the number of carriers rises far faster than the mobility falls, and resistance drops.

Q2. Distinguish between elemental semiconductors and compound semiconductors, giving two examples of each.

> **Question type:** Compare / distinguish · **Topic:** classes of semiconductor material · **Full notes:** Week 1 (Introduction to Semiconductor Materials)

What the examiner wants. Define each class, then set them side by side under the same headings (composition, position in the periodic table, crystal structure, bonding, band gap, processing, uses). Two correct examples of each are compulsory marks. End with a comparison table.

Key terms.

- **Elemental semiconductor:** a semiconductor made of a single chemical element.
- **Compound semiconductor:** a semiconductor made of two or more different elements in a fixed ratio.
- **Group IV, III–V, II–VI:** columns of the periodic table; e.g. a III–V compound joins a group III element (Ga, In, Al) with a group V element (As, P, N).
- **Band gap (E_g):** the energy (eV) needed to free an electron from a bond, i.e. to lift it from the valence band to the conduction band.
- **Direct / indirect band gap:** direct means an electron can drop across the gap and emit light without changing its momentum; indirect means a lattice vibration (phonon) must also take part, so light emission is weak.
- **Diamond cubic / zinc blende:** the crystal structures of Si and of GaAs; both have each atom bonded to four neighbours, but in zinc blende the two kinds of atom alternate.
- **Stoichiometry:** the exact ratio of the elements in a compound (1 : 1 for GaAs).
- **Band-gap engineering:** tuning E_g by alloying compounds, e.g. $Al_xGa_{1-x}As$.

Model answer.

Elemental semiconductors. These consist of atoms of one element, all from group IV of the periodic table, each with four valence electrons. They crystallise in the diamond-cubic structure, where every atom forms four identical covalent bonds, so the bonding is purely covalent. Their band gap is fixed by the element, and both important ones are indirect.

Examples: **silicon (Si)**, E_g = 1.12 eV; **germanium (Ge)**, E_g = 0.66 eV.

Compound semiconductors. These are formed from two or more elements whose valence electrons average four per atom, for example a group III atom (3 electrons) with a group V atom (5 electrons). They crystallise in the zinc-blende (or wurtzite) structure. Because the two atoms have different electronegativities, the bonds are covalent with partial ionic character. Their band gap depends on the elements chosen and can be tuned continuously by alloying; many are direct-gap.

Examples: **gallium arsenide (GaAs)**, III–V, E_g = 1.42 eV direct; **indium phosphide (InP)**, III–V. Others: gallium nitride (GaN, 3.4 eV), silicon carbide (SiC, IV–IV), cadmium telluride (CdTe, II–VI).

Main points of difference.

1. **Composition:** one element versus two or more elements in a fixed ratio.
2. **Bonding:** purely covalent versus covalent with some ionic character.
3. **Band gap:** fixed and indirect (Si, Ge) versus adjustable and often direct, so compounds can emit light.
4. **Mobility:** compounds such as GaAs can have much higher electron mobility (≈ 0.85 $m^2/V{\cdot}s$ against 0.135 for Si), suiting very high frequencies.
5. **Processing:** elemental crystals are easier to grow and dope, cheaper and more mature; compounds need careful control of stoichiometry, are more brittle, costly, and sometimes toxic (arsenic).
6. **Applications:** elemental for integrated circuits, power devices and solar cells; compound for LEDs, lasers, microwave/RF and high-power wide-gap devices.

Comparison table.

Feature	Elemental	Compound
Composition	One element	Two or more elements
Periodic-table groups	Group IV	III–V, II–VI, IV–IV
Crystal structure	Diamond cubic	Zinc blende or wurtzite
Bonding	Purely covalent	Covalent, partly ionic
Band gap	Fixed; Si and Ge indirect	Tunable by alloying; many direct
Light emission	Very poor	Efficient (direct-gap ones)
Cost and processing	Cheap, mature (especially Si)	Costly, stoichiometry must be controlled
Typical uses	ICs, power devices, solar cells	LEDs, lasers, RF, high-power
Examples	**Si, Ge**	**GaAs, InP** (also GaN, SiC, CdTe)

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definition of elemental (single element, group IV)
- ✓ definition of compound (two or more elements, e.g. III–V)
- ✓ two elemental examples: Si, Ge
- ✓ two compound examples, e.g. GaAs, InP (or GaN, SiC)
- ✓ difference in bonding (purely covalent vs partly ionic)
- ✓ difference in band gap (fixed/indirect vs tunable/often direct) and its consequence for light emission
- ✓ difference in cost/processing and in applications
- ✓ comparison table

Common mistakes.

- Giving SiO_2 or diamond as a "compound semiconductor". SiO_2 is an insulator; diamond is elemental (carbon).
- Calling SiGe or SiC "elemental" because they contain silicon. They contain two elements, so they are compound (IV–IV).

- Listing only examples with no actual point of difference. The question says *distinguish*: give at least four contrasting features.
- Stating that all compound semiconductors are direct-gap. GaP and SiC are indirect.

Remember it. "One element, one gap; two elements, choose your gap."

Try a variant. Classify the following as elemental or compound and state the group combination: Ge, GaN, CdTe, SiC. *Answer:* Ge, elemental (group IV); GaN, compound III–V; CdTe, compound II–VI; SiC, compound IV–IV.

Q3. Explain why silicon is widely used in semiconductor device fabrication.

> **Question type:** Explanation · **Topic:** properties of silicon · **Full notes:** Week 1 (Introduction to Semiconductor Materials)

What the examiner wants. A list of distinct reasons, each explained in one or two sentences with a supporting fact or number (abundance, native oxide, band gap, crystal growth, doping, mechanical and thermal properties). Six or seven well-explained reasons, plus one limitation, make a full-mark answer.

Key terms.

- **Native oxide:** an oxide that grows directly on the material by heating it in oxygen or steam; for silicon it is silicon dioxide (SiO_2).
- **Band gap (E_g):** the energy needed to free an electron from a bond; 1.12 eV for Si at 300 K.
- **Intrinsic carrier concentration (n_i):** the number of electrons (equal to holes) per m^3 created by heat in pure material; 1×10^{16} m^{-3} for Si at 300 K.
- **Leakage current:** the small unwanted current through a reverse-biased junction, caused by thermally generated carriers.
- **Czochralski (CZ) method:** growth of a large single crystal by slowly pulling a seed crystal from molten silicon.
- **Interface trap density (D_{it}):** the number of unwanted electron states at the Si/SiO_2 boundary per m^2 per eV.
- **Dopant:** an impurity added on purpose to control conductivity, e.g. boron (B) or phosphorus (P).

Model answer.

Silicon is used for more than 90 % of all semiconductor devices because it combines the following advantages.

1. Abundant and cheap. Silicon is about 28 % of the Earth's crust, in the form of SiO_2 (sand, quartz). The raw material costs little and the refining industry is enormous, so wafers are inexpensive.

2. An excellent native oxide. Heating silicon in oxygen grows SiO_2, which is:

- an excellent insulator ($E_g \approx 9$ eV, breakdown field $\approx 10^9$ V/m, i.e. 1 GV/m);
- chemically stable and insoluble in water;
- joined to silicon by a very clean interface ($D_{it} \approx 10^{14}$ $m^{-2}eV^{-1}$).

It is used as the MOSFET gate insulator, a mask against dopant diffusion, surface passivation, and isolation between devices. No other semiconductor has an oxide this good; this one property is why MOS technology and VLSI are built on silicon.

3. A well-suited band gap. E_g = 1.12 eV is large enough to keep n_i small (1×10^{16} m^{-3} at 300 K), so leakage is low and devices work up to about 150 °C (germanium, with 0.66 eV, fails above about 75 °C). It is also small enough that doped

silicon conducts well.

4. Mature crystal growth. Very large (300 mm diameter), highly pure (impurities below 1 part per billion) and nearly defect-free single crystals are grown routinely by the Czochralski method.

5. Easy, predictable doping. Boron (p-type) and phosphorus, arsenic and antimony (n-type) can be introduced accurately by diffusion and ion implantation, using SiO_2 as a mask.

6. Good mechanical and thermal properties. Silicon is hard and strong enough to be handled as thin wafers and conducts heat about three times better than GaAs, which helps remove heat from dense ICs and power devices.

7. Elemental and non-toxic. As a single element it has no stoichiometry to control and no toxic constituents (unlike the arsenic in GaAs), which simplifies processing.

Limitation. Silicon has an **indirect** band gap, so it emits light very poorly, and its electron mobility (0.135 $m^2/V{\cdot}s$) is lower than that of GaAs (≈ 0.85). LEDs, lasers and the highest-frequency circuits therefore use compound semiconductors instead.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ abundance and low cost (sand, ≈ 28 % of crust)
- ✓ stable native oxide SiO_2 and its uses (gate dielectric, mask, passivation)
- ✓ band gap 1.12 eV: low leakage, operation to ≈ 150 °C (compared with Ge)
- ✓ large, pure single crystals by the CZ method
- ✓ controllable doping by diffusion and ion implantation
- ✓ mechanical strength and thermal conductivity
- ✓ one limitation (indirect gap, poor light emitter)

Common mistakes.

- Writing "silicon has the highest mobility". It does not; Ge and GaAs are higher. Silicon wins on oxide, cost and maturity, not speed.
- Listing reasons without explaining them ("good oxide" alone earns little; say what the oxide is used for).
- Forgetting the oxide altogether. It is the single most important reason and examiners expect it.

Remember it. "Sand, oxide, gap": cheap sand, a perfect oxide, and a gap that is just right.

Try a variant. Explain why gallium arsenide, not silicon, is used to make laser diodes. *Answer:* GaAs has a direct band gap of 1.42 eV, so an electron dropping from the conduction band to the valence band can give its energy out as a photon without needing a phonon; emission is efficient, at $\lambda = 1.24/1.42 = 0.87$ μm (infrared). Silicon's gap is indirect, so recombination needs a phonon as well and the energy is lost mainly as heat.

Q4. Compare the properties and applications of silicon (Si), germanium (Ge), and gallium arsenide (GaAs).

> **Question type:** Compare · **Topic:** Si, Ge and GaAs · **Full notes:** Week 1 (Introduction to Semiconductor Materials)

What the examiner wants. A side-by-side comparison of the three materials under the same headings (type, structure, band gap and its nature, mobility, thermal behaviour, oxide, cost), a short explanation of what each difference means for

devices, and the main applications of each. A comparison table is expected.

Key terms.

- **Band gap (E_g):** the energy (eV) between the top of the valence band and the bottom of the conduction band.
- **Direct / indirect band gap:** direct means an electron can recombine and emit a photon without changing momentum (efficient light emission); indirect needs a lattice vibration (phonon) as well (poor emission).
- **Electron mobility (μ_n):** drift velocity per unit electric field, $m^2/V{\cdot}s$; higher mobility means faster devices.
- **Intrinsic carrier concentration (n_i):** thermally generated electrons per m^3 in pure material; smaller for wider gaps.
- **Diamond cubic / zinc blende:** crystal structures in which every atom has four tetrahedral neighbours; in zinc blende the neighbours are of the other element.
- **HEMT / MESFET:** high-speed field-effect transistors made in GaAs for microwave circuits.
- **SiGe HBT:** a fast bipolar transistor with a silicon–germanium base.

Model answer.

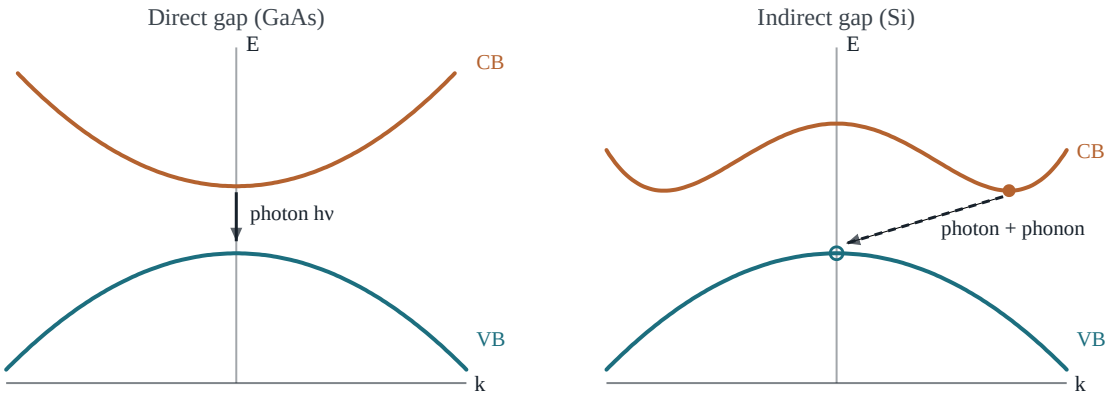
Comparison of properties (300 K).

Property	Si	Ge	GaAs
Type	Elemental (group IV)	Elemental (group IV)	Compound (III–V)
Crystal structure	Diamond cubic	Diamond cubic	Zinc blende
Lattice constant	0.543 nm	0.565 nm	0.565 nm
Bonding	Covalent	Covalent	Covalent, partly ionic
Band gap	1.12 eV	0.66 eV	1.42 eV
Nature of gap	Indirect	Indirect	**Direct**
Electron mobility ($m^2/V{\cdot}s$)	0.135	0.39	0.85
Max. operating temperature	≈ 150 °C	≈ 75 °C	High (wide gap)
Native oxide	Excellent (SiO_2)	Poor, water-soluble	None useful
Light emission	Very poor	Very poor	Efficient
Cost / maturity	Lowest, most mature	Higher	Highest; brittle, toxic As

What the differences mean.

- **Band gap and temperature.** The order is Ge (0.66) < Si (1.12) < GaAs (1.42 eV). A wider gap gives a smaller n_i, so lower leakage current and a higher maximum working temperature. Germanium leaks most and fails first when hot; this is why silicon replaced it.
- **Direct versus indirect.** Only GaAs is direct, so only GaAs emits light efficiently; the emitted wavelength is $\lambda = 1.24/E_g = 1.24/1.42 = 0.87$ μm (near infrared).
- **Mobility and speed.** GaAs electrons are about six times faster than silicon's, and Ge about three times, so GaAs (and Ge) suit very high-frequency work. GaAs hole mobility is poor, however, which makes GaAs CMOS impractical.
- **Oxide.** Only silicon has a stable, high-quality native oxide; this is why MOS transistors and very large-scale integration (VLSI) are silicon technologies.

- **Cost and processing.** Si is the cheapest and most mature; Ge is costlier; GaAs is the most expensive, brittle, and contains toxic arsenic.



Energy–momentum (E–k) diagrams: in GaAs the conduction-band minimum lies directly above the valence-band maximum (direct gap); in Si they are at different momentum (indirect gap).

Sketch in the exam: two small E–k curves, one with the band edges aligned (GaAs, label "photon") and one offset (Si, label "phonon needed").

Applications.

- **Silicon:** microprocessors and memories, all digital and analogue ICs, MOSFETs and BJTs, power devices (IGBTs, thyristors), solar cells, sensors and MEMS.
- **Germanium:** infrared detectors and lenses, SiGe high-speed transistors, substrates for multi-junction space solar cells; historically the first transistors (1947).
- **Gallium arsenide:** microwave and RF amplifiers in mobile phones, radar and satellite links (HEMTs, MESFETs), LEDs and laser diodes for optical-fibre links, high-efficiency solar cells for satellites.

Conclusion. Silicon wins on cost, oxide and thermal stability, so it dominates integrated circuits; germanium survives where its high mobility or narrow gap (infrared) helps; GaAs is chosen where speed or light emission matters more than cost.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ band gaps: Si 1.12, Ge 0.66, GaAs 1.42 eV
- ✓ direct (GaAs) versus indirect (Si, Ge) and its consequence for light emission
- ✓ electron mobilities in the right order (GaAs > Ge > Si) with values
- ✓ crystal structures: diamond cubic (Si, Ge), zinc blende (GaAs)
- ✓ thermal stability / leakage linked to band gap
- ✓ silicon's native oxide as its key advantage
- ✓ cost and processing comparison
- ✓ at least two applications for each material

Common mistakes.

- Mixing up the band gaps (e.g. giving Ge a larger gap than Si). Learn the order Ge < Si < GaAs.
- Saying GaAs is better than Si "in every way". It has no useful oxide, poor hole mobility and is costly.
- Writing only a table with no explanation. *Compare* asks what the differences mean, not just what they are.

- Forgetting applications: half of the question is about them.

Remember it. "Ge leaks, Si thinks, GaAs shines": germanium has the narrowest gap, silicon runs the computers, GaAs makes the light.

Try a variant. Which of Si, Ge and GaAs would you choose for (a) a photodetector for 1.55 μm fibre-optic light, (b) a red/infrared laser, (c) a microprocessor? Justify. *Answer:* The photon energy at 1.55 μm is E = 1.24/1.55 = 0.80 eV. (a) **Ge**: its gap (0.66 eV) is below 0.80 eV so it absorbs the light; Si (1.12 eV) is transparent to it. (b) **GaAs**: direct gap, emits at 0.87 μm. (c) **Si**: excellent oxide for MOSFETs, cheap, mature, low leakage.

Q5. What is a crystal lattice? Explain the significance of crystal structure in semiconductor materials.

> **Question type:** Definition and explanation · **Topic:** crystal structure · **Full notes:** Week 1 (Introduction to Semiconductor Materials)

What the examiner wants. A precise definition of a crystal lattice (with unit cell and basis), the crystal structures of Si and GaAs as illustrations, and then several distinct reasons why the arrangement of atoms matters for semiconductor behaviour and processing, each explained in a sentence or two.

Key terms.

- **Crystal lattice:** an infinite, periodic three-dimensional array of points in which every point has identical surroundings.
- **Basis:** the atom or group of atoms attached to each lattice point; crystal = lattice + basis.
- **Unit cell:** the smallest block which, repeated by translation in three directions, rebuilds the whole crystal.
- **Lattice constant (a):** the edge length of a cubic unit cell; Si 0.543 nm (1 nm = 10^{-9} m).
- **Single crystal:** a solid with the same lattice and orientation throughout; *polycrystalline* material is made of many small crystals (grains), *amorphous* has no long-range order.
- **Diamond cubic structure:** two interpenetrating face-centred cubic (FCC) lattices offset by a quarter of the body diagonal; each atom has four tetrahedral neighbours (the structure of Si and Ge).
- **Mobility (μ):** drift velocity per unit electric field ($m^2/V{\cdot}s$); it falls when carriers are scattered.
- **Epitaxy:** growth of a thin single-crystal layer that copies the crystal arrangement of the substrate.

Model answer.

Crystal lattice. A crystal lattice is a regular, infinitely repeating three-dimensional array of points in space, arranged so that every point has exactly the same surroundings. The smallest repeating block, the **unit cell**, is described by its edge lengths a, b, c and angles α, β, γ; repeating it in three directions rebuilds the whole crystal. Placing a **basis** (one or more atoms) on each lattice point gives the real crystal:

$$\text{crystal structure} = \text{lattice} + \text{basis}$$

Where:

- lattice = the geometric array of repeating points
- basis = the group of atoms attached to every lattice point

Examples. Silicon has an FCC lattice with a two-atom basis, giving the **diamond-cubic** structure: 8 atoms per cubic cell, each bonded covalently to 4 neighbours at 109.5°, a = 0.543 nm, atom density $8/a^3 = 5.0\times10^{28}$ m^{-3}. GaAs has the same arrangement, but the two basis atoms are Ga and As, giving the **zinc-blende** structure.

Sketch in the exam: a cube labelled with edge a, showing the corner and face-centre atoms (FCC), and note that the diamond structure adds four atoms inside the cube.

Significance of crystal structure.

1. It creates the energy bands. The periodic arrangement of atoms produces a periodic potential, which spreads the sharp atomic energy levels into allowed bands separated by a forbidden gap. The size of the gap and whether it is direct or indirect depend on the structure and atomic spacing.

2. It sets carrier mobility. In a perfectly periodic lattice electrons move without being scattered. Any break in periodicity (lattice vibrations, impurities, defects, grain boundaries) scatters them. Single crystals therefore have much higher mobility than polycrystalline or amorphous material, which is why devices are made on single-crystal wafers.

3. Properties depend on direction (anisotropy). Surface atom density, oxidation rate, etch rate and the density of interface traps all depend on which crystal plane is exposed. MOS chips use (100) wafers because they give the lowest Si/SiO_2 interface-trap density; KOH etches (100) much faster than (111), giving V-grooves used in MEMS.

4. Doping and diffusion. A dopant becomes electrically active only when it occupies a substitutional lattice site. The open diamond structure (only 34 % of space filled by atoms) also allows atoms to diffuse through interstitial spaces.

5. Epitaxy and heterostructures. A defect-free layer can be grown only on a substrate with the same structure and almost the same lattice constant (e.g. AlGaAs on GaAs, both ≈ 0.565 nm).

6. Defects control lifetime and leakage. Departures from the ideal structure (vacancies, dislocations, grain boundaries) introduce energy states in the gap that trap carriers and act as recombination centres, shortening carrier lifetime and increasing junction leakage.

7. Mechanical behaviour. Cleavage planes (e.g. {111} in Si, {110} in GaAs), hardness and how wafers are sawn and diced follow from the structure and bond directions.

Conclusion. The crystal structure is not just geometry: it fixes the band structure, how fast carriers move, how the material can be doped, grown and processed, and how reliable the final device is.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definition: periodic 3-D array of points, identical surroundings
- ✓ unit cell and lattice constant mentioned
- ✓ crystal = lattice + basis
- ✓ example: Si diamond cubic (two FCC lattices, 4 neighbours, a = 0.543 nm) and/or GaAs zinc blende
- ✓ periodic potential gives energy bands and the band gap
- ✓ single crystal gives high mobility; defects and grain boundaries scatter carriers
- ✓ anisotropy and wafer orientation ((100) for MOS)
- ✓ at least two further points (doping sites, epitaxy/lattice matching, defects, cleavage)

Common mistakes.

- Defining a lattice as "a group of atoms". A lattice is an array of *points*; the atoms are the basis placed on the points.
- Confusing "lattice" with "unit cell". The unit cell is the smallest repeating block; the lattice is the whole infinite array.

- Giving only one vague reason ("structure affects properties"). Name specific effects: bands, mobility, anisotropy, epitaxy, defects.
- Saying silicon is "simple cubic" or "FCC". It is diamond cubic: an FCC lattice with a two-atom basis.

Remember it. "Order makes bands, breaks in order make trouble."

Try a variant. Calculate the number of atoms per m^3 in silicon, given that its diamond-cubic unit cell (a = 0.543 nm) contains 8 atoms. *Answer:* Atoms per cell = 8 corners × 1/8 + 6 faces × 1/2 + 4 inside = 8. Volume of cell = a^3 = $(5.43\times10^{-10}\ m)^3 = 1.60\times10^{-28}\ m^3$. Atom density = 8 / 1.60×10^{-28} = **5.0×10^{28} atoms/m^3**.

Q6. Define Miller indices and explain how they are used to identify crystal planes.

> **Question type:** Definition and explanation · **Topic:** Miller indices · **Full notes:** Week 1 (Introduction to Semiconductor Materials)

What the examiner wants. A precise definition (smallest integers proportional to the reciprocals of the intercepts), the step-by-step procedure for finding them, the rules for zero and negative indices, the bracket notation, and a worked example. Stating why they matter in semiconductor processing completes the answer.

Key terms.

- **Crystal plane:** a flat sheet passing through lattice points of a crystal.
- **Intercept:** the distance from the origin at which a plane cuts a crystal axis, measured in units of the lattice constants a, b, c.
- **Reciprocal:** one divided by a number; the reciprocal of 2 is 1/2, and of ∞ is 0.
- **Lattice constant (a, b, c):** the edge lengths of the unit cell; in a cubic crystal a = b = c.
- **Lowest common multiple (LCM):** the smallest number that all the denominators divide into exactly.
- **Interplanar spacing (d_{hkl}):** the perpendicular distance between neighbouring parallel planes of the same (h k l) set.
- **Wafer orientation:** the crystal plane parallel to the wafer surface, e.g. a (100) wafer.

Model answer.

Definition. Miller indices are a set of three smallest whole numbers (h k l), proportional to the **reciprocals of the intercepts** a plane makes on the x, y and z crystal axes (measured in units of the lattice constants a, b, c). They specify the orientation of that plane and of every plane parallel to it.

Procedure for finding them.

1. Choose an origin at a lattice point that does not lie on the plane, and find where the plane cuts the x, y and z axes, in units of a, b and c. Call these intercepts p, q and r.
2. Take the reciprocals 1/p, 1/q, 1/r.
3. Multiply all three by the LCM of the denominators to clear the fractions.
4. Reduce to the smallest set of integers with the same ratio and write them in round brackets with no commas: (h k l).

In symbols:

(h k l) = smallest integers in the ratio 1/p : 1/q : 1/r

Where:

- h, k, l = Miller indices for the x, y, z axes (integers)
- p, q, r = intercepts on the x, y, z axes, in units of a, b, c

Rules.

- A plane **parallel** to an axis meets it at infinity; 1/∞ = 0, so that index is **zero**. For example, a plane cutting x at a and parallel to y and z is (1 0 0).
- A **negative** intercept gives a negative index, written with a bar over the number, e.g. ($\bar{1}$ 1 0).
- A large index means the plane cuts that axis close to the origin.

Worked example. Intercepts a, 2b, 3c → p, q, r = 1, 2, 3 → reciprocals 1, 1/2, 1/3 → multiply by 6 → **(6 3 2)**.

Notation.

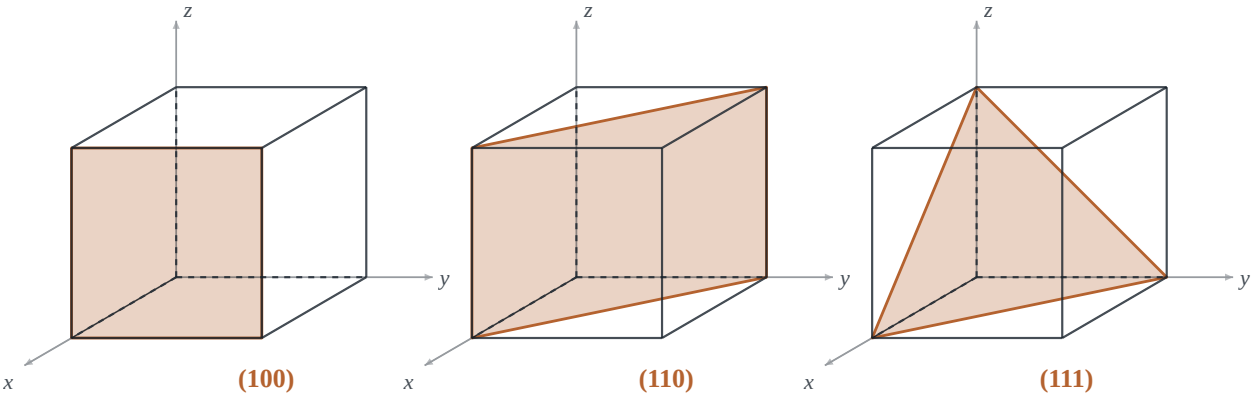
Notation	Meaning
(h k l)	One particular plane (and all planes parallel to it)
{h k l}	A family of planes equivalent by symmetry, e.g. {100} = (100), (010), (001) and their negatives
[u v w]	A direction in the crystal
⟨u v w⟩	A family of equivalent directions

In a cubic crystal the direction [h k l] is perpendicular to the plane (h k l), and the spacing of the planes is

$$d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$$

Where:

- d_{hkl} = spacing between neighbouring (h k l) planes (nm)
- a = lattice constant of the cubic cell (nm); Si 0.543 nm
- h, k, l = the Miller indices



The (100), (110) and (111) planes of a cubic unit cell, the three planes most used in semiconductor work.

Sketch in the exam: a cube with x, y, z axes and the (100), (110) and (111) planes shaded, each labelled.

How they are used in semiconductors.

- **Wafer specification:** wafers are sold as (100) or (111). MOS ICs use (100) because it has the lowest Si/SiO_2 interface-trap density; bipolar and some power devices use (111).

- **Processing:** oxidation rate, etch rate and surface atom density depend on the plane; KOH etches (100) much faster than (111), forming 54.7° V-grooves.
- **Cleavage and dicing:** Si cleaves along {111}, GaAs along {110}.
- **Ion implantation:** wafers are tilted about 7° away from low-index directions to avoid channelling.
- **X-ray diffraction:** each diffraction peak is labelled by the (h k l) planes that produce it, which is how crystal orientation is checked.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definition: smallest integers proportional to the reciprocals of the intercepts
- ✓ intercepts measured in units of the lattice constants
- ✓ four-step procedure (intercepts, reciprocals, clear fractions, reduce)
- ✓ zero index for a plane parallel to an axis; bar for a negative index
- ✓ notation (h k l), {h k l}, [u v w], ⟨u v w⟩
- ✓ worked example with the correct answer
- ✓ $d_{hkl} = a/\sqrt{(h^2 + k^2 + l^2)}$ or [h k l] ⊥ (h k l) in cubic crystals
- ✓ at least one use (wafer orientation (100)/(111), etching, cleavage, XRD)

Common mistakes.

- Using the intercepts directly instead of their reciprocals: intercepts 1, 2, 3 do **not** give (1 2 3).
- Writing a plane parallel to an axis as "∞" in the indices. The index is 0.
- Mixing brackets: (100) is a plane, [100] is a direction, {100} is a family of planes.
- Leaving a common factor, e.g. writing (2 2 0) instead of (1 1 0).

Remember it. "Intercept, invert, clear, reduce": flip the intercepts, clear the fractions, bracket the smallest whole numbers.

Try a variant. Find the spacing of the (111) planes in silicon (a = 0.543 nm). *Answer:* $d_{111} = a/\sqrt{(1^2 + 1^2 + 1^2)} = 0.543/\sqrt{3}$ = **0.314 nm** (0.3135 nm). For comparison d_{100} = 0.543 nm.

Q7. Determine the Miller indices of a crystal plane having intercepts of 2a, 3b, and 1c.

> **Question type:** Calculation · **Topic:** Miller indices · **Full notes:** Week 1 (Introduction to Semiconductor Materials)

What the examiner wants. The standard procedure laid out step by step: write the intercepts in lattice units, take reciprocals, clear fractions with the lowest common multiple, check no common factor remains, and state the answer in round brackets. A sketch of the plane is a bonus.

Key terms.

- **Miller indices (h k l):** the smallest integers proportional to the reciprocals of a plane's intercepts on the crystal axes; they name the plane's orientation.
- **Intercept:** where the plane cuts an axis, in units of the lattice constants a, b, c.
- **Lattice constants (a, b, c):** the edge lengths of the unit cell along x, y and z.
- **Reciprocal:** 1 divided by the number.

- **Lowest common multiple (LCM):** the smallest number into which all the denominators divide exactly.
- **Interplanar spacing (d_{hkl}):** perpendicular distance between neighbouring parallel planes; for a cubic crystal $d_{hkl} = a/\sqrt{(h^2 + k^2 + l^2)}$.

Model answer.

Given: intercepts x = 2a, y = 3b, z = 1c, i.e. p = 2, q = 3, r = 1 in lattice units.

Find: the Miller indices (h k l).

Formula:

(h k l) = smallest integers in the ratio 1/p : 1/q : 1/r

Where:

- h, k, l = Miller indices for the x, y, z axes
- p, q, r = intercepts on the x, y, z axes in units of a, b, c

Solution.

Step 1. Write the intercepts in lattice units: 2, 3, 1.

Step 2. Take reciprocals: 1/2, 1/3, 1/1.

Step 3. The denominators are 2, 3 and 1; their LCM is 6. Multiply each reciprocal by 6: 6 × 1/2 = 3, 6 × 1/3 = 2, 6 × 1 = 6.

Step 4. Check for a common factor: 3, 2 and 6 share none (the highest common factor is 1), so no reduction is needed.

Step	x	y	z
Intercepts (units of a, b, c)	2	3	1
Reciprocals	1/2	1/3	1
× LCM (6)	3	2	6

Answer: Miller indices = (3 2 6), read "three two six".

Check: going back, 1/3 : 1/2 : 1/6 = 2 : 3 : 1 when multiplied by 6, which reproduces the original intercepts.

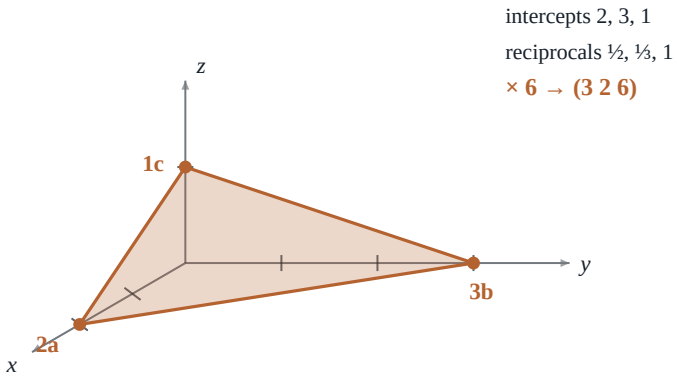
Spacing (if the crystal is cubic):

$d_{326} = a / \sqrt{(3^2 + 2^2 + 6^2)} = a / \sqrt{49} = a/7$

Where:

- d_{326} = spacing between neighbouring (3 2 6) planes
- a = lattice constant of the cubic cell

What this means: the plane cuts the z-axis closest to the origin (at 1c), so z receives the largest index (6); the y intercept is farthest (3b), so y receives the smallest (2). Every plane parallel to this one carries the same name, (3 2 6). Such a high-index plane is steeply tilted and its planes are closely spaced (a/7, against a for (100)), unlike the low-index (100) and (111) planes used for wafers.



The (3 2 6) plane, which cuts the x-axis at 2a, the y-axis at 3b and the z-axis at 1c.

Sketch in the exam: x, y, z axes with marks at 2a on x, 3b on y and 1c on z, joined to form a shaded triangle, labelled (3 2 6).

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ intercepts written in lattice units: 2, 3, 1
- ✓ reciprocals: 1/2, 1/3, 1
- ✓ LCM 6 used to clear fractions
- ✓ check that no common factor remains
- ✓ answer (3 2 6) in round brackets
- ✓ neat step table or numbered steps
- ✓ optional: sketch of the plane, or d = a/7 for a cubic crystal

Common mistakes.

- Writing the intercepts as the answer, (2 3 1). Miller indices come from the **reciprocals**.
- Reciprocating correctly but then inverting the order, giving (2 3 6) or (6 2 3). Keep the order x, y, z throughout.
- Multiplying by 2 or 3 only, leaving a fraction (e.g. 1, 2/3, 2). Use the LCM of all denominators.
- Using square brackets [3 2 6]: that is a direction, not a plane.

Remember it. "Small intercept, big index": the axis the plane hits nearest the origin gets the largest number.

Try a variant. A plane has intercepts 3a, 2b and is parallel to the z-axis. Find its Miller indices. *Answer:* Intercepts 3, 2, ∞ → reciprocals 1/3, 1/2, 0 → × 6 → **(2 3 0)**. The zero shows the plane is parallel to z.

Q8. Explain the difference between covalent, ionic, and metallic bonding and identify the dominant bonding mechanism in silicon.

> **Question type:** Explain and identify · **Topic:** bonding in solids · **Full notes:** Week 1 (Introduction to Semiconductor Materials)

What the examiner wants. "Explain the difference" means you describe each bond by how the valence electrons are held, then state the consequences (directional or not, free electrons or not, electrical behaviour) and give an example of

each. "Identify" needs a clear one-word answer for silicon (covalent), backed by a reason: four valence electrons, four bonds, tetrahedral lattice.

Key terms.

- **Valence electrons:** the electrons in an atom's outermost shell; they are the ones that form bonds. Silicon has four.
- **Octet (full outer shell):** eight electrons in the outer shell, the stable arrangement that atoms reach by bonding.
- **Electronegativity:** how strongly an atom attracts electrons towards itself. A large difference between two atoms favours ionic bonding.
- **Ion:** an atom that has lost electrons (positive ion, e.g. Na^+) or gained them (negative ion, e.g. Cl^-).
- **Directional bond:** a bond that points along fixed directions, so it fixes the angles between neighbouring atoms.
- **Delocalised electrons:** electrons not tied to any one atom, free to move through the whole crystal.
- **Electron–hole pair:** a free electron plus the empty bond (hole) it leaves behind when a covalent bond breaks.

Model answer.

1. Covalent bonding. Neighbouring atoms **share** pairs of valence electrons, one electron from each atom, so that each atom completes its outer shell (eight electrons). The shared pair sits between the two atoms, so the bond is strong and **directional**: it fixes the crystal geometry. At low temperature the electrons are locked in the bonds, but a modest amount of energy can free some of them, so covalent solids are **semiconductors or insulators**. They are hard with high melting points. Examples: Si, Ge, diamond (C).

2. Ionic bonding. One atom **transfers** one or more electrons to another atom of very different electronegativity. This creates a positive ion and a negative ion, which are held together by electrostatic (Coulomb) attraction. The attraction acts equally in all directions, so the bond is **non-directional**. The electrons are tightly held by the ions, so there are no free carriers: ionic solids are **insulators** with a large band gap, hard and brittle, and conduct only when molten or dissolved. Example: Na^+Cl^-.

3. Metallic bonding. Each atom releases its valence electrons into a common **"sea" of delocalised electrons** shared by the whole crystal. The sea holds the lattice of positive ion cores together. The bond is **non-directional**, and the free electron sea gives very high electrical and thermal **conductivity**, lustre and ductility. Examples: Cu, Al, Na.

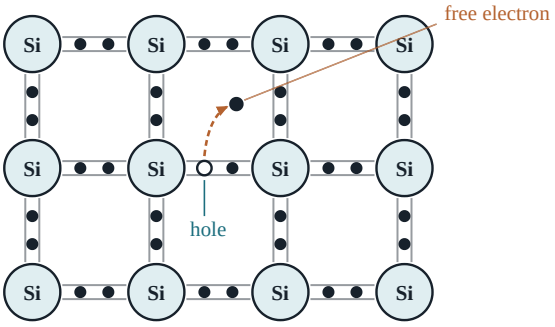
4. The key differences can be summed up in a table:

	Covalent	**Ionic**	**Metallic**
What happens to valence electrons	Shared between pairs of neighbours	Transferred from one atom to another	Shared by all atoms (electron sea)
What holds the solid together	Shared electron pairs	Attraction of + and − ions	Attraction of ion cores to the electron sea
Directional?	Yes	No	No
Free electrons	Few (made thermally)	None	Very many
Electrical behaviour	Semiconductor or insulator	Insulator	Conductor
Example	Si, Ge, C	NaCl, MgO	Cu, Al

5. Dominant bonding in silicon: covalent.

- Silicon (atomic number 14, shells 2, 8, 4) has **four valence electrons**.

- In the crystal (diamond-cubic lattice) each Si atom sits at the centre of a tetrahedron and forms **four covalent bonds** with four nearest neighbours, at **109.5°** to each other (sp^3 bonds). It shares one electron pair with each neighbour, so every atom is surrounded by eight shared electrons, a full octet.
- All the atoms are identical, so there is no electronegativity difference and **no ionic character**; and the electrons stay in the bonds, so there is no electron sea.
- The bonds are only moderately strong. At 0 K every electron is in a bond and pure Si is an insulator. At room temperature thermal energy breaks a few bonds (the energy needed to free an electron is the band gap, E_g = 1.12 eV), creating **electron–hole pairs**. This is exactly what makes silicon a semiconductor.



Silicon drawn flat: each atom shares one electron pair with each of its four neighbours. One bond has been broken by heat, leaving a free electron and a hole.

Sketch in the exam: a 3 × 3 grid of Si atoms, each joined to four neighbours by double lines (two shared electrons per bond), with one broken bond showing a free electron and a hole. Label "Si", "shared electron pair", "free electron", "hole".

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ covalent = sharing of electron pairs; directional; example (Si, Ge, diamond)
- ✓ ionic = electron transfer, Coulomb attraction between + and − ions; non-directional; example (NaCl)
- ✓ metallic = delocalised electron sea binding positive ion cores; non-directional; example (Cu)
- ✓ consequence for electrical behaviour of each (semiconductor/insulator, insulator, conductor)
- ✓ silicon identified as covalent
- ✓ reason: four valence electrons, four bonds to four neighbours, full octet
- ✓ tetrahedral (diamond-cubic) arrangement, 109.5°
- ✓ link to semiconduction: thermal breaking of bonds creates electron–hole pairs (E_g = 1.12 eV)

Common mistakes.

- Saying silicon has "free electrons like a metal". It does not: at 0 K every electron is in a bond; free electrons appear only when bonds are broken.
- Describing ionic bonding as "sharing". Ionic is *transfer*; covalent is *sharing between two atoms*; metallic is *sharing by all atoms*. Keep the three verbs distinct.
- Forgetting to give examples and the "directional / non-directional" property: both are easy marks.
- Writing that silicon is "partly ionic" like GaAs. Si–Si bonds join identical atoms, so silicon is purely covalent.

Remember it. Covalent **shares** with a neighbour, ionic **gives** to a neighbour, metallic **gives to everyone**; silicon shares, four bonds each.

Try a variant. Identify the bonding in gallium arsenide and explain how it differs from that in silicon. *Answer:* GaAs is **mainly covalent with partial ionic character**. Ga (group III, 3 valence electrons) and As (group V, 5 valence electrons) together supply an average of four electrons per atom, so each atom still forms four tetrahedral bonds, as in Si. But As attracts electrons more strongly than Ga (higher electronegativity), so the shared pairs sit closer to As, giving As a small negative and Ga a small positive charge. Silicon, with identical atoms, has no such charge transfer and is purely covalent.

Q9. Describe the major stages involved in converting metallurgical-grade silicon (MG-Si) into electronic-grade silicon (EG-Si).

> **Question type:** Describe · **Topic:** silicon purification · **Full notes:** Week 2 (Material Growth Techniques)

What the examiner wants. A sequence of stages in the right order, each with its chemical reaction, approximate temperature and purpose. A complete answer starts from MG-Si (briefly saying how it is made), goes through hydrochlorination, distillation and the Siemens process, and states the purity reached.

Key terms.

- **Metallurgical-grade silicon (MG-Si):** silicon about 98 % pure, made by reducing quartz with carbon in an arc furnace; too impure for devices.
- **Electronic-grade silicon (EG-Si):** polycrystalline silicon of 99.9999999 % ("9N") purity or better, i.e. at most about one impurity atom per billion silicon atoms.
- **Trichlorosilane ($SiHCl_3$):** a silicon compound that is liquid at room temperature (boils at 32 °C), used as the carrier that can be purified.
- **Fractional distillation:** separating liquids by repeatedly boiling and condensing them, using their different boiling points.
- **Chemical vapour deposition (CVD):** growing a solid from gases that react on a hot surface.
- **Siemens process:** the CVD step in which purified $SiHCl_3$ and hydrogen deposit pure silicon on heated silicon rods.
- **Polycrystalline silicon (polysilicon):** solid silicon made of many small crystal grains pointing in different directions.

Model answer.

The idea. A solid cannot easily be purified to parts per billion, but a liquid can, by distillation. So silicon is turned into a liquid compound, purified, and then turned back into solid silicon.

$$\text{MG-Si} \rightarrow SiHCl_3 \text{ (liquid)} \rightarrow \text{distilled } SiHCl_3 \rightarrow \text{EG-Si polysilicon}$$

Starting material: MG-Si. Quartzite (SiO_2) is reduced with carbon (coke, coal, wood chips) in an electric arc furnace at about 2000 °C:

$$SiO_2 + 2C \rightarrow Si + 2CO$$

The product is about 98 % pure; the main impurities are Fe, Al, Ca, C, B and P. Since silicon has 5×10^{28} atoms per m^3, 2 % impurity is of order 10^{27} impurity atoms per m^3, millions of times too many for devices.

Stage 1: Hydrochlorination (making a liquid). MG-Si is ground to a powder and reacted with dry hydrogen chloride gas at about 300 °C (usually in a fluidised bed with a catalyst):

$$Si + 3HCl \rightarrow SiHCl_3 + H_2$$

Silicon leaves as trichlorosilane, a liquid that boils at 32 °C. The impurities form their own chlorides ($FeCl_3$, BCl_3, PCl_3 …), which boil at different temperatures.

Stage 2: Purification by fractional distillation. The crude $SiHCl_3$ is distilled repeatedly. Because each impurity chloride boils at a different temperature from $SiHCl_3$, it is separated off. The electrically active impurities (especially B and P) are reduced to a few parts per billion.

Stage 3: Reduction and deposition (Siemens process). The purified $SiHCl_3$ is mixed with high-purity hydrogen and passed over thin silicon rods heated electrically to about 1100 °C in a CVD reactor:

$$SiHCl_3 + H_2 \rightarrow Si + 3HCl$$

Where (for all the reactions above):

- SiO_2 = silicon dioxide (quartzite); C = carbon; CO = carbon monoxide
- HCl = hydrogen chloride; H_2 = hydrogen
- $SiHCl_3$ = trichlorosilane

Silicon deposits on the hot rods, which grow over many hours into thick bars of **electronic-grade polysilicon**, 9N to 11N pure (an impurity level of order 10^{19} m^{-3} or less). The HCl and unreacted gases are collected and recycled.

Stage 4 (to make wafers): single-crystal growth. EG-Si is pure but polycrystalline and undoped. It is broken into chunks, melted with the required dopant and grown into a single-crystal ingot by the Czochralski or float-zone method, then sliced and polished into wafers.

Summary table.

Stage	Reaction	Temperature	Result
Arc furnace	$SiO_2 + 2C \rightarrow Si + 2CO$	≈ 2000 °C	MG-Si, 98 %
1. Hydrochlorination	$Si + 3HCl \rightarrow SiHCl_3 + H_2$	≈ 300 °C	Liquid $SiHCl_3$
2. Fractional distillation	Physical separation	$SiHCl_3$ boils at 32 °C	ppb-pure $SiHCl_3$
3. Siemens CVD	$SiHCl_3 + H_2 \rightarrow Si + 3HCl$	≈ 1100 °C	EG-Si polysilicon, 9N+

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ MG-Si origin: $SiO_2 + 2C \rightarrow Si + 2CO$, ≈ 2000 °C, ≈ 98 % pure
- ✓ hydrochlorination: $Si + 3HCl \rightarrow SiHCl_3 + H_2$ at ≈ 300 °C
- ✓ reason for the liquid: $SiHCl_3$ boils at 32 °C, can be distilled
- ✓ fractional distillation separates impurity chlorides (different boiling points)
- ✓ Siemens process: $SiHCl_3 + H_2 \rightarrow Si + 3HCl$ on heated Si rods at ≈ 1100 °C
- ✓ product: polycrystalline EG-Si, 9N (99.9999999 %) or better
- ✓ stages in the correct order
- ✓ (bonus) HCl recycled; EG-Si then grown into a single crystal by CZ/FZ

Common mistakes.

- Writing that the Siemens process produces a single crystal. It produces **polycrystalline** EG-Si; the single crystal comes later from CZ or FZ growth.

- Leaving out the reason for making $SiHCl_3$. Say it: a liquid can be purified by distillation, a solid cannot.
- Getting the reaction direction mixed up: HCl is a *reactant* in stage 1 and a *product* in stage 3.
- Listing stages without temperatures or reactions: these carry most of the marks.

Remember it. "Furnace, Fizz, Fractionate, Freeze-out": arc furnace → HCl reaction → distillation → Siemens deposition.

Try a variant. Why is fractional distillation of $SiHCl_3$ such an effective purification step, and how pure must EG-Si be compared with MG-Si? *Answer:* Each impurity (Fe, B, P, Al …) forms its own chloride with a boiling point different from that of $SiHCl_3$ (32 °C), so repeated boiling and condensing leaves those impurities behind. MG-Si contains about 2 % impurity (≈ 10^{27} m^{-3} in 5×10^{28} Si atoms/m^3); EG-Si at 9N contains at most one impurity per 10^9 atoms, $5\times10^{28} \times 10^{-9}$ = **5×10^{19} m^{-3}**, about **2×10^7 times purer**.

Q10. Explain the principle and operation of the Czochralski (CZ) crystal growth method.

> **Question type:** Explain · **Topic:** Czochralski crystal growth · **Full notes:** Week 2 (Material Growth Techniques)

What the examiner wants. Two parts. The **principle**: why pulling a seed from a melt gives one single crystal with the seed's orientation. The **operation**: the equipment and the growth steps in order (load, melt, dip, neck, shoulder, body, tail, cool), with typical values. A labelled sketch of the puller earns marks.

Key terms.

- **Single crystal:** a solid whose atoms form one regular pattern with the same orientation throughout.
- **Seed crystal:** a small, perfect single crystal cut to the wanted orientation (e.g. ⟨100⟩) that starts the growth.
- **Melt:** the molten silicon in the crucible, held just above silicon's melting point of 1414 °C.
- **Ingot (boule):** the large cylindrical single crystal that is pulled; later sliced into wafers.
- **Dislocation:** a line defect, a row of atoms out of place; it causes leakage and must be removed.
- **Dash necking:** pulling a thin neck (≈ 3 mm wide) quickly at the start so that dislocations grow out of the sides.
- **Dopant:** an impurity added on purpose (boron for p-type; phosphorus, arsenic or antimony for n-type).

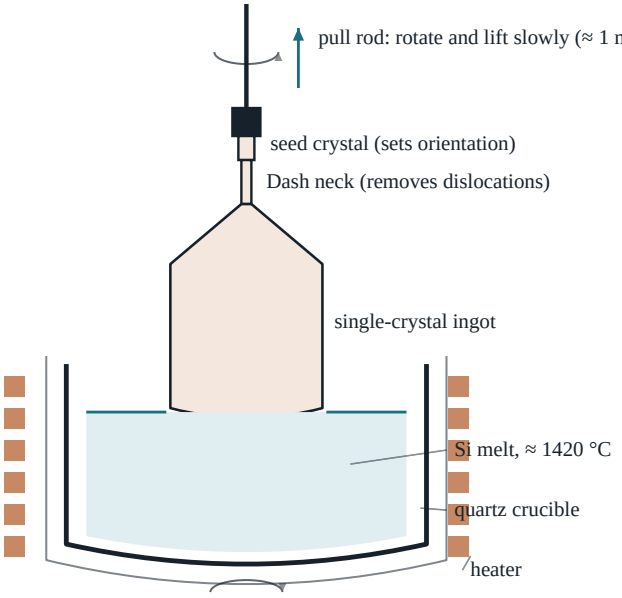
Model answer.

1. Principle. CZ growth is crystal **pulling from the melt**. A seed crystal is dipped into molten silicon kept just above its melting point (1414 °C) and slowly withdrawn. Heat flows up through the seed, so the liquid touching it cools and freezes. As each atom freezes, it takes up the position that continues the lattice of the solid beneath it. The growing solid therefore copies the seed, and the result is one large single crystal with **the same orientation as the seed**. The diameter is controlled by the pull rate and melt temperature: pulling faster or hotter gives a thinner crystal, slower or cooler a thicker one.

2. Equipment.

- **Furnace:** a fused-quartz (SiO_2) crucible holding the melt, supported in a graphite susceptor, heated by RF induction or resistance heaters.
- **Crystal-pulling mechanism:** a seed holder on a shaft that rotates and rises slowly; the crucible also rotates (opposite direction) and can be raised.
- **Ambient control:** argon gas at reduced pressure, so hot silicon does not react with air and evaporated SiO is swept away.

- **Control system:** sensors and feedback that hold the temperature and diameter constant.



A Czochralski puller: the rotating seed is lifted slowly from the melt in a quartz crucible, which rotates the other way; the frozen crystal copies the seed's orientation.

Sketch in the exam: crucible with melt, heater coils on both sides, the ingot above the melt with the seed and pull rod at the top, curved arrows showing opposite rotations. Label "quartz crucible", "graphite susceptor", "Si melt (1414 °C)", "heater", "seed", "neck", "ingot", "pull rod", "argon in/out".

3. Operation (steps in order).

1. **Load:** chunks of EG-Si polysilicon and a carefully weighed amount of dopant go into the crucible.
2. **Melt:** the chamber is purged with argon and the charge is melted (≈ 1420 °C).
3. **Dip:** the seed is lowered until it just touches the melt surface, and the temperature is adjusted so it neither melts away nor grows quickly.
4. **Neck (Dash technique):** the seed is pulled up fast to grow a thin neck about 3 mm across. Dislocations caused by the thermal shock of dipping run out of the sides, so the crystal below is dislocation-free.
5. **Shoulder:** the pull rate and temperature are lowered so the crystal widens to the target diameter (up to 300 mm).
6. **Body:** the crystal is grown at constant diameter, pulled at about **1 mm/min**, while seed and crucible **counter-rotate** (10–30 rpm) to stir the melt and keep temperature and doping even.
7. **Tail:** the pull rate is raised so the diameter narrows to a cone before the crystal leaves the melt; a sudden break would send dislocations back up the crystal.
8. **Cool:** the ingot is lifted clear and cooled slowly to avoid thermal stress.

4. After growth. The ingot is checked for orientation, resistivity and defects, ground to an exact diameter, given a flat or notch to mark its orientation, sliced into wafers, lapped, etched and polished.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ principle: seed dipped into melt and slowly pulled; melt freezes onto seed and copies its orientation
- ✓ melting point 1414 °C; melt held just above it
- ✓ equipment: quartz crucible, graphite susceptor, heater, rotating pull rod, argon ambient

- ✓ labelled sketch of the puller
- ✓ Dash necking explained (removes dislocations)
- ✓ shoulder, constant-diameter body, tail in order
- ✓ counter-rotation of seed and crucible, and why (stirring, uniformity)
- ✓ typical values: pull rate ≈ 1 mm/min, diameter up to 300 mm
- ✓ diameter control by pull rate and temperature

Common mistakes.

- Describing only the steps and forgetting the **principle** (why the crystal copies the seed). The question asks for both.
- Leaving out Dash necking, or saying it is "to make the crystal thinner". Its purpose is to **remove dislocations**.
- Saying seed and crucible rotate the same way: they rotate in **opposite** directions.
- Mixing up CZ with float-zone growth: CZ uses a crucible; FZ has none.

Remember it. "Dip, Neck, Shoulder, Body, Tail": the crystal copies the seed it is pulled from.

Try a variant. A 300 mm CZ ingot body 1.5 m long is pulled at 1.2 mm/min. How long does the body take to grow, and what is its mass (density of Si 2330 kg/m^3)? *Answer:* t = L/v = 1.5 m ÷ 1.2×10^{-3} m/min = 1250 min ≈ **20.8 h**. m = density × πr^{2}L = 2330 × π × 0.15^2 × 1.5 = **247 kg**, so the crucible must be charged with well over 250 kg of silicon.

Q11. State four advantages and two limitations of the Czochralski method.

> **Question type:** State · **Topic:** Czochralski crystal growth · **Full notes:** Week 2 (Material Growth Techniques)

What the examiner wants. "State" means a short, clear list: exactly four advantages and two limitations, each in one or two sentences with a reason or number. No long description of the process is needed.

Key terms.

- **Czochralski (CZ) method:** growing a single crystal by slowly pulling a rotating seed from molten silicon held in a quartz crucible.
- **Float-zone (FZ) method:** the crucible-free alternative, in which a molten zone is moved along a polysilicon rod.
- **Segregation coefficient (k_0):** the ratio of dopant concentration in the freezing solid to that in the liquid, $k_0 = C_s/C_l$; for most dopants $k_0 < 1$ (B 0.8, P 0.35).
- **Resistivity (ρ):** how strongly the material opposes current (Ω·m); set mainly by the doping.
- **Intrinsic gettering:** trapping of metal impurities by oxygen precipitates deep in the wafer, away from the devices at the surface.
- **Dislocation:** a line defect in the crystal that causes leakage.

Model answer.

Advantages.

1. **Large diameters.** Ingots of 200 mm and 300 mm diameter are routine, so many chips fit on each wafer and the cost per chip is low.
2. **Dislocation-free crystals of chosen orientation.** The ingot copies the seed, so (100) or (111) material is obtained at will, and Dash necking removes dislocations.

3. **Controlled doping.** Dopant added to the melt sets the conductivity type and resistivity of the crystal.
4. **High throughput and low cost.** The process is mature and automated, and fast enough that about 95 % of all silicon wafers are CZ-grown.

(A fifth acceptable advantage: the **oxygen** picked up from the crucible strengthens the wafer against warping and slip and allows **intrinsic gettering**.)

Limitations.

1. **Contamination from the crucible.** The hot quartz crucible slowly dissolves, adding oxygen ($\approx 10^{24}$ m^{-3}), and graphite parts add carbon. Oxygen forms thermal donors and precipitates and limits the resistivity to about 1 Ω·m or less, so CZ silicon is unsuitable for high-voltage power devices and detectors (FZ is used instead).
2. **Non-uniform doping along the ingot (segregation).** Because $k_0 < 1$, the crystal rejects dopant into the melt, which becomes steadily richer. The doping therefore rises from the seed end to the tail end:

   $$C_s = k_0 \, C_0 \, (1 - f)^{k_0 - 1}$$

Where:

- C_s = dopant concentration in the crystal at the point being grown (m^{-3})
- k_0 = segregation coefficient (dimensionless)
- C_0 = dopant concentration in the starting melt (m^{-3})
- f = fraction of the melt already frozen (0 at the seed end)

So resistivity varies along the ingot; melt convection also causes radial variations (growth striations).

(Also acceptable: the crucible is used up each run; high energy use; batch process.)

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ exactly four advantages listed, each with a reason
- ✓ large diameter (200–300 mm) → low cost per chip
- ✓ dislocation-free single crystal with the seed's orientation
- ✓ controlled doping / resistivity
- ✓ high throughput, mature, low cost (or oxygen strengthening and gettering)
- ✓ limitation: oxygen ($\approx 10^{24}$ m^{-3}) and carbon contamination from crucible; limits resistivity
- ✓ limitation: axial doping non-uniformity from segregation, $k_0 < 1$

Common mistakes.

- Giving more or fewer items than asked, or mixing advantages and limitations in one list. Use two numbered lists.
- Listing "oxygen" only as a limitation without noticing it is also an advantage (strength, gettering). Either is fine if you explain it; do not contradict yourself.
- Vague points such as "good quality". Say what is good: dislocation-free, chosen orientation, large diameter.
- Writing that doping *falls* along the ingot. For $k_0 < 1$ it **rises** towards the tail.

Remember it. CZ is **Big, Perfect, Doped and Cheap**, but **Oxygen-laden and Uneven**.

Try a variant. An arsenic-doped CZ ingot ($k_0 = 0.3$) is grown. By what factor is the arsenic concentration at the half-way point (f = 0.5) higher than at the seed end? *Answer:* $C_s(f)/C_s(0) = (1 - f)^{k_0 - 1} = (0.5)^{-0.7} =$ **1.62**, so the doping is 62 % higher and the resistivity about 38 % lower half-way along: a clear example of limitation 2.

Q12. Explain the principles of vapour-phase epitaxy (VPE), liquid-phase epitaxy (LPE), and molecular-beam epitaxy (MBE).

> **Question type:** Explain · **Topic:** epitaxial growth · **Full notes:** Week 2 (Material Growth Techniques)

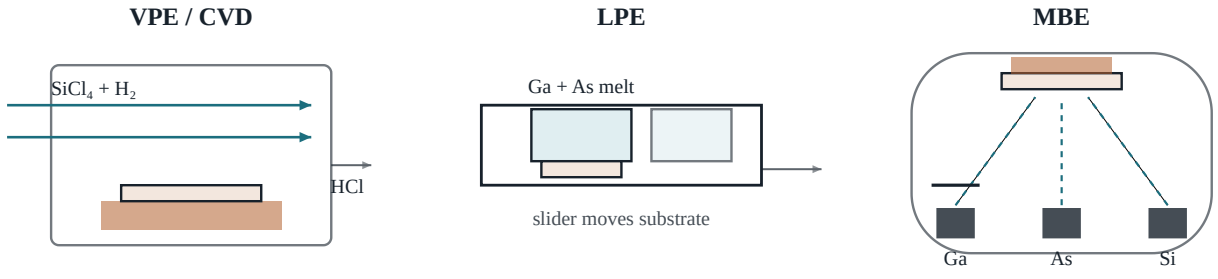
What the examiner wants. First define epitaxy. Then, for each of the three methods, explain where the atoms come from and why they settle in the crystal, with an example reaction or material, the typical temperature and pressure, and how doping is done. A labelled sketch of each set-up earns marks.

Key terms.

- **Epitaxy:** growing a thin single-crystal layer on a single-crystal wafer so that the layer continues the wafer's crystal structure and orientation.
- **Substrate:** the wafer on which the layer is grown.
- **Homoepitaxy / heteroepitaxy:** growing the same material (Si on Si) / a different material (AlGaAs on GaAs).
- **Supersaturated solution:** a liquid holding more dissolved material than it can keep at that temperature, so the excess comes out as solid.
- **Ultra-high vacuum (UHV):** pressure of about 10^{-10} Torr, so low that atoms travel in straight lines without colliding.
- **Effusion cell:** a small heated oven that evaporates one element to make an atomic beam (MBE).
- **RHEED (reflection high-energy electron diffraction):** an electron beam bounced off the growing surface; its pattern shows the surface order and counts atomic layers.
- **MOCVD:** metal-organic chemical vapour deposition, the version of VPE used for III–V compounds.

Model answer.

Epitaxy. Epitaxy (Greek *epi*, "upon", *taxis*, "arrangement") is the growth of a thin single-crystal layer on a single-crystal substrate so that the layer takes the substrate's crystal structure and orientation. It is used to make layers with different doping or a different material from the substrate, with sharp boundaries. The three methods differ in the **phase from which the atoms arrive**: gas, liquid or atomic beam.



The three epitaxy methods: reacting gases on a hot wafer (VPE), a cooling saturated melt on the wafer (LPE), and atomic beams in ultra-high vacuum (MBE).

Vapour-phase epitaxy (VPE)

Principle: growth from a gas by chemical reaction at a hot surface. A gas containing the semiconductor, carried by hydrogen, flows over a heated substrate in a reactor. At the hot surface the gas decomposes or is reduced, leaving

semiconductor atoms behind. These atoms move over the surface until they reach lattice sites, so the layer continues the substrate crystal. For silicon:

$SiCl_4 + 2H_2 \rightarrow Si + 4HCl$ (≈ 1150–1250 °C)

$SiH_4 \rightarrow Si + 2H_2$ (≈ 600–1000 °C)

Where:

- $SiCl_4$ = silicon tetrachloride; SiH_4 = silane (silicon sources)
- H_2 = hydrogen (carrier and reducing gas); HCl = hydrogen chloride (by-product)

Doping: a little dopant gas is added: diborane (B_2H_6) for p-type, phosphine (PH_3) or arsine (AsH_3) for n-type. **Rate control:** at high temperature the rate is limited by how fast gas reaches the surface (mass-transport limited); at lower temperature by the surface reaction (reaction-rate limited). Pressure is atmospheric or reduced. For III–V compounds the metal-organic form, MOCVD, is used, e.g. $(CH_3)_3Ga + AsH_3 \rightarrow GaAs + 3CH_4$ at 600–750 °C.

Liquid-phase epitaxy (LPE)

Principle: growth by precipitation from a cooling saturated solution. The semiconductor is dissolved in a molten metal solvent until the solution is saturated, e.g. GaAs in liquid gallium at about 700–900 °C. The substrate is brought into contact with the solution and the whole is cooled slowly (≈ 0.1–1 °C/min). As the temperature falls the solution can hold less, becomes **supersaturated**, and the excess semiconductor crystallises onto the substrate, continuing its lattice. Growth takes place **close to thermodynamic equilibrium**, so the crystal is of high quality. A **sliding graphite boat** moves the substrate under a series of melts of different composition or doping to build up multilayers. Dopants are added to the melt.

Molecular-beam epitaxy (MBE)

Principle: physical deposition from atomic beams in ultra-high vacuum. In a chamber at about 10^{-10} Torr, each element (Ga, As, Al, dopants such as Si or Be) is heated in its own **effusion cell**. Evaporated atoms travel as collision-free **beams** in straight lines to a heated substrate (≈ 500–650 °C for GaAs). They adsorb, migrate on the surface and are incorporated into lattice sites. Each cell has a **shutter** that opens or closes in a fraction of a second, so composition and doping can be switched abruptly. Growth is slow, about **one atomic layer per second** (≈ 1 µm/h), and is watched in real time by **RHEED**, so thickness is controlled to a single atomic layer. Growth is **far from equilibrium** and controlled by the beam fluxes.

In one sentence each. VPE: gas reacts on a hot wafer. LPE: a saturated melt is cooled so solid comes out onto the wafer. MBE: beams of atoms in vacuum stick to the wafer one layer at a time.

Sketch in the exam: (a) VPE: horizontal tube, gas inlet ($SiCl_4 + H_2$), wafers on a heated susceptor, exhaust (HCl). (b) LPE: graphite boat with wells of melt, sliding slider carrying the substrate, furnace. (c) MBE: vacuum chamber, heated substrate facing several effusion cells with shutters, RHEED gun and screen.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definition of epitaxy (layer continues substrate structure and orientation)
- ✓ VPE: gas-phase source, reaction on hot surface; $SiCl_4 + 2H_2 \rightarrow Si + 4HCl$ at ≈ 1200 °C
- ✓ VPE doping gases (B_2H_6, PH_3, AsH_3); mention of MOCVD for III–V
- ✓ LPE: saturated solution in molten metal, cooled → supersaturation → precipitation on substrate
- ✓ LPE example: GaAs in Ga, 700–900 °C, sliding graphite boat
- ✓ MBE: UHV ≈ 10^{-10} Torr, effusion cells, collision-free beams, shutters

- ✓ MBE: ≈ 1 monolayer/s, atomic-layer control, RHEED monitoring
- ✓ labelled sketches (any of the three)

Common mistakes.

- Explaining MBE as a chemical reaction. MBE is **physical** deposition of elemental atoms; VPE is the chemical method.
- Forgetting *why* LPE grows: cooling makes the solution supersaturated. "The wafer is dipped in a melt" alone does not explain the principle.
- Not defining epitaxy, or confusing it with simply depositing any film. An epitaxial layer must be **single-crystal and aligned** with the substrate.
- Quoting the MBE vacuum as "low vacuum". It is **ultra-high** vacuum, which is what keeps the beams collision-free.

Remember it. Vapour **R**eacts, **L**iquid **P**recipitates, **M**olecular **B**eams **B**uild layer by layer.

Try a variant. Why must MBE be carried out in ultra-high vacuum? *Answer:* Two reasons. (1) The atoms must reach the wafer as beams without colliding with gas molecules: the mean free path of a gas molecule is roughly $\lambda \approx 5\times10^{-5}/P$ m (P in Torr), so at 10^{-10} Torr $\lambda \approx 5\times10^{5}$ m (about 500 km), far longer than the ≈ 0.2 m cell-to-wafer distance. (2) At one atomic layer per second the surface is easily contaminated, so the background gas must be extremely low to keep the layer pure. UHV also allows electron-beam monitoring (RHEED) during growth.

Q13. Compare VPE, LPE, and MBE in terms of growth mechanism, temperature, precision, and applications.

> **Question type:** Compare · **Topic:** epitaxial growth · **Full notes:** Week 2 (Material Growth Techniques)

What the examiner wants. A point-by-point comparison under the four headings named (growth mechanism, temperature, precision, applications), with numbers where possible, ending with a comparison table and a one-line conclusion on when each method is chosen. Extra rows (growth rate, cost, purity) add strength but the four named headings must be there.

Key terms.

- **Epitaxy:** growth of a thin single-crystal layer that continues the crystal structure and orientation of the substrate below it.
- **VPE (vapour-phase epitaxy):** epitaxy from gases that react on a hot wafer; MOCVD is its metal-organic form for III–V compounds.
- **LPE (liquid-phase epitaxy):** epitaxy by precipitation from a saturated solution in molten metal as it cools.
- **MBE (molecular-beam epitaxy):** epitaxy from beams of atoms in ultra-high vacuum.
- **Monolayer:** one atomic layer of the crystal, about 0.28 nm thick for GaAs.
- **Interface abruptness:** how sharply the composition or doping changes from one layer to the next.
- **Autodoping / out-diffusion:** unwanted transfer of dopant from the substrate into the growing layer, by evaporation and re-deposition / by diffusion at high temperature.
- **Quantum well / heterostructure:** a very thin layer (a few nm) of one semiconductor sandwiched between another; it needs atomically sharp interfaces.

Model answer.

1. Growth mechanism.

- **VPE:** chemical. Gas-phase compounds ($SiCl_4$, SiH_4, or metal-organics in MOCVD) react or decompose at the hot substrate surface and leave atoms behind, e.g. $SiCl_4 + 2H_2 \rightarrow Si + 4HCl$. Rate is limited by mass transport or surface reaction.
- **LPE:** precipitation from a liquid. A saturated solution (GaAs in molten Ga) is cooled, becomes supersaturated, and deposits the semiconductor on the substrate. Growth is close to equilibrium.
- **MBE:** physical. Atoms evaporated from effusion cells travel as beams in ultra-high vacuum ($\approx 10^{-10}$ Torr), adsorb and migrate on the substrate. Growth is far from equilibrium, set by the beam fluxes.

2. Temperature.

- **VPE:** highest. Silicon 1000–1250 °C ($SiCl_4 \approx 1200$ °C); silane can work lower; MOCVD 600–750 °C for GaAs.
- **LPE:** intermediate, about 700–900 °C for GaAs.
- **MBE:** lowest, about 500–650 °C for GaAs. Low temperature means little diffusion, which keeps interfaces sharp.

3. Precision (thickness control and interfaces).

- **VPE:** good, about tens of nm, with good uniformity over many wafers; interfaces moderately sharp, blurred by autodoping and out-diffusion at high temperature. Growth rate 0.1–5 μm/min.
- **LPE:** poorest, about 0.1 μm; interfaces graded and surfaces rough, though crystal quality and purity are very high. Growth rate 0.1–1 μm/min.
- **MBE:** best, a single atomic layer (≈ 0.28 nm for GaAs); atomically abrupt interfaces controlled by shutters and monitored by RHEED. Growth rate slow, ≈ 1 μm/h (≈ 1 monolayer/s).

4. Applications.

- **VPE:** silicon epitaxial layers for bipolar transistors, CMOS and power devices; in MOCVD form, mass production of LEDs, lasers, solar cells and HEMTs (GaAs, InP, GaN).
- **LPE:** LEDs, early heterostructure lasers, photodetectors, thick high-quality III–V layers.
- **MBE:** quantum wells, superlattices, high-electron-mobility transistors (HEMTs), heterojunction bipolar transistors (HBTs), quantum-cascade and VCSEL lasers, and research.

Comparison table.

Feature	VPE	LPE	MBE
Source phase	Gas ($SiCl_4$, SiH_4, metal-organics)	Liquid solution in molten metal	Atomic/molecular beams from solid sources
Growth mechanism	Chemical reaction at hot surface	Precipitation from supersaturated solution on cooling; near equilibrium	Physical deposition, adsorption and surface migration; far from equilibrium
Pressure	Atmospheric or reduced	Atmospheric (H_2 or inert)	Ultra-high vacuum ≈ 10^{-10} Torr
Temperature	Highest: Si 1000–1250 °C (MOCVD 600–750 °C)	Medium: 700–900 °C	Lowest: 500–650 °C
Growth rate	0.1–5 µm/min	0.1–1 µm/min	≈ 1 µm/h (≈ 1 monolayer/s)
Thickness precision	Good, tens of nm	Poor, ≈ 0.1 µm	Excellent, one atomic layer
Interfaces	Moderately sharp	Graded	Atomically abrupt
In-situ monitoring	Limited	None	RHEED
Throughput and cost	High throughput, moderate cost	Low throughput, low cost	Low throughput, very high cost
Applications	Si epi for BJTs, CMOS, power; MOCVD LEDs, lasers, solar cells	LEDs, early lasers, thick layers, photodetectors	Quantum wells, HEMTs, HBTs, advanced lasers, research

Conclusion. VPE is the industrial workhorse for volume production; LPE is simple and cheap and gives high-quality thick layers but poor control; MBE gives atomic-scale control for advanced heterostructure devices at high cost and low throughput.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ mechanism for each: chemical reaction (VPE), precipitation on cooling (LPE), physical beam deposition in UHV (MBE)
- ✓ temperature ranking with values: VPE highest (Si ≈ 1000–1250 °C), LPE 700–900 °C, MBE lowest 500–650 °C
- ✓ precision ranking: MBE (one atomic layer) > VPE (tens of nm) > LPE (≈ 0.1 µm)
- ✓ interface abruptness linked to temperature / shutters
- ✓ growth rates with units
- ✓ at least two applications per method
- ✓ comparison table covering the four named headings
- ✓ conclusion on when each method is chosen

Common mistakes.

- Writing three separate descriptions with no comparison. Compare *under each heading*, then give the table.
- Ranking temperatures wrongly. MBE is the **lowest**, not the highest, despite its sophisticated equipment.
- Confusing growth rate units: VPE and LPE are in µm per **minute**, MBE in µm per **hour**, a factor of about 60–300 slower.

- Calling LPE "imprecise and low quality". Its thickness control is poor, but its **crystal quality is very high** (near-equilibrium growth).

Remember it. Hotter, faster, coarser → VPE; cooler, slower, finer → MBE; LPE sits in the middle on temperature but last on precision.

Try a variant. A 10 nm GaAs quantum well is to be grown. Which method would you choose, and how long does the well take at 1 monolayer per second (monolayer thickness a/2, a = 0.5653 nm)? *Answer:* **MBE**, because only it gives single-atomic-layer control and abrupt interfaces at this thickness. Monolayer = 0.5653/2 = 0.283 nm, so 10 nm needs 10/0.283 ≈ 35 monolayers, taking **≈ 35 s** (the growth rate is 0.283 nm/s ≈ 1.0 μm/h). LPE (± 0.1 μm = 100 nm) could not even define the layer.

Questions 14–21 — Impurity Control and Doping

Module 1, part 2: why and how semiconductors are doped: donors and acceptors, diffusion and Fick's laws, the diffusion profile, and ion implantation. Each question stands on its own: its key terms are defined with it, so you can start anywhere.

Symbols and abbreviations used on this page

Every symbol, constant and abbreviation on this page is listed here, even if another page already defined it, so you can study this page on its own.

Constants and units

Symbol	Meaning	Units / typical value
q	Magnitude of the electronic charge	1.6×10^{-19} C
k	Boltzmann constant	8.617×10^{-5} eV/K = 1.381×10^{-23} J/K
kT	Thermal energy	0.0259 eV (often 0.02585 eV) at 300 K
T	Absolute temperature	kelvin (K); room temperature = 300 K
h	Planck constant	6.626×10^{-34} J·s
eV	Electron-volt, a unit of energy	1 eV = 1.6×10^{-19} J
µm	Micrometre (micron)	1 µm = 10^{-6} m

Symbols

Symbol	Meaning	Units / typical value
E_c (E_C)	Energy of the bottom of the conduction band	eV
E_v (E_V)	Energy of the top of the valence band	eV
E_F	Fermi level: the energy at which a state has a 50 % chance of being occupied	eV
E_i	Intrinsic Fermi level, close to mid-gap	eV
E_D	Donor energy level, just below E_c	eV; $E_c - E_D \approx 0.045$ eV for P in Si
E_A	Acceptor energy level, just above E_v	eV; $E_A - E_v \approx 0.045$ eV for B in Si
E_a	Activation energy of dopant diffusion	eV; about 3–4 eV
n	Electron concentration (free electrons per m^3)	m^{-3}
p	Hole concentration (holes per m^3)	m^{-3}
n_i	Intrinsic carrier concentration ($n = p = n_i$ in pure material)	m^{-3}; Si $\approx 1\times10^{16}$ at 300 K
N_D	Donor concentration	m^{-3}
N_A	Acceptor concentration	m^{-3}
N_D^+, N_A^-	Concentrations of ionised donors and acceptors	m^{-3}
C_s	Surface concentration of a diffusion (or solid concentration when freezing)	m^{-3}
C(x, t)	Dopant concentration at depth x and time t	m^{-3}
D	Diffusion coefficient (dopants in a furnace, or carriers)	m^2/s
D_0	Diffusion constant in $D = D_0 e^{-E_a/kT}$	m^2/s
F	Particle flux in Fick's law	particles/(m^2·s)
Q	Dose: dopant atoms per unit area (in Weeks 13–14, Q also means charge)	atoms/m^2 (C or C/m^2 for charge)
erfc	Complementary error function, erfc(u) = 1 − erf(u); read from a table	dimensionless
x_j	Junction depth: where the dopant concentration equals the background	µm
N_B (C_B)	Background doping of the wafer (in Week 13, N_B is the base doping)	m^{-3}
R_p	Projected range: average depth at which implanted ions stop	µm
ΔR_p	Straggle: spread of the implanted ions' stopping depths	µm
N_p	Peak concentration of an implanted profile	m^{-3}
µ, μ_n, μ_p	Mobility: drift velocity per unit field (electrons, holes)	$m^2/(V\cdot s)$; Si μ_n 0.135, μ_p 0.048
σ	Conductivity, $q(n\mu_n + p\mu_p)$ (in Week 8, σ is also a capture cross-section)	S/m (siemens per metre)
ρ	Resistivity, 1/σ	Ω·m
R	Recombination rate	$m^{-3}s^{-1}$
V_{bi}	Built-in potential of a junction or Schottky contact	V; typically 0.5–1 V
I	Current	A
A (area)	Device or contact area	m^2

Symbol	Meaning	Units / typical value
G_n, G_p, R_n, R_p	Generation and recombination rates of electrons and holes	$m^{-3}s^{-1}$
erfc(u), $erfc^{-1}$	Complementary error function, 1 − erf(u), and its inverse (read from a table)	dimensionless
t	Time (or a thickness, where the text says so)	s (or m)

Abbreviations

Abbreviation	Meaning
Si	Silicon
GaAs	Gallium arsenide
SiO_2	Silicon dioxide (silica), the oxide insulator
RTA	Rapid thermal annealing
LDD	Lightly doped drain
MOSFET	Metal–oxide–semiconductor field-effect transistor
CMOS	Complementary MOS (n- and p-channel MOSFETs together)
BJT	Bipolar junction transistor
VLSI	Very-large-scale integration
LED	Light-emitting diode
n^+	Very heavily doped n-type (similarly p^+)

Questions on this page

Q	Question
14	What is semiconductor doping? Explain why doping is necessary in semiconductor device fabrication.
15	Differentiate between n-type and p-type semiconductors.
16	Explain the difference between donor impurities and acceptor impurities, giving suitable examples.
17	Describe the mechanism of dopant diffusion in a semiconductor.
18	State and explain Fick's first and second laws of diffusion.
19	A semiconductor is doped by diffusion with a constant surface concentration of 1×10^{20} cm^{-3}. Explain how the impurity concentration varies with depth.
20	Explain the principle of ion implantation and identify its major advantages over thermal diffusion.
21	Compare diffusion and ion implantation as semiconductor doping techniques.

Model answers

Q14. What is semiconductor doping? Explain why doping is necessary in semiconductor device fabrication.

> **Question type:** Definition and explanation · **Topic:** doping · **Full notes:** Week 3 (Impurity Control and Doping)

What the examiner wants. A precise definition of doping (deliberate, controlled, small amounts of chosen impurities) with the two dopant groups and their effect, then a set of distinct reasons why devices cannot be made without it. A number showing how much doping changes the conductivity earns extra credit.

Key terms.

- **Intrinsic semiconductor:** a pure semiconductor whose free electrons and holes come only from breaking bonds by heat, so $n = p = n_i$.
- **Intrinsic carrier concentration (n_i):** the number of free electrons (and holes) per m^3 in pure material; 1×10^{16} m^{-3} for Si at 300 K.
- **Dopant:** an impurity atom added on purpose to change the electrical behaviour of the semiconductor.
- **Donor:** a Group V atom (P, As, Sb) that gives a free electron to silicon, making it n-type.
- **Acceptor:** a Group III atom (B, Al, Ga, In) that takes an electron from a bond, creating a hole and making silicon p-type.
- **Extrinsic semiconductor:** a doped semiconductor whose carrier density is set by the dopants, not by n_i.
- **Resistivity (ρ):** how strongly a material opposes current, in Ω·m; $\rho = 1/\sigma$, where σ is conductivity.

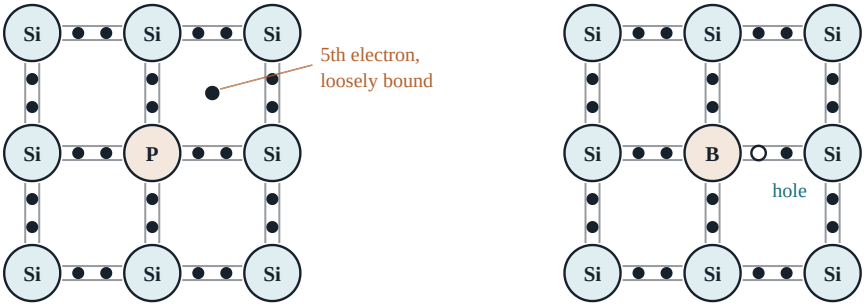
Model answer.

Definition

Doping is the deliberate introduction of small, carefully controlled quantities of selected impurity atoms (dopants) into a pure (intrinsic) semiconductor in order to change its conductivity and to decide whether current is carried mainly by electrons or by holes. Typical doping levels in silicon are 10^{20} to 10^{26} atoms/m^3, i.e. roughly one dopant atom per 5×10^{8} down to one per 500 silicon atoms (Si has 5×10^{28} atoms/m^3).

- **Group V donors** (phosphorus, arsenic, antimony) have five valence electrons; the fifth is loosely bound (≈ 0.045 eV) and becomes a free electron, so the silicon becomes **n-type**.
- **Group III acceptors** (boron, aluminium, gallium, indium) have three valence electrons; the missing bond electron is supplied by a neighbour, leaving a mobile **hole**, so the silicon becomes **p-type**.

Doping is done by **thermal diffusion** or **ion implantation**, using oxide or photoresist masks so that only chosen areas are doped.



n-type: phosphorus donor **p-type: boron acceptor**

A phosphorus atom in the silicon lattice donates its fifth electron (n-type); a boron atom leaves one bond incomplete, creating a hole (p-type).

Why doping is necessary

1. **Pure silicon conducts far too little.** With $n = p = n_i = 1\times10^{16}$ m^{-3}, the intrinsic resistivity is about 3.4×10^{3} Ω·m (worked below). Adding just 10^{22} m^{-3} phosphorus (one atom in 5 million) lowers it to about 4.6×10^{-3} Ω·m, roughly 700 000 times lower.
2. **It sets the carrier type.** Every device needs regions where electrons dominate (n) and regions where holes dominate (p). Without them there is no p-n junction, so no diode, bipolar transistor (BJT), MOSFET, solar cell or LED.
3. **It makes the conductivity stable with temperature.** In pure silicon n_i rises exponentially with temperature, so the current would drift wildly. In doped silicon the carrier density is fixed by N_D or N_A over a wide range (the extrinsic range, about −50 to +150 °C), so circuits behave predictably.
4. **It sets device parameters.** The doping levels fix the built-in voltage and depletion width of a junction, the breakdown voltage, the MOSFET threshold voltage, the BJT current gain, the value of diffused resistors and the quality of Ohmic contacts (heavily doped n^+ and p^+ regions).
5. **It allows local, selective control.** Through mask windows, different regions of one chip receive different doping. This is what makes integrated circuits possible.

Supporting calculation.

$$\rho = 1 / [\, q\,(n\,\mu_n + p\,\mu_p)\,]$$

Where:

- ρ = resistivity (Ω·m)
- q = electronic charge, 1.6×10^{-19} C
- n, p = electron and hole concentrations (m^{-3})
- μ_n, μ_p = electron and hole mobilities, 0.135 and 0.048 m^2/V·s in Si

Intrinsic: $\rho = 1/(1.6\times10^{-19} \times 10^{16} \times (0.135 + 0.048)) =$ **3.4×10^{3} Ω·m**.
Doped with $N_D = 10^{22}$ m^{-3} ($n \approx N_D$, holes negligible): $\rho = 1/(1.6\times10^{-19} \times 10^{22} \times 0.135) =$ **4.6×10^{-3} Ω·m**.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definition: deliberate, controlled addition of small amounts of impurity to an intrinsic semiconductor
- ✓ Group V donors (P, As, Sb) give n-type; Group III acceptors (B, Ga, Al, In) give p-type
- ✓ typical concentration range 10^{20}–10^{26} m^{-3}

- ✓ reason: intrinsic conductivity too low ($n_i = 10^{16}$ m^{-3}), with a number
- ✓ reason: creates n and p regions, hence p-n junctions and all devices
- ✓ reason: temperature-stable carrier density (extrinsic region)
- ✓ reason: sets device parameters (V_{bi}, breakdown, threshold voltage, contacts)
- ✓ reason: selective doping through masks makes integrated circuits possible

Common mistakes.

- Saying doping "adds charge" to the crystal. A doped crystal stays electrically neutral: each free carrier is balanced by a fixed ionised dopant.
- Giving only one reason ("to increase conductivity"). The question says *why necessary*: give at least four distinct reasons.
- Mixing up the groups: boron (Group III) is an acceptor, phosphorus (Group V) a donor.

Remember it. Doping turns silicon from a near-insulator into a material whose type, conductivity and junctions the designer chooses.

Try a variant. Calculate the resistivity of silicon doped with 1×10^{23} m^{-3} boron (take $\mu_p = 0.048$ $m^2/V\cdot s$) and compare it with intrinsic silicon. *Answer:* $p \approx N_A = 10^{23}$ m^{-3}, $\rho = 1/(qN_A\mu_p) = 1/(1.6\times10^{-19} \times 10^{23} \times 0.048) = \mathbf{1.3\times10^{-3}\ \Omega\cdot m}$, about 2.6 million times lower than 3.4×10^{3} Ω·m. (At this doping the real mobility is somewhat lower, so the true ρ is a little higher.)

Q15. Differentiate between n-type and p-type semiconductors.

> **Question type:** Compare / distinguish · **Topic:** n-type and p-type semiconductors · **Full notes:** Week 3 (Impurity Control and Doping)

What the examiner wants. A point-by-point contrast: how each type is made, what the dopant does, which carriers are majority and minority, the fixed ion left behind, the energy level and Fermi-level position, and the carrier-density formulas. End with a comparison table; a band sketch for each type earns marks.

Key terms.

- **n-type semiconductor:** a semiconductor doped mainly with donors, so free electrons outnumber holes.
- **p-type semiconductor:** a semiconductor doped mainly with acceptors, so holes outnumber free electrons.
- **Majority carrier:** the carrier present in larger number (electrons in n-type, holes in p-type).
- **Minority carrier:** the carrier present in smaller number (holes in n-type, electrons in p-type).
- **Hole:** an empty place in a covalent bond (a missing valence electron) that moves through the crystal like a positive charge +q.
- **Fermi level (E_F):** the energy at which a state has a 50 % chance of being occupied; its position shows whether electrons or holes dominate.
- **Mass-action law:** in equilibrium $np = n_i^2$, so raising one carrier lowers the other.

Model answer.

1. n-type semiconductor

An **n-type** semiconductor is made by adding a **pentavalent** (Group V) impurity, such as phosphorus, arsenic or antimony, to silicon. Four of the five valence electrons bond with neighbouring Si atoms; the fifth is bound by only ≈ 0.045 eV and is freed at room temperature. The impurity is a **donor** and is left as an immobile positive ion N_D^+. Electrons are the **majority carriers** and holes the **minority carriers**:

$$n \approx N_D, \quad p = n_i^2 / N_D$$

Where:

- n, p = electron and hole concentrations (m^{-3})
- N_D = donor concentration (m^{-3})
- n_i = intrinsic carrier concentration, 1×10^{16} m^{-3} for Si at 300 K

The donor level E_D lies just below the conduction-band edge E_c, and the Fermi level moves **up** from mid-gap towards E_c.

2. p-type semiconductor

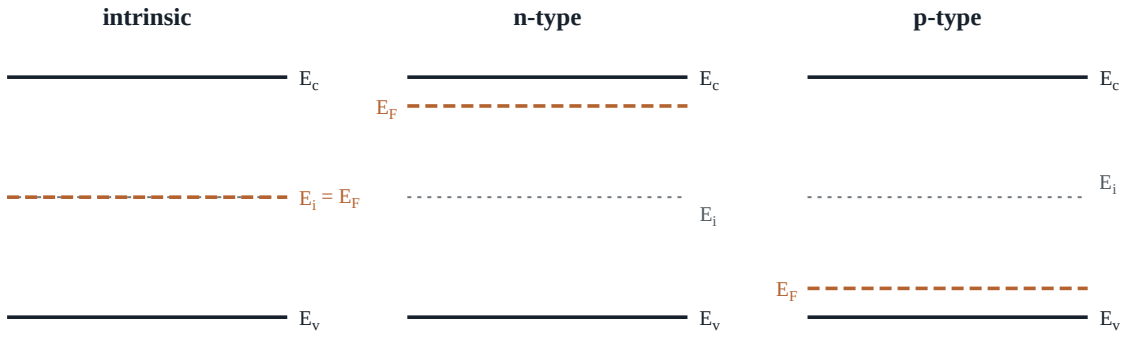
A **p-type** semiconductor is made by adding a **trivalent** (Group III) impurity, such as boron, aluminium, gallium or indium. The impurity completes only three bonds; an electron from a neighbouring bond fills the gap (needing ≈ 0.045 eV for B), leaving a mobile **hole** in the valence band. The impurity is an **acceptor** and becomes an immobile negative ion N_A^-. Holes are the majority carriers and electrons the minority carriers:

$$p \approx N_A, \quad n = n_i^2 / N_A$$

Where:

- N_A = acceptor concentration (m^{-3}); other symbols as above

The acceptor level E_A lies just above the valence-band edge E_v, and the Fermi level moves **down** towards E_v.



$E_F - E_i = kT \ln(N_D/n_i)$ for n-type; $E_i - E_F = kT \ln(N_A/n_i)$ for p-type

The Fermi level sits at mid-gap (E_i) in intrinsic silicon, near the conduction band in n-type and near the valence band in p-type.

Sketch in the exam: two band diagrams side by side. For n-type, draw E_c and E_v, a dashed donor level E_D just below E_c with "+" ions, and E_F above mid-gap. For p-type, draw a dashed acceptor level E_A just above E_v with "−" ions, and E_F below mid-gap.

3. Worked illustration

For Si with $N_D = 10^{22}$ m^{-3}: $n = 10^{22}$ m^{-3} and $p = (10^{16})^2/10^{22} = \mathbf{10^{10}}$ $\mathbf{m^{-3}}$, so electrons outnumber holes by 10^{12}. For p-type with $N_A = 10^{22}$ m^{-3} the numbers are simply swapped.

Both types are **electrically neutral** overall: every free electron is balanced by a fixed N_D^+ ion, every hole by a fixed N_A^- ion.

4. Comparison table

Feature	n-type	p-type
Impurity added	Pentavalent, Group V: P, As, Sb	Trivalent, Group III: B, Al, Ga, In
Name of impurity	Donor	Acceptor
What the impurity does	Gives a free electron to the conduction band	Takes an electron from a bond, creating a hole
Fixed ion left behind	Positive, N_D^+	Negative, N_A^-
Majority carriers	Electrons	Holes
Minority carriers	Holes	Electrons
Carrier densities	$n \approx N_D$, $p = n_i^2/N_D$	$p \approx N_A$, $n = n_i^2/N_A$
Impurity energy level	E_D just below E_c	E_A just above E_v
Fermi level	Above mid-gap, towards E_c	Below mid-gap, towards E_v
Conductivity	$\sigma \approx qN_D\mu_n$	$\sigma \approx qN_A\mu_p$
Hall coefficient sign	Negative	Positive

Because μ_n (0.135) is larger than μ_p (0.048 $m^2/V{\cdot}s$), n-type silicon conducts about 2.8 times better than p-type silicon at the same doping.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ dopant group and examples for each type (P, As, Sb; B, Al, Ga, In)
- ✓ donor gives an electron / acceptor creates a hole, with the mechanism
- ✓ majority and minority carriers correctly named for each type
- ✓ sign of the fixed ion (N_D^+, N_A^-) and overall neutrality
- ✓ $n \approx N_D$, $p = n_i^2/N_D$ and the p-type equivalents
- ✓ positions of E_D, E_A and E_F (band sketch)
- ✓ comparison table

Common mistakes.

- Saying a p-type semiconductor is positively charged. It is neutral; "p" refers to the sign of the majority carrier, not of the material.
- Writing that n-type has "no holes". It has a few (minority) holes, $p = n_i^2/N_D$.
- Placing the donor level near the valence band. Donors sit near E_c, acceptors near E_v.

Remember it. "N for Negative carriers (electrons), P for Positive carriers (holes); the crystal itself stays neutral."

Try a variant. A silicon sample is doped with 5×10^{21} m^{-3} boron. State its type and find the majority and minority carrier concentrations at 300 K. *Answer:* p-type; $p \approx N_A = \mathbf{5\times10^{21}\ m^{-3}}$ (holes, majority), $n = n_i^2/N_A = 10^{32}/(5\times10^{21}) = \mathbf{2\times10^{10}\ m^{-3}}$ (electrons, minority).

Q16. Explain the difference between donor impurities and acceptor impurities, giving suitable examples.

> **Question type:** Explain / distinguish · **Topic:** donors and acceptors · **Full notes:** Week 3 (Impurity Control and Doping)

What the examiner wants. For each kind of impurity: its valence compared with the host, what happens to the extra or missing electron, the charge of the ion left behind, where its energy level sits in the band gap, and examples (named elements, in Si and ideally in GaAs). A band diagram and a summary table complete the answer.

Key terms.

- **Impurity (dopant):** a foreign atom sitting on a lattice site of the semiconductor in place of a host atom (a substitutional impurity).
- **Valence electrons:** the outer electrons an atom uses for bonding; Si has four.
- **Donor:** an impurity with one more valence electron than the host atom it replaces; it donates an electron.
- **Acceptor:** an impurity with one fewer valence electron than the host atom; it accepts an electron and creates a hole.
- **Ionisation energy:** the small energy needed to free the donor's electron ($E_c - E_D$) or to fill the acceptor's empty bond ($E_A - E_v$); about 0.04–0.05 eV in Si.
- **Conduction band (E_c) and valence band (E_v):** the band of free-electron states and the band of bonding-electron states; E_c and E_v are their edges.
- **Ionised impurity:** a dopant that has given up or taken an electron and so carries a fixed charge (N_D^+ or N_A^-).

Model answer.

Donor impurities

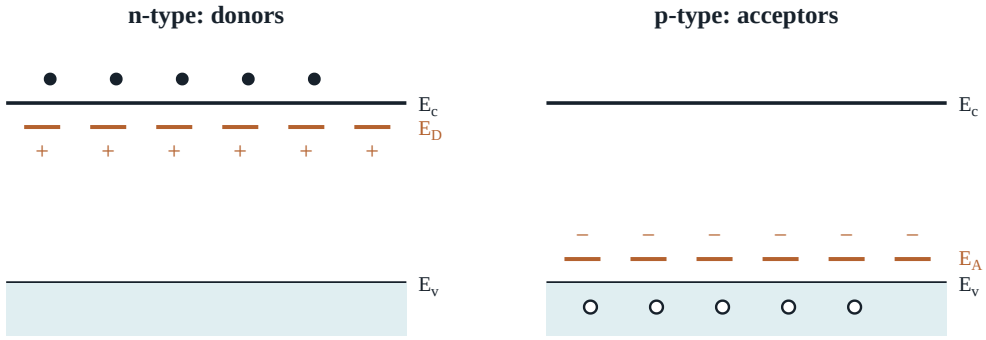
A **donor** has **one more** valence electron than the atom it replaces. In silicon (Group IV), the donors are Group V elements: **phosphorus (P), arsenic (As) and antimony (Sb)**.

- Four of the donor's five electrons form covalent bonds with the four neighbouring Si atoms.
- The fifth electron is held only by the weak attraction of the extra nuclear charge, about **0.045 eV** for P (about 0.04–0.05 eV for As and Sb). Thermal energy at room temperature (kT = 0.0259 eV) frees almost all of them.
- The freed electron enters the conduction band; the donor becomes a **fixed positive ion** (N_D^+).
- In the band diagram the donor gives a localised level **E_D just below E_c**.
- Donors make the material **n-type**.

Acceptor impurities

An **acceptor** has **one fewer** valence electron than the atom it replaces. In silicon the acceptors are Group III elements: **boron (B), aluminium (Al), gallium (Ga) and indium (In)**.

- Only three bonds can be completed; one bond is missing an electron.
- An electron from a neighbouring Si–Si bond moves in with very little energy (about **0.045 eV** for B). The acceptor becomes a **fixed negative ion** (N_A^-) and the empty bond left behind is a mobile **hole** in the valence band.
- In the band diagram the acceptor gives a level **E_A just above E_v**.
- Acceptors make the material **p-type**.



Donor levels just below the conduction band release electrons and become positive ions; acceptor levels just above the valence band take electrons, become negative ions and leave holes.

Sketch in the exam: an energy-band diagram with E_c at the top and E_v at the bottom; a dashed line E_D about 0.045 eV below E_c with "+" signs and arrows taking electrons up into the conduction band; a dashed line E_A about 0.045 eV above E_v with "−" signs and arrows lifting electrons from the valence band, leaving holes (○) behind.

Examples in a compound semiconductor (GaAs)

In GaAs the rule is applied site by site: an atom on an **arsenic (Group V) site** that has one more electron, such as **S, Se or Te (Group VI)**, is a donor; an atom on a **gallium (Group III) site** with one fewer electron, such as **Zn, Be or Mg (Group II)**, is an acceptor. Silicon (Group IV) is a donor on a Ga site and an acceptor on an As site; carbon on an As site is an acceptor.

Summary table

Feature	Donor impurity	Acceptor impurity
Valence electrons (in Si)	5 (one more than Si)	3 (one fewer than Si)
Periodic group (for Si)	Group V	Group III
Action	Gives (donates) an electron	Takes (accepts) an electron, creating a hole
Charge after ionisation	Positive, N_D^+ (fixed)	Negative, N_A^- (fixed)
Energy level	E_D just below E_c	E_A just above E_v
Ionisation energy	≈ 0.045 eV (P)	≈ 0.045 eV (B)
Type of material produced	n-type	p-type
Examples in Si	P, As, Sb	B, Al, Ga, In
Examples in GaAs	S, Se, Te on As site; Si on Ga site	Zn, Be, Mg on Ga site; C on As site

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ donor: one extra valence electron, Group V in Si, with at least two named examples
- ✓ acceptor: one missing valence electron, Group III in Si, with at least two named examples
- ✓ small ionisation energy (≈ 0.045 eV) compared with kT, so nearly all are ionised at 300 K
- ✓ donor becomes a fixed positive ion; acceptor becomes a fixed negative ion and creates a mobile hole
- ✓ E_D just below E_c; E_A just above E_v (labelled band diagram)
- ✓ donors give n-type, acceptors give p-type
- ✓ summary table

Common mistakes.

- Saying the acceptor "creates a positive ion". The acceptor ion is negative; the *hole* it releases behaves as a positive carrier.
- Calling the ion a carrier. The ionised dopant is fixed in the lattice and does not carry current.
- Giving Group VI or Group II elements as dopants for silicon. For Si use Group V (donors) and Group III (acceptors); Group VI and II are the GaAs dopants.

Remember it. "Donors donate (5 electrons, +ion, level near E_c); acceptors accept (3 electrons, −ion, level near E_v)."

Try a variant. Explain why a donor with ionisation energy 0.045 eV is almost fully ionised at 300 K, while a deep impurity with 0.4 eV is not. *Answer:* the chance of ionisation depends on the ratio of the binding energy to kT. For 0.045 eV the ratio is 0.045/0.0259 ≈ 1.7, so thermal energy easily frees the electron and nearly every donor is ionised. For 0.4 eV the ratio is about 15, and $e^{-15} \approx 3\times10^{-7}$, so very few are ionised; such deep levels act as traps and recombination centres rather than dopants.

Q17. Describe the mechanism of dopant diffusion in a semiconductor.

> **Question type:** Describe · **Topic:** dopant diffusion mechanisms · **Full notes:** Week 3 (Impurity Control and Doping)

What the examiner wants. An account of *how* dopant atoms move: the driving force (random thermal jumps plus a concentration gradient), the atomic mechanisms (vacancy, interstitial, interstitialcy), the temperature dependence $D = D_0 e^{-E_a/kT}$, and the macroscopic description by Fick's laws with the two practical stages (predeposition and drive-in). A sketch of the atomic jumps earns marks.

Key terms.

- **Diffusion:** the net movement of atoms from a region of high concentration to one of low concentration, caused by their random thermal motion.
- **Vacancy:** an empty lattice site where an atom is missing.
- **Interstitial site:** a space between normal lattice sites where a small atom can sit.
- **Substitutional impurity:** a dopant that occupies a normal lattice site in place of a silicon atom.
- **Diffusion coefficient (D):** a measure of how fast a dopant spreads, in m^2/s.
- **Activation energy (E_a):** the energy barrier an atom must overcome to make one jump, in eV.
- **Predeposition / drive-in:** the first diffusion step with a constant surface source, and the second step that spreads a fixed dose deeper.

Model answer.

1. What drives diffusion

At furnace temperatures (900–1200 °C for silicon) every atom vibrates about its lattice site and occasionally jumps to a neighbouring position. Each jump is random in direction. Where the dopant is crowded (at the surface) more atoms jump towards the sparse region than come back, so there is a **net flow down the concentration gradient**. No force pushes the atoms; the flow is purely statistical. At room temperature jumps are so rare that dopants do not move.

2. Atomic mechanisms

(a) Vacancy (substitutional) diffusion. The common dopants B, P, As and Sb sit on lattice sites. A dopant can move only when a **neighbouring site is vacant**: it jumps into the vacancy, and the vacancy appears where the atom was. The rate depends on how many vacancies exist (which rises steeply with temperature) and on the jump probability. The activation energy is high, about **3–4 eV**, so D is small: 10^{-19} to 10^{-16} m^2/s at processing temperatures.

(b) Interstitial diffusion. Small atoms, or atoms that do not bond into the lattice (Au, Cu, Fe, Na, Li, H), squeeze through the **open spaces between lattice sites** of the diamond structure without needing a vacancy. The activation energy is low (about 0.5–1.5 eV), so they diffuse many orders of magnitude faster. These are usually unwanted contaminants, which is why furnace cleanliness matters.

(c) Interstitialcy (kick-out) mechanism. A silicon self-interstitial knocks a substitutional dopant off its site; the dopant travels as an interstitial and then drops back into a lattice site. This explains why B and P diffuse faster during oxidation (**oxidation-enhanced diffusion**), because oxidation injects silicon interstitials.

Sketch in the exam: a square grid of Si atoms. Show (i) a dopant atom next to an empty square (vacancy) with an arrow into it, labelled "vacancy mechanism"; (ii) a small atom between grid points with a zig-zag arrow through the gaps, labelled "interstitial mechanism".

3. Temperature dependence

$$D = D_0 \, e^{-E_a/kT}$$

Where:

- D = diffusion coefficient (m^2/s)
- D_0 = diffusion constant for the dopant–silicon pair (m^2/s), e.g. 7.6×10^{-5} m^2/s for B
- E_a = activation energy (eV), e.g. 3.46 eV for B
- k = Boltzmann constant, 8.617×10^{-5} eV/K
- T = absolute temperature (K)

For boron, D ≈ 1.5×10^{-18} m^2/s at 1000 °C and 1.5×10^{-17} m^2/s at 1100 °C: **100 °C hotter makes D ten times larger** (D roughly doubles every 30 °C). Furnace temperature must therefore be controlled to about ±0.5 °C.

4. Macroscopic description

The flux follows **Fick's first law**, $F = -D\,\partial C/\partial x$ (F = atoms crossing 1 m^2 per second, C = dopant concentration in m^{-3}, x = depth in m), and the profile changes in time by **Fick's second law**, $\partial C/\partial t = D\,\partial^2 C/\partial x^2$. Depth grows as $\sqrt{(Dt)}$.

5. The practical two-step process

1. **Predeposition:** the dopant source (gas such as $POCl_3$ or B_2H_6, or a doped glass) keeps the surface at a constant concentration C_s, usually the solid solubility. The profile is a complementary error function (erfc).
2. **Drive-in:** the source is removed and the wafer is heated again; the fixed dose spreads deeper and the surface concentration falls, giving a Gaussian profile.

Doping is confined to windows in a silicon dioxide mask, because dopants diffuse much more slowly in SiO_2. Under the mask edge the dopant also spreads **sideways** by about 0.8 of the vertical depth (lateral diffusion).

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ diffusion = net movement from high to low concentration by random thermal jumps, at 900–1200 °C

- ✓ vacancy mechanism for substitutional dopants (B, P, As, Sb), high E_a, slow
- ✓ interstitial mechanism for small/metal atoms (Au, Cu, Fe, Na), low E_a, fast
- ✓ interstitialcy/kick-out mechanism (oxidation-enhanced diffusion) mentioned
- ✓ $D = D_0 e^{-E_a/kT}$ with symbols defined and the strong temperature effect
- ✓ link to Fick's laws and depth $\propto \sqrt{(Dt)}$
- ✓ predeposition (erfc) and drive-in (Gaussian); oxide masking and lateral diffusion
- ✓ labelled sketch of vacancy and interstitial jumps

Common mistakes.

- Describing diffusion as atoms being "pushed" by a force or field. The flow comes only from random jumps and a concentration gradient.
- Saying the common dopants move interstitially. B, P, As and Sb move mainly by vacancies (and interstitialcy); the fast interstitial diffusers are metals.
- Writing $D = D_0 e^{+E_a/kT}$. The exponent is negative: a larger barrier gives a smaller D.

Remember it. "Big dopants wait for a vacancy; small metals slip between the gaps; heat makes both go faster."

Try a variant. Phosphorus in silicon has $D_0 = 3.85\times10^{-4}$ m^2/s and $E_a = 3.66$ eV. Find D at 1000 °C. *Answer:* T = 1273 K, kT = $8.617\times10^{-5} \times 1273 = 0.1097$ eV, $E_a/kT = 33.37$, $D = 3.85\times10^{-4} \times e^{-33.37} =$ **1.2×10^{-18} m^2/s**.

Q18. State and explain Fick's first and second laws of diffusion.

> **Question type:** State and explain (with short derivation) · **Topic:** Fick's laws · **Full notes:** Week 3 (Impurity Control and Doping)

What the examiner wants. Each law stated in words and as an equation, every symbol defined with units, the meaning of the minus sign, how the second law follows from the first plus conservation of atoms, and what each law tells you physically. Naming the two standard solutions (erfc and Gaussian) completes the answer.

Key terms.

- **Flux (F):** the number of atoms crossing unit area per unit time, atoms/(m^2·s).
- **Concentration (C):** the number of dopant atoms per unit volume, m^{-3}.
- **Concentration gradient ($\partial C/\partial x$):** the rate at which concentration changes with depth, m^{-4}.
- **Diffusion coefficient (D):** the proportionality constant between flux and gradient, m^2/s; $D = D_0 e^{-E_a/kT}$.
- **Continuity (conservation of atoms):** atoms are neither created nor destroyed, so any change in the number stored in a slice equals inflow minus outflow.
- **Partial derivative (∂):** a rate of change with respect to one variable while the other (x or t) is held fixed.

Model answer.

Fick's first law

Statement: the diffusion flux of atoms is proportional to the concentration gradient and directed from high to low concentration.

$$F = -D\,\partial C/\partial x$$

Where:

- F = flux: atoms crossing 1 m^2 per second (atoms/m^2·s)
- D = diffusion coefficient (m^2/s)
- C = C(x, t), dopant concentration (atoms/m^3)
- x = depth below the surface (m)
- $\partial C/\partial x$ = concentration gradient (atoms/m^4)

Explanation.

- The **minus sign** means the flow is "downhill": where C falls with depth ($\partial C/\partial x$ negative), F is positive, i.e. atoms move deeper.
- If the concentration is uniform ($\partial C/\partial x = 0$) there is **no net flux**, even though atoms still jump randomly.
- A steeper gradient gives a larger flux; a larger D (hotter furnace) gives a larger flux for the same gradient.
- The first law gives the flux at one instant. It does not say how the profile changes with time, so it is used directly only in steady state.

Fick's second law

Statement: the rate of change of concentration at a point is proportional to the curvature (second derivative) of the concentration profile at that point.

Derivation. Assume D is constant (independent of C and x) and no atoms are generated or lost.

Step 1. Take a thin slice between x and x + dx, of unit area. Atoms enter at x with flux F(x) and leave at x + dx with flux F(x + dx).

Step 2. The increase in atoms stored per second is $(\partial C/\partial t)dx$, and it must equal inflow minus outflow:

$$(\partial C/\partial t)\, dx = F(x) - F(x + dx) = -(\partial F/\partial x)\, dx$$

Step 3. Cancel dx to get the continuity equation:

$$\partial C/\partial t = -\partial F/\partial x$$

Step 4. Substitute the first law $F = -D\, \partial C/\partial x$, with D constant:

$$\partial C/\partial t = D\, \partial^2 C/\partial x^2$$

This is Fick's second law (the diffusion equation): $\partial C/\partial t = D\, \partial^2 C/\partial x^2$.

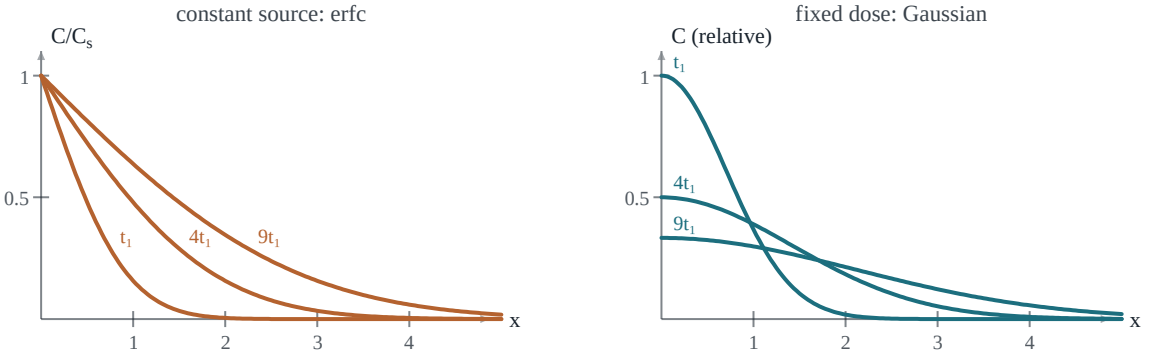
Where:

- $\partial C/\partial t$ = rate of change of concentration at a fixed depth (atoms/m^3·s)
- $\partial^2 C/\partial x^2$ = curvature of the profile (atoms/m^5)
- t = time (s); other symbols as above

Explanation.

- Where the profile curves **upwards** ($\partial^2 C/\partial x^2 > 0$, a hollow), concentration rises; where it curves **downwards** (a hump), it falls. Diffusion therefore always **smooths out** a profile.
- The equation contains D and t only as the product Dt, so the depth scale of any profile is $\sqrt{(Dt)}$: four times the time only doubles the depth.

- Solving it needs boundary and initial conditions. With a **constant surface concentration** C_s it gives the predeposition profile $C = C_s \operatorname{erfc}(x/2\sqrt{(Dt)})$; with a **fixed dose** Q it gives the drive-in Gaussian $C = (Q/\sqrt{(\pi Dt)})\, e^{-x^2/4Dt}$.



The two solutions of Fick's second law: constant-source erfc profiles (left) deepen with time at fixed surface value; fixed-dose Gaussian profiles (right) deepen while their surface value falls.

	First law	**Second law**
Relates	Flux to gradient	Rate of change of C to curvature
Equation	$F = -D\,\partial C/\partial x$	$\partial C/\partial t = D\,\partial^2 C/\partial x^2$
Use	Flux at an instant, steady state	Profile evolution with time

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ first law stated in words and as $F = -D\,\partial C/\partial x$
- ✓ every symbol defined with units (F, D, C, x, t)
- ✓ meaning of the minus sign (flow from high to low)
- ✓ continuity equation $\partial C/\partial t = -\partial F/\partial x$ from conservation of atoms in a slice
- ✓ second law $\partial C/\partial t = D\,\partial^2 C/\partial x^2$, with the assumption D constant
- ✓ physical meaning: curvature drives change, profiles smooth out, depth $\propto \sqrt{(Dt)}$
- ✓ the erfc and Gaussian solutions named with their boundary conditions

Common mistakes.

- Dropping the minus sign in the first law: without it atoms would flow up the gradient.
- Writing the second law as $\partial C/\partial t = D\,\partial C/\partial x$ (first derivative). It is the **second** derivative in x and the **first** in t.
- Stating the laws without explaining them. "Explain" means saying what each term does physically.

Remember it. "First law: flux follows the slope. Second law: the profile changes where it bends."

Try a variant. Near the surface of a wafer at 1000 °C the boron concentration falls linearly from 5×10^{25} m^{-3} to zero over 1 µm, and $D = 2\times10^{-18}$ m^2/s. Find the diffusion flux. *Answer:* $\partial C/\partial x = -5\times10^{25}/(1\times10^{-6}\text{ m}) = -5\times10^{31}$ m^{-4}; $F = -D\,\partial C/\partial x = 2\times10^{-18}\times 5\times10^{31} =$ **1×10^{14} atoms/m^2·s**, directed into the wafer.

Q19. A semiconductor is doped by diffusion with a constant surface concentration of 1×10^{20} cm^{-3}. Explain how the impurity concentration varies with depth.

> **Question type:** Explanation with calculation · **Topic:** constant-source (erfc) diffusion profile · **Full notes:** Week 3 (Impurity Control and Doping)

What the examiner wants. Recognise the constant-source (predeposition) case, state the boundary conditions, give the solution of Fick's second law $C(x, t) = C_s \operatorname{erfc}(x/2\sqrt{(Dt)})$, and describe how C falls with depth and how the profile changes with time. Full marks need numbers: choose a realistic D and t, tabulate C against depth, find the junction depth in a stated background doping, give the dose, and describe (or draw) the sketch.

Key terms.

- **Constant-source diffusion (predeposition):** diffusion in which a dopant supply keeps the surface concentration fixed at C_s for the whole process.
- **Surface concentration (C_s):** the dopant concentration at x = 0, here 1×10^{26} m^{-3} (close to the solid solubility of B or P in Si).
- **Complementary error function (erfc):** erfc(u) = 1 − erf(u); a tabulated function that equals 1 at u = 0 and falls rapidly towards 0 as u grows.
- **Diffusion length ($\sqrt{(Dt)}$):** the characteristic distance dopant spreads in time t; the profile's depth scale is $2\sqrt{(Dt)}$.
- **Background doping (N_B):** the doping the wafer already has, of the opposite type to the diffused dopant.
- **Junction depth (x_j):** the depth where the diffused concentration equals N_B, so a p-n junction forms.
- **Dose (Q):** the total number of dopant atoms introduced per m^2 of surface (the area under the profile).

Model answer.

1. The boundary-value problem

The profile obeys Fick's second law, $\partial C/\partial t = D\, \partial^2 C/\partial x^2$, with

- $C(0, t) = C_s = 1\times10^{26}$ m^{-3} for all t (constant surface source)
- $C(x, 0) = 0$ for $x > 0$ (no dopant at the start)
- $C(\infty, t) = 0$ (the wafer is effectively infinitely thick)

Its solution is the **complementary error function** profile:

$$C(x, t) = C_s \operatorname{erfc}\left(x / 2\sqrt{(Dt)} \right) = 1\times10^{26} \operatorname{erfc}\left(x / 2\sqrt{(Dt)} \right) \quad \text{m}^{-3}$$

Where:

- C(x, t) = dopant concentration at depth x after time t (m^{-3})
- C_s = surface concentration, 1×10^{26} m^{-3}
- x = depth below the surface (m)
- D = diffusion coefficient of the dopant at the furnace temperature (m^2/s)
- t = diffusion time (s)
- erfc = complementary error function (dimensionless); $u = x/2\sqrt{(Dt)}$ is its argument

2. How the concentration varies with depth

- **At the surface** (x = 0): erfc(0) = 1, so C = 1×10^{26} m^{-3} at all times.
- **It falls monotonically with depth.** Near the surface the fall is gentle (almost linear); further in it falls faster than exponentially, because for large u erfc(u) ≈ $e^{-u^2}/(u\sqrt{\pi})$.
- **The depth scale is 2√(Dt).** At x = 2√(Dt) (u = 1) C has dropped to 15.7 % of C_s; at x = 4√(Dt) (u = 2) to 0.47 %; at x = 6√(Dt) (u = 3) to about 2×10^{-5} of C_s.
- **With time the profile pushes deeper** as √t, while the surface value stays pinned at 10^{26} m^{-3}: four times the time gives twice the depth.
- **The dose keeps growing** because the source keeps supplying atoms:

$$Q = (2/\sqrt{\pi})\, C_s \sqrt{(Dt)}$$

Where:

- Q = dose (atoms/m^2); other symbols as above
- **A junction forms** where $C(x_j) = N_B$, the background doping of opposite type:

$$x_j = 2\sqrt{(Dt)} \cdot \text{erfc}^{-1}(N_B / C_s)$$

Where:

- erfc^{-1} = inverse complementary error function: the value of u for which erfc(u) equals the number in brackets (read the erfc table backwards)

3. Worked numerical illustration

Given: boron diffused into n-type silicon at 1000 °C (T = 1273 K) for t = 1 h = 3600 s, with $C_s = 1\times10^{20}$ $cm^{-3} = 1\times10^{26}$ m^{-3}. For boron in Si, $D_0 = 7.6\times10^{-5}$ m^2/s and $E_a = 3.46$ eV. Background doping $N_B = 1\times10^{22}$ m^{-3} (phosphorus).

Find: D, the depth scale, C(x) at several depths, the junction depth x_j and the dose Q.

Formula: $D = D_0 e^{-E_a/kT}$; $C = C_s\, \text{erfc}(x/2\sqrt{(Dt)})$; $x_j = 2\sqrt{(Dt)}\, \text{erfc}^{-1}(N_B/C_s)$; $Q = (2/\sqrt{\pi})C_s\sqrt{(Dt)}$.

Solution.

Step 1. Diffusion coefficient:

$$kT = 8.617\times10^{-5} \times 1273 = 0.1097 \text{ eV}, \quad E_a/kT = 3.46/0.1097 = 31.54$$
$$D = 7.6\times10^{-5} \times e^{-31.54} = 1.5\times10^{-18} \text{ m}^2/\text{s}$$

Step 2. Depth scale:

$$Dt = 1.5\times10^{-18} \times 3600 = 5.4\times10^{-15} \text{ m}^2$$
$$\sqrt{(Dt)} = 7.35\times10^{-8} \text{ m} = 0.0735 \text{ μm}, \quad 2\sqrt{(Dt)} = 1.47\times10^{-7} \text{ m} = 0.147 \text{ μm}$$

Step 3. Profile (x = u × 0.147 μm):

u = x/2√(Dt)	Depth x (μm)	erfc(u)	C (m^{-3})
0	0	1.000	1.0×10^{26}
0.5	0.074	0.480	4.8×10^{25}
1.0	0.147	0.157	1.6×10^{25}
1.5	0.220	0.0339	3.4×10^{24}
2.0	0.294	4.7×10^{-3}	4.7×10^{23}
2.5	0.367	4.1×10^{-4}	4.1×10^{22}
2.75	0.404	1.0×10^{-4}	1.0×10^{22} (junction)
3.0	0.441	2.2×10^{-5}	2.2×10^{21}

Step 4. Junction depth: $\text{erfc}(u_j) = N_B/C_s = 10^{22}/10^{26} = 1\times10^{-4}$, which gives $u_j = 2.75$.

$$x_j = 2.75 \times 1.47\times10^{-7}\ \text{m} = 4.04\times10^{-7}\ \text{m}\ (0.404\ \mu\text{m})$$

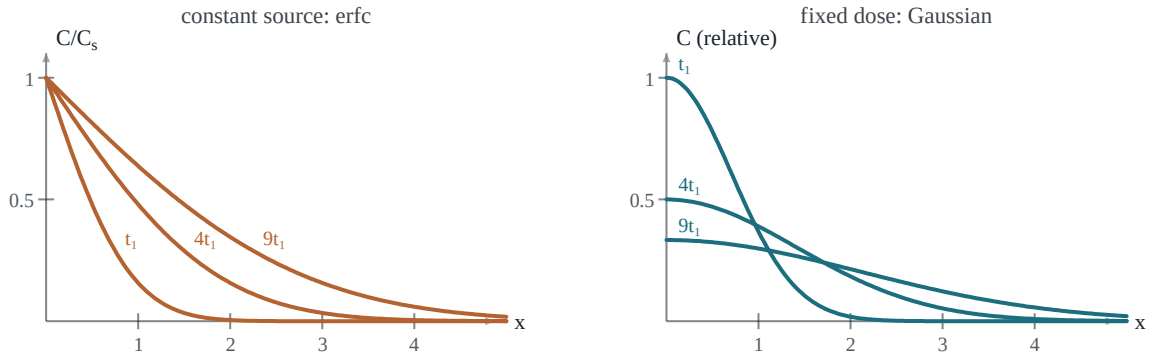
Step 5. Dose:

$$Q = (2/\sqrt{\pi}) \times 1\times10^{26} \times 7.35\times10^{-8} = 1.128 \times 7.35\times10^{18} = 8.3\times10^{18}\ \text{atoms/m}^2$$

Answer: $C(x) = 1\times10^{26}\ \text{erfc}(x/1.47\times10^{-7}\ \text{m})\ \text{m}^{-3}$; it falls from $10^{26}\ \text{m}^{-3}$ at the surface to $1.6\times10^{25}\ \text{m}^{-3}$ at 0.15 μm and $4.7\times10^{23}\ \text{m}^{-3}$ at 0.29 μm, reaching the background $10^{22}\ \text{m}^{-3}$ at $x_j \approx 4.0\times10^{-7}$ m (0.40 μm); dose $Q \approx 8.3\times10^{18}\ \text{m}^{-2}$.

What this means: in one hour at 1000 °C the boron forms a heavily doped p^+ layer only about 0.4 μm thick: concentration drops four orders of magnitude in 0.4 μm. Diffusing for 4 h would double x_j to 0.81 μm (and the dose to $1.7\times10^{19}\ \text{m}^{-2}$) while the surface would stay at $10^{26}\ \text{m}^{-3}$.

4. Sketch



Left panel: constant-source erfc profiles at times t_1, $4t_1$ and $9t_1$. All start from the same surface value C_s; longer times push the profile deeper.

Sketch in the exam: axes of log C (from 10^{21} to $10^{27}\ \text{m}^{-3}$) against depth x (μm). Draw three erfc curves for $t_1 < t_2 < t_3$, all starting at the same point $C_s = 10^{26}\ \text{m}^{-3}$ on the vertical axis, flat-ish near the surface then bending steeply down; the later curves lie further to the right. Draw a horizontal dashed line at $N_B = 10^{22}\ \text{m}^{-3}$ and mark x_j where each curve crosses it; label the region above the line "p-type ($B > N_B$)" and below "n-type substrate".

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ identifies constant-source (predeposition) diffusion and states the boundary conditions

- ✓ $C(x, t) = C_s \operatorname{erfc}(x/2\sqrt{(Dt)})$ with every symbol defined
- ✓ $C = C_s = 10^{26}$ m^{-3} at x = 0, independent of time
- ✓ monotonic, steep fall with depth on the scale $2\sqrt{(Dt)}$; profile deepens as $\sqrt{t}$
- ✓ dose $Q = (2/\sqrt{\pi})C_s\sqrt{(Dt)}$ grows with time
- ✓ junction where $C(x_j) = N_B$, $x_j = 2\sqrt{(Dt)}\ \operatorname{erfc}^{-1}(N_B/C_s)$
- ✓ realistic numbers: $D \approx 1.5\times10^{-18}$ m^2/s for B at 1000 °C, $x_j \approx 0.4$ µm for 1 h into 10^{22} m^{-3}
- ✓ labelled sketch (log C against x, several times, N_B line, x_j)

Common mistakes.

- Using the Gaussian profile. A **constant** surface concentration means erfc; a **fixed dose** (source removed) means Gaussian.
- Saying the surface concentration rises or falls with time. In this case it is fixed at 1×10^{26} m^{-3}; only the depth and the dose grow.
- Forgetting to convert units: $\sqrt{(Dt)}$ comes out in m; multiply by 10^6 for µm.
- Plotting on a linear scale only: the interesting part (10^{22}–10^{24} m^{-3} near the junction) is invisible unless C is on a log axis.

Remember it. "Constant source: surface stays put, the erfc tail creeps in as $\sqrt{(Dt)}$."

Try a variant. Repeat the junction-depth calculation for the same boron diffusion (1000 °C, 1 h) into a more lightly doped substrate, $N_B = 1\times10^{21}$ m^{-3} ($\operatorname{erfc}^{-1}(10^{-5}) = 3.12$). *Answer:* $x_j = 3.12 \times 1.47\times10^{-7}$ m = **4.6×10^{-7} m (0.46 µm)**. Ten times lighter background moves the junction only 14 % deeper, because the erfc tail is so steep.

Q20. Explain the principle of ion implantation and identify its major advantages over thermal diffusion.

> **Question type:** Explain and list · **Topic:** ion implantation · **Full notes:** Week 3 (Impurity Control and Doping)

What the examiner wants. The principle: ions generated, mass-selected, accelerated to a known energy, scanned over the wafer, stopping at a depth set by energy, with a Gaussian profile at R_p, dose from beam charge, then annealing. Then a list of distinct advantages, each tied to a contrast with diffusion. A labelled block diagram and a profile sketch earn marks.

Key terms.

- **Ion implantation:** doping by firing accelerated dopant ions into the wafer.
- **Ion energy (E):** the kinetic energy gained by an ion of charge q through accelerating voltage V, E = qV (keV to MeV).
- **Dose (Q or Φ):** the number of implanted ions per m^2 of wafer.
- **Projected range (R_p):** the average depth at which the ions stop.
- **Straggle (ΔR_p):** the standard deviation of the stopping depths about R_p.
- **Nuclear and electronic stopping:** energy loss by collisions with silicon nuclei (causes damage) and with electrons (little damage).
- **Annealing:** heating after implantation to repair lattice damage and place dopants on lattice sites where they become electrically active.

- **Channelling:** ions travelling unusually deep along open crystal directions; prevented by tilting the wafer about 7°.

Model answer.

1. Principle

Dopant atoms are **ionised, accelerated through a high voltage and fired into the wafer**, where they slow down by collisions and come to rest below the surface. The depth is set by the ion energy, the amount by the beam charge.

ion source → extraction → mass analyser (magnet) → accelerator → focusing and scanning → wafer (in vacuum)

- **Ion source:** a gas such as BF_3, PH_3 or AsH_3 is ionised in a plasma to give positive ions.
- **Extraction and mass analysis:** an electric field pulls the ions out; a magnet bends each ion by an amount that depends on its mass-to-charge ratio, so only the chosen species (e.g. $^{11}B^+$) passes the exit slit.
- **Acceleration:** the ions gain kinetic energy $E = qV$ (a singly charged ion through 100 kV has 100 keV).
- **Scanning:** the beam is swept across the wafer for uniform doping. The whole path is in high vacuum.
- **Stopping:** inside the silicon each ion loses energy by **nuclear stopping** (collisions with Si nuclei, which displace atoms and damage the lattice) and **electronic stopping** (interaction with electrons). It stops at a random depth; the distribution is approximately **Gaussian**:

$$C(x) = [\, Q / (\sqrt{(2\pi)}\, \Delta R_p) \,] \exp[\, -(x - R_p)^2 / (2\Delta R_p{}^2) \,]$$

Where:

- $C(x)$ = implanted concentration at depth x (m^{-3})
- Q = dose (ions/m^2)
- R_p = projected range (m), increasing with energy and decreasing with ion mass
- ΔR_p = straggle (m)

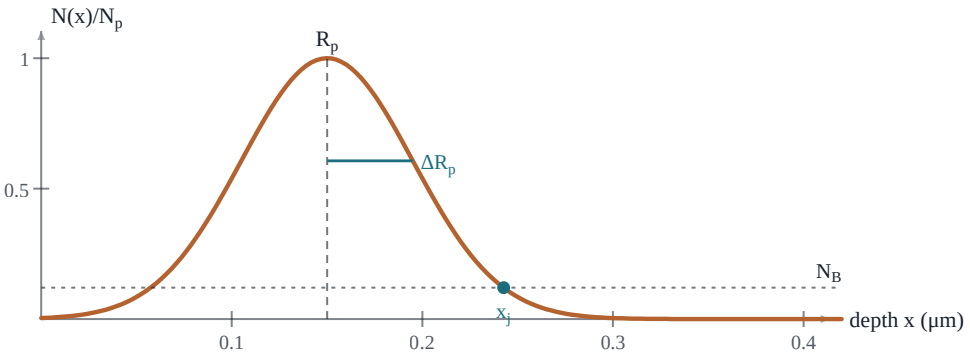
The peak is **below the surface** at $x = R_p$, with height $N_p = Q/(\sqrt{(2\pi)}\Delta R_p) \approx 0.4Q/\Delta R_p$.

- **Dose measurement:** the ion current reaching the wafer is collected (Faraday cup) and integrated:

$$Q = I\, t / (q\, A)$$

Where:

- I = beam current (A); t = implant time (s); $q = 1.6\times10^{-19}$ C per singly charged ion; A = implanted area (m^2)
- **Annealing:** furnace annealing or rapid thermal annealing (RTA, about 1000 °C for seconds) repairs the damage and activates the dopants. Channelling is avoided by tilting the wafer about 7° or using a thin screen oxide.



The implanted profile is a bell curve centred at the projected range R_p below the surface, with width set by the straggle ΔR_p; the junction forms where it falls to the background doping N_B.

Sketch in the exam: (i) the block diagram above; (ii) C against x with a Gaussian peak at R_p, width $\pm\Delta R_p$ marked, a dashed N_B line and x_j at the crossing.

2. Major advantages over thermal diffusion

1. **Precise dose control** (about ±1 %) over 10^{15} to 10^{20} ions/m^2, because the dose is counted from measured charge. In diffusion the dose depends on solid solubility, temperature and time and is far less accurate.
2. **Depth and dose are independent:** depth set by energy, dose by charge. In diffusion both depend on the same Dt product.
3. **Low process temperature:** implantation is done near room temperature, so earlier dopant profiles are not disturbed and the thermal budget is small.
4. **Very shallow junctions** with low-energy implants, essential for modern MOSFETs.
5. **Buried and retrograde profiles:** the peak can be placed below the surface; a diffused profile always has its maximum at the surface.
6. **Photoresist can be used as a mask**, which simplifies processing; diffusion needs oxide or nitride masks that survive 1000 °C.
7. **Less lateral spread** under mask edges (about ΔR_p, compared with 0.8 × depth for diffusion), allowing smaller, self-aligned devices.
8. **High purity:** mass analysis passes only the chosen ion.
9. **Not limited by solid solubility**, and good uniformity and reproducibility across large wafers.

Limitations (for balance): expensive equipment, lattice damage that must be annealed, channelling, and lower throughput (one or a few wafers at a time).

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ principle: ionise, mass-analyse, accelerate (E = qV), scan, implant in vacuum
- ✓ stopping by nuclear and electronic collisions; depth set by energy and ion mass
- ✓ Gaussian profile with R_p and ΔR_p; peak below the surface
- ✓ dose Q = It/qA measured from beam charge
- ✓ annealing to repair damage and activate dopants; channelling and 7° tilt
- ✓ at least five advantages, each contrasted with diffusion (dose precision, independent depth, low temperature, shallow/buried profiles, resist mask, small lateral spread)
- ✓ labelled block diagram and profile sketch

Common mistakes.

- Saying implantation needs no heating. It needs an anneal; the advantage is that the *doping step* is at room temperature and the anneal is short.
- Claiming the implanted peak is at the surface. It is at R_p below the surface.
- Listing advantages without saying *over diffusion*: each point should state what diffusion does instead.

Remember it. "Energy picks the depth, charge counts the dose, anneal heals the damage."

Try a variant. A 100 mm diameter wafer is implanted with singly charged P^+ ions using a beam current of 50 µA for 100 s. Find the dose. *Answer:* $A = \pi \times (0.05\ m)^2 = 7.85\times10^{-3}\ m^2$; $Q = It/(qA) = (50\times10^{-6} \times 100)/(1.6\times10^{-19} \times 7.85\times10^{-3})$ = **4.0×10^{18} ions/m^2**.

Q21. Compare diffusion and ion implantation as semiconductor doping techniques.

> **Question type:** Compare · **Topic:** diffusion versus ion implantation · **Full notes:** Week 3 (Impurity Control and Doping)

What the examiner wants. A short description of each technique (principle, profile equation), then a point-by-point comparison on the same headings: principle, temperature, profile shape, dose control, depth control, lateral spread, masking, damage, throughput and cost, typical uses. End with a comparison table and a one-paragraph conclusion.

Key terms.

- **Thermal diffusion:** doping in which dopant atoms move into a heated wafer (900–1200 °C) from a surface source by random thermal jumps.
- **Ion implantation:** doping in which accelerated dopant ions are fired into the wafer and stop by collisions.
- **Dose (Q):** dopant atoms introduced per m^2 of surface.
- **Profile:** the curve of dopant concentration C against depth x.
- **Projected range (R_p) and straggle (ΔR_p):** mean stopping depth of implanted ions and its spread.
- **Lateral diffusion:** sideways spread of dopant under the edge of a mask window.
- **Thermal budget:** the total amount of high-temperature processing (temperature and time) a wafer receives; more budget means existing profiles spread more.
- **Annealing:** post-implant heating that repairs damage and activates dopants.

Model answer.

1. Thermal diffusion

The wafer is heated to 900–1200 °C in a furnace with a dopant source (gas or doped glass) at the surface. Atoms move in by random thermal jumps, obeying Fick's laws. A **predeposition** at constant surface concentration C_s gives

$$C(x, t) = C_s \operatorname{erfc}(x / 2\sqrt{(Dt)})$$

and a **drive-in** of the fixed dose Q gives

$$C(x, t) = [\ Q / \sqrt{(\pi Dt)}\ ]\ e^{-x^2/4Dt}$$

Where:

- C(x, t) = concentration at depth x after time t (m^{-3})
- C_s = surface concentration (m^{-3}); Q = dose (m^{-2})
- D = diffusion coefficient (m^2/s); t = time (s); x = depth (m)

Both profiles have their **maximum at the surface**, and dose and depth are both governed by the single product Dt.

2. Ion implantation

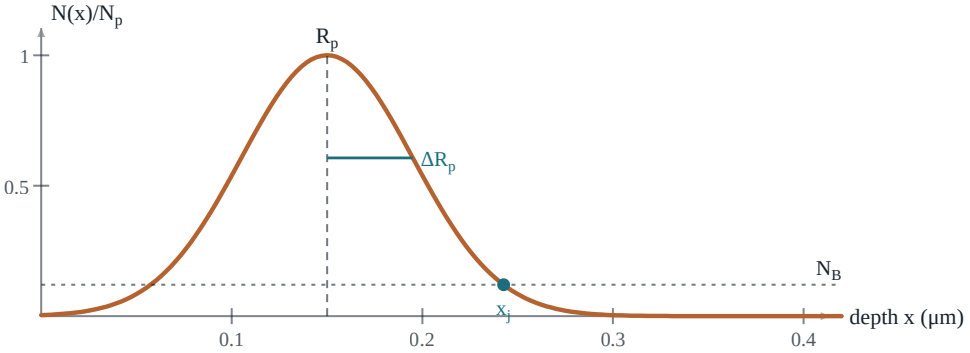
Dopant ions are mass-selected, accelerated to energy E = qV (keV–MeV) and scanned over the wafer at room temperature. The profile is approximately Gaussian with its **peak below the surface**:

$$C(x) = [\ Q / (\sqrt{(2\pi)}\ \Delta R_p)\ ]\ \exp[\ -(x - R_p)^2 / (2\Delta R_p^{\,2})\ ]$$

Where:

- R_p = projected range, set by ion energy (m); ΔR_p = straggle (m); Q = dose measured from beam charge, Q = It/qA

The lattice is damaged and the wafer must be **annealed**.



An implanted profile peaks at R_p below the surface; a diffused profile (erfc or Gaussian) always peaks at the surface.

Sketch in the exam: on one set of axes (log C against x), draw a diffused erfc curve falling from the surface and an implanted Gaussian peaking at R_p below the surface; add the N_B line and mark x_j for each.

3. Comparison table

Feature	Thermal diffusion	Ion implantation
Principle	Random thermal jumps down a concentration gradient (Fick's laws)	Kinetic energy of accelerated ions; stopping by nuclear and electronic collisions
Process temperature	High, 900–1200 °C	Near room temperature, plus anneal (RTA ≈ 1000 °C for seconds)
Profile shape	erfc (constant source) or Gaussian (fixed dose), maximum at surface	Gaussian centred at R_p, peak below surface; buried and retrograde profiles possible
Dose control	Moderate (± 5–10 %), set by solubility, T and t	Excellent (± 1 %), counted from beam charge
Depth control	Coupled to dose through Dt	Independent, set by ion energy
Shallow junctions	Difficult	Easy (low-energy implants)
Lateral spread	Large, about 0.8 × junction depth	Small, about ΔR_p
Masking material	SiO_2 or Si_3N_4 (must survive high T)	Photoresist, oxide, nitride, polysilicon
Crystal damage	Negligible	Significant; annealing essential; amorphisation at high doses
Special problem	Disturbs earlier profiles (large thermal budget)	Channelling (prevented by 7° tilt, screen oxide)
Concentration limit	Solid solubility	Can exceed solubility
Purity	Depends on source and furnace cleanliness	Mass-analysed, very pure
Throughput and cost	Batches of 100+ wafers, cheap furnace	One or a few wafers at a time, expensive implanter
Typical uses	Drive-in, deep wells, solar-cell emitters, older bipolar processes	Source/drain, threshold-voltage adjust, wells, LDD, channel stops in modern CMOS

4. Conclusion

Diffusion is simple, cheap, damage-free and handles many wafers at once, but its dose and depth are coupled, it needs high temperature, it spreads sideways and it cannot easily make very shallow junctions. Ion implantation gives precise, independent control of dose and depth at low temperature, allows buried profiles and resist masks, and is the standard method in modern VLSI, at the cost of expensive equipment and lattice damage that must be annealed. In practice the two are combined: **implant to set the dose, then a thermal drive-in (or anneal) to set the final depth**.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ brief principle of each technique
- ✓ profile equations: erfc / Gaussian for diffusion; Gaussian at R_p for implantation
- ✓ temperature contrast (900–1200 °C versus room temperature plus anneal)
- ✓ dose and depth control: coupled through Dt versus independent (charge and energy)
- ✓ lateral spread, masking materials and shallow-junction capability
- ✓ damage/annealing and channelling for implantation
- ✓ cost, throughput and typical applications
- ✓ comparison table and a concluding sentence (they are complementary)

Common mistakes.

- Writing two separate essays with no direct comparison. Use the same headings for both, ideally in a table.
- Saying diffusion gives a Gaussian "peaked below the surface". Diffused profiles always peak at the surface; only implanted profiles peak below it.
- Ignoring the disadvantages of implantation (damage, channelling, cost). A fair comparison scores higher.

Remember it. "Diffusion: hot, cheap, surface-peaked, coupled. Implantation: cold, precise, buried-peak, independent, but needs an anneal."

Try a variant. Boron is implanted with dose 1×10^{19} m^{-2} and straggle $\Delta R_p = 0.03$ μm (3×10^{-8} m). Find the peak concentration and say why such a high value could not be obtained this shallow by diffusion. *Answer:* $N_p = Q/(\sqrt{(2\pi)}\Delta R_p) = 1\times10^{19}/(2.507 \times 3\times10^{-8}$ m) = **1.3×10^{26} m^{-3}**, a few hundredths of a micron below the surface. Diffusion is limited by solid solubility and needs a long, hot cycle to put in this dose, during which the dopant spreads much deeper; implantation places it directly at the chosen depth.

Questions 22–34 — Energy Bands and Carrier Statistics

Module 2: the quantum model of electrons in a crystal, how energy bands form, direct and indirect gaps, intrinsic and extrinsic carriers, Fermi–Dirac statistics, the density of states and the Fermi level. Each question stands on its own: its key terms are defined with it, so you can start anywhere.

Symbols and abbreviations used on this page

Every symbol, constant and abbreviation on this page is listed here, even if another page already defined it, so you can study this page on its own.

Constants and units

Symbol	Meaning	Units / typical value
q	Magnitude of the electronic charge	1.6×10^{-19} C
k	Boltzmann constant	8.617×10^{-5} eV/K = 1.381×10^{-23} J/K
kT	Thermal energy	0.0259 eV (often 0.02585 eV) at 300 K
T	Absolute temperature	kelvin (K); room temperature = 300 K
h	Planck constant	6.626×10^{-34} J·s
ħ	Reduced Planck constant, h/2π	1.055×10^{-34} J·s
m_0	Free-electron mass	9.11×10^{-31} kg
eV	Electron-volt, a unit of energy	1 eV = 1.6×10^{-19} J
μm	Micrometre (micron)	1 μm = 10^{-6} m

Symbols

Symbol	Meaning	Units / typical value
E_c (E_C)	Energy of the bottom of the conduction band	eV
E_v (E_V)	Energy of the top of the valence band	eV
E_g	Band-gap energy, $E_c - E_v$	eV; Si 1.12, Ge 0.66, GaAs 1.42 at 300 K
E_F	Fermi level: the energy at which a state has a 50 % chance of being occupied	eV
E_i	Intrinsic Fermi level, close to mid-gap	eV
E_D	Donor energy level, just below E_c	eV; $E_c - E_D \approx 0.045$ eV for P in Si
E_A	Acceptor energy level, just above E_v	eV; $E_A - E_v \approx 0.045$ eV for B in Si
E_n	Allowed energy of level n of a particle in a box	J or eV
f(E)	Fermi–Dirac function: probability that a state at energy E is occupied	dimensionless, 0 to 1
g(E), g_c, g_v	Density of states: allowed states per unit energy per unit volume	states/(eV·m^3)
N_c	Effective density of states in the conduction band	m^{-3}; Si 2.8×10^{25} at 300 K
N_v	Effective density of states in the valence band	m^{-3}; Si 1.04×10^{25} at 300 K
n	Electron concentration (free electrons per m^3)	m^{-3}
p	Hole concentration (holes per m^3)	m^{-3}
n_i	Intrinsic carrier concentration (n = p = n_i in pure material)	m^{-3}; Si $\approx 1\times10^{16}$ at 300 K
n_0, p_0	Equilibrium electron and hole concentrations	m^{-3}
N_D	Donor concentration	m^{-3}
N_A	Acceptor concentration	m^{-3}
$m*$, m_n*, m_p*	Effective mass of electrons or holes	kg, or as a multiple of m_0
ψ	Wave function ($\lvert\psi\rvert^2$ is the probability density)	—
V(x)	Potential energy in the Schrödinger equation	J or eV
k (wave vector)	Wave vector, 2π/λ; measures the electron's momentum, p = ħk	m^{-1}
α (Kronig–Penney)	√(2mE)/ħ, set by the electron energy	m^{-1}
P (Kronig–Penney)	Barrier strength $mV_0ba/\hbar^2$	dimensionless
V_0	Height of a potential barrier	J or eV
λ	Wavelength	μm or nm; λ(μm) = 1.24/E(eV)
ν, hν	Frequency of light, and photon energy	Hz; eV
a	Lattice constant: edge length of the cubic unit cell	nm; Si 0.543, Ge and GaAs 0.565
f (growth)	Fraction of the melt already solidified	0 to 1
μ, μ_n, μ_p	Mobility: drift velocity per unit field (electrons, holes)	m^2/(V·s); Si μ_n 0.135, μ_p 0.048

Symbol	Meaning	Units / typical value
σ	Conductivity, $q(n\mu_n + p\mu_p)$ (in Week 8, σ is also a capture cross-section)	S/m (siemens per metre)
V, V_a	Applied voltage (positive for forward bias)	V
α	Common-base current gain, I_C/I_E	dimensionless; 0.98–0.999
β	Common-emitter current gain, I_C/I_B (also h_{FE})	dimensionless; 50–500
λ_{max}	Longest wavelength a material absorbs, $1.24/E_g$ (μm, E_g in eV)	μm
$\psi_k(x)$, $u_k(x)$	Bloch wavefunction of an electron, and its part that repeats with the lattice	—

Abbreviations

Abbreviation	Meaning
Si	Silicon
Ge	Germanium
GaAs	Gallium arsenide
GaN	Gallium nitride
InP	Indium phosphide
GaP	Gallium phosphide
SiO_2	Silicon dioxide (silica), the oxide insulator
CB	Conduction band
VB	Valence band
MOS	Metal–oxide–semiconductor
IC	Integrated circuit
LED	Light-emitting diode
n^+	Very heavily doped n-type (similarly p^+)

Questions on this page

Q	Question
22	Explain the basic assumptions of the quantum mechanical model of electrons in a semiconductor.
23	State the time-independent Schrödinger equation and explain the physical meaning of its terms.
24	Explain the basic principle of the Kronig-Penney model and how it leads to the formation of energy bands.
25	Distinguish between valence band, conduction band, and forbidden energy gap.
26	Explain the difference between a direct band-gap and an indirect band-gap semiconductor.
27	What is meant by an intrinsic semiconductor? Derive the expression for intrinsic carrier concentration.
28	Explain the difference between intrinsic and extrinsic carrier concentrations.
29	Silicon has an intrinsic carrier concentration of approximately 1.0×10^{10} cm^{-3} at a specified temperature. If the electron and hole concentrations are equal, determine the electron and hole concentrations.
30	Explain the Fermi-Dirac distribution function and its significance in semiconductor physics.
31	Write the Fermi-Dirac distribution equation and explain what happens to the distribution at absolute zero.
32	Define the density of states in a semiconductor and explain its importance in determining carrier concentration.
33	Explain how the Fermi energy level changes with doping in n-type and p-type semiconductors.
34	An n-type semiconductor has a donor concentration of 5×10^{16} cm^{-3}. Assuming complete ionization, determine the approximate majority-carrier concentration.

Model answers

Q22. Explain the basic assumptions of the quantum mechanical model of electrons in a semiconductor.

> **Question type:** Explanation · **Topic:** quantum model of electrons in a crystal · **Full notes:** Week 4 (Energy Bands and Carrier Concentrations)

What the examiner wants. "Explain the assumptions" means list each simplifying assumption the model makes AND say in a sentence why it is made or what it allows. A full answer names about six assumptions, from the wave nature of the electron to the statistics that fill the bands, and ends by saying what the model predicts (energy bands).

Key terms.

- **Wave function (ψ):** the mathematical wave that describes an electron; $|\psi|^2$ gives the probability of finding it at a point.
- **Schrödinger equation:** the wave equation whose solutions give the allowed wave functions and energies of an electron.
- **Lattice:** the regular, repeating arrangement of atoms in a crystal; its repeat distance is the lattice period a.
- **Periodic potential:** a potential energy V(x) that repeats with the lattice period, $V(x + a) = V(x)$.
- **Phonon:** a quantum (packet) of lattice vibration.
- **Effective mass ($m*$):** the mass a carrier appears to have when pushed by a force inside the crystal.

- **Fermi–Dirac distribution f(E):** the probability that an electron state of energy E is occupied in thermal equilibrium.

Model answer.

A crystal contains about 5×10^{28} atoms per m^3 and even more electrons, all interacting. This many-body problem cannot be solved exactly, so the quantum mechanical model makes the following assumptions.

1. Electrons behave as waves. Each electron is described by a wave function ψ that obeys the Schrödinger equation. Its momentum is written $p = \hbar k$, where $k = 2\pi/\lambda$ is the wave vector and $\hbar = h/2\pi = 1.055\times10^{-34}$ J·s. This replaces the classical "small ball" picture, which cannot explain why some solids conduct and others do not.

2. The nuclei are fixed (adiabatic or Born–Oppenheimer approximation). A silicon nucleus is about 50 000 times heavier than an electron and moves far more slowly, so the ion cores are treated as stationary at their lattice sites. Their vibrations (phonons) are added back later as a small disturbance that scatters carriers.

3. The crystal is perfect and infinite. The ion cores form a perfectly regular lattice that extends forever, so the potential energy seen by an electron is exactly periodic:

$$V(x + a) = V(x)$$

Where:

- $V(x)$ = potential energy of an electron at position x (eV)
- a = lattice period, the spacing of the repeating unit (m; for Si $a = 0.543$ nm)

Surfaces, impurities and defects are ignored at first and treated separately later (as donor, acceptor or trap levels).

4. Each electron moves independently (one-electron approximation). Instead of tracking every electron–electron repulsion, each electron is assumed to move in an *average* periodic potential created by all the ion cores and all the other electrons. The many-body problem becomes a single-particle problem that can be solved.

5. Pauli exclusion principle. Each quantum state (one value of k and one spin) can hold at most one electron. The states are therefore filled from the lowest energy upwards, which is why a band can be "full" or "empty".

6. Thermal-equilibrium statistics. At temperature T the electrons are shared among the available states according to the Fermi–Dirac function f(E). This decides how many electrons sit in the conduction band and how many holes are left in the valence band.

7. Effective-mass approximation. Near the edge of a band, an electron (or hole) responds to an applied force as if it were a free particle with an effective mass $m*$. The whole effect of the crystal potential is packed into this one number, so device equations can treat carriers as simple particles.

What the assumptions lead to. Assumptions 3 and 4 give **Bloch's theorem**: the electron wave in a periodic potential is a travelling wave multiplied by a function with the period of the lattice, $\psi_k(x) = u_k(x)e^{ikx}$. Solving the Schrödinger equation with this form (for example in the Kronig–Penney model) gives **allowed energy bands separated by forbidden gaps**, the basis of all semiconductor physics.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ electron described by a wave function ψ obeying the Schrödinger equation
- ✓ nuclei treated as fixed (Born–Oppenheimer); vibrations added later as phonons
- ✓ perfect, infinite periodic lattice with $V(x + a) = V(x)$
- ✓ one-electron (independent-electron, average-potential) approximation

- ✓ Pauli exclusion principle fills states from the bottom up
- ✓ Fermi–Dirac statistics in thermal equilibrium
- ✓ effective-mass approximation near band edges
- ✓ conclusion: Bloch waves and energy bands with gaps

Common mistakes.

- Listing the assumptions with no explanation. The command word is "explain": add one sentence on why each is made.
- Calling Bloch's theorem an assumption. It is a *result* of assuming a periodic potential; say so.
- Confusing "periodic potential" with "periodic wave function". The potential repeats exactly; the Bloch wave function repeats only up to a phase factor e^{ika}.

Remember it. "Wave, Fixed, Perfect, Alone, Pauli, Fermi, Effective": the electron is a wave, nuclei are fixed, the crystal is perfect, each electron is alone in an average field, Pauli fills the states, Fermi sets the occupation, and m∗ hides the crystal.

Try a variant. Why does an electron in a perfect crystal travel without being scattered by the atoms, and what does scatter it? *Answer:* in a perfectly periodic potential the solutions are Bloch waves, $\psi_k = u_k(x)e^{ikx}$, which are travelling waves that extend through the whole crystal; the atoms only shape the wave (through u_k) and do not deflect it. Scattering comes only from departures from perfect periodicity: lattice vibrations (phonons), ionised impurities, defects and surfaces. This is why mobility falls when temperature or doping rises.

Q23. State the time-independent Schrödinger equation and explain the physical meaning of its terms.

> **Question type:** Statement and explanation · **Topic:** Schrödinger equation · **Full notes:** Week 4 (Energy Bands and Carrier Concentrations)

What the examiner wants. Write the equation correctly (one-dimensional form is enough, the 3-D form earns credit), then explain every term: the wave function, the kinetic-energy term, the potential-energy term and the energy E. Mention the conditions ψ must satisfy and how quantisation follows.

Key terms.

- **Wave function (ψ):** the amplitude of the electron's probability wave; $|\psi|^2\,dx$ is the probability of finding it between x and x + dx.
- **Stationary state:** a state whose energy and probability distribution do not change with time.
- **Hamiltonian ($\hat{H}$):** the total-energy operator, kinetic plus potential.
- **Eigenvalue:** an allowed value of E for which the equation has an acceptable solution.
- **Reduced Planck constant (ħ):** $\hbar = h/2\pi = 1.055\times10^{-34}$ J·s.

Model answer.

The equation. In one dimension, for an electron of mass m in a potential energy V(x) that does not change with time:

$$-(\hbar^2/2m)\,d^2\psi(x)/dx^2 + V(x)\,\psi(x) = E\,\psi(x)$$

Where:

- $\psi(x)$ = wave function of the electron (for a 1-D problem its unit is $m^{-1/2}$)
- $\hbar$ = reduced Planck constant, $h/2\pi = 1.055\times10^{-34}$ J·s
- m = electron mass, 9.11×10^{-31} kg (in a crystal, the effective mass $m*$)
- x = position (m)
- V(x) = potential energy of the electron at x (J)
- E = total energy of the electron (J)

In three dimensions, $-(\hbar^2/2m)\nabla^2\psi(r) + V(r)\psi(r) = E\psi(r)$, written compactly as $\hat{H}\psi = E\psi$.

Physical meaning of each term. The equation is an energy balance: **kinetic energy + potential energy = total energy**.

- **$\psi(x)$, the wave function.** ψ itself cannot be measured. Its square, $|\psi(x)|^2\,dx$, is the probability of finding the electron between x and x + dx. Where $|\psi|^2$ is large, the electron is likely to be found.
- **$-(\hbar^2/2m)\,d^2\psi/dx^2$, the kinetic-energy term.** It comes from the classical kinetic energy $p^2/2m$ with the momentum replaced by the operator $p = -i\hbar\,d/dx$. The second derivative measures how sharply ψ curves: a wave that wiggles tightly (short wavelength) has high momentum and high kinetic energy.
- **$V(x)\psi$, the potential-energy term.** V(x) describes the forces acting on the electron: the attraction of the ion cores in a crystal, a barrier, or a well. In a crystal V(x) is periodic.
- **E, the total energy.** A constant, because the equation describes stationary states. Acceptable solutions exist only for certain values of E, the **eigenvalues**. This is how energy quantisation (and, in a crystal, energy bands) appears.
- **The left-hand side as a whole** is the Hamiltonian operator $\hat{H}$ acting on ψ.

Conditions on ψ. ψ must be finite, single-valued and continuous, with a continuous slope $d\psi/dx$ wherever V is finite, and normalised, $\int|\psi|^2\,dx = 1$ (the electron is certainly somewhere). These conditions are what pick out the allowed energies.

Link to time. The full wave function is $\Psi(x, t) = \psi(x)\,e^{-iEt/\hbar}$. Because $|e^{-iEt/\hbar}|^2 = 1$, the probability $|\Psi|^2$ does not change with time, which is why these are called stationary states.

Two quick illustrations.

- Free electron (V = 0): $\psi = Ae^{ikx}$ and $E = \hbar^2k^2/2m$. Any energy is allowed.
- Electron in a box of width L with impassable walls: ψ must be zero at both walls, so only $E_n = n^2\pi^2\hbar^2/2mL^2$ (n = 1, 2, 3 …) are allowed. Confinement produces quantisation.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ correct equation $-(\hbar^2/2m)d^2\psi/dx^2 + V\psi = E\psi$ (and/or $\hat{H}\psi = E\psi$)
- ✓ ψ explained, with $|\psi|^2$ as probability density
- ✓ first term identified as kinetic energy, linked to $p = -i\hbar\,d/dx$
- ✓ $V(x)\psi$ identified as potential energy, with an example (periodic crystal potential)
- ✓ E is the constant total energy; only eigenvalues allowed, giving quantisation
- ✓ conditions on ψ (finite, continuous, normalised)
- ✓ "time-independent" explained: V independent of t, stationary states

Common mistakes.

- Dropping the minus sign or writing $\hbar^2/m$ instead of $\hbar^2/2m$. Check by the free-electron result: $\psi = e^{ikx}$ must give $E = \hbar^2k^2/2m$.
- Saying ψ is the probability. It is $|\psi|^2$ that is the probability density.
- Writing E as a variable that changes with x. E is a single constant for each solution.

Remember it. "Curvature is kinetic, V is potential, E is the total": KE + PE = E, written for waves.

Try a variant. Find the lowest three energy levels of an electron confined in a one-dimensional box of width 1 nm. *Answer:* $E_n = n^2\pi^2\hbar^2/2mL^2$. $E_1 = \pi^2 \times (1.055\times10^{-34})^2 / (2 \times 9.11\times10^{-31} \times (10^{-9})^2) = 6.03\times10^{-20}$ J = **0.377 eV**; $E_2 = 4E_1$ = **1.51 eV**; $E_3 = 9E_1$ = **3.39 eV**. The levels are discrete and spread further apart as n rises; halving the box to 0.5 nm makes E_1 four times larger (1.51 eV).

Q24. Explain the basic principle of the Kronig-Penney model and how it leads to the formation of energy bands.

> **Question type:** Explanation · **Topic:** Kronig–Penney model and energy bands · **Full notes:** Week 4 (Energy Bands and Carrier Concentrations)

What the examiner wants. Two parts: (1) the principle, meaning the idealised periodic potential the model uses and how it is solved (Schrödinger equation, boundary conditions, Bloch's theorem); (2) how bands follow, meaning the final condition and the argument that cos(ka) must lie between −1 and +1. A labelled sketch of the potential and of the condition plotted against αa earns marks.

Key terms.

- **Periodic potential:** a potential energy that repeats with the lattice period.
- **Bloch's theorem:** in a periodic potential the wave function satisfies $\psi(x + L) = e^{ikL}\psi(x)$, where L is the period; it is a travelling wave times a periodic function.
- **Wave vector (k):** $k = 2\pi/\lambda$; it measures the electron's momentum, $p = \hbar k$.
- **Allowed band:** a range of energies for which electron states exist.
- **Forbidden gap:** a range of energies for which no electron states exist.
- **Brillouin-zone boundary:** the values $k = \pm n\pi/a$ where the electron wave is Bragg-reflected by the lattice and band edges occur.
- **Barrier strength (P):** a dimensionless number measuring how tightly each atom holds its electrons.

Model answer.

1. The principle. The real potential in a crystal is complicated. Kronig and Penney (1931) replaced it with the simplest *periodic* potential that can be solved exactly: an infinite one-dimensional row of **rectangular wells and barriers**.

- Each **well** (V = 0, width a) represents the region near an ion core, where the electron is attracted.
- Each **barrier** (height V_0, width b) represents the region between ions.
- The pattern repeats with period a + b.

Sketch: a square-wave potential V(x) against x, wells of width a at V = 0, barriers of height V_0 and width b, period (a + b) marked.

2. The method.

- Solve the Schrödinger equation in a well, where the solution oscillates, with $\alpha^2 = 2mE/\hbar^2$.
- Solve it in a barrier, where the solution grows or decays exponentially, with $\beta^2 = 2m(V_0 - E)/\hbar^2$.
- Make ψ and $d\psi/dx$ continuous at every edge.
- Apply Bloch's theorem: the solution in one period equals that in the next multiplied by $e^{ik(a+b)}$.
- Take the limit of very thin, very high barriers ($b \to 0$, $V_0 \to \infty$, with $V_0 b$ finite). A non-trivial solution then exists only if

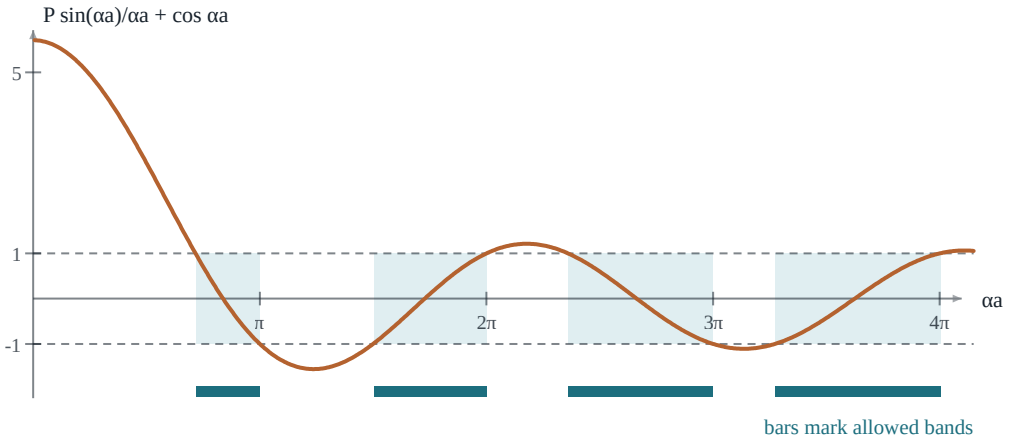
$$P \cdot \sin(\alpha a)/(\alpha a) + \cos(\alpha a) = \cos(ka), \quad \alpha = \sqrt{(2mE)}/\hbar, \quad P = mV_0 ba/\hbar^2$$

Where:

- α = a quantity that rises with electron energy E (m^{-1})
- a = well width, effectively the atomic spacing (m)
- k = electron wave vector (m^{-1})
- P = barrier strength (dimensionless)
- m = electron mass, 9.11×10^{-31} kg; E = electron energy (J)
- V_0 = barrier height (J); b = barrier width (m)
- $\hbar$ = reduced Planck constant, 1.055×10^{-34} J·s

3. How bands form. The right-hand side, cos(ka), is a cosine, so for any real k it must lie **between −1 and +1**. The left-hand side, f(αa), depends only on the energy. Plot f(αa) against αa (that is, against energy):

- it starts at 1 + P (above +1) at αa = 0 and then oscillates like a fading cosine;
- where **$-1 \le f(\alpha a) \le +1$**, a real k exists: these energies are **allowed bands**;
- where **$|f(\alpha a)| > 1$**, no real k exists, so no electron wave can travel: these energies are **forbidden gaps**.



The left-hand side of the Kronig–Penney condition against αa for P = 3π/2. Shaded strips, where the curve stays between −1 and +1, are the allowed bands; where it leaves the strip there are no states and a forbidden gap appears.

Sketch: f(αa) against αa with horizontal lines at +1 and −1; shade the allowed bands and label the gaps.

4. What the model shows.

- The energy spectrum is a series of **allowed bands separated by forbidden gaps**.
- Band edges occur at ka = nπ, i.e. k = ±nπ/a, the **Brillouin-zone boundaries**. There the electron wave is Bragg-reflected by the rows of atoms, standing waves form, and the energy splits to open a gap.
- Allowed bands get **wider at higher energy**, because the sin(αa)/(αa) term fades as αa grows.

- If **P** → ∞ (electrons tightly bound), each band shrinks to a single sharp level, as in an isolated atom.
- If **P** → **0** (no barriers), the gaps close and the free-electron result $E = \hbar^2k^2/2m$ returns.

So the model shows that bands and gaps are a direct result of an electron wave moving through a periodic potential. Whether the highest occupied band is full or partly full, and how wide the gap above it is, then decides whether a solid is a metal, semiconductor or insulator.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ periodic array of rectangular wells (width a) and barriers (V_0, width b) described or sketched
- ✓ Schrödinger equation solved in each region; continuity of ψ and dψ/dx
- ✓ Bloch's theorem applied
- ✓ final condition $P\,\sin(\alpha a)/(\alpha a) + \cos(\alpha a) = \cos(ka)$, with α and P defined
- ✓ argument: $|\cos ka| \le 1$, so only energies with $|f(\alpha a)| \le 1$ are allowed
- ✓ sketch of $f(\alpha a)$ with ±1 lines, bands and gaps labelled
- ✓ gaps at $k = n\pi/a$ (Bragg reflection)
- ✓ limits $P \to \infty$ (atomic levels) and $P \to 0$ (free electron)

Common mistakes.

- Stating the equation without the key argument. The marks are in "cos(ka) cannot exceed 1 in magnitude, so some energies have no solution".
- Plotting f against k instead of against αa (energy). The allowed/forbidden test is made along the energy axis.
- Mixing up the limits: large P means tight binding (narrow bands), not free electrons.

Remember it. "A cosine can't beat one": wherever the left-hand side goes beyond ±1, there is a gap.

Try a variant. Explain what happens to the band structure in the Kronig–Penney model as the barrier strength P is increased from zero to a very large value. *Answer:* at P = 0 the condition becomes $\cos(\alpha a) = \cos(ka)$, so $\alpha = k$ and $E = \hbar^2k^2/2m$ for every energy: no gaps, free electron. As P increases, the curve $f(\alpha a)$ rises above +1 and below −1 over wider ranges, so the forbidden gaps widen and the allowed bands narrow. As $P \to \infty$, the bands become allowed only where $\sin(\alpha a) = 0$, i.e. $\alpha a = n\pi$, giving discrete levels $E_n = n^2\pi^2\hbar^2/2ma^2$, the levels of an isolated well (electrons tightly bound to each atom).

Q25. Distinguish between valence band, conduction band, and forbidden energy gap.

> **Question type:** Distinction · **Topic:** energy bands · **Full notes:** Week 4 (Energy Bands and Carrier Concentrations)

What the examiner wants. Define each of the three clearly, then contrast them point by point: occupancy at 0 K, band edge, role in conduction, carriers and typical values. A labelled band diagram and a comparison table secure full marks.

Key terms.

- **Energy band:** a range of closely spaced allowed electron energies, formed when the levels of many atoms split as they come together.
- **Electron-volt (eV):** the energy an electron gains across 1 V; $1\text{ eV} = 1.6\times10^{-19}$ J.
- **E_c, E_v:** the energy of the bottom of the conduction band and the top of the valence band (eV).

- **Band gap (E_g):** $E_g = E_c - E_v$.
- **Hole:** an empty electron state in the valence band that behaves as a positive carrier.
- **Covalent bond:** a pair of electrons shared between neighbouring atoms (four per Si atom).

Model answer.

How the bands arise. An isolated silicon atom has sharp energy levels. When N atoms form a crystal, each level splits into N closely spaced levels, which merge into a band. In silicon the outer 3s and 3p levels (4 electrons per atom) split into a lower band and an upper band, each with room for 4 electrons per atom, separated by an energy range with no states.

Valence band. The highest energy band that is **completely filled** with electrons at absolute zero (0 K). Its top edge is E_v. Its electrons are the bonding electrons that hold the crystal together. A completely full band cannot carry current, because for every electron moving one way there is another moving the other way. When some electrons leave it, the empty states (holes) let the band conduct.

Conduction band. The next band above the valence band, **empty** at 0 K. Its bottom edge is E_c. An electron that reaches it is no longer tied to a bond and can move freely through the crystal, carrying current.

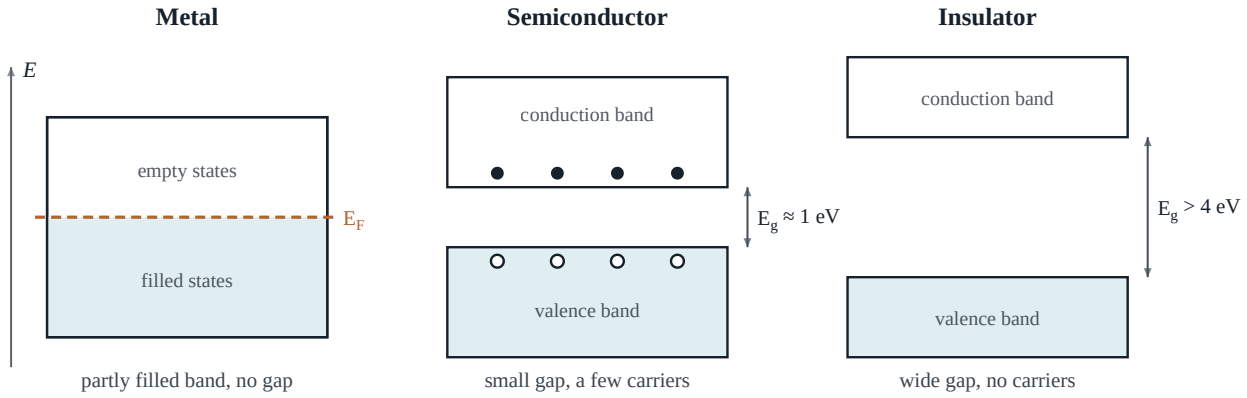
Forbidden energy gap. The range of energy between E_v and E_c in which a perfect crystal has **no allowed electron states**. Its width is

$$E_g = E_c - E_v$$

Where:

- E_g = band-gap energy (eV): Si 1.12, Ge 0.66, GaAs 1.42 eV at 300 K
- E_c = bottom of the conduction band (eV)
- E_v = top of the valence band (eV)

An electron needs at least E_g (from heat or a photon) to jump from the valence band to the conduction band. Each jump creates **two** carriers: a free electron in the conduction band and a hole in the valence band. Impurities can place a few levels inside the gap (donor and acceptor levels), but a perfect crystal has none.



Band pictures of a metal, a semiconductor and an insulator. In the semiconductor a small forbidden gap separates the full valence band from the nearly empty conduction band, so a few electrons cross by thermal excitation.

Sketch: two horizontal shaded bands; label the lower "valence band (full at 0 K)" with top E_v, the upper "conduction band (empty at 0 K)" with bottom E_c, and the space between "forbidden gap E_g". Show one electron in the conduction band and the hole it left.

Comparison.

	Valence band	Conduction band	Forbidden energy gap
Definition	Highest band completely full at 0 K	Lowest band above the valence band, empty at 0 K	Energy range between E_v and E_c with no allowed states
Edge	Top at E_v	Bottom at E_c	Width $E_g = E_c - E_v$
Electrons at 0 K	All states full	None	No states to hold any
Role of electrons	Bonding electrons, held in covalent bonds	Free electrons that move through the crystal	—
Carrier that conducts	Holes	Electrons	None (only impurity or defect levels in real crystals)
Origin	Bonding combinations of atomic orbitals	Antibonding combinations	Splitting between bonding and antibonding states
Values	—	—	Si 1.12 eV, Ge 0.66 eV, GaAs 1.42 eV, $SiO_2 \approx 9$ eV

Why the gap matters. Its width classifies solids: no gap (bands overlap or partly full) gives a metal; a small gap (about 0.1–3 eV) gives a semiconductor; a wide gap (above about 4 eV) gives an insulator.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ valence band defined as the highest band full at 0 K, top E_v
- ✓ conduction band defined as the band above it, empty at 0 K, bottom E_c
- ✓ forbidden gap defined as the range with no states, $E_g = E_c - E_v$
- ✓ conduction in the conduction band by electrons and in the valence band by holes
- ✓ an electron needs at least E_g to cross; creates an electron–hole pair
- ✓ values: Si 1.12, Ge 0.66, GaAs 1.42 eV
- ✓ labelled band diagram
- ✓ comparison table

Common mistakes.

- Saying electrons "sit in the gap". In a perfect crystal there are no states in the gap at all.
- Saying the valence band never conducts. A full band does not, but once holes exist it does.
- Defining the bands only for room temperature. "Full" and "empty" refer to 0 K; at 300 K a few electrons have crossed.

Remember it. "Valence is the full house below, conduction the empty floor above, and the gap is the staircase that isn't there."

Try a variant. What is the longest wavelength of light that can lift an electron from the valence band to the conduction band in germanium ($E_g = 0.66$ eV)? *Answer:* the photon energy must be at least E_g, so λ_{max} (μm) = 1.24/E_g (eV) = 1.24/0.66 = **1.88 μm**. Longer wavelengths pass through without creating electron–hole pairs.

Q26. Explain the difference between a direct band-gap and an indirect band-gap semiconductor.

> **Question type:** Comparison · **Topic:** direct and indirect band gaps · **Full notes:** Week 4 (Energy Bands and Carrier Concentrations)

What the examiner wants. Explain the difference using the E–k diagram (where the conduction-band minimum and valence-band maximum lie in k), explain why momentum conservation makes the phonon necessary in an indirect material, give the consequences for light emission and absorption, name examples and applications, and end with a comparison table. A sketch of both E–k diagrams earns marks.

Key terms.

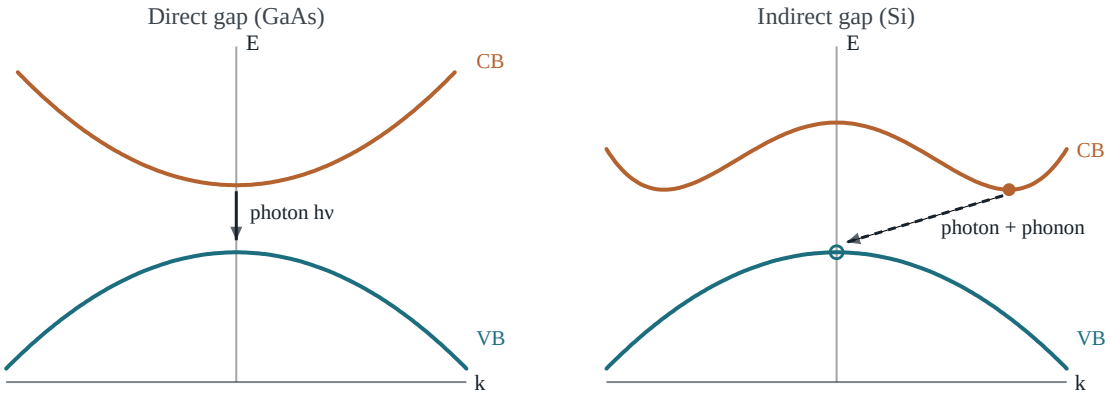
- **Wave vector (k):** measures the electron's crystal momentum, $p = \hbar k$.
- **E–k diagram (band structure):** a graph of allowed electron energy E against k.
- **Conduction-band minimum / valence-band maximum:** the lowest point of the conduction band and the highest point of the valence band on the E–k diagram.
- **Photon:** a quantum of light; it carries energy hν but almost no momentum compared with an electron in a crystal.
- **Phonon:** a quantum of lattice vibration; little energy but large momentum.
- **Radiative recombination:** an electron falls into a hole and the energy leaves as a photon.

Model answer.

The basis of the distinction. When an electron moves between the conduction band and the valence band it must conserve both **energy** and **momentum**. A photon can supply or carry away energy of order E_g, but its momentum is tiny, so a photon alone can only cause a **vertical** transition on the E–k diagram (no change in k).

Direct band gap. The conduction-band minimum lies **at the same k** as the valence-band maximum (usually at $k = 0$, the Γ point). An electron at the bottom of the conduction band can drop straight down into a hole, emitting a photon of energy $\approx E_g$. Only two particles are involved (electron and photon), so the process is fast and probable. Examples: **GaAs (1.42 eV)**, InP, GaN.

Indirect band gap. The conduction-band minimum lies **at a different k** from the valence-band maximum (in Si it is near the zone edge along the ⟨100⟩ direction). An electron dropping across the gap must also change its momentum, which a photon cannot supply. A **phonon** must be emitted or absorbed at the same instant. This three-particle event (electron, photon, phonon) is much less likely, so the electron usually loses its energy as heat through traps instead. Examples: **Si (1.12 eV)**, Ge (0.66 eV), GaP.



E–k diagrams. In GaAs (direct) the conduction-band minimum lies straight above the valence-band maximum, so an electron drops vertically and emits a photon. In Si (indirect) they are at different k, so a phonon must supply the momentum change.

Sketch: two E–k diagrams side by side. Direct: conduction-band parabola directly above the valence-band peak at k = 0, vertical arrow labelled "photon, hν = E_g". Indirect: conduction-band minimum shifted sideways; arrow with a vertical part (photon) and a horizontal part (phonon, Δk).

The emitted (or absorbed) wavelength.

$$\lambda\ (\mu m) = 1.24 / E_g\ (eV)$$

Where:

- λ = wavelength of light emitted or just absorbed (μm)
- E_g = band-gap energy (eV)
- 1.24 = hc expressed in eV·μm

GaAs: 1.24/1.42 = 0.87 μm (near infrared). Si: 1.24/1.12 = 1.11 μm.

Consequences.

- **Light emission:** direct-gap materials emit efficiently, so LEDs and laser diodes are made from them. Silicon is a very poor emitter; there is no practical silicon laser.
- **Absorption:** direct-gap materials absorb strongly just above E_g (absorption coefficient about 10^6 m^{-1}), so a few micrometres absorb most light. Indirect materials absorb weakly near E_g, so silicon solar cells must be much thicker.
- **Carrier lifetime:** radiative lifetime in direct-gap materials is about nanoseconds; in indirect-gap materials recombination is slow and mostly via traps, with lifetimes from microseconds to milliseconds.

Comparison.

	Direct band gap	Indirect band gap
Band edges on E–k diagram	CB minimum and VB maximum at the same k	CB minimum at a different k from VB maximum
Transition across the gap	Vertical; photon alone	Needs a change of k; photon plus phonon
Particles involved	Two (electron, photon)	Three (electron, photon, phonon)
Probability of radiative recombination	High	Very low
Light emission	Efficient	Poor; energy lost mainly as heat
Absorption near E_g	Strong and sharp	Weak and gradual
Minority-carrier lifetime	Short (≈ ns)	Long (µs–ms)
Examples	GaAs (1.42 eV), InP, GaN	Si (1.12 eV), Ge (0.66 eV), GaP
Typical uses	LEDs, lasers, fast photodetectors, thin solar cells	ICs, transistors, power devices, Si photodiodes and solar cells

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definitions in terms of k-position of the CB minimum and VB maximum
- ✓ energy and momentum must both be conserved; photon carries negligible momentum
- ✓ phonon needed in an indirect transition
- ✓ two labelled E–k sketches
- ✓ consequence for light emission (LEDs/lasers use direct-gap materials)
- ✓ consequence for absorption and/or lifetime
- ✓ examples of each (GaAs; Si, Ge) with E_g values
- ✓ $\lambda = 1.24/E_g$ and a worked value (GaAs 0.87 µm)
- ✓ comparison table

Common mistakes.

- Defining "direct" as "small gap" or "indirect" as "large gap". The difference is about k-position, not the size of E_g.
- Saying silicon cannot absorb light. It absorbs well above E_g (it is used in solar cells); it is emission that is poor.
- Saying a phonon supplies the energy. It mainly supplies the momentum; the photon carries almost all the energy.

Remember it. "Direct drops straight down; indirect needs a phonon to push it sideways."

Try a variant. An LED is needed at about 0.87 µm. Which of Si, Ge and GaAs is suitable, and why? *Answer:* the required gap is $E_g = 1.24/0.87 = 1.42$ eV, which matches GaAs. GaAs is also direct-gap, so it emits efficiently. Si (1.12 eV, emits near 1.11 µm) and Ge (0.66 eV, near 1.88 µm) have the wrong gap and are indirect, so they emit very weakly. **GaAs is the choice.**

Q27. What is meant by an intrinsic semiconductor? Derive the expression for intrinsic carrier concentration.

> **Question type:** Definition and derivation · **Topic:** intrinsic carrier concentration · **Full notes:** Week 4 (Energy

Bands and Carrier Concentrations); Week 5 (Carrier Statistics)

What the examiner wants. First a precise definition of an intrinsic semiconductor (pure, carriers only from thermal band-to-band excitation, $n = p = n_i$). Then a derivation: state the assumptions, start from the expressions for n and p, number every step, show that E_F cancels, and box the final result $n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT}$. A numerical check for silicon and a comment on the temperature and band-gap dependence complete the answer.

Key terms.

- **Intrinsic semiconductor:** a pure crystal with no significant impurities, in which free carriers come only from thermal excitation across the gap.
- **Intrinsic carrier concentration (n_i):** the equal electron and hole concentration of an intrinsic semiconductor (m^{-3}); 1×10^{16} m^{-3} for Si at 300 K.
- **Fermi level (E_F):** the energy at which a state has a 50 % chance of being occupied.
- **Effective density of states (N_c, N_v):** the number of states per m^3 that, placed at the band edge, give the same carrier count as the whole band; Si at 300 K: $N_c = 2.8\times10^{25}$, $N_v = 1.04\times10^{25}$ m^{-3}.
- **Boltzmann approximation:** replacing the Fermi–Dirac function by an exponential when E_F is more than about 3kT from the band edge.
- **Thermal energy (kT):** Boltzmann constant times absolute temperature; 0.0259 eV at 300 K.
- **Band gap (E_g):** $E_g = E_c - E_v$.

Model answer.

Part 1: Intrinsic semiconductor

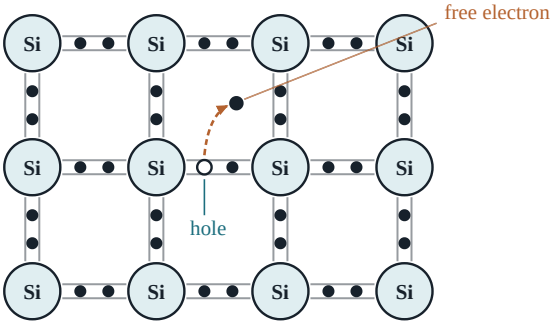
An **intrinsic semiconductor** is a pure, undoped crystal (such as perfectly pure silicon) in which the number of impurity atoms is negligible. Its free carriers are produced **only** by thermal energy breaking covalent bonds, which lifts electrons from the valence band to the conduction band. Each electron that crosses leaves exactly one hole behind, so

$$n = p = n_i$$

Where:

- n = concentration of electrons in the conduction band (m^{-3})
- p = concentration of holes in the valence band (m^{-3})
- n_i = intrinsic carrier concentration (m^{-3})

Its Fermi level, called the intrinsic level E_i, lies very close to the middle of the band gap.



Two-dimensional picture of the silicon lattice. Thermal energy has broken one covalent bond, producing a free electron and a hole together; in an intrinsic crystal this is the only source of carriers, so n = p.

Part 2: Derivation

Assumptions.

- A. The semiconductor is in **thermal equilibrium** at absolute temperature T (no light, no applied voltage).
- B. It is **non-degenerate**: E_F lies more than about 3kT below E_c and above E_v, so the Boltzmann approximation holds. This gives the standard results

$$n = N_c\, e^{-(E_c - E_F)/kT} \qquad p = N_v\, e^{-(E_F - E_v)/kT}$$

Where:

- N_c, N_v = effective densities of states in the conduction and valence bands (m^{-3})
- E_c = bottom of the conduction band (eV)
- E_v = top of the valence band (eV)
- E_F = Fermi level (eV)
- k = Boltzmann constant, 8.617×10^{-5} eV/K; T = absolute temperature (K)

(These come from integrating density of states × occupation probability over each band; $N_c = 2(2\pi m_n{*}kT/h^2)^{3/2}$ and $N_v = 2(2\pi m_p{*}kT/h^2)^{3/2}$, where $m_n{*}$, $m_p{*}$ are the electron and hole effective masses and h is Planck's constant.)

- C. The material is **intrinsic**: every free electron has a matching hole, so $n = p = n_i$.
- D. The band gap is $E_g = E_c - E_v$, and N_c, N_v and E_g are taken at the same temperature T.

Step 1. Write the electron concentration. From assumption B:

$$n = N_c \exp[\, -(E_c - E_F)/kT \,]$$

Step 2. Write the hole concentration. From assumption B:

$$p = N_v \exp[\, -(E_F - E_v)/kT \,]$$

Step 3. Multiply the two. Multiply n by p and combine the exponentials ($e^A \times e^B = e^{A+B}$):

$$n\,p = N_c\, N_v \exp[\, -(E_c - E_F)/kT - (E_F - E_v)/kT \,]$$

Step 4. Simplify the exponent: E_F cancels. Inside the bracket, $-(E_c - E_F) - (E_F - E_v) = -E_c + E_F - E_F + E_v = -(E_c - E_v)$. So

$$n\,p = N_c\, N_v \exp[\, -(E_c - E_v)/kT \,]$$

Step 5. Introduce the band gap. Using assumption D, $E_c - E_v = E_g$:

$$n\,p = N_c\, N_v \exp(\, -E_g/kT \,)$$

The product np depends only on the material (N_c, N_v, E_g) and the temperature, not on where the Fermi level is.

Step 6. Apply the intrinsic condition. By assumption C, $n = p = n_i$, so $n\,p = n_i \times n_i = n_i^2$:

$$n_i^2 = N_c\, N_v \exp(\, -E_g/kT \,)$$

Step 7. Take the square root. $\sqrt{(e^{-E_g/kT})} = e^{-E_g/2kT}$, so:

$$n_i = \sqrt{(N_c\, N_v)} \cdot \exp(\, -E_g / 2kT \,)$$

Final result: $n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT}$

Where:

- n_i = intrinsic carrier concentration (m^{-3})
- N_c, N_v = effective densities of states (m^{-3})
- E_g = band gap (eV)
- kT = thermal energy (eV); 0.0259 eV at 300 K

Step 8 (explicit temperature dependence). Substituting N_c and N_v from assumption B:

$$n_i = 2\,(2\pi kT/h^2)^{3/2}\,(m_n{*}\,m_p{*})^{3/4}\,\exp(-E_g/2kT) \quad \propto \quad T^{3/2}\exp(-E_g/2kT)$$

The exponential dominates: n_i rises very steeply with temperature and falls very steeply as E_g increases.

Step 9 (position of the intrinsic Fermi level). Setting n = p in Steps 1 and 2 and solving for $E_F = E_i$:

$$E_i = (E_c + E_v)/2 + (kT/2)\ln(N_v/N_c)$$

For Si: (0.0259/2) × ln(1.04/2.8) = −0.0128 eV, so E_i lies only about 13 meV below mid-gap.

Part 3: Numerical check for silicon at 300 K

Given: $N_c = 2.8\times10^{25}$ m^{-3}, $N_v = 1.04\times10^{25}$ m^{-3}, $E_g = 1.12$ eV, kT = 0.0259 eV.

Find: n_i.

Solution.

Step 1. $\sqrt{(N_cN_v)} = \sqrt{(2.8\times10^{25}\times1.04\times10^{25})} = \sqrt{(2.91\times10^{50})} = 1.71\times10^{25}$ m^{-3}.

Step 2. Exponent: $E_g/2kT = 1.12/(2\times0.0259) = 1.12/0.0518 = 21.62$, so $\exp(-21.62) = 4.07\times10^{-10}$.

Step 3. $n_i = 1.71\times10^{25}\times4.07\times10^{-10} = 6.95\times10^{15}$ m^{-3}.

Answer: $n_i \approx 7\times10^{15}$ m^{-3}, the same order as the accepted value 1×10^{16} m^{-3}.

What this means: the formula gives the right size of n_i from just three material numbers. The small difference from 1×10^{16} m^{-3} comes from the exact values of E_g and the effective masses; because E_g sits in an exponent, a change of only 0.02 eV (to $E_g \approx 1.10$ eV) would give exactly 1×10^{16} m^{-3}.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definition: pure crystal, carriers only from thermal excitation, $n = p = n_i$
- ✓ assumptions stated (equilibrium, non-degenerate/Boltzmann, intrinsic n = p)
- ✓ $n = N_c\,e^{-(E_c-E_F)/kT}$ and $p = N_v\,e^{-(E_F-E_v)/kT}$, symbols defined
- ✓ product np formed and E_F shown to cancel
- ✓ $E_c - E_v$ replaced by E_g
- ✓ $n = p = n_i$ used to write $n_i^2 = N_cN_v\,e^{-E_g/kT}$
- ✓ final boxed result $n_i = \sqrt{(N_cN_v)}\,e^{-E_g/2kT}$
- ✓ $T^{3/2}\,e^{-E_g/2kT}$ dependence and/or E_i near mid-gap
- ✓ numerical check for Si giving ≈ 7×10^{15} (order 10^{16}) m^{-3}

Common mistakes.

- Losing the factor 2 in the exponent: the final exponent is $-E_g/\mathbf{2}kT$, because of the square root. Using $-E_g/kT$ gives 2.8×10^{6} m^{-3} for Si, which is absurd.

- Adding instead of multiplying n and p. Only the product makes E_F cancel.
- Sign errors in the exponents: n uses $(E_c - E_F)$ and p uses $(E_F - E_v)$, both positive in a non-degenerate semiconductor.
- Mixing units: if E_g is in eV, kT must be in eV (0.0259 eV), not in joules.

Remember it. "Multiply n and p, the Fermi level flees, and half the gap stays in the exponent."

Try a variant. For GaAs at 300 K, $N_c = 4.7\times10^{23}$ m^{-3}, $N_v = 7.0\times10^{24}$ m^{-3} and $E_g = 1.42$ eV. Calculate n_i and compare with silicon. *Answer:* $\sqrt{(N_c N_v)} = \sqrt{(4.7\times10^{23} \times 7.0\times10^{24})} = 1.81\times10^{24}$ m^{-3}; $E_g/2kT = 1.42/0.0518 = 27.41$, $\exp(-27.41) = 1.24\times10^{-12}$; $n_i = 1.81\times10^{24} \times 1.24\times10^{-12} =$ **2.3×10^{12} m^{-3}**, about 3000 times smaller than in Si. The wider gap (by 0.30 eV) alone reduces the exponential by a factor $e^{0.30/0.0518} \approx 330$, and the smaller N_c (light electron mass) accounts for the rest.

Q28. Explain the difference between intrinsic and extrinsic carrier concentrations.

> **Question type:** Comparison · **Topic:** intrinsic and extrinsic carrier concentrations · **Full notes:** Week 4 (Energy Bands and Carrier Concentrations)

What the examiner wants. Explain what each concentration is, where the carriers come from, how each is calculated, how it depends on temperature, and how the two are linked by the mass-action law. Typical numbers for silicon and a closing comparison table secure full marks.

Key terms.

- **Intrinsic carrier concentration (n_i):** the equal electron and hole concentration in a pure semiconductor (m^{-3}).
- **Extrinsic semiconductor:** a doped semiconductor whose carrier concentrations are set mainly by its dopants.
- **Donor / acceptor:** an impurity that gives an electron to the conduction band (e.g. P in Si; concentration N_D) / takes an electron from the valence band, creating a hole (e.g. B in Si; concentration N_A).
- **Complete ionisation:** every dopant atom has given up its electron (donor) or hole (acceptor); true for Si at room temperature.
- **Majority / minority carrier:** the carrier present in larger / smaller numbers (electrons / holes in n-type).
- **Mass-action law:** $np = n_i^2$ in thermal equilibrium, whatever the doping.
- **Fermi level (E_F) and intrinsic level (E_i):** the 50 %-occupancy energy, and its value in the pure material (near mid-gap).

Model answer.

1. Intrinsic carrier concentration. In a pure semiconductor, carriers come only from thermal excitation: heat breaks covalent bonds and lifts electrons from the valence band into the conduction band, each leaving a hole. Electrons and holes are therefore always equal:

$$n = p = n_i = \sqrt{(N_c\, N_v)}\; e^{-E_g/2kT}$$

Where:

- n, p = electron and hole concentrations (m^{-3})
- n_i = intrinsic carrier concentration (m^{-3})
- N_c, N_v = effective densities of states in the conduction and valence bands (m^{-3}); Si: 2.8×10^{25} and 1.04×10^{25}

- E_g = band gap (eV); Si 1.12 eV
- kT = thermal energy, 0.0259 eV at 300 K

n_i is fixed by the material (through E_g) and the temperature; the designer cannot change it. For Si at 300 K, $n_i = 1\times10^{16}$ m^{-3}, so only about one atom in 5×10^{12} provides a carrier, and pure silicon is a poor conductor ($\sigma = qn_i(\mu_n + \mu_p) = 1.6\times10^{-19} \times 10^{16} \times 0.183 \approx 2.9\times10^{-4}$ S/m). n_i rises exponentially with temperature.

2. Extrinsic carrier concentration. In a doped semiconductor most carriers come from ionised dopant atoms, not from broken bonds. Electrons and holes are no longer equal. With complete ionisation and doping much greater than n_i:

$$\text{n-type: } n \approx N_D, \quad p = n_i^2/N_D \qquad \text{p-type: } p \approx N_A, \quad n = n_i^2/N_A$$

Where:

- N_D = donor concentration (m^{-3})
- N_A = acceptor concentration (m^{-3})
- n_i = intrinsic carrier concentration (m^{-3})

The majority-carrier concentration is set by the designer through the doping (typically 10^{20}–10^{26} m^{-3} in Si) and stays nearly constant over a wide temperature range (the extrinsic range, roughly 150–450 K for moderately doped Si). At very low temperature the dopants are not all ionised (freeze-out); at very high temperature n_i overtakes the doping and the material behaves intrinsically again.

3. The link: the mass-action law. In thermal equilibrium,

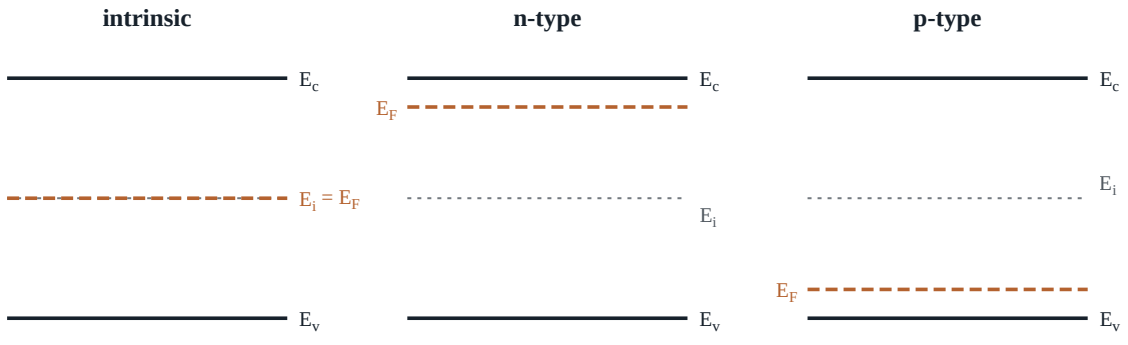
$$n\,p = n_i^2$$

Where:

- n, p = equilibrium electron and hole concentrations (m^{-3})
- n_i = intrinsic carrier concentration (m^{-3})

Doping raises one carrier type and lowers the other so that the product is unchanged. Example, Si with $N_D = 1\times10^{22}$ m^{-3}: $n = 1\times10^{22}$ m^{-3} and $p = (10^{16})^2/10^{22} = 1\times10^{10}$ m^{-3}. The electron concentration is 10^6 times n_i and the hole concentration 10^6 times smaller. Conductivity rises to $\sigma = qn\mu_n = 1.6\times10^{-19} \times 10^{22} \times 0.135 \approx 2.2\times10^2$ S/m, about 7×10^5 times the intrinsic value.

4. Position of the Fermi level. In intrinsic material $E_F = E_i$, near mid-gap. In n-type material E_F moves up towards E_c; in p-type it moves down towards E_v.



$E_F - E_i = kT \ln(N_D/n_i)$ for n-type; $E_i - E_F = kT \ln(N_A/n_i)$ for p-type

Fermi level position: at mid-gap (E_i) in intrinsic material, close to E_c in n-type and close to E_v in p-type.

Sketch: three band diagrams side by side (intrinsic, n-type, p-type), with E_F drawn at mid-gap, near E_c and near E_v.

Comparison.

	Intrinsic carrier concentration	Extrinsic carrier concentration
Material	Pure (undoped) semiconductor	Doped semiconductor (n-type or p-type)
Source of carriers	Thermal breaking of bonds (band-to-band excitation) only	Mainly ionised donors or acceptors; small thermal contribution
Electrons vs holes	Equal: $n = p = n_i$	Unequal: majority and minority carriers
Formula	$n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT}$	$n \approx N_D$, $p = n_i^2/N_D$ (n-type); $p \approx N_A$, $n = n_i^2/N_A$ (p-type)
Typical value, Si, 300 K	1×10^{16} m^{-3}	Majority 10^{20}–10^{26} m^{-3}
Set by	Material (E_g) and temperature	Doping level, chosen by the designer
Temperature dependence	Very strong, exponential	Almost constant in the extrinsic range; intrinsic again at high T
Fermi level	At E_i, near mid-gap	Towards E_c (n-type) or E_v (p-type)
Conductivity	Low (Si $\approx 3\times10^{-4}$ S/m)	High, controlled by doping

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ intrinsic: pure material, carriers from thermal excitation, $n = p = n_i$
- ✓ formula $n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT}$ with symbols defined; Si value 1×10^{16} m^{-3}
- ✓ extrinsic: doped material, carriers mainly from ionised dopants, $n \neq p$
- ✓ $n \approx N_D$, $p = n_i^2/N_D$ (and the p-type equivalent)
- ✓ mass-action law $np = n_i^2$ linking the two
- ✓ temperature dependence contrasted (exponential vs nearly constant, then intrinsic at high T)
- ✓ Fermi-level positions
- ✓ a numerical example and a comparison table

Common mistakes.

- Thinking doping leaves the minority carriers unchanged. Doping *reduces* the minority concentration ($np = n_i^2$ stays fixed).
- Writing $n = N_D + n_i$ as if the intrinsic electrons simply add. For $N_D \gg n_i$, $n \approx N_D$ and the thermal electrons are negligible; the exact form needs the charge-neutrality formula.
- Saying extrinsic concentration "does not depend on temperature at all". It is constant only over the extrinsic range; freeze-out and intrinsic behaviour occur at the extremes.

Remember it. "Intrinsic is nature's choice, extrinsic is the designer's choice, and $np = n_i^2$ keeps them honest."

Try a variant. Silicon at 300 K is doped with $N_A = 1\times10^{23}$ m^{-3} boron atoms. Find the hole and electron concentrations and say which is the majority carrier. *Answer:* $p \approx N_A = \mathbf{1\times10^{23}\ m^{-3}}$ (majority); $n = n_i^2/N_A = (1\times10^{16})^2/10^{23} = \mathbf{1\times10^{9}\ m^{-3}}$ (minority). The material is p-type, and its Fermi level lies close to E_v.

Q29. Silicon has an intrinsic carrier concentration of approximately 1.0×10^{10} cm^{-3} at a specified temperature. If the electron and hole concentrations are equal, determine the electron and hole concentrations.

> **Question type:** Calculation · **Topic:** intrinsic carrier concentration and the mass-action law · **Full notes:** Week 4 (Energy Bands and Carrier Concentrations)

What the examiner wants. A short calculation. Marks come from recognising that n = p means the silicon is intrinsic, writing the mass-action law $np = n_i^2$, solving it to get $n = p = n_i$, and giving the value with its unit (m^{-3}). A check and a one-line physical comment earn the final marks.

Key terms.

- **Electron concentration (n):** the number of free electrons in the conduction band per m^3 (m^{-3}).
- **Hole concentration (p):** the number of holes (empty bond states in the valence band) per m^3 (m^{-3}).
- **Intrinsic semiconductor:** a pure semiconductor whose free carriers come only from thermally broken bonds, so each broken bond gives one electron and one hole and n = p.
- **Intrinsic carrier concentration (n_i):** the common value of n and p in intrinsic material; for Si at 300 K, $n_i = 1\times10^{16}$ m^{-3}.
- **Mass-action law:** in thermal equilibrium the product np equals n_i^2 at a given temperature, whatever the doping.
- **Thermal equilibrium:** the steady state of a sample at fixed temperature with no applied voltage, light or current.

Model answer.

Given: $n_i = 1.0\times10^{10}$ $cm^{-3} = 1.0\times10^{16}$ m^{-3}; n = p.

Find: the electron concentration n and hole concentration p.

Formula: the mass-action law

$$n\,p = n_i^2$$

Where:

- n = electron concentration (m^{-3})
- p = hole concentration (m^{-3})
- n_i = intrinsic carrier concentration (m^{-3})

Solution.

Step 1. Set n = p in the mass-action law:

$$n \times n = n_i^2 \quad \Rightarrow \quad n^2 = n_i^2 \quad \Rightarrow \quad n = n_i$$ (the positive root, since a concentration cannot be negative)

Step 2. Substitute:

$$n = p = 1.0\times10^{16}\ m^{-3}$$

Step 3. Check: $np = (1.0\times10^{16})(1.0\times10^{16}) = 1.0\times10^{32}\ m^{-6} = n_i^2$. ✓

Answer: $n = p = n_i = 1.0\times10^{16}$ m^{-3}.

What this means: equal electron and hole concentrations are the signature of intrinsic (undoped) silicon, where every free electron came from a broken bond that also left a hole. Silicon has about 5×10^{28} atoms per m^3, so only about one atom in 5×10^{12} has lost an electron. Pure silicon is therefore a poor conductor: $\sigma = qn_i(\mu_n + \mu_p) = 1.6\times10^{-19} \times 10^{16} \times (0.135 + 0.048) \approx 2.9\times10^{-4}$ S/m, a resistivity of about 3.4×10^{3} Ω·m.

Sketch (optional, for a fuller answer): a band diagram with E_c at the top and E_v at the bottom, a few electrons (dots) just above E_c and the same number of holes (circles) just below E_v, and $E_F = E_i$ drawn as a dashed line near mid-gap.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ n = p identified as the intrinsic condition
- ✓ mass-action law $np = n_i^2$ stated, with symbols defined
- ✓ algebra $n^2 = n_i^2 \Rightarrow n = n_i$ shown
- ✓ answer $n = p = 1.0\times10^{16}$ m^{-3}, with the unit
- ✓ check that $np = 1.0\times10^{32}$ m^{-6}
- ✓ one-line physical comment (each broken bond gives one electron and one hole; very few carriers)

Common mistakes.

- Writing $n = p = n_i/2$ (thinking the carriers "share" n_i). n_i is the concentration of each carrier type, not the total; the total is $2n_i$.
- Writing $n = n_i^2 = 10^{32}$ m^{-3}. That is the product np, in m^{-6}, not a concentration.
- Leaving out the unit, or writing m^3 instead of m^{-3} (per cubic metre).

Remember it. "Pure means equal": in intrinsic material $n = p = n_i$, because every broken bond makes one of each.

Try a variant. Germanium at 300 K has $n_i = 2.4\times10^{19}$ m^{-3}. If it is undoped, find n, p and the product np. *Answer:* $n = p = n_i =$ **2.4×10^{19} m^{-3}**; $np = (2.4\times10^{19})^2 =$ **5.76×10^{38} m^{-6}**. Germanium has far more carriers than silicon because its band gap (0.66 eV) is smaller than silicon's (1.12 eV).

Q30. Explain the Fermi-Dirac distribution function and its significance in semiconductor physics.

> **Question type:** Explain · **Topic:** Fermi–Dirac statistics · **Full notes:** Week 5 (Carrier Statistics)

What the examiner wants. "Explain" asks for the equation with every symbol defined, what the function means physically (a probability of occupation), its key properties, and then why it matters: how it is used to find carrier concentrations and why the Fermi level it defines is so important. A labelled sketch of f(E) earns marks.

Key terms.

- **Fermion:** a particle, such as the electron, that obeys the Pauli exclusion principle.
- **Pauli exclusion principle:** no two electrons can occupy the same quantum state (at most one electron per state, counting spin).
- **Allowed state:** an energy level that an electron may occupy in the crystal (in the conduction or valence band, not in the gap).
- **Fermi level (E_F):** the energy at which a state has exactly a 50 % chance of being occupied.
- **Thermal energy (kT):** Boltzmann's constant times absolute temperature; 0.0259 eV at 300 K.

- **Boltzmann approximation:** the simpler form $f(E) \approx e^{-(E - E_F)/kT}$, valid when $E - E_F$ is more than about 3kT.
- **Carrier concentration (n, p):** the number of free electrons (n) or holes (p) per m^3.

Model answer.

1. The function

Electrons are fermions: they obey the Pauli exclusion principle, so each allowed state can hold at most one electron. In thermal equilibrium at temperature T, the probability that an allowed state of energy E is occupied by an electron is the Fermi–Dirac distribution function:

$$f(E) = 1 / (1 + e^{(E - E_F)/kT})$$

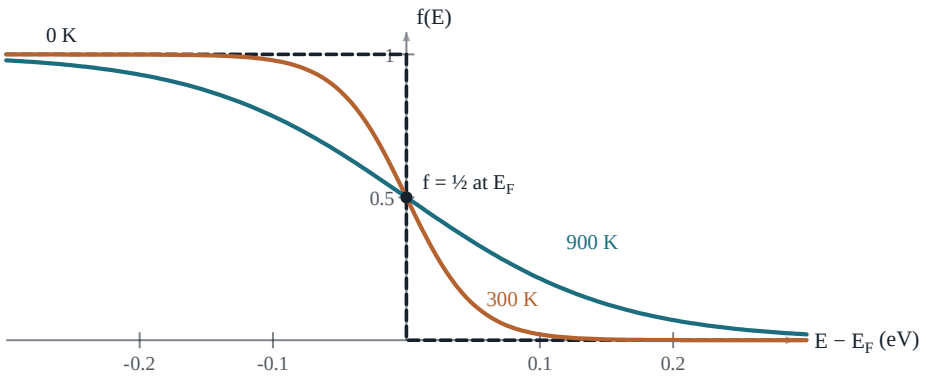
Where:

- f(E) = probability that a state at energy E is occupied (dimensionless, 0 to 1)
- E = energy of the state (eV)
- E_F = Fermi level (eV)
- k = Boltzmann constant, 8.617×10^{-5} eV/K
- T = absolute temperature (K); kT = 0.0259 eV at 300 K

The probability that the state is **empty** is 1 − f(E). An empty state in the valence band is a hole, so 1 − f(E) is the probability of finding a hole.

2. Properties

- **Bounded:** $0 \le f(E) \le 1$, because a state holds at most one electron.
- **Half at E_F:** at $E = E_F$ the exponent is zero, so $f = 1/(1 + 1) = ½$ at every temperature.
- **Symmetric about E_F:** a state ΔE above E_F is as likely to be full as a state ΔE below E_F is to be empty: $f(E_F + \Delta E) = 1 - f(E_F - \Delta E)$.
- **Width of a few kT:** f changes from nearly 1 to nearly 0 over about ±3kT (±0.08 eV at 300 K). For example, at 300 K a state 0.1 eV above E_F has f = 0.021, and one 0.1 eV below has f = 0.979.
- **Boltzmann tail:** for $E - E_F > 3kT$, $f(E) \approx e^{-(E - E_F)/kT}$, accurate to within 5 %. Most semiconductor calculations use this form.
- **At 0 K** it becomes a sharp step: f = 1 below E_F and f = 0 above.



The Fermi–Dirac function f(E): a sharp step at E_F at 0 K, smoothed over a few kT at higher temperatures, and always equal to ½ at E_F.

Sketch: f(E) on the horizontal axis (0 to 1) against E on the vertical axis; mark E_F where f = ½; draw the 0 K step and a smooth 300 K curve crossing it at E_F.

3. Significance in semiconductor physics

1. **It gives the carrier concentrations.** The number of electrons is "available states × probability occupied", summed over the conduction band; holes use 1 − f(E) over the valence band (see the equations below). Without f(E), n, p and n_i cannot be calculated.
2. **It defines the Fermi level.** The position of E_F between E_v and E_c tells at once whether the material is intrinsic (E_F near mid-gap), n-type (E_F near E_c) or p-type (E_F near E_v), and how heavily it is doped.
3. **Constant E_F in equilibrium.** In any system in equilibrium E_F is the same everywhere. This rule produces the band bending and built-in voltage of p-n junctions, metal–semiconductor contacts and MOS structures.
4. **It explains temperature effects:** more electrons in the high-energy tail as T rises, hence more carriers, intrinsic behaviour at high T and freeze-out at low T.
5. **Degeneracy:** when E_F comes within 3kT of a band edge (very heavy doping), the Boltzmann form fails and the full Fermi–Dirac function must be used.
6. **Out of equilibrium** (under bias or light) the idea is extended to separate quasi-Fermi levels for electrons and holes.

The carrier-concentration integrals are:

$$n = \int g_c(E)\, f(E)\, dE, \qquad p = \int g_v(E)\, [1 - f(E)]\, dE$$

Where:

- $g_c(E)$, $g_v(E)$ = density of states in the conduction and valence bands: allowed states per unit energy per unit volume ($m^{-3}eV^{-1}$)
- the integrals run over the conduction band (for n) and the valence band (for p)

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ $f(E) = 1/(1 + e^{(E - E_F)/kT})$ written correctly, every symbol defined
- ✓ stated as the probability that a state at energy E is occupied; based on the Pauli principle
- ✓ $f(E_F) = ½$ at any temperature
- ✓ 1 − f(E) = probability of a hole
- ✓ step at 0 K, smoothed over a few kT at higher T (labelled sketch)
- ✓ Boltzmann approximation for $E - E_F > 3kT$
- ✓ use in $n = \int g_c\, f\, dE$ and $p = \int g_v(1 - f) dE$
- ✓ E_F position indicates n-type, p-type or intrinsic; E_F is flat in equilibrium

Common mistakes.

- Saying f(E) is the number of electrons at energy E. It is only a probability; the number also needs the density of states g(E).
- Writing the exponent with the wrong sign, $e^{(E_F - E)/kT}$, which makes high-energy states full. Check: large E must give $f \to 0$.
- Defining E_F as "the highest filled level" without saying this is true only at 0 K. At T > 0 the correct definition is the energy where f = ½.

Remember it. "Half full at the Fermi level": f(E) is a probability that falls from 1 to 0 across E_F over a few kT.

Try a variant. At 300 K, what is the probability that a state 3kT above the Fermi level is occupied, and how good is the Boltzmann approximation there? *Answer:* $f = 1/(1 + e^3) = 1/(1 + 20.09) =$ **0.047**; Boltzmann gives $e^{-3} =$ **0.050**, about 5

% too high, which is why 3kT is taken as the limit of the approximation.

Q31. Write the Fermi-Dirac distribution equation and explain what happens to the distribution at absolute zero.

> **Question type:** State and explain · **Topic:** Fermi–Dirac function at T = 0 K · **Full notes:** Week 5 (Carrier Statistics)

What the examiner wants. Write the equation and define every symbol, then show mathematically (by taking the limit $kT \to 0$ for E below and above E_F) that f(E) becomes a step, and state what that means physically for a semiconductor: full valence band, empty conduction band, no free carriers. A sketch of the step and a remark on how it softens as T rises complete the answer.

Key terms.

- **Fermi–Dirac function (f(E)):** the probability that an allowed electron state at energy E is occupied in thermal equilibrium.
- **Fermi level (E_F):** the energy at which f = ½; at 0 K, the boundary between filled and empty states.
- **Absolute zero (0 K):** the lowest possible temperature, where thermal energy kT = 0.
- **Valence band (E_v):** the band of bond electrons, full at 0 K; E_v is its top edge.
- **Conduction band (E_c):** the band of free electrons, empty at 0 K; E_c is its bottom edge.
- **Free carrier:** an electron in the conduction band or a hole in the valence band, able to carry current.

Model answer.

The equation

$$f(E) = 1 / (1 + e^{(E - E_F)/kT})$$

Where:

- f(E) = probability that a state at energy E is occupied (dimensionless)
- E = energy of the state (eV)
- E_F = Fermi level (eV)
- k = Boltzmann constant, 8.617×10^{-5} eV/K
- T = absolute temperature (K)

At absolute zero

At T = 0 K, $kT \to 0$, so the exponent $(E - E_F)/kT$ goes to $\pm\infty$ depending on the sign of $E - E_F$.

Case 1, states below E_F ($E < E_F$). $E - E_F$ is negative, so the exponent $\to -\infty$ and $e^{-\infty} = 0$:

$$f(E) = 1/(1 + 0) = 1$$

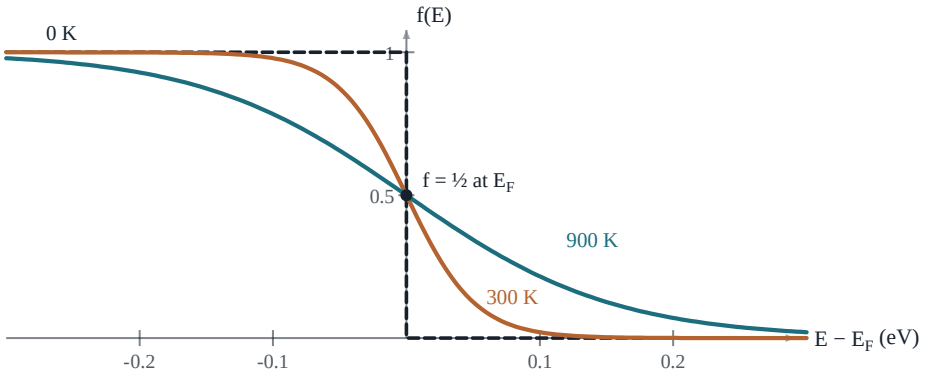
Every state below E_F is certainly filled.

Case 2, states above E_F ($E > E_F$). $E - E_F$ is positive, so the exponent $\to +\infty$ and $e^{+\infty} \to \infty$:

$$f(E) = 1/(1 + \infty) = 0$$

Every state above E_F is certainly empty.

So at 0 K the distribution is a **sharp rectangular step** at $E = E_F$. The Fermi level is then exactly the boundary between completely filled and completely empty states, which is why it is sometimes described as the highest occupied energy at 0 K. (At $E = E_F$ itself the step is conventionally given the value ½.)



At 0 K the Fermi–Dirac function is a step: every state below E_F is full and every state above is empty. At higher temperatures the step is rounded over a few kT but still passes through ½ at E_F.

Sketch: E on the vertical axis, f(E) from 0 to 1 on the horizontal axis; draw the 0 K curve as a vertical-sided step at E_F, and a smooth S-shaped curve for T > 0 crossing ½ at E_F.

Physical meaning for a semiconductor

In a semiconductor E_F lies inside the band gap. At 0 K every valence-band state (below E_F) is full and every conduction-band state (above E_F) is empty. There are no electrons in the conduction band and no holes in the valence band, so there are **no free carriers**: a pure semiconductor is a **perfect insulator at absolute zero**.

As the temperature rises

The step softens over about ±3kT around E_F: a few states just below E_F empty and a few just above fill. The high-energy tail reaches the conduction band and supplies free electrons; the matching empty states in the valence band are holes. That is why a semiconductor conducts at room temperature. $f(E_F) = ½$ at every temperature.

$E - E_F$	f at 0 K	f at 300 K (kT = 0.0259 eV)
−0.1 eV	1	0.979
0	½ (edge of step)	0.500
+0.1 eV	0	0.021
+0.3 eV	0	9.3×10^{-6}

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ $f(E) = 1/(1 + e^{(E - E_F)/kT})$ with every symbol defined
- ✓ limit for $E < E_F$: exponent → −∞, f = 1
- ✓ limit for $E > E_F$: exponent → +∞, f = 0
- ✓ "step function at E_F" stated, with a labelled sketch
- ✓ E_F is the boundary between filled and empty states at 0 K
- ✓ semiconductor at 0 K: valence band full, conduction band empty, no free carriers, insulator

- ✓ as T rises the step is smoothed over a few kT; $f(E_F) = ½$

Common mistakes.

- Only stating "f is a step" without showing the two limits; the explanation marks are for the mathematics ($e^{-\infty} = 0$, $e^{+\infty} = \infty$).
- Saying all electrons are "at" E_F at 0 K. They fill every state up to E_F; E_F is the top of the filled region, not where the electrons sit.
- Concluding that a semiconductor at 0 K conducts like a metal. The Fermi level is in the gap, where there are no states, so the filled band is completely full and cannot conduct.

Remember it. "At zero, the Fermi function is a cliff": everything below E_F full, everything above empty.

Try a variant. At 300 K, find the probability that a state 0.05 eV above the Fermi level is occupied, and the value at 0 K. *Answer:* $(E - E_F)/kT = 0.05/0.0259 = 1.93$; $f = 1/(1 + e^{1.93}) = 1/(1 + 6.89) =$ **0.127** at 300 K; at 0 K the state is above E_F, so **f = 0**.

Q32. Define the density of states in a semiconductor and explain its importance in determining carrier concentration.

> **Question type:** Define and explain · **Topic:** density of states · **Full notes:** Week 5 (Carrier Statistics)

What the examiner wants. A precise definition with units, the formulas for the conduction and valence bands (square-root dependence, zero in the gap), and an explanation of how g(E) combines with the Fermi–Dirac function to give n and p, leading to the effective densities of states N_c and N_v. A sketch of g(E) against E earns marks.

Key terms.

- **Allowed state:** an electron energy level permitted in the crystal; each can hold one electron (two per orbital, one for each spin).
- **Density of states (g(E)):** the number of allowed states per unit energy per unit volume at energy E ($m^{-3}eV^{-1}$).
- **Effective mass (m_n*, m_p*):** the mass an electron or hole appears to have when it moves through the crystal; it describes the curvature of the band.
- **Fermi–Dirac function (f(E)):** the probability that a state at energy E is occupied.
- **Effective density of states (N_c, N_v):** the number of states which, if all placed at the band edge, would give the same carrier concentration as the real band; Si at 300 K: $N_c = 2.8\times10^{25}$, $N_v = 1.04\times10^{25}$ m^{-3}.
- **Band gap ($E_g = E_c - E_v$):** the energy range between the top of the valence band E_v and the bottom of the conduction band E_c, where there are no allowed states.

Model answer.

Definition

The **density of states** g(E) is the number of allowed electron energy states (counting both spins) per unit energy interval, per unit volume of crystal, at energy E. Its unit is $m^{-3}eV^{-1}$ (or $m^{-3}J^{-1}$). So g(E)dE is the number of states per m^3 with energies between E and E + dE.

For parabolic bands:

$$g_c(E) = (4\pi\,(2m_n{*})^{3/2} / h^3)\,\sqrt{(E - E_c)} \quad \text{for } E \geq E_c$$
$$g_v(E) = (4\pi\,(2m_p{*})^{3/2} / h^3)\,\sqrt{(E_v - E)} \quad \text{for } E \leq E_v$$
$$g(E) = 0 \quad \text{for } E_v < E < E_c$$

Where:

- $g_c(E)$, $g_v(E)$ = density of states in the conduction and valence bands ($m^{-3}J^{-1}$ as written with SI values of $m{*}$ and h; multiply by 1.6×10^{-19} J/eV to get $m^{-3}eV^{-1}$)
- $m_n{*}$, $m_p{*}$ = density-of-states effective masses of electrons and holes (kg)
- h = Planck constant, 6.626×10^{-34} J·s
- E_c = bottom of the conduction band, E_v = top of the valence band (eV)

In words: there are no states in the gap; just inside each band the number of states starts at zero and grows as the square root of the distance into the band. A heavier effective mass (a flatter band) gives more states per unit energy.

Where it comes from (briefly). The allowed electron wave vectors k in a crystal of volume V are evenly spaced, one per $(2\pi)^3/V$ of k-space. Counting the states inside a sphere of radius k (times 2 for spin), and using $E - E_c = \hbar^2k^2/2m_n{*}$ ($\hbar = h/2\pi$), gives a number of states that grows as $(E - E_c)^{3/2}$; differentiating with respect to E gives the $\sqrt{(E - E_c)}$ law.

Sketch: E on the vertical axis; $g_c(E)$ as a half-parabola opening upward from zero at E_c; $g_v(E)$ opening downward from zero at E_v; g = 0 in the gap between them. Optionally add f(E) beside it and shade the product g_c f (electrons, a narrow peak just above E_c) and $g_v(1 - f)$ (holes, just below E_v).

Importance in determining carrier concentration

- **Carriers = states × probability.** The Fermi function gives only the probability that a state is occupied; g(E) gives how many states there are. The carrier concentrations are their product, summed over each band:

  $$n = \int g_c(E)\, f(E)\, dE, \qquad p = \int g_v(E)\, [1 - f(E)]\, dE$$

Where:

- n, p = electron and hole concentrations (m^{-3})
- f(E) = Fermi–Dirac occupation probability; 1 − f(E) = probability a state is empty (a hole)
- **It produces N_c and N_v.** Doing the integral with the Boltzmann approximation gives

  $$n = N_c\, e^{-(E_c - E_F)/kT}, \qquad p = N_v\, e^{-(E_F - E_v)/kT}$$
  $$N_c = 2\,(2\pi\, m_n{*}\, kT / h^2)^{3/2}, \quad N_v = 2\,(2\pi\, m_p{*}\, kT / h^2)^{3/2}$$

Where:

- N_c, N_v = effective densities of states (m^{-3}); Si at 300 K: 2.8×10^{25} and 1.04×10^{25} m^{-3}
- E_F = Fermi level (eV); kT = thermal energy (0.0259 eV at 300 K; in J inside the N_c formula)

These appear in every carrier-concentration formula, including $n_i = \sqrt{(N_cN_v)}\, e^{-E_g/2kT}$.

- **It explains the "forbidden" gap.** f(E) is not zero in the gap, but g(E) is, so no carriers can exist there.
- **It sets the temperature dependence.** N_c and N_v grow as $T^{3/2}$, which forms part of the temperature dependence of n_i.
- **It explains differences between materials.** GaAs has a small electron effective mass ($0.067m_0$), so its N_c (4.7×10^{23} m^{-3}) is about 60 times smaller than silicon's. For the same doping its Fermi level sits closer to E_c, and it

becomes degenerate at lower doping.

- **It fixes the intrinsic level.** Because $N_v \neq N_c$ in Si, the intrinsic Fermi level E_i sits slightly below mid-gap.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definition: number of states per unit energy per unit volume, unit $m^{-3}eV^{-1}$
- ✓ $g_c \propto \sqrt{(E - E_c)}$ and $g_v \propto \sqrt{(E_v - E)}$; $g = 0$ in the gap
- ✓ dependence on effective mass, $(m*)^{3/2}$
- ✓ labelled sketch of g(E) against E
- ✓ $n = \int g_c\, f\, dE$ and $p = \int g_v(1 - f)dE$
- ✓ leads to N_c, N_v and $n = N_c\, e^{-(E_c - E_F)/kT}$, with Si values
- ✓ at least two further points of importance (gap has no carriers, $T^{3/2}$, material differences)

Common mistakes.

- Confusing g(E) with the number of electrons. g(E) counts available places; most conduction-band states are empty.
- Drawing g(E) as non-zero in the band gap, or as linear rather than square-root in shape.
- Mixing up g(E) (per unit energy, $m^{-3}eV^{-1}$) with N_c (a plain concentration, m^{-3}).

Remember it. "Seats times chance": carriers = how many seats (g) × chance each seat is taken (f).

Try a variant. For silicon at 300 K the Fermi level is 0.25 eV below E_c. Using the effective density of states, find the electron concentration. *Answer:* $n = N_c\, e^{-(E_c - E_F)/kT} = 2.8\times10^{25} \times e^{-0.25/0.0259} = 2.8\times10^{25} \times e^{-9.65} = \mathbf{1.8\times10^{21}\ m^{-3}}$.

Q33. Explain how the Fermi energy level changes with doping in n-type and p-type semiconductors.

> **Question type:** Explain · **Topic:** Fermi level and doping · **Full notes:** Week 5 (Carrier Statistics)

What the examiner wants. Start from the intrinsic case ($E_F = E_i$ near mid-gap), then show with equations that donors move E_F up towards E_c and acceptors move it down towards E_v, that the shift grows logarithmically with doping, and what happens at very heavy doping (degeneracy). A band diagram with three cases and a short numerical example earn full marks.

Key terms.

- **Fermi level (E_F):** the energy where a state has a 50 % chance of being occupied; its position fixes n and p.
- **Intrinsic level (E_i):** the Fermi level of undoped material, close to mid-gap (about 0.56 eV below E_c in Si).
- **Donor (N_D):** an impurity (P, As, Sb in Si) that gives an electron to the conduction band; N_D is donors per m^3.
- **Acceptor (N_A):** an impurity (B, Al, Ga in Si) that takes an electron from the valence band, creating a hole; N_A is acceptors per m^3.
- **Effective density of states (N_c, N_v):** Si at 300 K: 2.8×10^{25} and 1.04×10^{25} m^{-3}.
- **Degenerate semiconductor:** one so heavily doped that E_F is within 3kT of a band edge or inside a band.
- **Compensation:** partial cancelling when both donors and acceptors are present.

Model answer.

Starting point: intrinsic material

In pure material $n = p = n_i$ and $E_F = E_i$, very close to the middle of the gap. The carrier concentrations are linked to the Fermi level by

$$n = n_i\, e^{(E_F - E_i)/kT}, \qquad p = n_i\, e^{(E_i - E_F)/kT}$$

Where:

- n, p = electron and hole concentrations (m^{-3})
- n_i = intrinsic carrier concentration (1×10^{16} m^{-3} for Si at 300 K)
- $E_F - E_i$ = height of the Fermi level above the intrinsic level (eV)
- kT = thermal energy (0.0259 eV at 300 K)

So any change in n or p caused by doping must move E_F.

n-type: E_F moves up, towards E_c

Donors add electrons; with complete ionisation and $N_D \gg n_i$, $n \approx N_D$. Rearranging:

$$E_F - E_i = kT \ln(N_D/n_i), \qquad E_c - E_F = kT \ln(N_c/N_D)$$

Where:

- N_D = donor concentration (m^{-3})
- $E_c - E_F$ = distance from E_F up to the conduction-band edge (eV)
- N_c = effective density of states in the conduction band (m^{-3})

More donors → larger $\ln(N_D/n_i)$ → E_F rises further above E_i and closer to E_c.

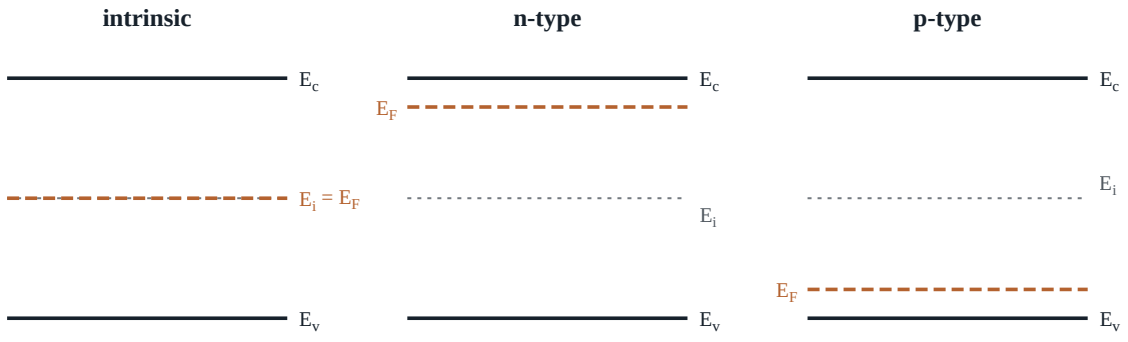
p-type: E_F moves down, towards E_v

Acceptors add holes; $p \approx N_A$, so

$$E_i - E_F = kT \ln(N_A/n_i), \qquad E_F - E_v = kT \ln(N_v/N_A)$$

Where:

- N_A = acceptor concentration (m^{-3})
- $E_F - E_v$ = height of E_F above the valence-band edge (eV)
- N_v = effective density of states in the valence band (m^{-3})



$E_F - E_i = kT \ln(N_D/n_i)$ for n-type; $E_i - E_F = kT \ln(N_A/n_i)$ for p-type

Fermi-level position in silicon: at E_i near mid-gap when intrinsic, raised towards E_c by donors (n-type), lowered towards E_v by acceptors (p-type).

Sketch: three band diagrams side by side (intrinsic, n-type, p-type), each with E_c, E_v and a dashed E_i at mid-gap; draw E_F on E_i, just below E_c (with donor levels E_D marked as short dashes just under E_c), and just above E_v (with acceptor levels E_A just above E_v).

Key features

- **Logarithmic shift.** Each tenfold increase in doping moves E_F by $kT \ln 10 \approx 0.06$ eV at 300 K, so large doping changes produce modest shifts.
- **Heavy doping → degeneracy.** As N_D approaches N_c (or N_A approaches N_v), E_F reaches the band edge or enters the band (n^+ or p^+ material); the Boltzmann formulas above then fail.
- **Compensation.** With both dopants, E_F is set by the net doping: $N_D - N_A$ (n-type) or $N_A - N_D$ (p-type).
- **Temperature.** As T rises, n_i grows and E_F moves back towards E_i; at high enough T every semiconductor becomes intrinsic. At very low T (freeze-out) the donors are not all ionised and E_F rises to between E_D and E_c.
- **Equilibrium.** E_F is constant through a device in equilibrium, so where n and p regions meet the bands bend; this gives the built-in voltage of a p-n junction.

Numerical example (Si, 300 K, $n_i = 10^{16}$ m^{-3})

Doping	$E_F - E_i$ (eV)	E_F distance from nearer band edge (eV)
Intrinsic	0	about 0.56 from each edge
$N_D = 10^{21}$ m^{-3}	+0.298	$E_c - E_F = 0.265$
$N_D = 10^{23}$ m^{-3}	+0.417	$E_c - E_F = 0.146$
$N_A = 10^{22}$ m^{-3}	−0.358	$E_F - E_v = 0.180$

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ intrinsic: $E_F = E_i$ near mid-gap
- ✓ n-type: E_F moves towards E_c; $E_F - E_i = kT \ln(N_D/n_i)$ or $E_c - E_F = kT \ln(N_c/N_D)$
- ✓ p-type: E_F moves towards E_v; $E_i - E_F = kT \ln(N_A/n_i)$ or $E_F - E_v = kT \ln(N_v/N_A)$
- ✓ labelled band diagram of the three cases
- ✓ shift is logarithmic (≈ 0.06 eV per decade at 300 K)
- ✓ very heavy doping → degenerate, E_F at or inside the band
- ✓ temperature or compensation effect mentioned
- ✓ numerical example

Common mistakes.

- Saying E_F moves "linearly" with doping or "to the donor level". The shift is logarithmic, and at room temperature E_F normally lies below E_D in non-degenerate n-type silicon.
- Getting the direction wrong for p-type: acceptors pull E_F **down** towards E_v.
- Using N_D in $E_F - E_i = kT \ln(N_D/n_i)$ when the material is compensated; use the net doping $N_D - N_A$.

Remember it. "Donors push up, acceptors pull down, by 60 meV per decade."

Try a variant. Silicon at 300 K is doped with $N_A = 1\times10^{23}$ m^{-3}. Where is the Fermi level? *Answer:* $E_i - E_F = 0.0259 \ln(10^{23}/10^{16}) =$ **0.417 eV below E_i**; $E_F - E_v = 0.0259 \ln(1.04\times10^{25}/10^{23}) = 0.0259 \times 4.64 =$ **0.120 eV above E_v**.

Q34. An n-type semiconductor has a donor concentration of 5×10^{16} cm^{-3}. Assuming complete ionization, determine the approximate majority-carrier concentration.

> **Question type:** Calculation · **Topic:** carrier concentrations in doped semiconductors · **Full notes:** Week 4 (Energy Bands and Carrier Concentrations); Fermi level in Week 5 (Carrier Statistics)

What the examiner wants. Identify the majority carrier (electrons in n-type), state that with complete ionisation and $N_D \gg n_i$ each donor gives one electron so $n_0 \approx N_D$, and give the value with its unit. Examiners often follow up with the minority-carrier concentration (mass-action law) and the Fermi-level position, so both are worked below, taking the material as silicon at 300 K.

Key terms.

- **n-type semiconductor:** one doped with donors, so electrons outnumber holes.
- **Donor concentration (N_D):** number of donor atoms per m^3; each ionised donor gives one free electron.
- **Complete ionisation:** every donor has given up its electron (true for Si between about 150 K and 450 K).
- **Majority carrier:** the more numerous carrier (electrons in n-type); concentration n_0 (subscript 0 means thermal equilibrium).
- **Minority carrier:** the less numerous carrier (holes in n-type); concentration p_0.
- **Mass-action law:** $n_0 p_0 = n_i^2$ in thermal equilibrium.
- **Fermi level (E_F) and intrinsic level (E_i):** E_F is the energy with a 50 % occupation probability; E_i is its position in undoped material, near mid-gap.

Model answer.

Given: $N_D = 5\times10^{16}$ cm^{-3} = 5×10^{22} m^{-3}, complete ionisation. For the follow-ups: silicon at 300 K, $n_i = 1\times10^{16}$ m^{-3}, kT = 0.0259 eV, $N_c = 2.8\times10^{25}$ m^{-3}.

Find: the majority-carrier concentration n_0; also the minority-carrier concentration p_0 and the Fermi-level position.

Formula:

$$n_0 \approx N_D \qquad \text{(complete ionisation, } N_D \gg n_i\text{)}$$

$$p_0 = n_i^2 / n_0 \qquad \text{(mass-action law)}$$

$$E_F - E_i = kT \ln(n_0/n_i), \qquad E_c - E_F = kT \ln(N_c/n_0)$$

Where:

- n_0 = equilibrium electron (majority) concentration (m^{-3})
- p_0 = equilibrium hole (minority) concentration (m^{-3})
- N_D = donor concentration (m^{-3})
- n_i = intrinsic carrier concentration (m^{-3})
- kT = thermal energy (eV)
- N_c = effective density of states in the conduction band (m^{-3})
- E_c = conduction-band edge, E_F = Fermi level, E_i = intrinsic level (eV)

Solution.

Step 1. Check the approximation: $N_D = 5\times10^{22} \gg n_i = 10^{16}\ m^{-3}$ (by a factor of 5×10^{6}), so thermally generated electrons are negligible and every donor supplies one electron:

$$n_0 \approx N_D = 5\times10^{22}\ m^{-3}$$

Step 2 (follow-up). Minority holes from the mass-action law:

$$p_0 = n_i^2/n_0 = (1\times10^{16})^2 / (5\times10^{22}) = 1\times10^{32} / 5\times10^{22} = 2\times10^{9}\ m^{-3}$$

Step 3 (follow-up). Fermi level relative to the intrinsic level:

$$E_F - E_i = 0.0259 \times \ln(5\times10^{22} / 1\times10^{16}) = 0.0259 \times \ln(5\times10^{6}) = 0.0259 \times 15.42 = 0.40\ eV$$

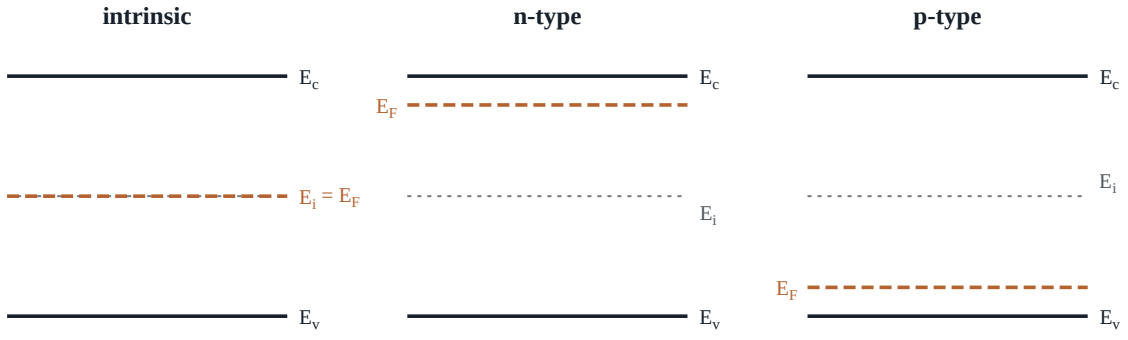
Step 4 (follow-up). Fermi level relative to the conduction-band edge:

$$E_c - E_F = 0.0259 \times \ln(2.8\times10^{25} / 5\times10^{22}) = 0.0259 \times \ln(560) = 0.0259 \times 6.33 = 0.164\ eV$$

Check: $(E_F - E_i) + (E_c - E_F) = 0.40 + 0.164 = 0.56\ eV \approx E_g/2$, as expected since E_i is near mid-gap. $E_c - E_F = 0.164\ eV$ is more than 3kT (0.078 eV), so the material is non-degenerate and the Boltzmann formulas used are valid.

Answer: Majority (electron) concentration $n_0 \approx 5\times10^{22}\ m^{-3}$. Follow-ups (Si, 300 K): **minority (hole) concentration $p_0 = 2\times10^{9}\ m^{-3}$; Fermi level 0.40 eV above E_i, i.e. about 0.16 eV below E_c.**

What this means: doping has raised the electron concentration by a factor of 5 million above the intrinsic value and, through $np = n_i^2$, cut the hole concentration by the same factor; there are about 2.5×10^{13} electrons for every hole. The Fermi level has moved most of the way from mid-gap up to the conduction band, which is the band-diagram picture of a strongly n-type sample.



In n-type silicon the Fermi level lies above the intrinsic level, between E_i and E_c; for $N_D = 5\times10^{22}\ m^{-3}$ it is 0.40 eV above E_i and 0.16 eV below E_c.

Sketch: a band diagram with E_c and E_v, a dashed E_i near mid-gap, and E_F drawn 0.40 eV above E_i (0.16 eV below E_c), labelled with these values.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ majority carriers identified as electrons
- ✓ $n_0 \approx N_D$ justified by complete ionisation and $N_D \gg n_i$
- ✓ answer $5\times10^{22}\ m^{-3}$ with the unit
- ✓ (follow-up) $p_0 = n_i^2/n_0 = 2\times10^{9}\ m^{-3}$
- ✓ (follow-up) $E_F - E_i = kT \ln(n_0/n_i) = 0.40\ eV$, Fermi level above E_i, towards E_c
- ✓ (follow-up) $E_c - E_F = kT \ln(N_c/n_0) = 0.16\ eV$

- ✓ assumption stated: silicon at 300 K, $n_i = 10^{16}$ m^{-3}

Common mistakes.

- Adding n_i to N_D ("$5\times10^{22} + 10^{16}$") as if it mattered, or, worse, giving n_i as the answer. n_i is negligible here; $n_0 \approx N_D$.
- Calculating $p_0 = n_i/N_D$ or N_D/n_i instead of n_i^2/N_D; check the units: $(m^{-3})^2/m^{-3} = m^{-3}$.
- Using $\log_{10}$ instead of the natural log in $E_F - E_i = kT \ln(n_0/n_i)$: $\log_{10}(5\times10^6) = 6.70$ would give 0.17 eV, which is wrong.
- Placing E_F below E_i for n-type material; donors raise E_F.

Remember it. "Majority = dopant, minority = n_i^2 over dopant, and E_F climbs kT $\ln(N_D/n_i)$ above mid-gap."

Try a variant. p-type silicon at 300 K has $N_A = 2\times10^{21}$ m^{-3} (complete ionisation). Find the majority and minority concentrations and the Fermi-level position. *Answer:* $p_0 \approx N_A =$ **2×10^{21} m^{-3}** (holes); $n_0 = n_i^2/p_0 = 10^{32}/2\times10^{21} =$ **5×10^{10} m^{-3}** (electrons); $E_i - E_F = 0.0259 \ln(2\times10^5) =$ **0.316 eV below E_i**, i.e. $E_F - E_v = 0.0259 \ln(1.04\times10^{25}/2\times10^{21}) =$ **0.222 eV above E_v**.

Questions 35–46 — Carrier Transport and Injection

Module 3: how carriers move (drift, mobility, scattering, diffusion, the Einstein relation, high fields) and how excess carriers are generated, recombine and flow (SRH, lifetime, continuity, surfaces). Each question stands on its own: its key terms are defined with it, so you can start anywhere.

Symbols and abbreviations used on this page

Every symbol, constant and abbreviation on this page is listed here, even if another page already defined it, so you can study this page on its own.

Constants and units

Symbol	Meaning	Units / typical value
q	Magnitude of the electronic charge	1.6×10^{-19} C
k	Boltzmann constant	8.617×10^{-5} eV/K = 1.381×10^{-23} J/K
kT	Thermal energy	0.0259 eV (often 0.02585 eV) at 300 K
kT/q	Thermal voltage	0.0259 V (about 26 mV) at 300 K
T	Absolute temperature	kelvin (K); room temperature = 300 K
h	Planck constant	6.626×10^{-34} J·s
m_0	Free-electron mass	9.11×10^{-31} kg
eV	Electron-volt, a unit of energy	1 eV = 1.6×10^{-19} J
µm	Micrometre (micron)	1 µm = 10^{-6} m

Symbols

Symbol	Meaning	Units / typical value
E_c (E_C)	Energy of the bottom of the conduction band	eV
E_v (E_V)	Energy of the top of the valence band	eV
E_g	Band-gap energy, $E_c - E_v$	eV; Si 1.12, Ge 0.66, GaAs 1.42 at 300 K
E_F	Fermi level: the energy at which a state has a 50 % chance of being occupied	eV
E_i	Intrinsic Fermi level, close to mid-gap	eV
E_t	Energy level of a trap (recombination centre) in the gap	eV
n	Electron concentration (free electrons per m^3)	m^{-3}
p	Hole concentration (holes per m^3)	m^{-3}
n_i	Intrinsic carrier concentration ($n = p = n_i$ in pure material)	m^{-3}; Si $\approx 1\times10^{16}$ at 300 K
n_0, p_0	Equilibrium electron and hole concentrations	m^{-3}
N_D	Donor concentration	m^{-3}
N_A	Acceptor concentration	m^{-3}
Δn, Δp	Excess electron and hole concentrations above equilibrium	m^{-3}
$m*$, m_n*, m_p*	Effective mass of electrons or holes	kg, or as a multiple of m_0
ν, $h\nu$	Frequency of light, and photon energy	Hz; eV
C(x, t)	Dopant concentration at depth x and time t	m^{-3}
D	Diffusion coefficient (dopants in a furnace, or carriers)	m^2/s
F	Particle flux in Fick's law	particles/(m^2·s)
Q	Dose: dopant atoms per unit area (in Weeks 13–14, Q also means charge)	atoms/m^2 (C or C/m^2 for charge)
R_p	Projected range: average depth at which implanted ions stop	µm
E (field)	Electric field (on band diagrams, E is energy instead)	V/m
μ, μ_n, μ_p	Mobility: drift velocity per unit field (electrons, holes)	$m^2/(V\cdot s)$; Si μ_n 0.135, μ_p 0.048
μ_L, μ_I	Mobility limited by lattice scattering, by ionised-impurity scattering	$m^2/(V\cdot s)$
v_d	Drift velocity	m/s
v_{th}	Thermal velocity of carriers	$\approx 10^5$ m/s at 300 K
v_{sat}	Saturation velocity at high field	$\approx 10^5$ m/s in Si
τ_m	Mean free time between collisions	s; about 0.1–1 ps
σ	Conductivity, $q(n\mu_n + p\mu_p)$ (in Week 8, σ is also a capture cross-section)	S/m (siemens per metre)
ρ	Resistivity, $1/\sigma$	Ω·m
J, J_n, J_p	Current density (total, electrons, holes)	A/m^2

Symbol	Meaning	Units / typical value
D_n, D_p	Diffusion coefficients of electrons and holes	m^2/s; Si 3.5×10^{-3} and 1.24×10^{-3}
T_e	Electron temperature of hot carriers	K
G, G_L	Generation rate of electron–hole pairs (G_L: by light)	$m^{-3}s^{-1}$
R	Recombination rate	$m^{-3}s^{-1}$
U	Net recombination rate, R − G	$m^{-3}s^{-1}$
τ, τ_n, τ_p	Minority-carrier lifetime (τ_{eff}: effective, including surfaces)	s; Si 1 ns – 1 ms
N_t	Density of traps (recombination centres)	m^{-3}
σ_n, σ_p	Capture cross-section of a trap for electrons, holes	m^2
n_1, p_1	SRH parameters $n_{ie}^{(E_t - E_i)/kT}$ and $n_{ie}^{(E_i - E_t)/kT}$	m^{-3}
L, L_n, L_p	Diffusion length $\sqrt{(D\tau)}$: distance a minority carrier travels before recombining	µm or m
S	Surface recombination velocity	m/s
N_{st}	Density of surface traps	m^{-2}
U_s	Recombination rate per unit area at a surface	$m^{-2}s^{-1}$
W	Depletion-region width (in Week 13 W_B, W_E are base and emitter widths; in Week 14 W is channel width)	µm or m
I_s, J_s (I_S)	Saturation current (density) of a diode or transistor	A; A/m^2
A (area)	Device or contact area	m^2
α	Common-base current gain, I_C/I_E	dimensionless; 0.98–0.999
β	Common-emitter current gain, I_C/I_B (also h_{FE})	dimensionless; 50–500
N_E, W_E, D_E; N_B, W_B, D_B	Doping, width and minority diffusion coefficient of the emitter; of the base	m^{-3}; µm; m^2/s
f_T	Transition (cut-off) frequency	Hz
V_T (V_{th})	Threshold voltage of a MOSFET	V
V_{GS}, V_{DS}, V_{SB}, V_G	Gate–source, drain–source, source–body and gate voltages	V
σ_i, ρ_i	Conductivity and resistivity of intrinsic (undoped) material; σ_{doped} after doping	S/m; Ω·m
F_n, F_p	Particle flux of electrons and holes (carriers crossing 1 m^2 per second)	$m^{-2}s^{-1}$
B_z, I_x, V_H	Hall effect: magnetic field (z), current (x) and Hall voltage	T (tesla); A; V
v_{dn}, v_{dp}	Drift velocity of electrons and holes	m/s
μ_i, τ_i (μ_L, μ_I, N_I)	Mobility and scattering time for one scattering mechanism (L: lattice, I: ionised impurity); N_I = ionised-impurity density	$m^2/(V\cdot s)$; s; m^{-3}
J_{drift}, J_{diff}	Drift and diffusion parts of the current density	A/m^2
G_{th}, R_{th}	Thermal generation and recombination rates (equal in equilibrium)	$m^{-3}s^{-1}$

Symbol	Meaning	Units / typical value
G_n, G_p, R_n, R_p	Generation and recombination rates of electrons and holes	$m^{-3}s^{-1}$
f_t, e_n, e_p	Probability a trap is occupied by an electron; emission rates of electrons and holes from traps	dimensionless; s^{-1}
r_a, r_b, r_c, r_d	SRH rates: electron capture, electron emission, hole capture, hole emission	$m^{-3}s^{-1}$
τ_b (τ_{bulk}), $\tau_{surface}$	Lifetime from bulk recombination only; from surface recombination only	s
U_s, p_s, Δp_s	Surface recombination rate; hole density and excess at the surface	$m^{-2}s^{-1}$; m^{-3}
I_{gen}, J_{gen}	Generation current (density) from carriers generated in the depletion region	A; A/m^2
τ_0 (τ_g)	Effective generation lifetime in a depletion region	s
t	Time (or a thickness, where the text says so)	s (or m)

Abbreviations

Abbreviation	Meaning
Si	Silicon
GaAs	Gallium arsenide
SiO_2	Silicon dioxide (silica), the oxide insulator
SRH	Shockley–Read–Hall (trap-assisted) recombination
LDD	Lightly doped drain
MOS	Metal–oxide–semiconductor
MOSFET	Metal–oxide–semiconductor field-effect transistor
CMOS	Complementary MOS (n- and p-channel MOSFETs together)
BJT	Bipolar junction transistor
LED	Light-emitting diode
IGBT	Insulated-gate bipolar transistor
TCAD	Technology computer-aided design (device simulation software)

Questions on this page

Q	Question
35	Define carrier mobility and explain the factors that affect electron and hole mobility in semiconductors.
36	Explain the mechanisms responsible for carrier scattering in semiconductor materials.
37	Derive the expression for drift current density due to electrons and holes.
38	A semiconductor has an electron concentration of 2×10^{16} cm^{-3}, electron mobility of 1350 $cm^2/V \cdot s$, and an electric field of 10 V/cm. Calculate the electron drift current density.
39	Explain diffusion current in a semiconductor and state Fick's law.
40	State and explain the Einstein relation between carrier mobility and diffusion coefficient.
41	Explain velocity saturation and hot-carrier effects in semiconductors.
42	What is meant by generation and recombination of charge carriers?
43	Explain the Shockley-Read-Hall (SRH) recombination mechanism.
44	Define minority-carrier lifetime and explain its significance in semiconductor devices.
45	Explain the continuity equation for minority carriers and discuss its application to semiconductor devices.
46	What is surface recombination velocity? Explain how semiconductor surface states affect carrier lifetime.

Model answers

Q35. Define carrier mobility and explain the factors that affect electron and hole mobility in semiconductors.

> **Question type:** Definition and explanation · **Topic:** carrier mobility · **Full notes:** Week 7 (Drift and Diffusion Currents)

What the examiner wants. Two parts. First a precise definition of mobility, with its formula and unit. Then an explanation of each factor that changes mobility, saying *why* it does so (through the collision time or the effective mass), not just listing names.

Key terms.

- **Drift velocity (v_d):** the small average velocity carriers gain in the direction of an electric field (m/s).
- **Mobility (μ):** drift velocity per unit electric field, $\mu = v_d/E$ ($m^2/V \cdot s$).
- **Mean free time (τ_m):** the average time between two collisions of a carrier (about 0.1–1 ps).
- **Effective mass ($m*$):** the mass a carrier appears to have when it is accelerated inside the crystal.
- **Scattering:** any collision that randomises a carrier's direction and destroys the velocity it gained from the field.
- **Phonon:** a quantum of lattice vibration; its number grows with temperature.
- **Matthiessen's rule:** when several scattering mechanisms act, $1/\mu = 1/\mu_1 + 1/\mu_2 + \ldots$

Model answer.

Definition

Carrier mobility is the drift velocity a carrier gains per unit applied electric field. It measures how easily electrons or holes move through the crystal.

$$\mu = v_d / E = q\tau_m / m*$$

Where:

- μ = mobility ($m^2/V\cdot s$)
- v_d = drift velocity (m/s)
- E = electric field (V/m)
- q = electronic charge, 1.6×10^{-19} C
- τ_m = mean free time between collisions (s)
- $m*$ = effective mass of the carrier (kg)

Electrons and holes have separate mobilities, μ_n and μ_p. In silicon at 300 K, $\mu_n = 0.135$ and $\mu_p = 0.048$ $m^2/V\cdot s$. The formula shows that mobility is high when collisions are rare (long τ_m) and the carrier is light (small $m*$). Every factor below acts through one of these two quantities.

Factors affecting mobility

1. **Effective mass.** $\mu \propto 1/m*$. Electrons usually have a smaller effective mass than holes, so $\mu_n > \mu_p$ (Si 0.135 vs 0.048 $m^2/V\cdot s$; GaAs electrons, $m* = 0.067m_0$, reach about 0.85 $m^2/V\cdot s$).
2. **Temperature (lattice scattering).** Atoms vibrate more as T rises, so carriers collide with phonons more often and τ_m falls: $\mu_L \propto T^{-3/2}$. Near room temperature mobility falls as temperature rises.
3. **Temperature (impurity scattering).** Slow carriers are deflected more by charged ions. At higher T carriers move faster and are deflected less, so $\mu_I \propto T^{3/2}/N_I$ rises with T. This dominates at low temperature.
4. **Doping concentration.** More ionised donors and acceptors mean more Coulomb scattering centres, so mobility falls as $N_D + N_A$ rises (Si electrons: about 0.135 $m^2/V\cdot s$ at light doping, about 0.080 $m^2/V\cdot s$ at 10^{23} m^{-3}, about 0.010 $m^2/V\cdot s$ at 10^{26} m^{-3}). Compensated material has low mobility even if the net doping is small, because both ion types scatter.
5. **Electric field.** At low field μ is constant. At high field (above about 10^6 V/m in Si) carriers emit optical phonons, the drift velocity saturates near 10^5 m/s, and the effective mobility v_d/E falls.
6. **Crystal defects.** Dislocations, grain boundaries and vacancies add scattering; polycrystalline and amorphous material have much lower mobility than single crystal.
7. **Surface and interface scattering.** In a MOSFET channel, carriers are pressed against the rough Si/SiO_2 interface and its trapped charges; channel mobility is roughly half the bulk value.
8. **Orientation, strain and carrier density.** Mobility depends on crystal direction and can be raised by strain (strained silicon); at very high carrier densities carrier–carrier scattering lowers it.

The separate effects combine through Matthiessen's rule:

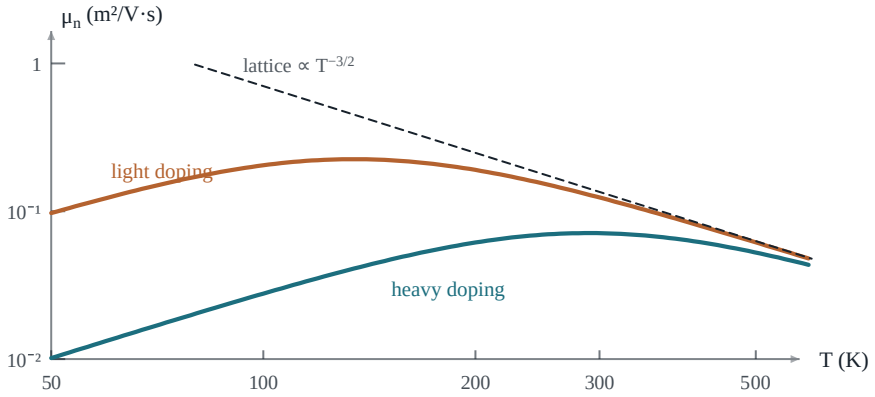
$$1/\mu = 1/\mu_L + 1/\mu_I + \dots$$

Where:

- μ = total mobility ($m^2/V\cdot s$)
- μ_L = mobility limited by lattice scattering alone ($m^2/V\cdot s$)

- μ_I = mobility limited by ionised-impurity scattering alone ($m^2/V{\cdot}s$)

The smallest term dominates, so mobility against temperature rises at low T (impurity-limited), peaks, then falls at high T (lattice-limited).



Mobility against temperature for light and heavy doping. Impurity scattering limits the low-temperature side, lattice scattering the high-temperature side; heavier doping lowers the curve and moves the peak to a higher temperature.

Sketch: log μ against log T, a curve rising as $T^{3/2}$ then falling as $T^{-3/2}$; label "impurity scattering" and "lattice scattering", and draw a lower curve labelled "higher N_I".

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definition $\mu = v_d/E$ with unit $m^2/V{\cdot}s$
- ✓ $\mu = q\tau_m/m*$, linking mobility to collision time and effective mass
- ✓ effective mass: why $\mu_n > \mu_p$, with Si values 0.135 and 0.048 $m^2/V{\cdot}s$
- ✓ lattice scattering, $\mu_L \propto T^{-3/2}$
- ✓ ionised-impurity scattering, $\mu_I \propto T^{3/2}/N_I$, and fall of μ with doping
- ✓ high-field effect (velocity saturation)
- ✓ at least one further factor: defects, surface scattering, strain
- ✓ Matthiessen's rule and/or the sketch of μ against T

Common mistakes.

- Saying "mobility increases with temperature" without qualification: near room temperature it decreases; it only increases at low T where impurity scattering dominates.
- Defining mobility as "the speed of a carrier": it is velocity *per unit field*, so its unit is $m^2/V{\cdot}s$, not m/s.
- Listing factors with no reason: always say whether the factor changes τ_m or $m*$.
- Adding mobilities directly: it is the reciprocals that add.

Remember it. "Mobility = q τ over m": anything that shortens the time between bumps, or makes the carrier heavier, slows it down.

Try a variant. In a silicon sample the lattice-limited mobility is 0.135 $m^2/V{\cdot}s$ and the impurity-limited mobility is 0.200 $m^2/V{\cdot}s$. Find the total electron mobility. *Answer:* $1/\mu = 1/0.135 + 1/0.200 = 7.41 + 5.00 = 12.41$ $V{\cdot}s/m^2$, so **$\mu \approx 0.0806$ $m^2/V{\cdot}s$**, lower than either value alone.

Q36. Explain the mechanisms responsible for carrier scattering in semiconductor materials.

> **Question type:** Explanation · **Topic:** carrier scattering · **Full notes:** Week 7 (Drift and Diffusion Currents)

What the examiner wants. Say what scattering is and why it happens (anything that breaks the perfect periodicity of the crystal), then explain each mechanism: its physical cause, when it dominates, and how it depends on temperature and doping. Finish with how the mechanisms combine.

Key terms.

- **Scattering:** a collision that changes a carrier's momentum and randomises the velocity it gained from a field.
- **Periodic potential:** the regularly repeating electric potential of a perfect crystal lattice; a carrier moves through it without scattering (Bloch's theorem).
- **Phonon:** a quantum of lattice vibration. *Acoustic* phonons are low-energy; *optical* phonons are high-energy (about 63 meV in Si).
- **Ionised impurity:** a donor (+) or acceptor (−) ion that has given up or taken an electron and so carries a charge.
- **Mean free time (τ_m):** the average time between collisions.
- **Mobility (μ):** drift velocity per unit field, $\mu = q\tau_m/m^*$ ($m^2/V\cdot s$).
- **Matthiessen's rule:** scattering rates add, so $1/\mu = \Sigma\ 1/\mu_i$.

Model answer.

Why scattering happens

In a perfectly periodic, perfectly still crystal an electron would travel freely with no collisions (Bloch's theorem). Scattering happens whenever something breaks that perfect periodicity. Each scattering event randomises the carrier's direction, so it limits the mean free time τ_m and hence the mobility, $\mu = q\tau_m/m^*$.

The mechanisms

1. **Lattice (phonon) scattering.** Above 0 K the atoms vibrate about their sites and distort the periodic potential. Carriers exchange energy and momentum with these vibrations (phonons). *Acoustic-phonon* scattering is nearly elastic and dominates in lightly doped Si at room temperature. More vibration at higher T means more scattering, giving $\mu_L \propto T^{-3/2}$. *Optical-phonon* scattering (phonon energy about 63 meV in Si) becomes important at high temperature and high field, and causes velocity saturation.
2. **Polar-optical and piezoelectric scattering.** In compound semiconductors such as GaAs the bonds are partly ionic, so vibrating atoms create electric fields that scatter carriers strongly.
3. **Ionised-impurity scattering.** The Coulomb field of each charged donor or acceptor ion deflects a passing carrier, like a charged particle passing a nucleus (Rutherford scattering). A slow carrier spends longer near the ion and is deflected more. It is therefore strongest at low temperature and high doping: $\mu_I \propto T^{3/2}/N_I$, where N_I is the ionised-impurity concentration.
4. **Neutral-impurity scattering.** Un-ionised dopant atoms distort the lattice potential slightly. This is weak and matters only at very low temperature, when dopants are frozen out.
5. **Carrier–carrier scattering.** Electrons and holes repel or attract one another. Important only at very high carrier densities (heavily doped or strongly injected regions).

6. **Defect scattering.** Dislocations, grain boundaries, stacking faults and vacancies disturb the lattice; dislocations in n-type material can also be charged lines.
7. **Surface and interface scattering.** Carriers confined near a surface (a MOSFET channel at the Si/SiO_2 interface) are scattered by surface roughness, interface traps and oxide charge; channel mobility is about half the bulk value.
8. **Alloy scattering.** In alloys such as SiGe or $Al_xGa_{1-x}As$ the random placement of different atoms makes the potential non-periodic.
9. **Intervalley scattering.** A carrier absorbs or emits a phonon and moves to another conduction-band minimum; in GaAs this causes negative differential mobility (the Gunn effect).

Combining the mechanisms

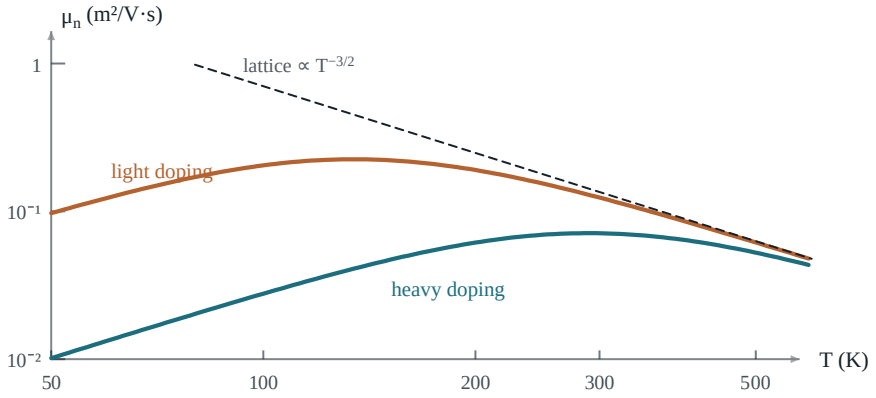
The scattering rates add, $1/\tau = \Sigma\ 1/\tau_i$, so the mobilities combine by Matthiessen's rule:

$$1/\mu = 1/\mu_L + 1/\mu_I + \dots$$

Where:

- μ = total mobility ($m^2/V{\cdot}s$)
- μ_L = lattice-limited mobility ($m^2/V{\cdot}s$)
- μ_I = ionised-impurity-limited mobility ($m^2/V{\cdot}s$)

The mechanism with the lowest mobility dominates. In lightly doped Si near 300 K this is lattice scattering; in heavily doped Si or at low temperature it is ionised-impurity scattering.



The two main mechanisms compete: impurity scattering sets the rising, low-temperature side of the mobility curve and lattice scattering the falling, high-temperature side.

Sketch: log μ against log T with the $T^{3/2}$ impurity branch and the $T^{-3/2}$ lattice branch, and the actual curve passing under both.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ scattering is caused by any departure from a perfect periodic lattice
- ✓ lattice (phonon) scattering, acoustic and optical, with $\mu_L \propto T^{-3/2}$
- ✓ ionised-impurity (Coulomb) scattering, with $\mu_I \propto T^{3/2}/N_I$
- ✓ neutral-impurity and carrier–carrier scattering, and when they matter
- ✓ defect and surface/interface scattering
- ✓ at least one of: polar, alloy, intervalley scattering
- ✓ Matthiessen's rule and which mechanism dominates where

Common mistakes.

- Saying the electron "collides with the atoms" of a perfect lattice: a perfect still lattice does not scatter; only vibrations and imperfections do.
- Getting the temperature dependence of impurity scattering backwards: it is *weaker* at high T, because fast carriers are deflected less.
- Giving only two mechanisms: the question says "mechanisms", so cover at least five with a reason for each.

Remember it. "Shake, charge, break": the lattice shakes (phonons), ions are charged (impurities), the crystal breaks (defects and surfaces).

Try a variant. Lattice-limited electron mobility in Si is 0.135 $m^2/V{\cdot}s$ at 300 K. Estimate it at 400 K, assuming $\mu_L \propto T^{-3/2}$. *Answer:* $\mu_L = 0.135 \times (400/300)^{-3/2} = 0.135 \times 0.650 = \approx$ **0.0877 $m^2/V{\cdot}s$**: heating by 100 K cuts the lattice-limited mobility by about a third.

Q37. Derive the expression for drift current density due to electrons and holes.

> **Question type:** Derivation · **Topic:** drift current · **Full notes:** Week 7 (Drift and Diffusion Currents)

What the examiner wants. A step-by-step derivation starting from "current density = charge × concentration × velocity", showing how the drift velocity follows from the field ($v_d = \mu E$), handling the signs for electrons and holes, and ending with $J = q(n\mu_n + p\mu_p)E$, the conductivity σ and the resistivity ρ. State assumptions first and box the final result.

Key terms.

- **Drift:** carrier motion caused by an electric field.
- **Drift velocity (v_d):** the average velocity carriers gain from the field, on top of their random thermal motion (m/s).
- **Mobility (μ_n, μ_p):** drift velocity per unit field ($m^2/V{\cdot}s$).
- **Current density (J):** charge crossing unit area per unit time (A/m^2).
- **Effective mass ($m*$):** the apparent mass of a carrier accelerated inside the crystal.
- **Mean free time (τ_m):** average time between collisions (s).
- **Conductivity (σ):** J/E, the material's ability to carry current (S/m); resistivity $\rho = 1/\sigma$ (Ω·m).

Model answer.

Sketch: a bar of semiconductor of cross-section A with the field E pointing in +x. Draw holes (⊕) drifting along +x and electrons (⊖) drifting along −x; draw both current arrows J_p and J_n pointing along +x.

Assumptions

- The semiconductor is uniform (n and p do not vary with position), so there is no diffusion current.
- Steady state, with a uniform field E in the +x direction.
- The field is low enough that carriers stay near thermal equilibrium, so v_d is proportional to E (no velocity saturation).
- Each carrier type has a constant effective mass $m*$ and mean free time τ_m.

Derivation

Step 1. Force on a carrier. An electron (charge −q) feels F = −qE; a hole (charge +q) feels F = +qE.

Step 2. Drift velocity of electrons. In steady state the momentum each electron gains from the field per unit time equals the momentum lost in collisions, which on average is $m_n{*}v_{dn}/\tau_{mn}$:

$$m_n{*}\, v_{dn} / \tau_{mn} = -qE \quad \Rightarrow \quad v_{dn} = -(q\tau_{mn} / m_n{*})\, E = -\mu_n\, E$$

Where:

- v_{dn} = electron drift velocity (m/s)
- $m_n{*}$ = electron effective mass (kg)
- τ_{mn} = electron mean free time (s)
- $\mu_n = q\tau_{mn}/m_n{*}$ = electron mobility ($m^2/V{\cdot}s$)
- E = electric field (V/m)

The minus sign shows electrons drift against the field.

Step 3. Drift velocity of holes. By the same argument with charge +q:

$$v_{dp} = +(q\tau_{mp} / m_p{*})\, E = +\mu_p\, E$$

Holes drift along the field.

Step 4. Current density from charge × velocity. Take a plane of area A across the bar. In time dt, every carrier within a distance $v_d{\cdot}dt$ behind the plane crosses it: that is a volume $v_d{\cdot}dt{\cdot}A$. If the concentration is c and each carrier has charge Q, the charge crossing is $Q{\cdot}c{\cdot}v_d{\cdot}dt{\cdot}A$. Dividing by A and dt:

$$J = Q\, c\, v_d$$

Where:

- J = current density (A/m^2)
- Q = charge per carrier (C)
- c = carrier concentration (m^{-3})
- v_d = drift velocity (m/s)

Step 5. Electron drift current. Put Q = −q, c = n, $v_d = -\mu_n E$:

$$J_n = (-q)(n)(-\mu_n\, E) = qn\mu_n\, E$$

The two minus signs cancel: electrons move against E, but because they are negative their current flows along E.

Step 6. Hole drift current. Put Q = +q, c = p, $v_d = +\mu_p E$:

$$J_p = (+q)(p)(+\mu_p\, E) = qp\mu_p\, E$$

Step 7. Total drift current. Both components point along E, so they add:

$$J_{drift} = J_n + J_p = qn\mu_n\, E + qp\mu_p\, E$$

Step 8. Conductivity and resistivity. Comparing with Ohm's law in the form $J = \sigma E$:

$$\sigma = q(n\mu_n + p\mu_p), \qquad \rho = 1/\sigma = 1 / [q(n\mu_n + p\mu_p)]$$

Result

$J_{drift} = q(n\mu_n + p\mu_p)E$, with $\sigma = q(n\mu_n + p\mu_p)$ and $\rho = 1/\sigma$.

Where:

- J_{drift} = total drift current density (A/m^2)
- q = electronic charge, 1.6×10^{-19} C
- n, p = electron and hole concentrations (m^{-3})
- μ_n, μ_p = electron and hole mobilities ($m^2/V\cdot s$)
- E = electric field (V/m)
- σ = conductivity (S/m); ρ = resistivity ($\Omega\cdot m$)

Special cases

- n-type ($n \gg p$): $\sigma \approx qN_D\mu_n$. For $N_D = 10^{22}$ m^{-3} in Si, $\sigma = 1.6\times10^{-19} \times 10^{22} \times 0.135 = 216$ S/m.
- p-type ($p \gg n$): $\sigma \approx qN_A\mu_p$.
- Intrinsic ($n = p = n_i$): $\sigma_i = qn_i(\mu_n + \mu_p) = 1.6\times10^{-19} \times 10^{16} \times 0.183 = 2.93\times10^{-4}$ S/m.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ assumptions stated (uniform sample, low field, steady state)
- ✓ $v_d = \mu E$ derived from force balance, with $\mu = q\tau_m/m*$
- ✓ correct directions: electrons against E, holes along E
- ✓ J = (charge)(concentration)(velocity), justified by counting carriers crossing a plane
- ✓ $J_n = qn\mu_nE$ with the sign cancellation explained
- ✓ $J_p = qp\mu_pE$
- ✓ boxed total $J = q(n\mu_n + p\mu_p)E$
- ✓ $\sigma = q(n\mu_n + p\mu_p)$ and $\rho = 1/\sigma$, with n-type, p-type or intrinsic special case
- ✓ labelled sketch of carrier and current directions

Common mistakes.

- Writing $J_n = -qn\mu_nE$ and concluding the electron and hole currents cancel: they add, because the electron's negative charge and negative velocity cancel.
- Skipping the step J = (charge)(concentration)(velocity): it carries marks, so show the carrier-counting argument.
- Using the thermal velocity (about 10^5 m/s) instead of the drift velocity: the random thermal motion averages to zero and carries no current.
- Forgetting to state the low-field assumption: $v_d = \mu E$ fails at high field.

Remember it. "Charge × count × speed": J = (q)(n)(μE) for each carrier, and the two minus signs of the electron cancel.

Try a variant. Find the conductivity and resistivity of intrinsic silicon at 300 K. *Answer:* $\sigma_i = qn_i(\mu_n + \mu_p) = 1.6\times10^{-19} \times 10^{16} \times (0.135 + 0.048) =$ **2.93×10^{-4} S/m**, so $\rho_i = 1/\sigma_i =$ **3.4×10^{3} Ω·m**.

Q38. A semiconductor has an electron concentration of 2×10^{16} cm^{-3}, electron mobility of 1350 $cm^2/V \cdot s$, and an electric field of 10 V/cm. Calculate the electron drift current density.

> **Question type:** Calculation · **Topic:** drift current · **Full notes:** Week 7 (Drift and Diffusion Currents)

What the examiner wants. A single-formula calculation. Marks come from writing the drift-current formula, substituting with units, and giving the answer with the correct unit (A/m^2). A one-line comment on what the number means shows understanding.

Key terms.

- **Drift:** the motion of charge carriers caused by an electric field.
- **Mobility (μ_n):** how fast an electron drifts per unit field, $v_d = \mu_n E$; unit $m^2/V \cdot s$.
- **Drift velocity (v_d):** the average velocity carriers gain from the field (m/s).
- **Current density (J):** current per unit cross-sectional area (A/m^2).
- **Saturation velocity (v_{sat}):** the maximum drift velocity at high field, about 10^5 m/s in Si.

Model answer.

Given: $n = 2\times10^{16}$ cm^{-3} = 2×10^{22} m^{-3}, μ_n = 1350 $cm^2/V \cdot s$ = 0.135 $m^2/V \cdot s$, E = 10 V/cm = 1000 V/m, $q = 1.6\times10^{-19}$ C.

Find: the electron drift current density J_n.

Formula:

$$J_n = q\, n\, \mu_n\, E$$

Where:

- J_n = electron drift current density (A/m^2)
- q = electronic charge, 1.6×10^{-19} C
- n = electron concentration (m^{-3})
- μ_n = electron mobility ($m^2/V \cdot s$)
- E = electric field (V/m)

Solution.

Step 1. Drift velocity: $v_d = \mu_n E = 0.135 \times 1000 = 135$ m/s.

Step 2. Current density:

$$J_n = 1.6\times10^{-19} \times 2\times10^{22} \times 0.135 \times 1000 = 4.32\times10^5 \text{ A/m}^2$$

Step 3. Units check: $C \times m^{-3} \times m^2/(V \cdot s) \times V/m = C/(s \cdot m^2) = A/m^2$.

Answer: $J_n = 4.32\times10^5$ A/m^2 (with $q = 1.602\times10^{-19}$ C, 4.33×10^5 A/m^2).

What this means: a field of only 1000 V/m drives 4.32×10^5 A through every square metre; the drift velocity (135 m/s) is still far below the saturation velocity (10^5 m/s), so the simple linear formula is valid.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ $J_n = qn\mu_nE$ written down, with each symbol defined

- ✓ correct substitution with units
- ✓ answer 4.32×10^{5} A/m^2, with the unit
- ✓ comment that the field is low enough for $v_d = \mu E$ to hold

Common mistakes.

- Mixing metres and centimetres: convert everything to SI first (m^{-3}, $m^2/V\cdot s$, V/m) and the answer comes out in A/m^2.
- Forgetting q, and giving 2.7×10^{24} as the "current".
- Giving the unit as A instead of A/m^2: this is a current density, not a current.

Remember it. "J = q n μ E": charge × how many × how mobile × how hard you push.

Try a variant. p-type silicon has $p = 5\times10^{21}$ m^{-3} and $\mu_p = 0.048$ $m^2/V\cdot s$ in a field of 10^4 V/m. Find the hole drift current density. *Answer:* $J_p = qp\mu_pE = 1.6\times10^{-19}\times5\times10^{21}\times0.048\times10^4 =$ **3.84×10^5 A/m^2**.

Q39. Explain diffusion current in a semiconductor and state Fick's law.

> **Question type:** Explanation and statement · **Topic:** diffusion current · **Full notes:** Week 7 (Drift and Diffusion Currents)

What the examiner wants. Explain what causes diffusion current (random thermal motion plus a concentration gradient, no field needed), state Fick's law with its minus sign, and turn it into the electron and hole diffusion current densities with the correct signs. A labelled sketch and an example of where diffusion matters earn extra marks.

Key terms.

- **Diffusion:** net movement of particles from high to low concentration caused by their random thermal motion.
- **Concentration gradient (dn/dx, dp/dx):** the rate at which carrier concentration changes with position (m^{-4}).
- **Flux (F):** number of particles crossing 1 m^2 per second ($m^{-2}s^{-1}$).
- **Diffusion coefficient (D_n, D_p):** how fast carriers spread out (m^2/s); in Si, $D_n = 3.5\times10^{-3}$ and $D_p = 1.24\times10^{-3}$ m^2/s.
- **Diffusion current density:** the current carried by the diffusion flux of charged carriers (A/m^2).
- **Minority carrier:** the less numerous carrier type (holes in n-type, electrons in p-type).

Model answer.

What diffusion current is

Carriers in a semiconductor move randomly at the thermal velocity (about 10^5 m/s), changing direction at every collision. If the concentration is the same everywhere, as many carriers cross any plane from left to right as from right to left, and there is no net flow. If the concentration is higher on one side, more carriers cross from that side simply because there are more of them. The result is a net flow from high to low concentration, with no electric field needed. Because the carriers are charged, this flow is an electric current: the **diffusion current**.

Diffusion current is important wherever carrier gradients exist:

- in the neutral regions of a forward-biased p-n junction, where injected minority carriers diffuse away from the junction;
- in the base of a bipolar transistor;

- under non-uniform illumination (solar cells, photodetectors) and in non-uniformly doped material.

Fick's first law

The particle flux is proportional to the concentration gradient and flows down the gradient:

$$F = -D\,dC/dx$$

Where:

- F = particle flux ($m^{-2}s^{-1}$)
- D = diffusion coefficient (m^2/s)
- C = particle concentration (m^{-3})
- dC/dx = concentration gradient (m^{-4})
- the minus sign = flow is from high to low concentration

For electrons and holes: $F_n = -D_n\,dn/dx$ and $F_p = -D_p\,dp/dx$.

Diffusion current densities

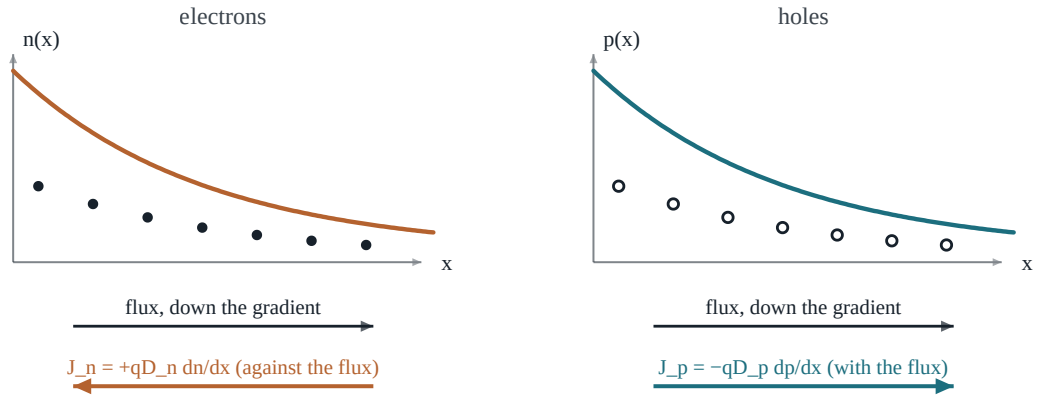
Multiply each flux by the carrier's charge (−q for electrons, +q for holes):

$$J_{n,diff} = (-q)(-D_n\,dn/dx) = +qD_n\,dn/dx$$
$$J_{p,diff} = (+q)(-D_p\,dp/dx) = -qD_p\,dp/dx$$

Where:

- $J_{n,diff}$, $J_{p,diff}$ = electron and hole diffusion current densities (A/m^2)
- q = electronic charge, 1.6×10^{-19} C
- D_n, D_p = electron and hole diffusion coefficients (m^2/s)
- n, p = electron and hole concentrations (m^{-3})

Both carriers flow down their gradient, but the electron current points *opposite* to the electron flow (negative charge), while the hole current points *with* the hole flow.



Electrons and holes both diffuse from high to low concentration; the electron current is opposite to the electron flow, the hole current is in the same direction as the hole flow.

Sketch: a concentration profile falling from left to right; draw the particle-flow arrows pointing right for both carriers, the hole current arrow pointing right and the electron current arrow pointing left.

Example. Electron concentration in Si falls linearly from 10^{23} m^{-3} to zero over 1 µm (10^{-6} m). Then dn/dx = -10^{29} m^{-4} and $J_n = qD_n\, dn/dx = 1.6\times10^{-19} \times 3.5\times10^{-3} \times (-10^{29}) = -5.6\times10^{7}$ A/m^2: a large current, flowing in −x, from a modest gradient.

When a field is also present, the total currents are $J_n = qn\mu_n E + qD_n\, dn/dx$ and $J_p = qp\mu_p E - qD_p\, dp/dx$.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ cause: random thermal motion plus a concentration gradient; no field needed
- ✓ Fick's law F = −D dC/dx with the minus sign explained
- ✓ D defined with unit m^2/s (Si values 3.5×10^{-3} and 1.24×10^{-3} m^2/s)
- ✓ $J_n = +qD_n\, dn/dx$ and $J_p = -qD_p\, dp/dx$ with correct signs
- ✓ direction of electron current opposite to electron flow
- ✓ where diffusion matters (forward-biased diode, BJT base)
- ✓ labelled sketch or numerical example
- ✓ total current = drift + diffusion

Common mistakes.

- Giving both current densities the same sign: the electron one is $+qD_n\, dn/dx$, the hole one $-qD_p\, dp/dx$.
- Saying diffusion needs a field: it happens with no field at all.
- Stating Fick's law without the minus sign, which would send particles uphill.
- Confusing the diffusion of carriers (D about 10^{-3} m^2/s) with dopant diffusion in a furnace (D about 10^{-17} m^2/s).

Remember it. "Ink in water": carriers spread from crowded to empty; the electron's minus sign flips its current.

Try a variant. In n-type silicon the hole concentration falls linearly from 10^{22} m^{-3} to zero over 10 µm. Find the hole diffusion current density ($D_p = 1.24\times10^{-3}$ m^2/s). *Answer:* $dp/dx = -10^{22}/10^{-5} = -10^{27}$ m^{-4}, so $J_p = -qD_p\, dp/dx = 1.6\times10^{-19} \times 1.24\times10^{-3} \times 10^{27} =$ **1.98×10^{5} A/m^2**, in the +x direction (with the hole flow).

Q40. State and explain the Einstein relation between carrier mobility and diffusion coefficient.

> **Question type:** Statement and explanation · **Topic:** Einstein relation · **Full notes:** Week 7 (Drift and Diffusion Currents)

What the examiner wants. State the relation exactly for both carriers, say what kT/q is and its value at 300 K, explain *why* D and µ are linked (same collisions), and justify it, ideally with the equilibrium drift-equals-diffusion argument. A numerical use (finding D from µ) completes the answer.

Key terms.

- **Mobility (µ):** drift velocity per unit field (m^2/V·s).
- **Diffusion coefficient (D):** constant in Fick's law, F = −D dC/dx (m^2/s).
- **Thermal voltage (kT/q):** Boltzmann constant × temperature ÷ electronic charge; 0.0259 V at 300 K.
- **Thermal equilibrium:** no external bias or illumination; no net current flows and the Fermi level is flat.
- **Fermi level (E_F):** the energy at which the probability of occupation is one half.

- **Intrinsic level (E_i):** the Fermi level of the undoped material; it follows the band edges and the electrostatic potential.
- **Non-degenerate semiconductor:** doped lightly enough that Boltzmann statistics apply (E_F more than about 3kT from the band edges).

Model answer.

Statement

For a non-degenerate semiconductor, the ratio of diffusion coefficient to mobility equals the thermal voltage, for both electrons and holes:

$$D_n / \mu_n = D_p / \mu_p = kT/q$$

Where:

- D_n, D_p = electron and hole diffusion coefficients (m^2/s)
- μ_n, μ_p = electron and hole mobilities ($m^2/V \cdot s$)
- k = Boltzmann constant, 8.617×10^{-5} eV/K
- T = absolute temperature (K)
- q = electronic charge, 1.6×10^{-19} C
- kT/q = thermal voltage, 0.0259 V at 300 K

Why μ and D are linked

Drift (response to a field) and diffusion (response to a gradient) are done by the same carriers and are limited by the same collisions. Mobility is $\mu = q\tau_m/m*$. Diffusion depends on how far a carrier travels between collisions, $D = v_{th}^2 \tau_m/3$, and with $(1/2)m* v_{th}^2 = (3/2)kT$ this is $D = kT\tau_m/m*$. Dividing, the collision time τ_m and the effective mass $m*$ cancel:

$$D/\mu = (kT\tau_m/m*) / (q\tau_m/m*) = kT/q$$

So a carrier that rarely collides both drifts easily and spreads quickly, and the ratio depends only on temperature.

Equilibrium proof

Take an n-type sample with non-uniform doping, in thermal equilibrium.

Step 1. Electrons diffuse from the high-doped side and uncover positive donor ions, creating an internal field E.

Step 2. That field drives a drift current back. With no external circuit the net current is zero:

$$J_n = qn\mu_n E + qD_n \, dn/dx = 0$$

Step 3. In equilibrium E_F is flat and $n = n_i \, e^{(E_F - E_i)/kT}$. E_i follows the electrostatic potential, $dE_i/dx = qE$, so

$$dn/dx = -(n/kT) \, dE_i/dx = -(qn/kT) \, E$$

Step 4. Substituting into the zero-current condition:

$$qn\mu_n E - qD_n(qn/kT)E = 0 \quad \Rightarrow \quad D_n = (kT/q) \, \mu_n$$

The same argument with holes gives $D_p = (kT/q)\mu_p$.

Use and significance

- Mobility is easy to measure (Hall effect, conductivity), so D follows: Si electrons $D_n = 0.0259 \times 0.135 = 3.50\times10^{-3}$ m^2/s; holes $D_p = 0.0259 \times 0.048 = 1.24\times10^{-3}$ m^2/s.
- It links the drift and diffusion terms of the current equations and is needed for diffusion lengths $L = \sqrt{(D\tau)}$ and diode saturation currents.
- In degenerate (very heavily doped) material Boltzmann statistics fail and D/μ is larger than kT/q.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ D/μ = kT/q stated for both electrons and holes
- ✓ kT/q named the thermal voltage, 0.0259 V (about 26 mV) at 300 K
- ✓ physical reason: the same scattering limits both drift and diffusion
- ✓ equilibrium argument: drift current cancels diffusion current
- ✓ use of flat E_F and Boltzmann statistics to get $dn/dx = -(qn/kT)E$
- ✓ numerical example: $D_n = 3.5\times10^{-3}$ m^2/s, $D_p = 1.24\times10^{-3}$ m^2/s for Si
- ✓ limitation to non-degenerate material

Common mistakes.

- Writing μ/D = kT/q (upside down): check units, D/μ has units $(m^2/s)/(m^2/V\cdot s) = V$.
- Using kT = 0.0259 J: it is 0.0259 eV, so kT/q = 0.0259 V.
- Forgetting the temperature dependence: kT/q scales with T, so at 350 K it is 0.0302 V, not 0.0259 V.
- Stating it without saying when it is valid (thermal equilibrium physics, non-degenerate doping).

Remember it. "D over μ is kT over q": spread over push equals thermal voltage, 26 mV at room temperature.

Try a variant. Electrons in a sample at 350 K have mobility 0.100 $m^2/V\cdot s$. Find their diffusion coefficient. *Answer:* $kT/q = 8.617\times10^{-5} \times 350 = 0.0302$ V, so $D_n = (kT/q)\mu_n = 0.0302 \times 0.100 = \mathbf{3.02\times10^{-3}\ m^2/s}$.

Q41. Explain velocity saturation and hot-carrier effects in semiconductors.

> **Question type:** Explanation · **Topic:** high-field effects · **Full notes:** Week 8 (High-Field Effects and Excess Carriers)

What the examiner wants. Two linked parts. Explain why drift velocity stops rising at high field (energy loss by optical-phonon emission), give v_{sat} for Si and sketch v_d against E. Then explain what hot carriers are and list their effects (impact ionisation, oxide injection, substrate current, velocity overshoot) with device consequences and cures.

Key terms.

- **Drift velocity (v_d):** average velocity carriers gain from a field; $v_d = \mu E$ at low field.
- **Saturation velocity (v_{sat}):** the limiting drift velocity at high field, about 10^5 m/s in Si at 300 K.
- **Optical phonon:** a high-energy lattice vibration (about 0.063 eV in Si) that can absorb a large amount of carrier energy in one collision.
- **Hot carrier:** a carrier whose average energy is far above the thermal value (3/2)kT; described by an electron temperature T_e much greater than the lattice temperature T.

- **Impact ionisation:** a hot carrier breaks a covalent bond and creates a new electron–hole pair.
- **Negative differential mobility:** a field range where more field gives less drift velocity (GaAs electrons).
- **LDD (lightly doped drain):** a MOSFET drain extension with lower doping that reduces the peak field.

Model answer.

Velocity saturation

At low field carriers lose the energy they gain from the field to acoustic phonons and stay close to thermal equilibrium, so the drift velocity is proportional to field: $v_d = \mu E$.

At high field (above about 10^6 V/m in Si) carriers gain energy faster than acoustic-phonon scattering can remove it. Once a carrier's extra energy reaches the optical-phonon energy (about 0.063 eV in Si), it emits an optical phonon and loses that energy in one collision. Each time it speeds up it is knocked back, so the drift velocity stops growing and levels off at the saturation velocity:

$$v_{sat} \approx 1\times10^5 \text{ m/s} \quad \text{(Si electrons and holes, 300 K)}$$

An empirical formula (Caughey–Thomas) joins the two regimes:

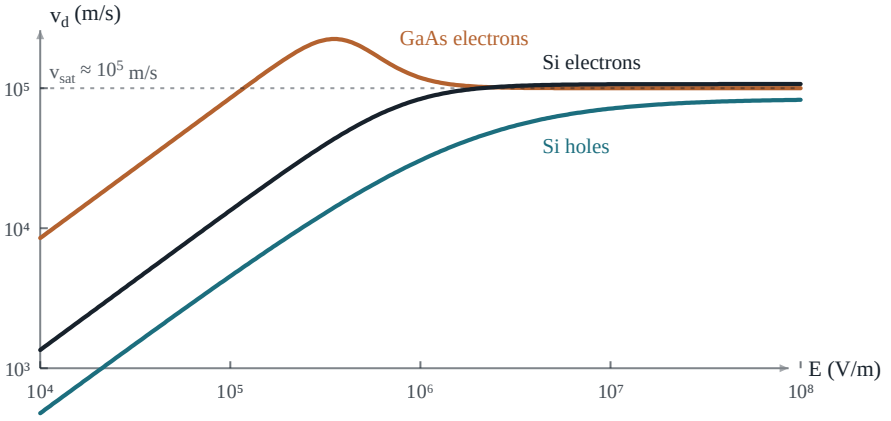
$$v_d = \mu E / [1 + (\mu E / v_{sat})^{\beta}]^{1/\beta}$$

Where:

- v_d = drift velocity (m/s)
- μ = low-field mobility ($m^2/V\cdot s$); Si $\mu_n = 0.135$, $\mu_p = 0.048$
- E = electric field (V/m)
- v_{sat} = saturation velocity (m/s)
- β = fitting exponent, about 2 for Si electrons and 1 for holes (dimensionless)

At low field this gives μE; at high field, v_{sat}. The effective mobility v_d/E therefore falls at high field.

GaAs behaves differently: electrons reach about 2×10^5 m/s near 3.5×10^5 V/m (350 kV/m), then slow down as they transfer to a heavier conduction-band valley (negative differential mobility, used in Gunn diodes).



Drift velocity against electric field: silicon carriers rise linearly then level off near 10^5 m/s; GaAs electrons peak near 2×10^5 m/s and then fall.

Sketch: v_d against E on log axes; a straight line of slope μ at low field bending over to a flat line at $v_{sat} = 10^5$ m/s; label the "ohmic region ($v = \mu E$)" and "saturation region".

Consequences. In short-channel MOSFETs carriers travel at v_{sat} over most of the channel, so drain current becomes proportional to $(V_{GS} - V_T)$ instead of $(V_{GS} - V_T)^2$, and the speed is limited to $f_T \approx v_{sat}/(2\pi L)$; for a channel length L = 100 nm this is about 160 GHz.

Hot-carrier effects

When the field is high, the carriers' average kinetic energy rises far above (3/2)kT; they are "hot", with an effective temperature T_e much higher than the lattice temperature. Hot carriers cause:

1. **Impact ionisation and avalanche breakdown.** A carrier with energy above about $1.5E_g$ knocks an electron out of a bond, creating an electron–hole pair. The new carriers are also accelerated and do the same, giving avalanche multiplication and junction breakdown (used deliberately in avalanche photodiodes).
2. **Hot-carrier injection into the gate oxide.** Near a MOSFET drain some electrons exceed the 3.1 eV Si/SiO_2 barrier, enter the oxide and are trapped or create interface traps. The threshold voltage drifts and transconductance falls over the device's life: a major reliability problem.
3. **Substrate and gate currents.** Holes from impact ionisation flow into the substrate; this can forward-bias source junctions and trigger parasitic bipolar action or CMOS latch-up.
4. **Velocity overshoot.** In very short channels carriers cross before scattering brings them to steady state, briefly exceeding v_{sat}.
5. **Extra noise and light emission** from the hot-carrier region.

Cures: lightly doped drain (LDD) structures to spread the voltage drop and lower the peak field, lower supply voltages, graded junctions, and hardened (nitrided) gate oxides.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ low field: $v_d = \mu E$, carriers near equilibrium
- ✓ mechanism of saturation: optical-phonon emission (about 0.063 eV in Si)
- ✓ $v_{sat} \approx 10^5$ m/s for Si, with a sketch of v_d against E
- ✓ GaAs peak and negative differential mobility (intervalley transfer)
- ✓ definition of hot carriers (energy $\gg$ (3/2)kT, $T_e \gg T$)
- ✓ impact ionisation and avalanche breakdown
- ✓ hot-carrier injection into the oxide and threshold-voltage drift
- ✓ device consequences (short-channel MOSFET current, f_T) and cures (LDD)

Common mistakes.

- Saying carriers "cannot go faster than v_{sat} because of relativity" or "because the mobility becomes zero": the cause is optical-phonon emission, and μ is only *effectively* reduced.
- Calling hot carriers "carriers in a hot crystal": the crystal can be at room temperature; only the carriers are hot.
- Confusing v_{sat} (about 10^5 m/s, the high-field drift limit) with the thermal velocity (also about 10^5 m/s, random motion that carries no current).
- Listing hot-carrier effects without device consequences or cures.

Remember it. "Push too hard and the lattice pushes back": optical phonons cap the speed at 10^5 m/s, and the excess energy makes carriers hot enough to break bonds and burrow into the oxide.

Try a variant. Using the Caughey–Thomas formula with β = 2, find the electron drift velocity in Si at $E = 10^6$ V/m and compare it with the low-field prediction ($\mu_n = 0.135$ m^2/V·s, $v_{sat} = 10^5$ m/s). *Answer:* $\mu E = 1.35\times10^5$ m/s, so v_d =

$1.35\times10^5/\sqrt{(1+1.35^2)} = 1.35\times10^5/1.68 =$ **8.0×10^4 m/s**, about 40 % below the linear value of 1.35×10^5 m/s. The linear formula already fails at this field (the crossover field v_{sat}/μ_n is only 7.4×10^5 V/m).

Q42. What is meant by generation and recombination of charge carriers?

> **Question type:** Definition / explanation · **Topic:** generation and recombination · **Full notes:** Week 8 (High-Field Effects and Excess Carriers)

What the examiner wants. "What is meant by" asks you to define both terms precisely, then show you understand them: name the mechanisms of each, state how they balance in equilibrium and what happens when they do not, and say why they matter in devices. A sketch of the recombination paths earns extra marks.

Key terms.

- **Conduction band / valence band:** the band of energies where electrons move freely / the band of energies occupied by the bonding electrons; they are separated by the band gap E_g (1.12 eV in Si).
- **Electron–hole pair:** a free electron in the conduction band together with the empty state (hole) it left in the valence band.
- **Generation rate (G):** number of electron–hole pairs created per unit volume per unit time ($m^{-3}s^{-1}$).
- **Recombination rate (R):** number of electron–hole pairs destroyed per unit volume per unit time ($m^{-3}s^{-1}$).
- **Trap:** a defect or impurity with an energy level inside the band gap that helps electrons and holes recombine.
- **Thermal equilibrium:** no light, no applied voltage, steady temperature; then $n_0p_0 = n_i^2$.
- **Minority-carrier lifetime (τ):** average time an excess minority carrier survives before it recombines.

Model answer.

Generation

Generation is any process that gives an electron enough energy to leave the valence band (or a level in the gap) and enter the conduction band, creating a free electron and a free hole. Its rate is G ($m^{-3}s^{-1}$). The mechanisms are:

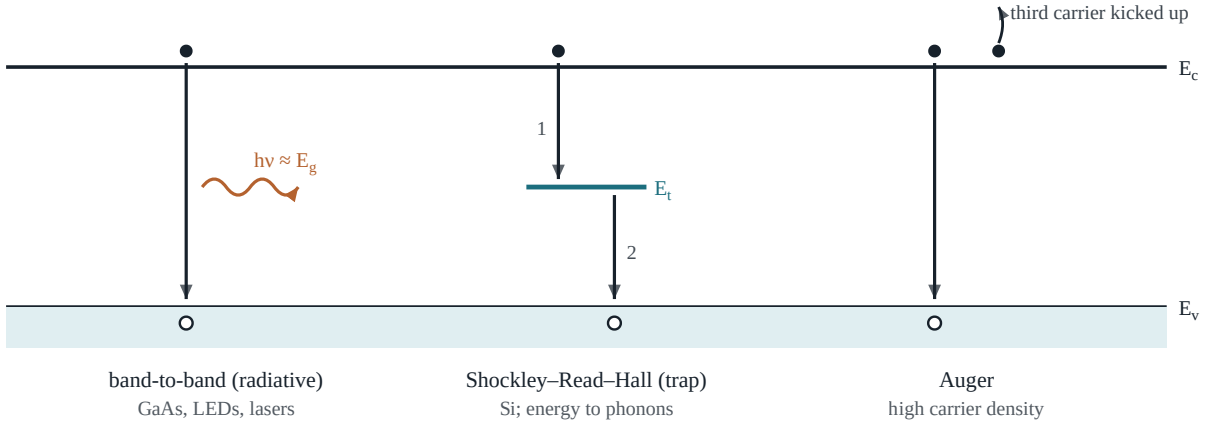
- **Thermal generation:** lattice vibrations (heat) supply the energy, either across the whole gap or in two steps through a trap level.
- **Optical generation:** a photon with energy $h\nu \geq E_g$ is absorbed and creates a pair (photodiodes, solar cells).
- **Impact ionisation:** a carrier accelerated by a strong field hits the lattice hard enough to break a bond (avalanche breakdown).
- **Tunnelling (Zener) generation:** at very high fields an electron tunnels straight from the valence band to the conduction band.

Recombination

Recombination is the reverse: a conduction-band electron falls back into an empty valence-band state, so one electron and one hole disappear together. The energy released (about E_g) leaves as a photon, as heat (phonons, i.e. lattice vibrations), or as kinetic energy of a third carrier. Its rate is R ($m^{-3}s^{-1}$). The mechanisms are:

- **Direct (band-to-band, radiative):** the electron drops straight across the gap and emits a photon; dominant in direct-gap materials such as GaAs (LEDs, lasers).

- **Indirect (trap-assisted, Shockley–Read–Hall, SRH):** the electron drops in two steps via a trap level and gives off heat; dominant in silicon.
- **Auger:** the energy is handed to a third carrier; important at very high carrier densities.
- **Surface recombination:** SRH recombination through states at the surface.



The three recombination paths: band-to-band (photon emitted), trap-assisted SRH (two steps via a gap level, heat released) and Auger (energy passed to a third carrier).

Sketch in the exam: E_c and E_v as two horizontal lines, a short dashed trap level E_t near mid-gap; draw a straight downward arrow (band-to-band, label "hν"), a two-step arrow via E_t (label "SRH, heat") and an upward arrow from E_v to E_c (label "generation").

Balance and imbalance

In thermal equilibrium thermal generation and recombination are equal, so the carrier densities stay constant:

$$G_{th} = R_{th} \quad \text{and} \quad n_0 p_0 = n_i^2$$

Where:

- G_{th}, R_{th} = thermal generation and recombination rates ($m^{-3}s^{-1}$)
- n_0, p_0 = equilibrium electron and hole concentrations (m^{-3})
- n_i = intrinsic carrier concentration, 1×10^{16} m^{-3} in Si at 300 K

When the balance is disturbed, the net rate $U = R - G$ always acts to restore equilibrium:

- **Excess carriers ($np > n_i^2$)**, e.g. from light or forward bias: $U > 0$, net recombination.
- **Carrier deficit ($np < n_i^2$)**, e.g. in a reverse-biased depletion region: $U < 0$, net generation.

For low-level injection (excess much smaller than the majority density) in n-type material the net rate is proportional to the excess:

$$U = \Delta p/\tau_p$$

Where:

- U = net recombination rate ($m^{-3}s^{-1}$)
- $\Delta p = p - p_0$ = excess hole (minority-carrier) concentration (m^{-3})
- τ_p = minority-carrier (hole) lifetime (s)

Why it matters

Generation and recombination set the reverse leakage current of diodes, the photocurrent of solar cells and photodetectors, the light output of LEDs, the current gain of bipolar transistors and the switching speed of p-n junctions.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ generation defined as creation of electron–hole pairs, with rate G in $m^{-3}s^{-1}$
- ✓ recombination defined as annihilation of a pair, energy released as photon, heat or to a third carrier
- ✓ at least three generation mechanisms (thermal, optical $h\nu \geq E_g$, impact ionisation, tunnelling)
- ✓ the recombination mechanisms: direct, SRH (trap-assisted), Auger, surface
- ✓ equilibrium $G_{th} = R_{th}$ with $n_0 p_0 = n_i^2$
- ✓ net recombination when $np > n_i^2$, net generation when $np < n_i^2$
- ✓ $U = \Delta p/\tau$ for low-level injection
- ✓ labelled band diagram and one or two device examples

Common mistakes.

- Saying generation "creates electrons" only: generation across the gap always creates electrons and holes in pairs.
- Writing that recombination stops in equilibrium: both processes continue at equal rates; only the *net* rate is zero.
- Claiming direct recombination dominates in silicon: Si has an indirect gap, so trap-assisted (SRH) recombination dominates.
- Forgetting the units: G, R and U are per unit volume per unit time ($m^{-3}s^{-1}$), not per second.

Remember it. Generation makes pairs, recombination breaks pairs, and nature always pushes np back towards n_i^2.

Try a variant. n-type silicon is uniformly illuminated, producing $G_L = 1\times10^{26}$ $m^{-3}s^{-1}$ pairs, and the hole lifetime is 1 µs. Find the steady-state excess hole concentration. *Answer:* in steady state generation equals net recombination, $G_L = \Delta p/\tau_p$, so $\Delta p = G_L\tau_p = 10^{26} \times 10^{-6} = \mathbf{1\times10^{20}\ m^{-3}}$.

Q43. Explain the Shockley-Read-Hall (SRH) recombination mechanism.

> **Question type:** Explanation · **Topic:** SRH (trap-assisted) recombination · **Full notes:** Week 8 (High-Field Effects and Excess Carriers)

What the examiner wants. Explain how a trap level in the band gap helps an electron and a hole recombine: the four capture and emission processes, the SRH rate equation with every symbol defined, and what it tells us (mid-gap traps are most effective, lifetime $\propto 1/N_t$, low-level and depletion-region limits). A labelled energy diagram and a device application complete the answer.

Key terms.

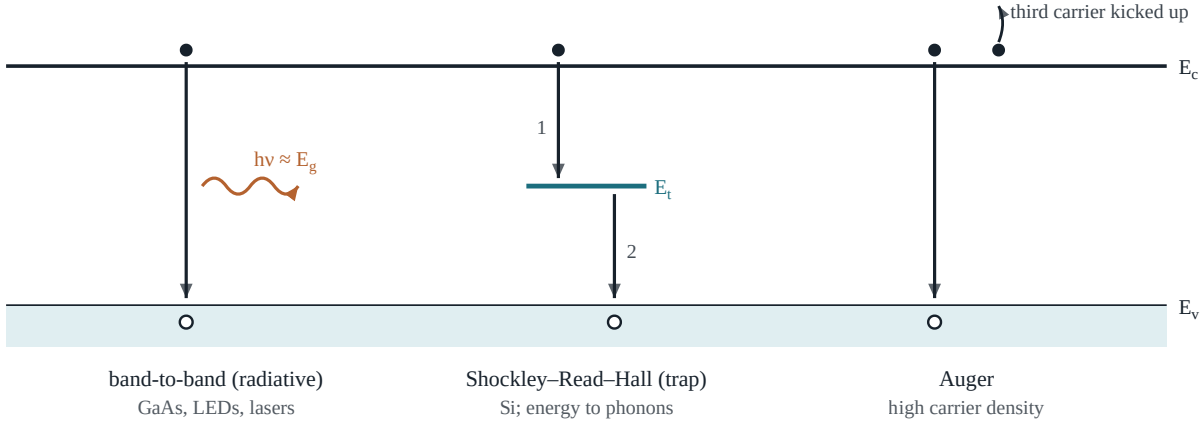
- **Trap (recombination centre):** a localised energy level E_t inside the band gap caused by a defect or impurity (Au, Fe, Cu, vacancies, dislocations); N_t is its density (m^{-3}).
- **Capture cross-section (σ_n, σ_p):** the effective target area a trap presents to an electron or a hole (m^2).
- **Thermal velocity (v_{th}):** the average random speed of carriers, about 1×10^5 m/s at 300 K.
- **Capture / emission:** a trap taking a carrier from a band / giving a carrier back to a band.

- **Intrinsic level (E_i):** the Fermi level of undoped material, close to mid-gap.
- **Phonon:** a quantum of lattice vibration; the energy given to the lattice as heat.
- **Low-level injection:** excess carrier density much smaller than the majority density.

Model answer.

1. Principle

In an indirect-gap semiconductor such as silicon an electron cannot easily drop straight across the gap, because it would have to change its momentum and emit a photon at the same time. Instead it recombines in two smaller steps through a trap level E_t in the gap, a process analysed by Shockley, Read and Hall. The energy is released as heat (phonons), not light.



SRH recombination (middle) uses a trap level in the gap as a stepping stone; band-to-band and Auger recombination are shown for comparison.

Sketch in the exam: E_c and E_v, a trap level E_t near mid-gap, and four arrows labelled (a) to (d) as in the table below.

2. The four processes at a trap

Process	What happens	Rate ($m^{-3}s^{-1}$)
(a) Electron capture	a conduction-band electron drops into an empty trap	$r_a = \sigma_n v_{th} n N_t(1 - f_t)$
(b) Electron emission	a trapped electron is thermally lifted back to the conduction band	$r_b = e_n N_t f_t$
(c) Hole capture	the trapped electron drops to the valence band and fills a hole	$r_c = \sigma_p v_{th} p N_t f_t$
(d) Hole emission	a valence electron jumps into an empty trap, leaving a hole	$r_d = e_p N_t(1 - f_t)$

Here f_t is the probability that a trap is occupied by an electron, and e_n, e_p are emission coefficients (s^{-1}).
Recombination is (a) followed by (c); generation is (d) followed by (b).

3. The SRH rate

In steady state the trap occupation does not change, so the net electron capture equals the net hole capture: $r_a - r_b = r_c - r_d = U$. Detailed balance in equilibrium gives $e_n = \sigma_n v_{th} n_1$ and $e_p = \sigma_p v_{th} p_1$. Solving for f_t and substituting gives the SRH net recombination rate:

$$U = (np - n_i^2) / [\ \tau_p(n + n_1) + \tau_n(p + p_1)\ ]$$

$$\tau_n = 1/(\sigma_n\, v_{th}\, N_t), \quad \tau_p = 1/(\sigma_p\, v_{th}\, N_t)$$

$$n_1 = n_i\, e^{(E_t - E_i)/kT}, \quad p_1 = n_i\, e^{(E_i - E_t)/kT}$$

Where:

- U = net recombination rate ($m^{-3}s^{-1}$); negative U means net generation
- n, p = actual electron and hole concentrations (m^{-3})
- n_i = intrinsic carrier concentration, 1×10^{16} m^{-3} in Si
- τ_n, τ_p = electron and hole lifetimes (s)
- σ_n, σ_p = capture cross-sections (m^2); v_{th} = thermal velocity (m/s); N_t = trap density (m^{-3})
- n_1, p_1 = the electron and hole densities there would be if E_F were at E_t (m^{-3})
- E_t, E_i = trap level and intrinsic level (eV); kT = 0.0259 eV at 300 K

4. What the equation tells us

- **Driving force ($np - n_i^2$).** Positive with excess carriers (recombination), negative when carriers are depleted (generation), zero in equilibrium.
- **Mid-gap traps are most effective.** If E_t is near a band edge, n_1 or p_1 becomes large, the denominator grows and U falls: a shallow trap just captures a carrier and re-emits it. Deep levels near E_i (such as gold in Si) are the efficient recombination centres.
- **$U \propto N_t$.** More traps mean shorter lifetime, $\tau = 1/(\sigma v_{th} N_t)$.
- **Low-level injection in n-type** ($n \gg p, n_1, p_1$): $U = \Delta p/\tau_p$. The rate is limited by the capture of the scarce minority carrier.
- **Reverse-biased depletion region** ($n, p \approx 0$, $E_t = E_i$): $U = -n_i/(\tau_n + \tau_p) = -n_i/2\tau_0$ with $\tau_0 = \tau_n = \tau_p$, a net generation that gives the generation leakage current $J_{gen} = qn_iW/2\tau_0$ (W = depletion width). For $W = 1$ µm (10^{-6} m) and $\tau_0 = 1$ µs, $J_{gen} = 1.6\times10^{-19} \times 10^{16} \times 10^{-6}/(2 \times 10^{-6}) = 8\times10^{-4}$ A/m^2.

5. Example and applications

With $N_t = 1\times10^{19}$ m^{-3}, $\sigma_p = 1\times10^{-19}$ m^2 and $v_{th} = 1\times10^5$ m/s: $\tau_p = 1/(10^{-19} \times 10^5 \times 10^{19}) = 1\times10^{-5}$ s = 10 µs.

- **Lifetime killing:** gold or platinum diffusion, or electron irradiation, adds mid-gap traps on purpose to cut τ to nanoseconds in fast-switching diodes and thyristors.
- **Lifetime preservation:** clean processing and gettering remove traps to keep τ long in solar cells, CCD imagers and low-leakage devices.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ recombination through a trap level E_t in the gap, two steps, energy released as phonons
- ✓ why it dominates in Si (indirect gap makes direct recombination unlikely)
- ✓ the four processes (electron capture/emission, hole capture/emission) with a labelled diagram
- ✓ SRH equation $U = (np - n_i^2)/[\tau_p(n + n_1) + \tau_n(p + p_1)]$ with symbols defined
- ✓ $\tau = 1/(\sigma v_{th} N_t)$
- ✓ mid-gap traps most effective; shallow traps ineffective

- ✓ low-level limit $U = \Delta p/\tau_p$; depletion-region generation $-n_i/2\tau_0$
- ✓ an application (gold doping for fast diodes, or gettering for long lifetime)

Common mistakes.

- Saying SRH emits light: the energy goes to the lattice as heat; photons come from band-to-band recombination.
- Claiming the shallowest traps are the most effective: it is the deep, mid-gap traps.
- Writing $\tau = \sigma v_{th} N_t$ instead of its inverse: more traps must give a *shorter* lifetime.
- Leaving out the generation case: the same equation with $np < n_i^2$ explains diode leakage current.

Remember it. An electron crossing a wide river uses a stepping stone in the middle; stones near the banks are no help.

Try a variant. n-type Si has $N_D = 1\times10^{21}$ m^{-3} and $\tau_n = \tau_p = 1$ µs, but the traps lie 0.3 eV above E_i instead of at mid-gap. Find the effective low-level lifetime. *Answer:* $n_1 = n_i\, e^{0.3/0.0259} = 10^{16} \times e^{11.58} = 1.07\times10^{21}$ m^{-3} ($p_1 \approx 9\times10^{10}$ m^{-3} is negligible). With $n \approx N_D \gg p$, $U \approx \Delta p \times N_D/[\tau_p(N_D + n_1)]$, so $\tau_{eff} = \tau_p(1 + n_1/N_D) = 1$ µs $\times$ (1 + 1.07) ≈ **2.1 µs**: moving the trap 0.3 eV off mid-gap roughly halves its effectiveness.

Q44. Define minority-carrier lifetime and explain its significance in semiconductor devices.

> **Question type:** Definition and explanation · **Topic:** minority-carrier lifetime · **Full notes:** Week 8 (High-Field Effects and Excess Carriers)

What the examiner wants. A precise definition (with the equation $U = \Delta p/\tau$ and the exponential decay), how lifetime depends on traps, and then a list of devices where lifetime controls performance, each with the reason. Quote a number (for example a diffusion length) to show depth.

Key terms.

- **Minority carrier:** the less numerous carrier type: holes in n-type, electrons in p-type material.
- **Excess carrier concentration (Δp, Δn):** the carrier density above its equilibrium value, $\Delta p = p - p_0$ (m^{-3}).
- **Low-level injection:** Δp much smaller than the majority-carrier density, so the majority density hardly changes.
- **Recombination rate (U):** net number of electron–hole pairs destroyed per m^3 per second.
- **Diffusion coefficient (D):** how quickly carriers spread by diffusion (m^2/s); $D_p = 1.24\times10^{-3}$ m^2/s for holes in Si.
- **Diffusion length (L):** average distance a minority carrier diffuses before recombining, $L = \sqrt{(D\tau)}$.
- **Trap density (N_t):** concentration of recombination centres in the gap (m^{-3}).

Model answer.

Definition

The minority-carrier lifetime is the average time an excess minority carrier survives in a semiconductor before it recombines with a majority carrier. Under low-level injection it is defined by the recombination rate:

$$U = \Delta p/\tau_p \quad \text{(holes in n-type)}; \quad U = \Delta n/\tau_n \quad \text{(electrons in p-type)}$$

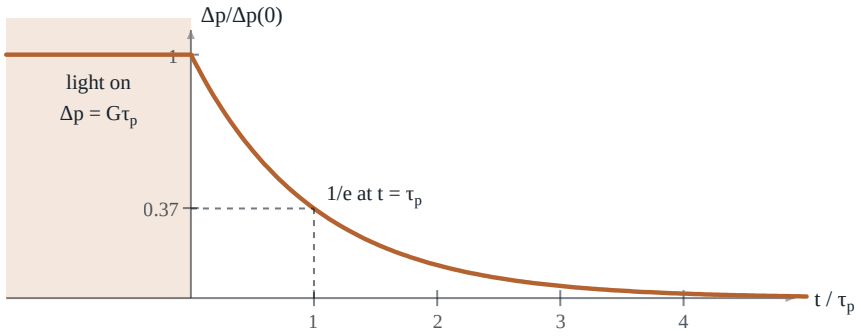
If the source of excess carriers (for example light) is removed at t = 0, the excess decays exponentially:

$$\Delta p(t) = \Delta p(0)\, e^{-t/\tau_p}$$

Where:

- U = net recombination rate ($m^{-3}s^{-1}$)
- $\Delta p(t)$ = excess hole concentration at time t (m^{-3}); $\Delta p(0)$ = its value at t = 0
- τ_p, τ_n = hole and electron minority-carrier lifetimes (s)
- t = time after the source is removed (s)

So τ is the time for the excess to fall to 1/e (37 %) of its starting value.



Under steady light the excess holds at $\Delta p = G_L\tau_p$; after the light is switched off it decays exponentially, falling to 1/e of its value in one lifetime τ_p.

Sketch in the exam: Δp against t, flat at $G_L\tau_p$, then an exponential fall after "light off", with τ_p marked where the curve reaches 0.37 of the start.

What sets τ. For trap-assisted (SRH) recombination, $\tau_p = 1/(\sigma_p\, v_{th}\, N_t)$, where σ_p is the trap capture cross-section (m^2), $v_{th} \approx 10^5$ m/s the thermal velocity and N_t the trap density. Lifetime is inversely proportional to the trap density, so it is a sensitive measure of crystal purity: about 1 ns to 1 ms in silicon.

Significance in devices

1. **Diffusion length.** Carriers spread a distance $L = \sqrt{(D\tau)}$ before recombining. For holes in Si with $D_p = 1.24\times10^{-3}$ m^2/s and $\tau_p = 1$ μs, $L_p = \sqrt{(1.24\times10^{-3} \times 10^{-6})} = 3.5\times10^{-5}$ m (35 μm). L decides how thick device regions can be.
2. **Diode current.** The saturation current $J_s = qn_i^2[D_p/(L_pN_D) + D_n/(L_nN_A)]$ depends on τ through L, and the depletion-region generation current $qn_iW/2\tau_0$ is inversely proportional to τ. Short lifetime means more leakage.
3. **Switching speed.** A forward-biased diode stores minority charge $Q = I\tau$, which must be removed before it can turn off, so the reverse-recovery time is of the order of τ. Fast-recovery diodes are gold- or platinum-doped to cut τ to nanoseconds.
4. **Bipolar transistor gain.** Electrons injected into the base must cross it before recombining. The base transport factor is about $1 - W_B^2/2L_n^2$ (W_B = base width), so a long lifetime ($L_n \gg W_B$) gives a high current gain β.
5. **Solar cells and photodetectors.** Photogenerated carriers must diffuse to the junction before they recombine; a longer L collects more of them, raising photocurrent and efficiency.
6. **LEDs and lasers.** The share of recombination that is radiative, and the maximum modulation speed (bandwidth roughly $\propto 1/\tau$), depend on lifetime.
7. **Power devices.** In p-i-n diodes and IGBTs a long lifetime gives conductivity modulation and a low on-state voltage but slow turn-off: a design trade-off.
8. **Material quality.** Because τ reacts strongly to metal contamination, it is measured (photoconductive decay) to monitor crystal growth and process cleanliness.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definition: average time an excess minority carrier exists before recombining
- ✓ $U = \Delta p/\tau_p$ under low-level injection
- ✓ $\Delta p(t) = \Delta p(0)e^{-t/\tau}$; τ is the 1/e time
- ✓ $\tau = 1/(\sigma v_{th} N_t)$: lifetime falls as trap density rises
- ✓ $L = \sqrt{(D\tau)}$ with a numerical example
- ✓ at least four devices explained: diode leakage, switching speed, BJT gain, solar cells
- ✓ the speed-versus-gain/efficiency trade-off (gold doping)

Common mistakes.

- Defining lifetime for majority carriers: it is the *minority* carriers whose excess decays with τ, because their relative change is large.
- Saying "long lifetime is always good": it is good for gain and efficiency but bad for switching speed.
- Writing $L = D\tau$ instead of $L = \sqrt{(D\tau)}$: check the unit, $\sqrt{(m^2/s \times s)} = m$.
- Stating that τ is the time for all excess carriers to disappear: after one τ, 37 % are still there.

Remember it. Lifetime is how long, diffusion length is how far: $L = \sqrt{(D\tau)}$.

Try a variant. Excess holes in n-type Si have $\tau_p = 1$ µs. How long after the light is switched off does Δp fall to 10 % of its initial value? *Answer:* $0.1 = e^{-t/\tau_p}$, so $t = \tau_p \ln 10 = 1\ \mu s \times 2.303 =$ **2.3 µs**.

Q45. Explain the continuity equation for minority carriers and discuss its application to semiconductor devices.

> **Question type:** Explanation with derivation · **Topic:** continuity equation · **Full notes:** Week 9 (Continuity Equation and Surface Phenomena)

What the examiner wants. State the continuity equation as carrier conservation, derive it from a thin slab, reduce it to the minority-carrier diffusion equation with the assumptions stated, then show how it is solved and used in real devices (diode, BJT, solar cell, lifetime measurement). Standard solutions with their equations earn the "discuss application" marks.

Key terms.

- **Continuity equation:** a book-keeping statement that carriers are conserved: change = net inflow + generation − recombination.
- **Current density (J_p, J_n):** hole and electron current per unit area (A/m^2).
- **Drift and diffusion:** carrier motion driven by an electric field E, and by a concentration gradient.
- **Excess carrier concentration (Δp):** $p - p_0$, the density above equilibrium (m^{-3}).
- **Quasi-neutral region:** a region outside the depletion layer where the net charge, and so the field, is nearly zero.
- **Lifetime (τ_p):** average survival time of an excess minority carrier (s).
- **Diffusion length (L_p):** $L_p = \sqrt{(D_p\tau_p)}$, how far minority carriers travel before recombining (m).

Model answer.

1. Statement

In any small volume, the rate at which the carrier density increases equals the net inflow due to current, plus generation, minus recombination. In one dimension:

$$\partial p/\partial t = -(1/q)\,\partial J_p/\partial x + G_p - R_p$$

$$\partial n/\partial t = +(1/q)\,\partial J_n/\partial x + G_n - R_n$$

Where:

- p, n = hole and electron concentrations (m^{-3}); t = time (s); x = position (m)
- J_p, J_n = hole and electron current densities (A/m^2)
- q = electronic charge, 1.6×10^{-19} C
- G_p, G_n = generation rates; R_p, R_n = recombination rates ($m^{-3}s^{-1}$)

The signs differ because electrons carry negative charge: electron flow is opposite to J_n.

2. Derivation (holes)

Take a slab of area A between x and x + dx.

- Holes flowing in at x per second: $J_p(x)A/q$. Holes flowing out at x + dx: $J_p(x + dx)A/q$.
- Holes created per second: $G_pA\,dx$; holes lost: $R_pA\,dx$.
- Rate of change of the number in the slab: $(\partial p/\partial t)A\,dx = (1/q)[J_p(x) - J_p(x + dx)]A + (G_p - R_p)A\,dx$.
- Using $J_p(x + dx) - J_p(x) = (\partial J_p/\partial x)dx$ and dividing by A dx gives the hole equation above.

3. Minority-carrier diffusion equation

For holes in uniformly doped n-type material, substitute the current $J_p = q\mu_{pp}E - qD_p\,\partial p/\partial x$ and, for low-level injection, $R_p - G_{th} = \Delta p/\tau_p$, with G_L the extra generation by light. Assume:

- low-level injection ($\Delta p \ll n_0$), so recombination is $\Delta p/\tau_p$;
- a quasi-neutral region, so $E \approx 0$ and drift is negligible;
- uniform doping, so p_0 does not vary with x.

Then:

$$\partial \Delta p/\partial t = D_p\,\partial^2\Delta p/\partial x^2 - \Delta p/\tau_p + G_L$$

Where:

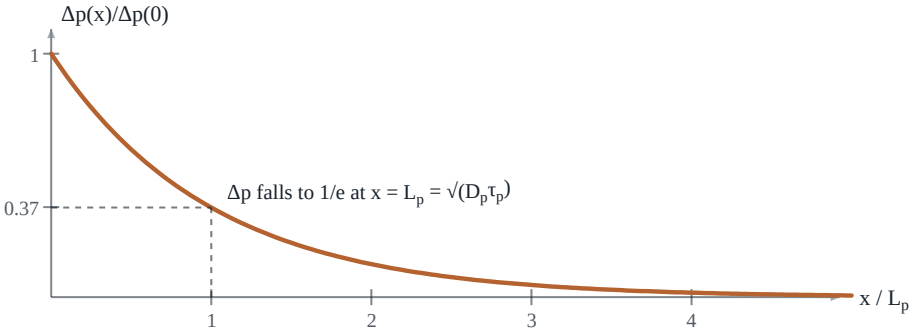
- D_p = hole diffusion coefficient (m^2/s), 1.24×10^{-3} m^2/s in Si
- τ_p = hole (minority-carrier) lifetime (s)
- G_L = generation rate by light ($m^{-3}s^{-1}$)

This is the minority-carrier diffusion equation. For electrons in p-type: $\partial \Delta n/\partial t = D_n\,\partial^2\Delta n/\partial x^2 - \Delta n/\tau_n + G_L$. Each term has a meaning: diffusion spreads carriers, recombination removes them, light creates them.

4. Standard solutions and applications

- **Uniform light switched off (photoconductive decay):** no x-dependence and $G_L = 0$ give $\Delta p(t) = \Delta p(0)e^{-t/\tau_p}$. Measuring the fall in conductivity gives the lifetime.

- **Steady uniform light:** $\partial/\partial t = 0$ and no gradient give $\Delta p = G_L \tau_p$ (photoconductors, light-dependent resistors).
- **p-n junction diode (long neutral region):** steady state, no light: $D_p \, d^2\Delta p/dx^2 = \Delta p/\tau_p$ gives $\Delta p(x) = \Delta p(0) e^{-x/L_p}$. The diffusion current at the depletion edge, $J_p = qD_p\Delta p(0)/L_p$, leads to the Shockley diode equation.
- **Bipolar transistor (short base, $W_B \ll L_n$):** recombination is negligible, the profile is almost a straight line, and its slope gives the collector current; the small recombination loss gives the base transport factor and current gain.
- **Solar cells and photodiodes:** with $G_L(x) = G_0 e^{-\alpha x}$ (α = absorption coefficient) the solution gives the collected photocurrent and shows that L must be longer than the absorption depth $1/\alpha$.
- **Haynes–Shockley experiment:** a pulse of injected carriers drifts and spreads as a moving Gaussian; its speed gives μ_p, its spreading gives D_p, its shrinking area gives τ_p, confirming the Einstein relation.
- **Switching transients:** the time-dependent equation gives the stored charge and the reverse-recovery time of diodes.
- **Device simulation:** with Poisson's equation and the current equations, the continuity equations are solved numerically in every TCAD simulator.



Steady-state solution for holes injected at x = 0 into a long n-type bar: the excess decays exponentially with characteristic length L_p.

Sketch in the exam: Δp against x, starting at $\Delta p(0)$ and decaying exponentially; mark L_p where it has fallen to 0.37 $\Delta p(0)$.

Worked number. Holes injected into a long n-type Si bar with $\Delta p(0) = 1\times10^{20}$ m^{-3}, $D_p = 1.24\times10^{-3}$ m^2/s, $\tau_p = 1$ µs: $L_p = \sqrt{(1.24\times10^{-3} \times 10^{-6})} = 3.52\times10^{-5}$ m (35.2 µm), and $J_p(0) = qD_p\Delta p(0)/L_p = 1.6\times10^{-19} \times 1.24\times10^{-3} \times 10^{20}/3.52\times10^{-5} = 5.6\times10^{2}$ A/m^2.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ physical statement: conservation of carriers (inflow + generation − recombination)
- ✓ $\partial p/\partial t = -(1/q)\partial J_p/\partial x + G - R$ and the electron form with the opposite sign
- ✓ slab derivation
- ✓ assumptions stated (low-level injection, $E \approx 0$, uniform doping)
- ✓ minority-carrier diffusion equation $\partial\Delta p/\partial t = D_p\partial^2\Delta p/\partial x^2 - \Delta p/\tau_p + G_L$
- ✓ standard solutions: $e^{-t/\tau}$, $G_L\tau$, e^{-x/L_p}, linear profile in a short region
- ✓ at least three device applications (diode, BJT, solar cell, lifetime measurement)
- ✓ $L_p = \sqrt{(D_p\tau_p)}$ defined

Common mistakes.

- Getting the sign of the current term wrong for electrons: it is $+(1/q)\partial J_n/\partial x$ because electrons are negative.
- Dropping the drift term without saying why: state that the region is quasi-neutral ($E \approx 0$).

- Writing the recombination term as $\Delta p \times \tau_p$ instead of $\Delta p/\tau_p$.
- Using the long-region exponential for a narrow BJT base: when $W \ll L$ the profile is a straight line.

Remember it. "What comes in, plus what is made, minus what dies, is what piles up."

Try a variant. For the same bar ($\Delta p(0) = 1\times10^{20}$ m^{-3}, $L_p = 35.2$ μm), find the excess hole concentration at $x = 2L_p$. *Answer:* $\Delta p = \Delta p(0)e^{-2} = 10^{20} \times 0.135 =$ **1.35×10^{19} m^{-3}**: after two diffusion lengths only 13.5 % of the injected holes remain.

Q46. What is surface recombination velocity? Explain how semiconductor surface states affect carrier lifetime.

> **Question type:** Definition and explanation · **Topic:** surface recombination · **Full notes:** Week 9 (Continuity Equation and Surface Phenomena)

What the examiner wants. Define S with its equation and unit, explain where surface states come from, then explain how they lower the effective lifetime, with the τ_{eff} formula and its limit of validity. A short number example and the ways of reducing S (passivation) complete a full-mark answer.

Key terms.

- **Surface states:** localised energy levels in the band gap at a surface, caused by broken (dangling) bonds, adsorbed atoms and oxide defects.
- **Dangling bond:** an unpaired bond left on a surface atom because the crystal ends there.
- **Surface recombination velocity (S):** the rate of recombination per unit area divided by the excess minority density at the surface (m/s).
- **Bulk lifetime (τ_{bulk}):** the lifetime set by recombination inside the volume alone (s).
- **Effective lifetime (τ_{eff}):** the lifetime actually measured, including both bulk and surface losses (s).
- **Diffusion coefficient (D):** how fast minority carriers spread by diffusion (m^2/s); $D_p = 1.2\times10^{-3}$ m^2/s for holes is used below.
- **Passivation:** treating a surface so that it has few active surface states.

Model answer.

1. Surface recombination velocity

At a surface, recombination takes place in a very thin layer, so it is counted per unit area rather than per unit volume:

$$U_s = S\,\Delta p_s$$

Where:

- U_s = surface recombination rate (m^{-2}s^{-1})
- S = surface recombination velocity (m/s)
- Δp_s = excess minority-carrier (hole) concentration at the surface (m^{-3})

S has the units of a velocity: it is the effective speed at which excess carriers flow into the surface and are annihilated. For surface traps of density N_{st} (m^{-2}) with capture cross-section σ_p (m^2):

$$S = \sigma_p \, v_{th} \, N_{st}$$

with $v_{th} \approx 1\times10^5$ m/s the thermal velocity. S appears in the diffusion equation as a boundary condition at the surface: carriers arriving by diffusion equal carriers recombining there, $D_p \, d\Delta p/dx = S\Delta p_s$.

- S = 0: an ideal surface, no recombination.
- $S \to \infty$: every arriving carrier recombines, $\Delta p_s = 0$ (like an Ohmic contact).
- Typical values: bare Si 10^2–10^4 m/s; thermally oxidised Si 0.1–10 m/s; well-passivated Si below 0.1 m/s.

2. Origin of surface states

The periodic crystal ends at the surface. Surface atoms have dangling bonds; there are also adsorbed impurities, oxide defects and contamination. Together they create a high density of localised levels spread through the band gap, many near mid-gap, which are exactly the levels that make efficient Shockley–Read–Hall (SRH, trap-assisted) recombination centres.

3. How surface states affect lifetime

- **An extra recombination path.** Carriers that reach the surface recombine there quickly, so the measured lifetime is shorter than the bulk lifetime. The rates add:

  $$1/\tau_{eff} = 1/\tau_{bulk} + 1/\tau_{surface}$$

- **Small S ($SW/2D \ll 1$).** Carriers reach the surfaces easily and the excess is nearly uniform across a wafer of thickness W with both faces recombining, so $1/\tau_{surface} = 2S/W$.
- **Large S (diffusion limit).** The surfaces swallow carriers faster than diffusion can bring them, so the loss is limited by the time to diffuse to a surface. A good approximation for any S is:

  $$\tau_{surface} \approx W/(2S) + W^2/(\pi^2 D)$$

Where:

- τ_{eff}, τ_{bulk}, $\tau_{surface}$ = effective, bulk and surface-limited lifetimes (s)
- W = wafer thickness (m)
- D = minority-carrier diffusion coefficient (m^2/s)
- $W/(2S)$ = time for the surfaces to remove carriers once they arrive; $W^2/(\pi^2 D)$ = time for carriers to diffuse to the surfaces, the shortest $\tau_{surface}$ can ever be

For small S the first term dominates and the formula reduces to $1/\tau_{surface} = 2S/W$.

- **Depletion of carriers near the surface.** Δp falls towards the surface, and the gradient keeps pulling carriers there, so even carriers generated deep inside are lost.
- **Surface band bending.** Charge in surface states bends the bands. Under depletion, electron and hole densities near the surface become similar, which maximises SRH recombination and generation.
- **Leakage, noise and instability.** Surface generation adds to junction leakage where the junction meets the surface; trapping and release of carriers gives 1/f noise; charged states shift MOS threshold voltages.
- **Device performance.** High S lowers solar-cell efficiency (especially for blue light absorbed near the surface), lowers BJT gain through recombination at the emitter edge, and weakens photodetector response.

4. Worked example

Given: n-type Si wafer, W = 300 μm = 3×10^{-4} m, τ_{bulk} = 100 μs, $D_p = 1.2\times10^{-3}$ m^2/s; oxidised faces S = 1 m/s; bare faces S = 10^3 m/s.

Find: τ_{eff} in each case.

Solution.

Step 1. Oxidised: $SW/2D = 1 \times 3\times10^{-4}/(2.4\times10^{-3}) = 0.13$ (small), so the simple formula applies.

$$1/\tau_{eff} = 1/10^{-4} + 2 \times 1/(3\times10^{-4}) = 1.0\times10^4 + 6.7\times10^3 = 1.67\times10^4\ s^{-1} \quad \Rightarrow \quad \tau_{eff} = 60\ \mu s$$

Step 2. Bare: $SW/2D = 10^3 \times 3\times10^{-4}/(2.4\times10^{-3}) = 125$ (not small), so use the full form. $W/(2S) = 3\times10^{-4}/(2 \times 10^3) = 0.15$ μs; $W^2/(\pi^2 D) = (3\times10^{-4})^2/(9.87 \times 1.2\times10^{-3}) = 7.6$ μs; $\tau_{surface} \approx 7.75$ μs.

$$1/\tau_{eff} = 1/100\ \mu s + 1/7.75\ \mu s \quad \Rightarrow \quad \tau_{eff} \approx 7.2\ \mu s$$

Answer: τ_{eff} = 60 μs (oxidised) and about 7.2 μs (bare). The simple formula would wrongly give 0.15 μs for the bare wafer.

What this means: a good oxide keeps most of the bulk lifetime; with bare surfaces the surfaces control the lifetime, and the loss is then limited by how fast holes can diffuse 150 μm to the nearest face.

5. Reducing surface recombination (passivation)

- Thermal SiO_2 followed by a forming-gas (H_2) anneal, which ties up dangling bonds.
- Silicon nitride or Al_2O_3 films whose fixed charge repels minority carriers (field-effect passivation).
- A heavily doped surface layer (back-surface field) that keeps minority carriers away from the surface.
- Clean, damage-free surface preparation.

Sketch in the exam: Δp against x across a wafer, flat-topped for small S and dropping towards both faces for large S; label the faces "S".

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ $U_s = S\Delta p_s$ with S in m/s, and $S = \sigma_{pvth} N_{st}$
- ✓ boundary condition $D_p\, d\Delta p/dx = S\Delta p_s$; limits S = 0 and $S \to \infty$
- ✓ origin of surface states: dangling bonds, adsorbates, oxide defects, levels near mid-gap
- ✓ $1/\tau_{eff} = 1/\tau_{bulk} + 1/\tau_{surface}$
- ✓ $1/\tau_{surface} = 2S/W$ only when $SW/2D \ll 1$; diffusion limit $\tau_{surface} \approx W/(2S) + W^2/(\pi^2 D)$ for large S
- ✓ band bending and leakage/noise effects
- ✓ a numerical example or typical values of S
- ✓ passivation methods

Common mistakes.

- Using $1/\tau_{eff} = 1/\tau_{bulk} + 2S/W$ for a bare surface: when $SW/2D$ is large it predicts absurdly short lifetimes; diffusion to the surface sets the limit $W^2/(\pi^2 D)$.
- Giving S the unit s^{-1} or $m^{-2}s^{-1}$: S is a velocity, m/s; U_s is the per-area rate.
- Adding lifetimes instead of rates: it is the *inverses* ($1/\tau$) that add.

- Treating surface states as single levels near the band edges: they are spread through the gap, with many near mid-gap, which is why they recombine so well.

Remember it. The surface is a carrier trapdoor: S says how wide it opens, diffusion says how fast carriers can reach it.

Try a variant. Repeat the 300 µm wafer (τ_{bulk} = 100 µs, $D_p = 1.2\times10^{-3}$ m^2/s) for a poorly passivated surface with S = 10^2 m/s. *Answer:* SW/2D = $10^2 \times 3\times10^{-4}/(2.4\times10^{-3}) = 12.5$, not small. $\tau_{surface} \approx 3\times10^{-4}/(2 \times 10^2) + 7.6$ µs = 1.5 µs + 7.6 µs = 9.1 µs; $1/\tau_{eff} = 1/100 + 1/9.1$ (µs^{-1}), so $\boldsymbol{\tau_{eff} \approx 8.3}$ **µs** (the simple formula would wrongly give 1.5 µs).

Questions 47–50 — Junctions, Interfaces and Characterisation

Module 4: the p-n junction and its diode equation, metal–semiconductor contacts, and how devices are measured with I-V and C-V characterisation. These are long questions: practise writing them in full. Each question stands on its own: its key terms are defined with it, so you can start anywhere.

Symbols and abbreviations used on this page

Every symbol, constant and abbreviation on this page is listed here, even if another page already defined it, so you can study this page on its own.

Constants and units

Symbol	Meaning	Units / typical value
q	Magnitude of the electronic charge	1.6×10^{-19} C
k	Boltzmann constant	8.617×10^{-5} eV/K = 1.381×10^{-23} J/K
kT	Thermal energy	0.0259 eV (often 0.02585 eV) at 300 K
kT/q	Thermal voltage	0.0259 V (about 26 mV) at 300 K
T	Absolute temperature	kelvin (K); room temperature = 300 K
ε_s (ε_{si})	Permittivity of silicon, $11.7\varepsilon_0$	1.04×10^{-10} F/m
eV	Electron-volt, a unit of energy	1 eV = 1.6×10^{-19} J
µm	Micrometre (micron)	1 µm = 10^{-6} m

Symbols

Symbol	Meaning	Units / typical value
E_c (E_C)	Energy of the bottom of the conduction band	eV
E_v (E_V)	Energy of the top of the valence band	eV
E_g	Band-gap energy, $E_c - E_v$	eV; Si 1.12, Ge 0.66, GaAs 1.42 at 300 K
E_F	Fermi level: the energy at which a state has a 50 % chance of being occupied	eV
E_i	Intrinsic Fermi level, close to mid-gap	eV
N_c	Effective density of states in the conduction band	m^{-3}; Si 2.8×10^{25} at 300 K
N_v	Effective density of states in the valence band	m^{-3}; Si 1.04×10^{25} at 300 K
n	Electron concentration (free electrons per m^3)	m^{-3}
p	Hole concentration (holes per m^3)	m^{-3}
n_i	Intrinsic carrier concentration ($n = p = n_i$ in pure material)	m^{-3}; Si $\approx 1\times10^{16}$ at 300 K
p_{n0}, n_{p0}	Equilibrium minority concentrations: holes on the n side, electrons on the p side	m^{-3}; $p_{n0} = n_i^2/N_D$
N_D	Donor concentration	m^{-3}
N_A	Acceptor concentration	m^{-3}
N_D^+, N_A^-	Concentrations of ionised donors and acceptors	m^{-3}
Δn, Δp	Excess electron and hole concentrations above equilibrium	m^{-3}
C(x, t)	Dopant concentration at depth x and time t	m^{-3}
R_s	Sheet resistance (or diode series resistance in Week 12)	Ω/□ (ohms per square); Ω
E (field)	Electric field (on band diagrams, E is energy instead)	V/m
μ, μ_n, μ_p	Mobility: drift velocity per unit field (electrons, holes)	$m^2/(V\cdot s)$; Si μ_n 0.135, μ_p 0.048
J, J_n, J_p	Current density (total, electrons, holes)	A/m^2
D_n, D_p	Diffusion coefficients of electrons and holes	m^2/s; Si 3.5×10^{-3} and 1.24×10^{-3}
G, G_L	Generation rate of electron–hole pairs (G_L: by light)	$m^{-3}s^{-1}$
τ, τ_n, τ_p	Minority-carrier lifetime (τ_{eff}: effective, including surfaces)	s; Si 1 ns – 1 ms
L, L_n, L_p	Diffusion length $\sqrt{(D\tau)}$: distance a minority carrier travels before recombining	µm or m
V_{bi}	Built-in potential of a junction or Schottky contact	V; typically 0.5–1 V
V, V_a	Applied voltage (positive for forward bias)	V
W	Depletion-region width (in Week 13 W_B, W_E are base and emitter widths; in Week 14 W is channel width)	µm or m
x_n, x_p	Parts of the depletion region on the n side and on the p side	µm
E_{max}	Peak electric field, at the junction	V/m
I_s, J_s (I_S)	Saturation current (density) of a diode or transistor	A; A/m^2
I	Current	A

Symbol	Meaning	Units / typical value
A (area)	Device or contact area	m^2
η	Ideality factor of a diode, 1 (ideal) to 2	dimensionless
C, C_j	Capacitance (junction or depletion capacitance)	F (pF = 10^{-12} F)
Φ_m	Work function of a metal	eV; Al 4.28, Au 5.1
Φ_s	Work function of a semiconductor, $\chi + (E_c - E_F)$	eV
χ	Electron affinity: energy from E_c up to the vacuum level	eV; Si 4.05
Φ_B	Schottky barrier height seen from the metal	eV (or V)
A^{**}	Effective Richardson constant	$A/(m^2 \cdot K^2)$; n-Si $\approx 1.1 \times 10^6$
ρ_c	Specific contact resistance	$\Omega \cdot m^2$
E_{vac}	Vacuum level: energy of an electron just free of the material	eV
n_{n0}, p_{p0}, p_{n0}, n_{p0}	Equilibrium densities: majority (n_{n0}, p_{p0}) and minority (p_{n0} holes on n side, n_{p0} electrons on p side)	m^{-3}
Δp_n, Δn_p	Excess minority carriers at the depletion edges (holes on n side, electrons on p side)	m^{-3}
R_c	Contact resistance	Ω (specific: $\Omega \cdot m^2$)
$\Phi_B n$, $\Phi_B p$	Schottky barrier height for electrons (n-type) and for holes (p-type)	eV
p_n, n_p	Minority-carrier density: holes on the n side, electrons on the p side	m^{-3}
N(x), N(W)	Doping concentration at depth x (or at the depletion edge W), from C-V profiling	m^{-3}
t	Time (or a thickness, where the text says so)	s (or m)

Abbreviations

Abbreviation	Meaning
Si	Silicon
Ge	Germanium
GaAs	Gallium arsenide
I-V	Current–voltage (measurement or characteristic)
C-V	Capacitance–voltage (measurement or characteristic)
LCR	Inductance–capacitance–resistance meter
MOSFET	Metal–oxide–semiconductor field-effect transistor
MESFET	Metal–semiconductor field-effect transistor
BJT	Bipolar junction transistor
RF	Radio frequency
DC	Direct current (steady)
AC	Alternating current (small signal)
n^+	Very heavily doped n-type (similarly p^+)

Questions on this page

Q	Question
47	Explain the formation of the depletion region and built-in potential in a p-n junction.
48	Derive the ideal Shockley diode equation and explain the effect of forward and reverse bias.
49	Differentiate between a Schottky barrier diode and an Ohmic contact. Explain their energy-band characteristics.
50	Explain the principles of I-V and C-V characterisation of semiconductor devices and describe how C-V measurements can be used to estimate doping concentration.

Model answers

Q47. Explain the formation of the depletion region and built-in potential in a p-n junction.

> **Question type:** Explanation · **Topic:** p-n junction electrostatics · **Full notes:** Week 10 (The p-n Junction Diode)

What the examiner wants. "Explain" means give the physical cause-and-effect chain: diffusion, uncovered ions, field, drift that opposes diffusion, equilibrium with a flat Fermi level. Then state what the built-in potential is, derive its formula from the Fermi-level positions, and back it up with a labelled sketch and a typical number.

Key terms.

- **p-n junction:** the boundary between a p-type and an n-type region in one continuous crystal.
- **Acceptor and donor ions (N_A^-, N_D^+):** dopant atoms that have taken or given up an electron; they are charged and fixed in the lattice, so they cannot move.
- **Diffusion:** movement of carriers from where they are many to where they are few.
- **Drift:** movement of carriers pushed by an electric field.
- **Depletion region (space-charge region), width W:** the layer around the junction emptied of mobile carriers, containing only fixed ion charge.
- **Fermi level (E_F):** the energy at which a state has a 50 % chance of being occupied; in equilibrium it is flat (the same) throughout a connected system.
- **Built-in potential (V_{bi}):** the voltage across the depletion region in equilibrium; the barrier majority carriers must climb to cross the junction.
- **Intrinsic carrier concentration (n_i):** carrier density of pure silicon, 1×10^{16} m^{-3} at 300 K.

Model answer.

1. Before equilibrium: diffusion. When p-type silicon (holes $p \approx N_A$) and n-type silicon (electrons $n \approx N_D$) are formed in one crystal, there is a huge concentration difference across the boundary. Holes therefore diffuse from the p side into the n side, and electrons diffuse from the n side into the p side. Carriers that cross recombine with the majority carriers on the other side and disappear.

2. Fixed charge is uncovered. Each hole that leaves the p side leaves behind a negative acceptor ion; each electron that leaves the n side leaves behind a positive donor ion. These ions are locked in the lattice. A layer next to the junction is therefore stripped of mobile carriers and holds only fixed charge: **negative on the p side, positive on the n side**. This is the **depletion region**.

3. Charge balance. The two layers carry equal and opposite charge:

$$N_A x_p = N_D x_n$$

Where:

- N_A, N_D = acceptor and donor concentrations (m^{-3})
- x_p, x_n = depth of the depletion region on the p side and on the n side (m)

So the region extends further into the **more lightly doped** side.

4. A field builds up and drift opposes diffusion. The positive and negative charge layers create an electric field pointing from the n side to the p side, largest at the junction. This field drives holes back to p and electrons back to n (a drift current), opposite to the diffusion current.

5. Equilibrium. The depletion region grows until, for holes and for electrons separately, drift exactly cancels diffusion. The net current is zero and the Fermi level is one flat line through the whole structure.

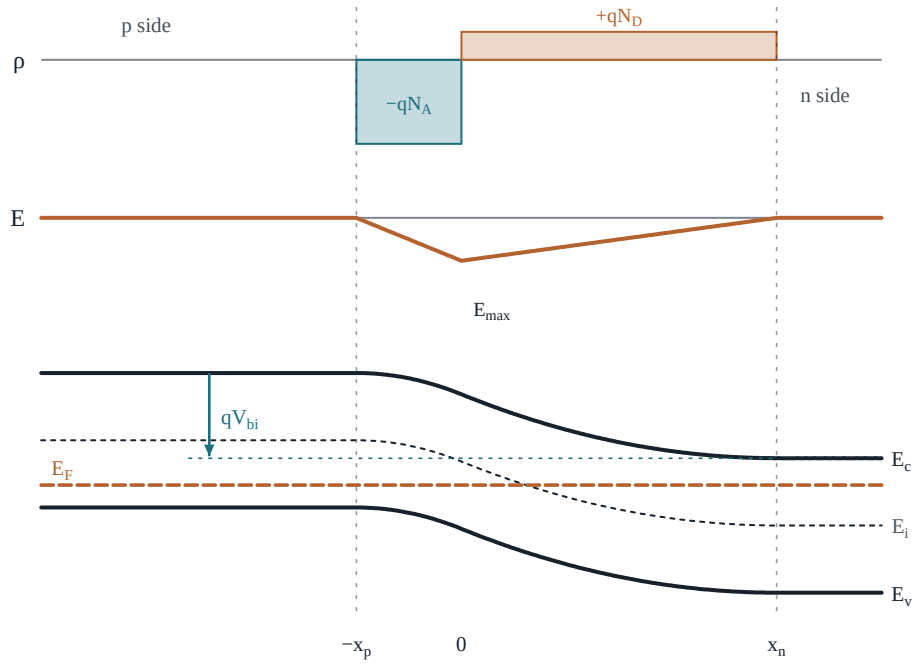
6. The built-in potential. The field across the depletion region corresponds to a potential difference V_{bi}, with the n side positive relative to the p side. In the band diagram the bands on the p side sit higher by qV_{bi}, a barrier that holds back the majority carriers. Because E_F must be flat, the bands bend by the sum of the Fermi-level offsets from the intrinsic level E_i on the two sides:

$$qV_{bi} = kT \ln(N_A/n_i) + kT \ln(N_D/n_i)$$

$$V_{bi} = (kT/q) \ln(N_A N_D / n_i^2)$$

Where:

- V_{bi} = built-in potential (V)
- q = electronic charge, 1.6×10^{-19} C
- kT = thermal energy, 0.0259 eV at 300 K, so kT/q = 0.0259 V
- N_A, N_D = acceptor and donor concentrations (m^{-3})
- n_i = intrinsic carrier concentration, 1×10^{16} m^{-3} for Si at 300 K



Abrupt p-n junction in equilibrium: fixed negative and positive ion charge either side of the junction, the triangular field peaking at the junction, and bands bent by qV_{bi} under a flat Fermi level.

Sketch in the exam: three stacked graphs against x. (a) Charge density: a negative block of height qN_A from $-x_p$ to 0, a positive block of height qN_D from 0 to x_n. (b) Field: a triangle, zero at both edges, peak E_{max} at x = 0. (c) Band diagram: E_c and E_v higher on the p side by qV_{bi}, a flat dashed E_F. Label W, x_p, x_n, qV_{bi}.

7. Width and field (depletion approximation: no free carriers inside W, neutral outside).

$$W = x_n + x_p = \sqrt{\left((2\varepsilon_s/q)\, V_{bi}\, (1/N_A + 1/N_D) \right)}, \qquad E_{max} = qN_D\, x_n/\varepsilon_s$$

Where:

- W = total depletion width (m)
- ε_s = permittivity of silicon, 1.04×10^{-10} F/m
- E_{max} = peak electric field, at the junction (V/m)
- other symbols as above

Typical numbers (Si, $N_A = 10^{23}$ m^{-3}, $N_D = 10^{22}$ m^{-3}, 300 K): $V_{bi} = 0.0259 \times \ln(10^{23} \times 10^{22}/10^{32}) = 0.0259 \times \ln(10^{13}) =$ **0.775 V**; $W = \sqrt{((2 \times 1.04\times10^{-10}/1.6\times10^{-19}) \times 0.775 \times (10^{-23} + 10^{-22}))} =$ **3.3×10^{-7} m (0.33 µm)** (3.0×10^{-7} m on the n side, 3.0×10^{-8} m on the p side); $E_{max} = 1.6\times10^{-19} \times 10^{22} \times 3.03\times10^{-7}/1.04\times10^{-10} \approx$ **4.7×10^{6} V/m**.

8. Remarks. V_{bi} cannot be read with a voltmeter: contact potentials at the metal leads cancel it exactly, so it drives no external current. Applied bias changes the barrier to $V_{bi} - V_a$ (V_a = applied voltage, positive for forward bias), which is the basis of rectification.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ diffusion of holes p → n and electrons n → p, driven by the concentration gradient
- ✓ immobile ions uncovered: negative acceptors on p side, positive donors on n side
- ✓ field from n to p; drift opposes diffusion; equilibrium when they balance
- ✓ flat Fermi level in equilibrium
- ✓ $V_{bi} = (kT/q) \ln(N_A N_D/n_i^2)$, with symbols defined

- ✓ $N_A x_p = N_D x_n$: wider on the lightly doped side
- ✓ labelled sketch (charge, field, band diagram)
- ✓ typical value (≈ 0.7–0.8 V for Si) and the point that it cannot be measured with a voltmeter

Common mistakes.

- Saying the depletion region contains "no charge". It contains no *mobile* carriers but a lot of fixed ion charge; that charge is the whole point.
- Getting the field direction backwards. The field points from the positive (n) side to the negative (p) side.
- Writing $V_{bi} = (kT/q) \ln(N_A N_D/n_i)$ (forgetting the square on n_i) or using $\log_{10}$ instead of ln.
- Claiming V_{bi} can be larger than the band gap for normal doping: it is always less than E_g/q (1.12 V for Si).

Remember it. "Carriers leave, ions stay, the field says stop": diffusion uncovers ions, the ions' field stops further diffusion, and the voltage they leave behind is V_{bi}.

Try a variant. Calculate V_{bi} and say on which side most of the depletion region lies for a silicon junction with $N_A = 1\times10^{24}$ m^{-3} and $N_D = 1\times10^{21}$ m^{-3} at 300 K. *Answer:* $V_{bi} = 0.0259 \times \ln(10^{24} \times 10^{21}/10^{32}) = 0.0259 \times \ln(10^{13}) =$ **0.775 V**; since $x_n/x_p = N_A/N_D = 1000$, **almost all of W lies in the lightly doped n side** (a one-sided p^+n junction).

Q48. Derive the ideal Shockley diode equation and explain the effect of forward and reverse bias.

> **Question type:** Derivation · **Topic:** ideal diode equation · **Full notes:** Week 10 (The p-n Junction Diode); minority-carrier diffusion in Week 9

What the examiner wants. A full, ordered derivation: list the assumptions, obtain the minority densities at the depletion edges (law of the junction), solve the minority-carrier diffusion equation in each neutral region, find the diffusion currents at the depletion edges, add them, and identify J_s. Then use the result to explain forward and reverse bias, ideally with the I-V sketch.

Key terms.

- **Minority carrier:** the less numerous carrier (holes in n-type, electrons in p-type). Equilibrium values: $p_{n0} = n_i^2/N_D$ on the n side, $n_{p0} = n_i^2/N_A$ on the p side.
- **Excess carrier density (Δp, Δn):** the amount by which a carrier density exceeds its equilibrium value.
- **Low-level injection:** injected minority densities remain much smaller than the majority density.
- **Lifetime (τ_p, τ_n):** average time an excess minority carrier survives before recombining (s).
- **Diffusion coefficient (D_p, D_n):** how quickly carriers spread down a concentration gradient (m^2/s).
- **Diffusion length (L_p, L_n):** average distance a minority carrier diffuses before recombining, $L = \sqrt{(D\tau)}$ (m).
- **Saturation current (I_s, J_s):** the small reverse current of the diode, and the scale factor of the exponential (A, or A/m^2 for J_s).
- **Quasi-equilibrium:** carriers in the depletion region stay so close to equilibrium that Boltzmann statistics still link the densities on either side of it.

Model answer.

Assumptions

1. Abrupt junction; **depletion approximation**: no mobile carriers inside the depletion region (from $x = -x_p$ to $x = x_n$), exact neutrality outside it.
2. Non-degenerate doping, so Boltzmann statistics apply.
3. **Low-level injection**: $\Delta p \ll N_D$ on the n side and $\Delta n \ll N_A$ on the p side, so majority densities are unchanged.
4. **No generation or recombination inside the depletion region**, so the electron and hole currents are each constant across it.
5. All the applied voltage V appears across the depletion region; the neutral regions carry no field (E = 0), so minority current there is pure diffusion.
6. Neutral regions much longer than the diffusion lengths, ending in ohmic contacts; steady state.

Take x = 0 at the junction, the n region for $x > x_n$ and the p region for $x < -x_p$. V > 0 means forward bias (p side positive).

Step 1: law of the junction

In equilibrium the Boltzmann relation links the electron densities on the two sides of the barrier:

$$n_{n0} = n_{p0}\, e^{qV_{bi}/kT}$$

Where:

- n_{n0} = equilibrium electron (majority) density in the n region $\approx N_D$ (m^{-3})
- n_{p0} = equilibrium electron (minority) density in the p region = n_i^2/N_A (m^{-3})
- V_{bi} = built-in potential (V); kT/q = 0.0259 V at 300 K

With bias V the barrier becomes $V_{bi} - V$. Under quasi-equilibrium the same relation holds at the depletion edges with the new barrier, and by low-level injection $n_n(x_n) \approx n_{n0}$:

$$n_p(-x_p) = n_{n0}\, e^{-q(V_{bi} - V)/kT} = n_{p0}\, e^{qV/kT}$$

The same argument for holes gives $p_n(x_n) = p_{n0}\, e^{qV/kT}$. The **excess** minority densities at the edges are therefore:

$$\Delta p_n(x_n) = p_{n0}\,(\, e^{qV/kT} - 1\,), \qquad \Delta n_p(-x_p) = n_{p0}\,(\, e^{qV/kT} - 1\,)$$

Where:

- $p_{n0} = n_i^2/N_D$ = equilibrium hole density in the n region (m^{-3})
- $\Delta p_n(x_n)$, $\Delta n_p(-x_p)$ = excess hole and electron densities at the n-side and p-side depletion edges (m^{-3})
- V = applied voltage (V)

Step 2: minority-carrier diffusion equation in the neutral n region

The hole continuity equation is $\partial p/\partial t = -(1/q)\, dJ_p/dx - \Delta p/\tau_p$. In steady state ($\partial p/\partial t = 0$) with no field, $J_p = -qD_p\, dp/dx$, so:

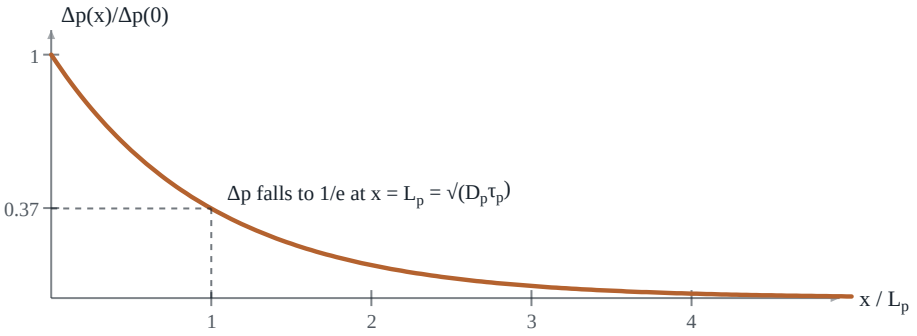
$$D_p\, d^2\Delta p_n/dx^2 - \Delta p_n/\tau_p = 0, \qquad \text{or} \qquad d^2\Delta p_n/dx^2 = \Delta p_n/L_p^2, \quad L_p = \sqrt{(D_p\tau_p)}$$

Where:

- D_p = hole diffusion coefficient (m^2/s); τ_p = hole lifetime (s); L_p = hole diffusion length (m)
- J_p = hole current density (A/m^2)

The general solution is a sum of $e^{-(x - x_n)/L_p}$ and $e^{+(x - x_n)/L_p}$. In a long region Δp_n must fall to zero far from the junction, so the growing term is dropped. Using Step 1 as the boundary value:

$$\Delta p_n(x) = \Delta p_n(x_n)\, e^{-(x - x_n)/L_p} \qquad (x \geq x_n)$$



Injected minority holes decay exponentially into the neutral n region with characteristic length L_p; the slope at the depletion edge sets the diffusion current.

Step 3: hole current at the n-side edge, $x = x_n$

$$J_p(x_n) = -qD_p\, d\Delta p_n/dx\,|_{x_n} = (qD_p/L_p)\,\Delta p_n(x_n) = (qD_p\, p_{n0}/L_p)(\, e^{qV/kT} - 1\,)$$

Where:

- $J_p(x_n)$ = hole diffusion current density at the n-side depletion edge (A/m^2), positive in the p → n direction
- $d\Delta p_n/dx = -\Delta p_n(x_n)/L_p$ = slope of the excess-hole profile at x_n (m^{-4})

Step 4: electron current at the p-side edge, $x = -x_p$

The identical argument in the p region gives $\Delta n_p(x) = \Delta n_p(-x_p)\, e^{(x + x_p)/L_n}$ for $x \leq -x_p$, and $J_n = +qD_n\, dn/dx$, so:

$$J_n(-x_p) = (qD_n\, n_{p0}/L_n)(\, e^{qV/kT} - 1\,), \qquad L_n = \sqrt{(D_n\tau_n)}$$

Where:

- D_n = electron diffusion coefficient (m^2/s); τ_n = electron lifetime (s); L_n = electron diffusion length (m)

Step 5: add the two currents

The total current density $J = J_n + J_p$ is the same at every x. By assumption 4, J_n does not change across the depletion region, so $J_n(x_n) = J_n(-x_p)$. Evaluating the total at $x = x_n$:

$$J = J_p(x_n) + J_n(-x_p) = (\, qD_p\, p_{n0}/L_p + qD_n\, n_{p0}/L_n\,)(\, e^{qV/kT} - 1\,)$$

Where:

- J = total current density (A/m^2), the same at every point of the diode
- $J_n(-x_p)$ = electron diffusion current density at the p-side depletion edge (A/m^2)

Step 6: identify J_s

Substituting $p_{n0} = n_i^2/N_D$ and $n_{p0} = n_i^2/N_A$, and multiplying by the area A:

$$\mathbf{J = J_s\,(\, e^{qV/kT} - 1\,),\; J_s = q\, n_i^2\,(\, D_p/(L_pN_D) + D_n/(L_nN_A)\,),\; I = A\,J = I_s\,(\, e^{qV/kT} - 1\,)}$$

Where:

- J, I = diode current density (A/m^2) and current (A), positive in the forward direction
- J_s, I_s = saturation current density (A/m^2) and saturation current (A), $I_s = AJ_s$
- A = junction area (m^2); $n_i = 1\times10^{16}\ m^{-3}$ for Si at 300 K
- N_A, N_D = acceptor and donor concentrations (m^{-3})

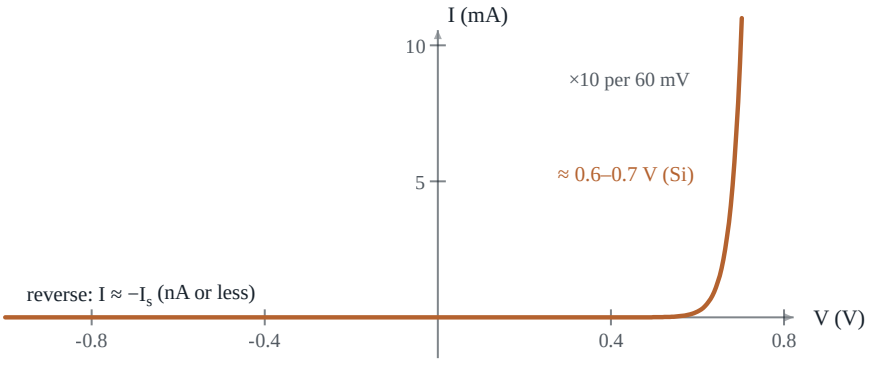
This is the **ideal Shockley diode equation**. Note that $J_s \propto n_i^2$, and the term of the *lightly* doped side dominates.

Effect of forward bias (V > 0, p side positive)

- The barrier falls to $V_{bi} - V$ and the depletion region narrows.
- Minority carriers are **injected**: edge densities rise by $e^{qV/kT}$.
- For V above about 0.1 V ($\gg$ kT/q) the −1 is negligible: $I \approx I_s\, e^{qV/kT}$. The current rises tenfold every $(kT/q)\ln 10 \approx$ **60 mV**.
- Because I_s is tiny, appreciable current needs about 0.6–0.7 V in Si (0.2–0.3 V in Ge).

Effect of reverse bias (V < 0, p side negative)

- The barrier rises to $V_{bi} + |V|$ and the depletion region widens.
- $e^{qV/kT} \to 0$, so the edge minority densities fall almost to zero: carriers are **extracted**, not injected.
- For |V| more than a few kT/q: $I \approx$ **$-I_s$**, small and independent of voltage, carried by minority carriers generated within a diffusion length of the junction.



Ideal diode characteristic: exponential forward current (a decade per 60 mV at 300 K) and a tiny constant reverse current $-I_s$.

Sketch in the exam: I against V; an exponential rising sharply near 0.6–0.7 V; a flat line just below the axis at $-I_s$ for V < 0; label I_s and the cut-in voltage.

Numerical check

Given: Si, $N_A = 10^{23}\ m^{-3}$, $N_D = 10^{22}\ m^{-3}$, $D_p = 1.24\times10^{-3}\ m^2/s$, $D_n = 3.5\times10^{-3}\ m^2/s$, $\tau_p = \tau_n = 1\ \mu s$, $A = 10^{-8}\ m^2$, 300 K.

Solution. $L_p = \sqrt{(1.24\times10^{-3} \times 10^{-6})} = 3.52\times10^{-5}$ m; $L_n = \sqrt{(3.5\times10^{-3} \times 10^{-6})} = 5.92\times10^{-5}$ m.

$$J_s = 1.6\times10^{-19} \times 10^{32} \times \left(1.24\times10^{-3}/(3.52\times10^{-5} \times 10^{22}) + 3.5\times10^{-3}/(5.92\times10^{-5} \times 10^{23}) \right)$$
$$= 1.6\times10^{13} \times (3.52\times10^{-21} + 5.92\times10^{-22}) = 6.6\times10^{-8}\ A/m^2$$

Answer: $I_s = AJ_s = 10^{-8} \times 6.6\times10^{-8} = 6.6\times10^{-16}$ A; $I(0.6\ V) \approx 6.6\times10^{-16} \times e^{23.17} \approx$ **7.6 µA**; $I(0.7\ V) \approx$ **0.36 mA**; $I(-5\ V) =$ **-6.6×10^{-16} A**.

What this means: 86 % of the current is holes injected into the lightly doped n side. At 0.6 V the injected hole density at x_n is 1.2×10^{20} m^{-3}, still $\ll N_D = 10^{22}$ m^{-3}, so low-level injection holds.

Real diodes depart from this: recombination in the depletion region (low forward bias), high-level injection and series resistance (high current), generation current and breakdown (reverse). These are lumped into $I = I_s(e^{qV/\eta kT} - 1)$, with ideality factor η between 1 and 2.

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ assumptions listed: depletion approximation, low-level injection, no G–R in depletion region, Boltzmann statistics, E = 0 in neutral regions, long diodes
- ✓ law of the junction: $p_n(x_n) = p_{n0}e^{qV/kT}$, $n_p(-x_p) = n_{p0}e^{qV/kT}$, derived from the barrier $V_{bi} - V$
- ✓ diffusion equation $D_p\, d^2\Delta p/dx^2 - \Delta p/\tau_p = 0$ and its decaying-exponential solution with $L_p = \sqrt{(D_p\tau_p)}$
- ✓ $J_p(x_n) = (qD_{ppn0}/L_p)(e^{qV/kT} - 1)$ and the matching electron term
- ✓ argument that the currents may be added at the depletion edges (J_n constant across W)
- ✓ boxed result with $J_s = qn_i^2(D_p/(L_pN_D) + D_n/(L_nN_A))$
- ✓ forward bias: barrier lowered, injection, $I \approx I_{se}^{qV/kT}$, 60 mV/decade
- ✓ reverse bias: barrier raised, extraction, $I \approx -I_s$
- ✓ labelled I-V sketch

Common mistakes.

- Using total (not excess) minority density in the current: the "−1" comes from $\Delta p = p_{n0}(e^{qV/kT} - 1)$; forgetting it gives zero reverse current and a wrong equation.
- Keeping the growing exponential e^{+x/L_p} in the long-diode solution; it must be dropped so $\Delta p \to 0$ far away.
- Not justifying the addition of J_n and J_p: say explicitly that no recombination in the depletion region makes each constant across it.
- Swapping the doping in J_s: holes injected into the n side carry $D_p/(L_pN_D)$, so the *lighter* doping dominates.

Remember it. "Inject, diffuse, add": the junction law injects, the diffusion equation spreads them over L, and the two slopes at the edges add up to $I_s(e^{qV/kT} - 1)$.

Try a variant. For a p^+n silicon diode ($N_D = 10^{22}$ m^{-3} on the n side, $D_p = 1.24\times10^{-3}$ m^2/s, $\tau_p = 1$ µs, $A = 10^{-8}$ m^2), find I_s and the forward voltage for I = 1 mA. *Answer:* the electron term is negligible, so $I_s \approx qAn_i^2D_p/(L_pN_D) = 1.6\times10^{-19} \times 10^{-8} \times 10^{32} \times 1.24\times10^{-3}/(3.52\times10^{-5} \times 10^{22}) =$ **5.6×10^{-16} A**; $V = (kT/q)\ln(I/I_s) = 0.0259 \times \ln(1.78\times10^{12}) =$ **0.73 V**.

Q49. Differentiate between a Schottky barrier diode and an Ohmic contact. Explain their energy-band characteristics.

> **Question type:** Compare and explain · **Topic:** metal–semiconductor contacts · **Full notes:** Week 11 (Metal–Semiconductor Interfaces)

What the examiner wants. Both are metal–semiconductor contacts; the answer must show what makes one rectifying and the other not. Marks are for the band diagrams (before contact, at equilibrium, under bias for the Schottky diode), the barrier formulas, the two ways of making an ohmic contact, the I-V behaviour, and a comparison table.

Key terms.

- **Vacuum level (E_{vac}):** energy of an electron at rest just outside the material; the common reference.
- **Work function (Φ_m, Φ_s):** energy from the Fermi level up to the vacuum level, for the metal and the semiconductor (eV).
- **Electron affinity (χ):** energy from the conduction-band edge E_c up to the vacuum level; 4.05 eV for Si, independent of doping.
- **Schottky barrier height (Φ_B):** barrier an electron in the metal must climb to enter the semiconductor (eV); fixed by the interface.
- **Built-in potential (V_{bi}):** barrier an electron in the semiconductor must climb to reach the metal (V).
- **Depletion / accumulation layer:** a surface layer with fewer / more majority carriers than the bulk; bands bend up / down in n-type.
- **Thermionic emission:** carriers crossing *over* a barrier using thermal energy.
- **Tunnelling (field emission):** carriers passing *through* a barrier only a few nanometres thick.

Model answer.

A **Schottky barrier diode** is a metal–semiconductor contact with a significant barrier to majority carriers, so it **rectifies**. An **ohmic contact** has no barrier that limits the current, so it conducts equally in both directions with a linear I-V ($V = IR_c$). Both are explained with band diagrams using one rule: **at equilibrium the Fermi level is flat**. The cases below are for a metal on **n-type** silicon.

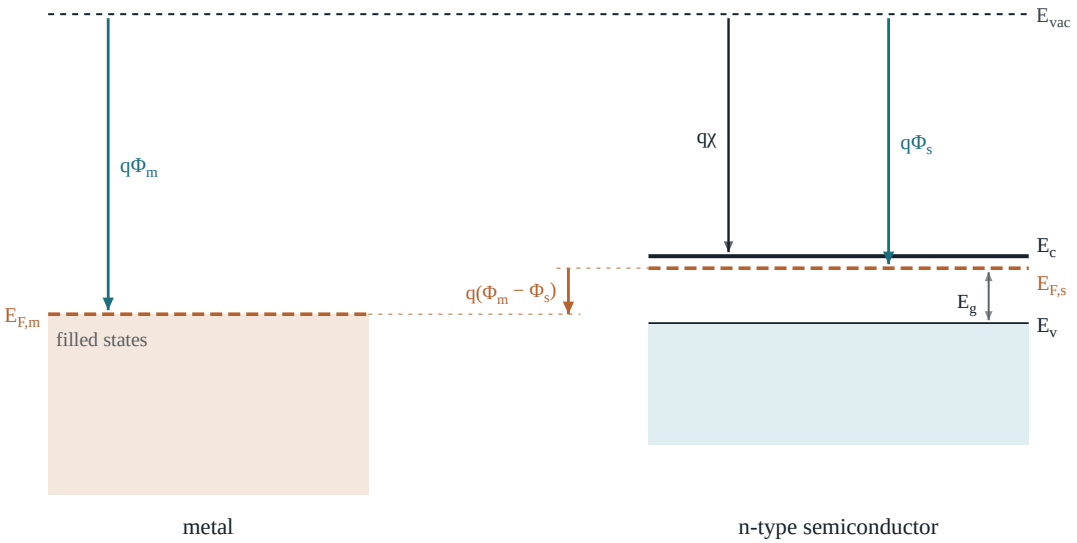
Before contact

Energies are measured down from the common vacuum level. The semiconductor work function depends on doping:

$$\Phi_s = \chi + (E_c - E_F), \qquad E_c - E_F = kT \ln(N_c/N_D)$$

Where:

- Φ_s = semiconductor work function (eV); χ = electron affinity, 4.05 eV (Si)
- $E_c - E_F$ = depth of the Fermi level below the conduction-band edge (eV)
- kT = 0.0259 eV at 300 K; $N_c = 2.8\times10^{25}$ m^{-3} (Si); N_D = donor concentration (m^{-3})



Metal and n-type semiconductor before contact. With $\Phi_m > \Phi_s$ the semiconductor's Fermi level lies higher by $\Phi_m - \Phi_s$, so electrons will flow into the metal on contact.

Whether E_F of the semiconductor lies above or below E_F of the metal decides which way electrons flow on contact, and hence the type of contact.

Schottky barrier ($\Phi_m > \Phi_s$, moderate doping)

At equilibrium.

1. On contact, electrons flow from the semiconductor (higher E_F) into the metal.
2. They leave behind positive donor ions: a **depletion region** forms in the semiconductor, with a thin sheet of negative charge on the metal surface.
3. The resulting field bends the bands **upward** towards the interface until the Fermi levels align.
4. Two barriers result:

$$\Phi_B = \Phi_m - \chi \qquad \text{(barrier seen from the metal; Schottky–Mott rule)}$$

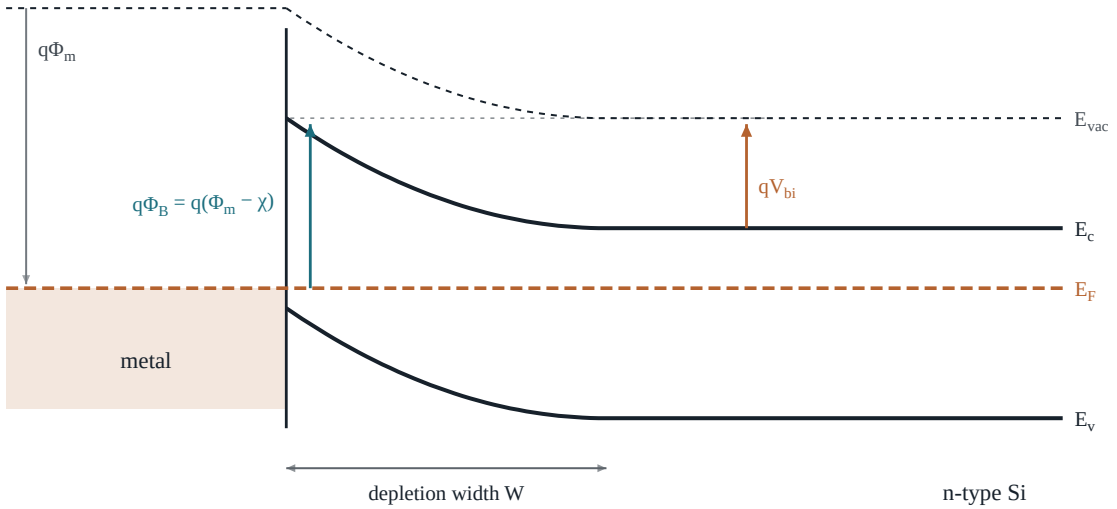
$$V_{bi} = \Phi_m - \Phi_s = \Phi_B - (E_c - E_F)/q \qquad \text{(barrier seen from the semiconductor)}$$

$$W = \sqrt{(\,2\varepsilon_s(V_{bi} - V)/qN_D\,)}$$

Where:

- Φ_m = metal work function (eV), e.g. Au 5.1 eV
- W = depletion width (m); $\varepsilon_s = 1.04\times10^{-10}$ F/m; $q = 1.6\times10^{-19}$ C
- V = applied voltage (V), positive for forward bias (metal positive)

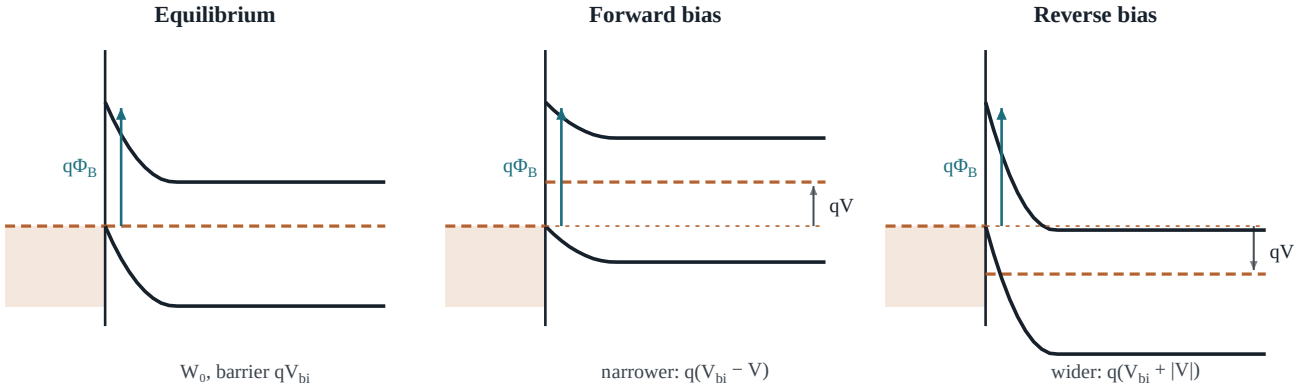
Example (Au on n-Si, $N_D = 10^{22}$ m^{-3}): $E_c - E_F = 0.0259 \ln(2.8\times10^{25}/10^{22}) = 0.0259 \ln(2800) = 0.21$ eV, $\Phi_s = 4.26$ eV, $\Phi_B = 5.1 - 4.05 =$ **1.05 eV**, $V_{bi} = 5.1 - 4.26 =$ **0.84 V**, $W = \sqrt{(2 \times 1.04\times10^{-10} \times 0.84/(1.6\times10^{-19} \times 10^{22}))} =$ **3.3×10^{-7} m (0.33 µm)**: far too thick to tunnel through. (Real interfaces are **Fermi-level pinned**, so Au on n-Si actually measures $\Phi_B \approx 0.8$ eV.)



Schottky contact at equilibrium: flat E_F, bands bent upward in the semiconductor, barrier $q\Phi_B$ seen from the metal and qV_{bi} seen from the semiconductor.

Under bias. $q\Phi_B$ is fixed by the interface and never changes; the applied voltage appears across the depletion region and changes only the semiconductor-side barrier.

- **Forward bias** (metal +): the semiconductor E_F rises by qV, the band bending falls to $q(V_{bi} - V)$, W narrows, and electrons pour over the barrier into the metal; current rises exponentially.
- **Reverse bias** (metal −): bending rises to $q(V_{bi} + |V|)$, W widens, the semiconductor-to-metal flow is cut off, and only the small metal-to-semiconductor flow over the fixed Φ_B remains: the saturation current.



The Schottky contact at equilibrium, forward bias and reverse bias: $q\Phi_B$ stays fixed while the semiconductor-side barrier falls to $q(V_{bi} - V)$ or rises to $q(V_{bi} + |V|)$.

Current is by **thermionic emission** of majority carriers:

$$J = J_s \left(e^{qV/\eta kT} - 1 \right), \qquad J_s = A^{**} T^2 e^{-q\Phi_B/kT}$$

Where:

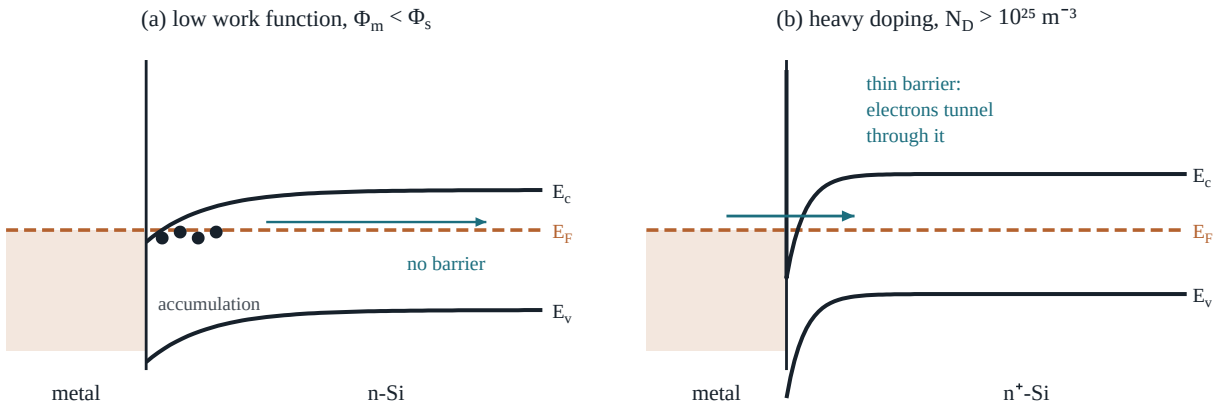
- J, J_s = current density and saturation current density (A/m^2)
- A^{**} = effective Richardson constant, about 1.1×10^6 A/m^2K^2 for n-Si
- T = temperature (K); η = ideality factor (1.0–1.1 for a good Schottky diode)

For $\Phi_B = 0.8$ eV: $J_s = 1.1\times10^6 \times 300^2 \times e^{-0.8/0.0259} = 3.8\times10^{-3}$ A/m^2 and 10^4 A/m^2 flows at only about $0.0259 \times \ln(10^4/3.8\times10^{-3})$ = **0.38 V**, against about 0.7 V for a Si p-n diode. No minority charge is stored, so it switches very fast.

Ohmic contact: the two mechanisms

(a) Low-barrier metal, accumulation ($\Phi_m < \Phi_s$ on n-type). Before contact the metal's E_F is higher, so on contact electrons flow from the metal **into** the semiconductor. They pile up in an **accumulation layer**; the bands bend **downward** at the interface. There is no barrier to electrons in either direction, so current flows freely both ways. (For p-type the condition is reversed: $\Phi_m > \Phi_s$.) In practice Fermi-level pinning makes suitable metals rare on Si and GaAs.

(b) Heavy doping, tunnelling (the practical method). The surface under the contact is doped n^+ (above about 10^{25} m^{-3}). The barrier height is unchanged, but since $W \propto 1/\sqrt{N_D}$ the barrier becomes only a few nanometres thick, and electrons **tunnel** through it in both directions. Example: for a 0.60 V barrier, $W = 2.79\times10^{-7}$ m (279 nm) at $N_D = 10^{22}$ m^{-3} (rectifying) but only **8.8×10^{-9} m (8.8 nm)** at 10^{25} m^{-3} (ohmic).



The two routes to an ohmic contact on n-type silicon: (a) $\Phi_m < \Phi_s$, downward band bending into an accumulation layer with no barrier; (b) heavy doping, a barrier so thin that electrons tunnel through it.

An ohmic contact is judged by its specific contact resistance $\rho_c = R_cA$ ($\Omega\cdot m^2$; R_c = contact resistance in Ω, A = area in m^2); good contacts reach 10^{-10}–10^{-12} $\Omega\cdot m^2$. Silicides ($TiSi_2$, NiSi) on Si and alloyed Au–Ge–Ni on n-GaAs are standard.

Sketch in the exam: for each case draw E_{vac}, metal E_F, and the semiconductor E_c, E_F, E_v; mark $q\Phi_m$, $q\Phi_s$, $q\chi$ before contact; after contact show the flat E_F, the upward bend with $q\Phi_B$ and qV_{bi} (Schottky), the downward bend (accumulation) and the thin barrier with a tunnelling arrow (heavy doping).

Comparison

Feature	Schottky barrier diode	Ohmic contact
Work-function condition (n-type)	$\Phi_m > \Phi_s$	$\Phi_m < \Phi_s$, or any metal on n^+
Doping under contact	Moderate (10^{21}–10^{23} m^{-3})	Heavy ($> 10^{25}$ m^{-3})
Band bending (n-type)	Upward: depletion layer	Downward (accumulation) or a very thin barrier
Barrier to majority carriers	Yes: $q\Phi_B$ and qV_{bi}	None, or thin enough to tunnel through
Transport	Thermionic emission over the barrier	Free flow, or tunnelling (field emission)
I-V	Exponential, rectifying	Linear, symmetric, $V = IR_c$
Resistance	Depends on bias polarity	Small and constant
Effect of bias on bands	Semiconductor-side barrier changes, $q\Phi_B$ fixed	Negligible: no barrier to modulate
Use	Fast rectifiers, RF detectors, clamps, MESFET gates	Connecting every device to the circuit

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ definitions: rectifying barrier vs linear, symmetric, low-resistance contact
- ✓ $\Phi_s = \chi + (E_c - E_F)$; comparison of Φ_m and Φ_s decides the contact type
- ✓ Schottky formation: electrons to metal, depletion region, upward bending, Fermi levels align
- ✓ $\Phi_B = \Phi_m - \chi$ and $V_{bi} = \Phi_m - \Phi_s$
- ✓ band diagrams before contact, at equilibrium and under forward/reverse bias, with $q\Phi_B$ fixed
- ✓ thermionic emission $J = A**T^2 e^{-q\Phi_B/kT}(e^{qV/\eta kT} - 1)$

- ✓ ohmic route 1: accumulation, downward bending, no barrier
- ✓ ohmic route 2: heavy doping, W ∝ $1/\sqrt{N_D}$, tunnelling
- ✓ comparison table

Common mistakes.

- Saying an ohmic contact "has no barrier". The practical (n^+) ohmic contact has a barrier; it is simply thin enough to tunnel through.
- Reversing the conditions for p-type: there a Schottky barrier needs $\Phi_m < \Phi_s$ and an ohmic contact $\Phi_m > \Phi_s$.
- Drawing Φ_B changing with bias. Only the semiconductor-side barrier ($V_{bi} \mp V$) changes.
- Mixing up Φ_B and V_{bi}: Φ_B is measured from the metal Fermi level, V_{bi} from the semiconductor conduction band; they differ by ($E_c - E_F$).

Remember it. "Schottky climbs over, ohmic walks through": over a fixed barrier by thermionic emission, or through a paper-thin one by tunnelling.

Try a variant. Platinum (Φ_m = 5.65 eV) is deposited on p-type silicon with $N_A = 1\times10^{22}$ m^{-3}. Is the ideal contact rectifying or ohmic? *Answer:* $E_F - E_v = 0.0259 \ln(N_v/N_A) = 0.0259 \ln(1.04\times10^{25}/10^{22}) = 0.0259 \ln(1040) = 0.18$ eV, so $\Phi_s = \chi + E_g - 0.18 = 4.05 + 1.12 - 0.18 =$ **4.99 eV**. $\Phi_m > \Phi_s$ on **p-type**, so electrons leave the semiconductor, holes accumulate at the surface and the bands bend upward with no hole barrier: the contact is **ohmic** (by contrast Al, 4.28 eV, gives a hole barrier $\Phi_{Bp} = 1.12 - (4.28 - 4.05) = 0.89$ eV, a Schottky contact).

Q50. Explain the principles of I-V and C-V characterisation of semiconductor devices and describe how C-V measurements can be used to estimate doping concentration.

> **Question type:** Explanation with calculation · **Topic:** device characterisation · **Full notes:** Week 12 (Interface Characterisation)

What the examiner wants. For each technique: what is applied, what is measured, which equation links them, and which plot turns the data into a straight line. Then a step-by-step description of extracting doping (and V_{bi}) from a Mott–Schottky plot, best shown with a worked numerical example and a note on limitations.

Key terms.

- **I-V characterisation:** measuring current as the DC voltage is stepped, to extract saturation current, ideality factor, series resistance, leakage, breakdown and barrier height.
- **Ideality factor (η):** 1 for pure diffusion (or thermionic emission) current, up to 2 when depletion-region recombination dominates.
- **Series resistance (R_s):** resistance of the bulk, contacts and leads in series with the junction (Ω).
- **C-V characterisation:** measuring small-signal capacitance as the DC reverse bias is stepped, to find doping and V_{bi}.
- **Depletion capacitance (C):** capacitance of the carrier-free depletion layer, $C = \varepsilon_s A/W$ (F).
- **One-sided junction:** p^+n, n^+p or Schottky diode; the depletion region lies almost entirely in the lightly doped side.
- **Mott–Schottky plot:** $1/C^2$ against V; straight for uniform doping.
- **Doping profile N(x):** doping concentration as a function of depth below the surface.

Model answer.

1. I-V characterisation

Principle. A source-measure unit applies a DC voltage in steps from reverse to forward bias and records the current (picoamps to amps, often on a probe station, sometimes at several temperatures). The data are compared with the diode law:

$$I = I_s \left(e^{qV/\eta kT} - 1 \right) \quad \Rightarrow \quad \ln I = \ln I_s + (q/\eta kT)\, V \qquad (V > \text{about } 0.1 \text{ V})$$

Where:

- I = diode current (A); I_s = saturation current (A)
- V = applied voltage, positive for forward bias (V)
- η = ideality factor (dimensionless); kT/q = 0.0259 V at 300 K

A **semi-log plot** of ln I against V is a straight line:

- intercept at V = 0 → $\mathbf{I_s}$;
- slope = q/(ηkT) → **η = (q/kT)/slope**;
- bending below the line at high current → **series resistance**, R_s = (extra voltage)/I;
- reverse branch → **leakage** and **breakdown voltage**;
- for a Schottky diode, **barrier height** $\Phi_B = (kT/q)\ln(A{**}AT^2/I_s)$, with A = area ($m^2$), $A{**} \approx 1.1\times10^6$ A/m^2K^2 for n-Si, T in K;
- a straight, symmetric line through the origin between two contacts confirms an **ohmic contact**.

Transistors are characterised the same way (threshold voltage and mobility from MOSFET curves, current gain from BJT curves).

2. C-V characterisation

Principle. An LCR meter superimposes a small AC signal (10–50 mV, about 1 MHz) on a DC **reverse** bias and measures the capacitance as the bias is stepped. A reverse-biased one-sided junction or Schottky diode is a parallel-plate capacitor whose "gap" is the depletion region:

$$C = \varepsilon_s A / W, \qquad W = \sqrt{\left(2\varepsilon_s (V_{bi} - V)/qN \right)}$$

Where:

- C = depletion capacitance (F); A = junction area (m^2)
- ε_s = permittivity of silicon, 1.04×10^{-10} F/m; q = 1.6×10^{-19} C
- W = depletion width (m); V_{bi} = built-in potential (V)
- V = applied voltage, negative for reverse bias (V); N = doping of the lightly doped side (m^{-3})

As reverse bias grows, W widens and C falls.

3. Estimating doping from C-V

Eliminating W and squaring:

$$1/C^2 = 2(V_{bi} - V) / (q\varepsilon_s N A^2)$$

So the procedure is:

- Measure C at a series of reverse biases (0, −1, … −5 V).

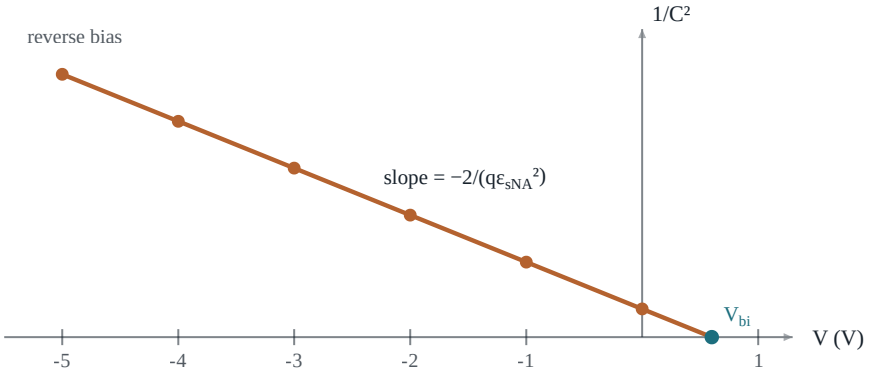
- Plot **$1/C^2$ against V**. For uniform doping the points lie on a straight line.
- The **slope** is $-2/(q\varepsilon_s N A^2)$, so:

  $$N = 2 / (q\varepsilon_s A^2 |\text{slope}|)$$

- The **voltage-axis intercept** (where $1/C^2 = 0$) is $\mathbf{V_{bi}}$. For a Schottky diode, $\Phi_B = V_{bi} + (kT/q)\ln(N_c/N)$.
- If the plot curves, the doping is non-uniform; use the local slope and the depth $W = \varepsilon_s A/C$ to obtain a **profile**: $N(W) = 2/(q\varepsilon_s A^2 |d(1/C^2)/dV|)$.

Where:

- |slope| = magnitude of the slope of the $1/C^2$ line ($F^{-2}V^{-1}$)
- N_c = conduction-band effective density of states, 2.8×10^{25} m^{-3} (Si)



Mott–Schottky plot: $1/C^2$ falls linearly towards the forward-bias side and meets the voltage axis at V_{bi}; the slope gives the doping.

Sketch in the exam: $1/C^2$ on the vertical axis, V on the horizontal with reverse bias to the left; a straight line falling to the right and cutting the V axis at $+V_{bi}$; label the slope $-2/(q\varepsilon_s N A^2)$.

4. Worked example: doping and V_{bi} from a $1/C^2$ line

Given: a Schottky diode on n-type Si, A = 1.0×10^{-7} m^2, 300 K, with readings:

V (V)	0	−1	−2	−3	−4	−5
C (pF)	37.2	22.8	17.9	15.2	13.4	12.2
$1/C^2$ (F^{-2})	7.2×10^{20}	1.92×10^{21}	3.12×10^{21}	4.33×10^{21}	5.57×10^{21}	6.72×10^{21}

Find: N_D, V_{bi}, the depletion depth at −5 V and the barrier height.

Formula: $N_D = 2/(q\varepsilon_s A^2|\text{slope}|)$; V_{bi} from $1/C^2 = 0$; $W = \varepsilon_s A/C$; $\Phi_B = V_{bi} + (kT/q)\ln(N_c/N_D)$.

Solution.

Step 1. The $1/C^2$ values rise by about 1.2×10^{21} F^{-2} per volt: the line is straight, so the doping is uniform. Slope from the end points:

$$\text{slope} = (6.72\times10^{21} - 7.2\times10^{20})/(-5 - 0) = -1.20\times10^{21}\ F^{-2}V^{-1}$$

Step 2. Doping:

$$N_D = 2 / (1.6\times10^{-19} \times 1.04\times10^{-10} \times (10^{-7})^2 \times 1.20\times10^{21}) = 2 / 2.0\times10^{-22} = 1.0\times10^{22}\ m^{-3}$$

Step 3. Intercept: at V = 0, $1/C^2 = 7.2\times10^{20}$, and it falls by 1.20×10^{21} per volt of forward bias, so it reaches zero at

$V_{bi} = 7.2\times10^{20} / 1.20\times10^{21} = 0.60$ V

Step 4. Depth probed at −5 V:

$W = \varepsilon_s A/C = 1.04\times10^{-10} \times 10^{-7} / 12.2\times10^{-12} = 8.5\times10^{-7}$ m = 0.85 μm

Step 5. Barrier height: $(kT/q)\ln(N_c/N_D) = 0.0259 \times \ln(2.8\times10^{25}/10^{22}) = 0.0259 \times \ln(2800) = 0.21$ V, so $\Phi_B = 0.60 + 0.21 = 0.81$ eV.

Answer: $N_D = 1.0\times10^{22}$ m^{-3}, $V_{bi} = 0.60$ V, $W(-5\text{ V}) = 8.5\times10^{-7}$ m (0.85 μm), $\Phi_B \approx 0.81$ eV.

What this means: the doping was measured electrically, without cutting or etching the wafer, and is uniform at least to 0.85 μm below the contact. A 0.81 eV barrier is typical of a good Schottky contact on n-Si.

5. Accuracy and limits

- Measure only in reverse bias (forward bias adds a large diffusion capacitance).
- Correct for stray (pad and probe) capacitance and series resistance; keep the AC signal small.
- Know the area accurately: $N \propto 1/A^2$, so a 10 % area error gives a 20 % doping error.
- C-V sees only the lightly doped side, only to depths reached before breakdown; deep traps can distort the result at the wrong frequency.

	I-V	**C-V**
Applied	DC voltage, reverse to forward	DC reverse bias + small AC signal
Measured	Current	Capacitance
Straight-line plot	ln I against V	$1/C^2$ against V
Slope gives	η	Doping N
Intercept gives	I_s (and Φ_B)	V_{bi} (and Φ_B)

Marking checklist. The points a marker looks for. Tick each one off in your own answer:

- ✓ I-V principle: voltage stepped, current measured, $I = I_s(e^{qV/\eta kT} - 1)$
- ✓ semi-log plot: intercept → I_s, slope → η; R_s, leakage, breakdown, Φ_B
- ✓ C-V principle: reverse bias + small AC signal, $C = \varepsilon_s A/W$, C falls with reverse bias
- ✓ derivation of $1/C^2 = 2(V_{bi} - V)/(q\varepsilon_s N A^2)$
- ✓ $N = 2/(q\varepsilon_s A^2|\text{slope}|)$ and V_{bi} from the voltage intercept
- ✓ profiling: local slope and $W = \varepsilon_s A/C$ for non-uniform doping
- ✓ worked numbers: $N \approx 1\times10^{22}$ m^{-3}, $V_{bi} = 0.60$ V
- ✓ labelled Mott–Schottky sketch and at least two practical limitations

Common mistakes.

- Plotting C or 1/C against V: only $1/C^2$ gives a straight line for uniform doping.
- Leaving the area unsquared, or mixing units (area in mm^2 or μm^2 with ε_s in F/m): keep everything in SI (A in m^2, ε_s in F/m), and remember A^2.
- Reading V_{bi} from the $1/C^2$ axis instead of the voltage axis, or giving it a negative sign: the line crosses at a positive (forward) voltage.

- Applying it to a symmetric p-n junction without saying so: then N is the combined $N_A N_D/(N_A + N_D)$, not a single doping.

Remember it. "Slope gives the dopants, intercept gives the barrier": $1/C^2$ against V is a straight line whose steepness is $2/(q\varepsilon_s N A^2)$ and whose zero is V_{bi}.

Try a variant. A one-sided Si junction of area 2×10^{-7} m^2 gives a $1/C^2$ plot with slope magnitude 4.0×10^{20} F^{-2}V^{-1}. Find the doping of the lightly doped side. *Answer:* $N = 2/(q\varepsilon_s A^2|\text{slope}|) = 2/(1.6\times10^{-19} \times 1.04\times10^{-10} \times 4\times10^{-14} \times 4.0\times10^{20}) =$ **7.5×10^{21} m^{-3}**. (Doubling the area quarters the slope for the same doping.)

REVISION TOOLS

Formula and Constants Sheet

Every formula the 50 questions need, grouped by topic, with the question numbers that use it. Learn the boxed results of the derivations (Q27, Q37, Q48) by heart; the rest you should be able to rebuild from them. Every symbol is explained in the table that follows. All values are in SI units; energies are in electron-volts (eV), which SI accepts for use alongside its units.

Symbols and abbreviations used on this page

Every symbol, constant and abbreviation on this page is listed here, even if another page already defined it, so you can study this page on its own.

Constants and units

Symbol	Meaning	Units / typical value
q	Magnitude of the electronic charge	1.6×10^{-19} C
k	Boltzmann constant	8.617×10^{-5} eV/K = 1.381×10^{-23} J/K
kT	Thermal energy	0.0259 eV (often 0.02585 eV) at 300 K
kT/q	Thermal voltage	0.0259 V (about 26 mV) at 300 K
h	Planck constant	6.626×10^{-34} J·s
ħ	Reduced Planck constant, h/2π	1.055×10^{-34} J·s
m_0	Free-electron mass	9.11×10^{-31} kg
ε_0	Permittivity of free space	8.854×10^{-12} F/m
ε_s (ε_{si})	Permittivity of silicon, $11.7\varepsilon_0$	1.04×10^{-10} F/m
ε_r	Relative permittivity (dielectric constant)	Si 11.7, SiO_2 3.9, GaAs 12.9
eV	Electron-volt, a unit of energy	1 eV = 1.6×10^{-19} J
Å	Ångström, a unit of length	1 Å = 10^{-10} m = 0.1 nm (not SI: this course uses nm)
µm	Micrometre (micron)	1 µm = 10^{-6} m

Symbols

Symbol	Meaning	Units / typical value
E_c (E_C)	Energy of the bottom of the conduction band	eV
E_v (E_V)	Energy of the top of the valence band	eV
E_g	Band-gap energy, $E_c - E_v$	eV; Si 1.12, Ge 0.66, GaAs 1.42 at 300 K
E_F	Fermi level: the energy at which a state has a 50 % chance of being occupied	eV
E_i	Intrinsic Fermi level, close to mid-gap	eV
E_a	Activation energy of dopant diffusion	eV; about 3–4 eV
f(E)	Fermi–Dirac function: probability that a state at energy E is occupied	dimensionless, 0 to 1
g(E), g_c, g_v	Density of states: allowed states per unit energy per unit volume	states/(eV·m^3)
N_c	Effective density of states in the conduction band	m^{-3}; Si 2.8×10^{25} at 300 K
N_v	Effective density of states in the valence band	m^{-3}; Si 1.04×10^{25} at 300 K
n	Electron concentration (free electrons per m^3)	m^{-3}
p	Hole concentration (holes per m^3)	m^{-3}
n_i	Intrinsic carrier concentration (n = p = n_i in pure material)	m^{-3}; Si ≈ 1×10^{16} at 300 K
p_{n0}, n_{p0}	Equilibrium minority concentrations: holes on the n side, electrons on the p side	m^{-3}; $p_{n0} = n_i^2/N_D$
N_D	Donor concentration	m^{-3}
N_A	Acceptor concentration	m^{-3}
Δn, Δp	Excess electron and hole concentrations above equilibrium	m^{-3}
m^*, m_n^*, m_p^*	Effective mass of electrons or holes	kg, or as a multiple of m_0
ψ	Wave function ($\lvert\psi\rvert^2$ is the probability density)	—
V(x)	Potential energy in the Schrödinger equation	J or eV
k (wave vector)	Wave vector, 2π/λ; measures the electron's momentum, p = ħk	m^{-1}
α (Kronig–Penney)	√(2mE)/ħ, set by the electron energy	m^{-1}
P (Kronig–Penney)	Barrier strength $mV_0ba/\hbar^2$	dimensionless
λ	Wavelength	μm or nm; λ(μm) = 1.24/E(eV)
a	Lattice constant: edge length of the cubic unit cell	nm; Si 0.543, Ge and GaAs 0.565
(h k l), {h k l}	Miller indices of a plane; family of equivalent planes	integers
d_{hkl}	Spacing between neighbouring (h k l) planes, $a/\sqrt{(h^2 + k^2 + l^2)}$	nm
k_0	Segregation coefficient, C_s/C_l at the freezing interface	dimensionless; B 0.8, P 0.35

Symbol	Meaning	Units / typical value
C_s	Surface concentration of a diffusion (or solid concentration when freezing)	m^{-3}
C_l, C_0	Concentration in the melt (liquid); C_0 at the start	m^{-3}
C(x, t)	Dopant concentration at depth x and time t	m^{-3}
D	Diffusion coefficient (dopants in a furnace, or carriers)	m^2/s
D_0	Diffusion constant in $D = D_0 e^{-E_a/kT}$	m^2/s
F	Particle flux in Fick's law	particles/(m^2·s)
Q	Dose: dopant atoms per unit area (in Weeks 13–14, Q also means charge)	atoms/m^2 (C or C/m^2 for charge)
Φ (dose)	Implanted dose (ions per m^2)	m^{-2}
erfc	Complementary error function, erfc(u) = 1 − erf(u); read from a table	dimensionless
x_j	Junction depth: where the dopant concentration equals the background	μm
N_B (C_B)	Background doping of the wafer (in Week 13, N_B is the base doping)	m^{-3}
R_p	Projected range: average depth at which implanted ions stop	μm
ΔR_p	Straggle: spread of the implanted ions' stopping depths	μm
E (field)	Electric field (on band diagrams, E is energy instead)	V/m
μ, μ_n, μ_p	Mobility: drift velocity per unit field (electrons, holes)	$m^2/(V\cdot s)$; Si μ_n 0.135, μ_p 0.048
μ_L, μ_I	Mobility limited by lattice scattering, by ionised-impurity scattering	$m^2/(V\cdot s)$
v_d	Drift velocity	m/s
v_{sat}	Saturation velocity at high field	≈ 10^5 m/s in Si
τ_m	Mean free time between collisions	s; about 0.1–1 ps
σ	Conductivity, $q(n\mu_n + p\mu_p)$ (in Week 8, σ is also a capture cross-section)	S/m (siemens per metre)
ρ	Resistivity, 1/σ	Ω·m
J, J_n, J_p	Current density (total, electrons, holes)	A/m^2
D_n, D_p	Diffusion coefficients of electrons and holes	m^2/s; Si 3.5×10^{-3} and 1.24×10^{-3}
G, G_L	Generation rate of electron–hole pairs (G_L: by light)	$m^{-3}s^{-1}$
R	Recombination rate	$m^{-3}s^{-1}$
U	Net recombination rate, R − G	$m^{-3}s^{-1}$
τ, τ_n, τ_p	Minority-carrier lifetime (τ_{eff}: effective, including surfaces)	s; Si 1 ns – 1 ms
N_t	Density of traps (recombination centres)	m^{-3}
n_1, p_1	SRH parameters $n_{ie}^{(E_t - E_i)/kT}$ and $n_{ie}^{(E_i - E_t)/kT}$	m^{-3}
L, L_n, L_p	Diffusion length √(Dτ): distance a minority carrier travels before recombining	μm or m
S	Surface recombination velocity	m/s
N_{st}	Density of surface traps	m^{-2}

Symbol	Meaning	Units / typical value
U_s	Recombination rate per unit area at a surface	$m^{-2}s^{-1}$
V_{bi}	Built-in potential of a junction or Schottky contact	V; typically 0.5–1 V
W	Depletion-region width (in Week 13 W_B, W_E are base and emitter widths; in Week 14 W is channel width)	µm or m
x_n, x_p	Parts of the depletion region on the n side and on the p side	µm
E_{max}	Peak electric field, at the junction	V/m
I_s, J_s (I_S)	Saturation current (density) of a diode or transistor	A; A/m^2
I	Current	A
η	Ideality factor of a diode, 1 (ideal) to 2	dimensionless
C, C_j	Capacitance (junction or depletion capacitance)	F (pF = 10^{-12} F)
Φ_m	Work function of a metal	eV; Al 4.28, Au 5.1
Φ_s	Work function of a semiconductor, $\chi + (E_c - E_F)$	eV
χ	Electron affinity: energy from E_c up to the vacuum level	eV; Si 4.05
Φ_B	Schottky barrier height seen from the metal	eV (or V)
A∗∗	Effective Richardson constant	$A/(m^2 \cdot K^2)$; n-Si ≈ 1.1×10^6
μ_i, τ_i (μ_L, μ_I, N_I)	Mobility and scattering time for one scattering mechanism (L: lattice, I: ionised impurity); N_I = ionised-impurity density	$m^2/(V \cdot s)$; s; m^{-3}
G_n, G_p, R_n, R_p	Generation and recombination rates of electrons and holes	$m^{-3}s^{-1}$
τ_b (τ_{bulk}), $\tau_{surface}$	Lifetime from bulk recombination only; from surface recombination only	s
U_s, p_s, Δp_s	Surface recombination rate; hole density and excess at the surface	$m^{-2}s^{-1}$; m^{-3}
n_{n0}, p_{p0}, p_{n0}, n_{p0}	Equilibrium densities: majority (n_{n0}, p_{p0}) and minority (p_{n0} holes on n side, n_{p0} electrons on p side)	m^{-3}
p_n, n_p	Minority-carrier density: holes on the n side, electrons on the p side	m^{-3}
erfc(u), $erfc^{-1}$	Complementary error function, 1 − erf(u), and its inverse (read from a table)	dimensionless
N(x), N(W)	Doping concentration at depth x (or at the depletion edge W), from C-V profiling	m^{-3}

Abbreviations

Abbreviation	Meaning
Si	Silicon
Ge	Germanium
GaAs	Gallium arsenide
SiO_2	Silicon dioxide (silica), the oxide insulator
MG-Si	Metallurgical-grade silicon (≈ 98 % pure)
EG-Si	Electronic-grade silicon (99.9999999 % or better)
CZ	Czochralski crystal growth
CVD	Chemical vapour deposition
VPE	Vapour-phase epitaxy
SRH	Shockley–Read–Hall (trap-assisted) recombination
I-V	Current–voltage (measurement or characteristic)
C-V	Capacitance–voltage (measurement or characteristic)

Physical constants

Quantity	Symbol	Value
Electronic charge	q	1.6×10^{-19} C
Boltzmann constant	k	8.617×10^{-5} eV/K = 1.381×10^{-23} J/K
Thermal energy at 300 K	kT	0.0259 eV
Thermal voltage at 300 K	kT/q	0.0259 V (about 26 mV)
Planck constant	h	6.626×10^{-34} J·s
Reduced Planck constant	$\hbar = h/2\pi$	1.055×10^{-34} J·s
Free-electron mass	m_0	9.11×10^{-31} kg
Permittivity of free space	ε_0	8.854×10^{-12} F/m
Electron-volt	1 eV	1.6×10^{-19} J
Photon wavelength	λ	1.24/E μm (1.24×10^{-6}/E m), with E in eV
Speed of light	c	3×10^{8} m/s

Material data at 300 K

	Si	Ge	GaAs
Crystal structure	Diamond	Diamond	Zinc blende
Lattice constant a (nm)	0.543	0.565	0.565
Band gap E_g (eV)	1.12, indirect	0.66, indirect	1.42, direct
n_i (m^{-3})	1×10^{16}	2.4×10^{19}	2×10^{12}
N_c (m^{-3})	2.8×10^{25}	1.04×10^{25}	4.7×10^{23}
N_v (m^{-3})	1.04×10^{25}	6.0×10^{24}	7.0×10^{24}
μ_n ($m^2/V\cdot s$)	0.135	0.39	0.85
μ_p ($m^2/V\cdot s$)	0.048	0.19	0.04
D_n, D_p (m^2/s)	3.5×10^{-3}, 1.24×10^{-3}	1.01×10^{-2}, 4.9×10^{-3}	2.2×10^{-2}, 1.0×10^{-3}
Relative permittivity ε_r	11.7	16.0	12.9
Permittivity $\varepsilon_s = \varepsilon_r\varepsilon_0$ (F/m)	1.04×10^{-10}	1.42×10^{-10}	1.14×10^{-10}
Atom density (m^{-3})	5.0×10^{28}	4.4×10^{28}	4.4×10^{28}
Electron affinity χ (eV)	4.05	4.0	4.07
Saturation velocity v_{sat} (m/s)	1×10^{5}	6×10^{4}	about 1×10^{5}
Melting point (°C)	1414	937	1238

Crystals (Q5–Q8)

Miller indices: intercepts → reciprocals → clear fractions → (h k l) (Q6, Q7)

$d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$ spacing of (h k l) planes, cubic crystal

Diamond lattice: 8 atoms per cell, 4 nearest neighbours, bond angle 109.5° (Q8)

Growth (Q9–Q13)

$SiO_2 + 2C \rightarrow Si + 2CO$ (MG-Si, about 98 % pure, arc furnace) (Q9)

$Si + 3HCl \rightarrow SiHCl_3 + H_2$ (trichlorosilane, about 300 °C)

$SiHCl_3 + H_2 \rightarrow Si + 3HCl$ (Siemens CVD, about 1100 °C → EG-Si)

$k_0 = C_s/C_l$; $C_s = k_0 C_0 (1 - f)^{k_0 - 1}$ (CZ segregation) (Q10, Q11)

$SiCl_4 + 2H_2 \rightarrow Si + 4HCl$ (VPE of silicon) (Q12)

Doping and diffusion (Q14–Q21)

$n \approx N_D$ (n-type), $p \approx N_A$ (p-type), complete ionisation (Q15, Q34)

Fick's first law: $F = -D\,\partial C/\partial x$ (Q18, Q39)

Fick's second law: $\partial C/\partial t = D\,\partial^2 C/\partial x^2$ (Q18)

$D = D_0\, e^{-E_a/kT}$ (Q17)

Constant source: $C(x,t) = C_s \operatorname{erfc}(x / 2\sqrt{Dt})$, $Q = 2C_s\sqrt{Dt/\pi}$ (Q19)

Limited source: $C(x,t) = (Q/\sqrt{\pi Dt})\, e^{-x^2/4Dt}$

Junction depth: $C(x_j) = N_B$

Implant: $N(x) = (\Phi/(\sqrt{2\pi}\Delta R_p))\, e^{-(x-R_p)^2/2\Delta R_p^2}$, $\Phi = It/qA$ (Q20)

Bands and carrier statistics (Q22–Q34)

Schrödinger: $-(\hbar^2/2m)\, d^2\psi/dx^2 + V(x)\psi = E\psi$ (Q23)

Kronig–Penney: $P \sin(\alpha a)/(\alpha a) + \cos(\alpha a) = \cos(ka)$ (Q24)

$E_g = E_c - E_v$; $\lambda = 1.24/E_g\ \mu m$ (Q25, Q26)

Fermi–Dirac: $f(E) = 1 / (1 + e^{(E-E_F)/kT})$ (Q30, Q31)

$g_c(E) = (4\pi(2m_n^*)^{3/2}/h^3)\sqrt{(E - E_c)}$ (Q32)

$n = N_c\, e^{-(E_c - E_F)/kT}$, $p = N_v\, e^{-(E_F - E_v)/kT}$ (Q27)

$n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT}$ (boxed result of Q27)

$np = n_i^2$ (mass-action law) (Q28, Q29, Q34)

$E_F - E_i = kT \ln(N_D/n_i)$ (n-type); $E_i - E_F = kT \ln(N_A/n_i)$ (p-type) (Q33, Q34)

$E_c - E_F = kT \ln(N_c/N_D)$

Transport (Q35–Q41)

$v_d = \mu E$; $\mu = q\tau_m/m^*$; $1/\mu = 1/\mu_L + 1/\mu_I$ (Matthiessen) (Q35, Q36)

$J = q(n\mu_n + p\mu_p)E$; $\sigma = q(n\mu_n + p\mu_p)$; $\rho = 1/\sigma$ (boxed result of Q37) (Q37, Q38)

$J_n = qn\mu_n E + qD_n\, dn/dx$; $J_p = qp\mu_p E - qD_p\, dp/dx$ (Q39)

Einstein: $D/\mu = kT/q$ (Q40)

High field: $v_d \to v_{sat} \approx 1\times10^5$ m/s (Q41)

Recombination and continuity (Q42–Q46)

SRH: $U = (np - n_i^2) / [\,\tau_p(n + n_1) + \tau_n(p + p_1)\,]$, $\tau = 1/(\sigma v_{th} N_t)$ (Q43)

Low-level, n-type: $U = \Delta p/\tau_p$; $\Delta p(t) = \Delta p(0)\, e^{-t/\tau_p}$ (Q42, Q44)

Continuity: $\partial\Delta p/\partial t = D_p\,\partial^2\Delta p/\partial x^2 - \mu_p E\,\partial\Delta p/\partial x + G_L - \Delta p/\tau_p$ (Q45)

Steady state, no field: $\Delta p(x) = \Delta p(0)\, e^{-x/L_p}$, $L_p = \sqrt{(D_p\tau_p)}$

Surface: $U_s = S\Delta p_s$, $S = \sigma v_{th} N_{st}$ (Q46)

$1/\tau_{eff} = 1/\tau_b + 2S/W$ (valid when $SW/2D \ll 1$); for large S, $\tau_{surface} \approx W/2S + W^2/(\pi^2 D)$

Junctions, contacts and characterisation (Q47–Q50)

$V_{bi} = (kT/q)\ln(N_A N_D / n_i^2)$ (Q47)

$W = \sqrt{(2\varepsilon_s/q)(V_{bi} - V)(1/N_A + 1/N_D)}$; $N_A x_p = N_D x_n$; $E_{max} = qN_D x_n/\varepsilon_s$

Law of the junction: $p_n(x_n) = p_{n0}\, e^{qV/kT}$ (Q48)

$I = I_s(e^{qV/kT} - 1)$, $J_s = qn_i^2(D_p/(L_p N_D) + D_n/(L_n N_A))$ (boxed result of Q48)

Schottky: $\Phi_B = \Phi_m - \chi$; $V_{bi} = \Phi_m - \Phi_s$; $J_s = A{**}T^2 e^{-q\Phi_B/kT}$ (Q49)

Ohmic by tunnelling: $N_D \gtrsim 10^{25}\ m^{-3}$, so $W = \sqrt{(2\varepsilon_s V_{bi}/qN_D)}$ is a few nm

I-V: $I = I_s(e^{qV/\eta kT} - 1)$; $\eta = (q/kT)\, dV/d(\ln I)$ (Q50)

C-V: $C = \varepsilon_s A/W$; $1/C^2 = 2(V_{bi} - V)/(q\varepsilon_s N A^2)$; $N = 2/(q\varepsilon_s A^2 |\text{slope}|)$; V_{bi} = V-axis intercept

Rules of thumb

- Each decade of doping moves E_F by 2.3kT ≈ 60 mV; each 60 mV of forward bias multiplies a diode's current by 10 (300 K).
- In silicon at room temperature, n_i roughly doubles for every 10 °C rise.
- $10^{22}\ m^{-3}$ of dopant is 1 atom in 5×10^6 silicon atoms (Si has 5×10^{28} atoms/m^3), yet it raises the conductivity nearly a million times (from 2.9×10^{-4} to 216 S/m, n-type).
- Diffusion distances go as √(Dt): four times the time gives twice the depth.

Converting from centimetre units

Textbooks, your lecturer's notes and some exam questions use centimetres. Convert to SI before you substitute:

Quantity	cm-based unit	SI unit	Multiply by	Example
Length	cm	m	10^{-2}	0.33 µm = 3.3×10^{-5} cm = 3.3×10^{-7} m
Concentration	cm^{-3}	m^{-3}	10^{6}	$1\times10^{16}\ cm^{-3} = 1\times10^{22}\ m^{-3}$
Dose, surface density	cm^{-2}	m^{-2}	10^{4}	$1\times10^{15}\ cm^{-2} = 1\times10^{19}\ m^{-2}$
Area	cm^2	m^2	10^{-4}	$1\times10^{-3}\ cm^2 = 1\times10^{-7}\ m^2$
Mobility	$cm^2/V{\cdot}s$	$m^2/V{\cdot}s$	10^{-4}	1350 $cm^2/V{\cdot}s$ = 0.135 $m^2/V{\cdot}s$
Diffusion coefficient	cm^2/s	m^2/s	10^{-4}	35 cm^2/s = 3.5×10^{-3} m^2/s
Velocity	cm/s	m/s	10^{-2}	10^7 cm/s = 10^5 m/s
Electric field	V/cm	V/m	10^{2}	10 V/cm = 1000 V/m
Current density	A/cm^2	A/m^2	10^{4}	43.2 A/cm^2 = 4.32×10^5 A/m^2
Permittivity	F/cm	F/m	10^{2}	1.04×10^{-12} F/cm = 1.04×10^{-10} F/m
Conductivity	S/cm	S/m	10^{2}	2.16 S/cm = 216 S/m
Resistivity	Ω·cm	Ω·m	10^{-2}	1 Ω·cm = 0.01 Ω·m
Capacitance per area	F/cm^2	F/m^2	10^{4}	3.45×10^{-7} F/cm^2 = 3.45×10^{-3} F/m^2
Lattice spacing	Å	nm	10^{-1}	5.43 Å = 0.543 nm

A quick check: a concentration in m^{-3} is always a million times the number in cm^{-3} (n_i of silicon is 1×10^{10} cm^{-3} = 1×10^{16} m^{-3}).

REVISION TOOLS

Practice Examination Papers

Three full practice papers built from the 50 questions. Each question combines two or three of the 50, the way exam questions often do, and some parts change the numbers so you cannot simply recall an answer. The layout (eight questions, answer any five, 3 hours) is a practice format only: follow the instructions printed on your real paper.

Symbols and abbreviations used on this page

Every symbol, constant and abbreviation on this page is listed here, even if another page already defined it, so you can study this page on its own.

Constants and units

Symbol	Meaning	Units / typical value
q	Magnitude of the electronic charge	1.6×10^{-19} C
k	Boltzmann constant	8.617×10^{-5} eV/K = 1.381×10^{-23} J/K
kT	Thermal energy	0.0259 eV (often 0.02585 eV) at 300 K
kT/q	Thermal voltage	0.0259 V (about 26 mV) at 300 K
eV	Electron-volt, a unit of energy	1 eV = 1.6×10^{-19} J
µm	Micrometre (micron)	1 µm = 10^{-6} m

Symbols

Symbol	Meaning	Units / typical value
E_g	Band-gap energy, $E_c - E_v$	eV; Si 1.12, Ge 0.66, GaAs 1.42 at 300 K
E_F	Fermi level: the energy at which a state has a 50 % chance of being occupied	eV
E_i	Intrinsic Fermi level, close to mid-gap	eV
f(E)	Fermi–Dirac function: probability that a state at energy E is occupied	dimensionless, 0 to 1
N_c	Effective density of states in the conduction band	m^{-3}; Si 2.8×10^{25} at 300 K
N_v	Effective density of states in the valence band	m^{-3}; Si 1.04×10^{25} at 300 K
n	Electron concentration (free electrons per m^3)	m^{-3}
p	Hole concentration (holes per m^3)	m^{-3}
n_i	Intrinsic carrier concentration ($n = p = n_i$ in pure material)	m^{-3}; Si $\approx 1\times10^{16}$ at 300 K
N_D	Donor concentration	m^{-3}
N_A	Acceptor concentration	m^{-3}
λ	Wavelength	μm or nm; λ(μm) = 1.24/E(eV)
μ, μ_n, μ_p	Mobility: drift velocity per unit field (electrons, holes)	$m^2/(V\cdot s)$; Si μ_n 0.135, μ_p 0.048
J, J_n, J_p	Current density (total, electrons, holes)	A/m^2
D_n, D_p	Diffusion coefficients of electrons and holes	m^2/s; Si 3.5×10^{-3} and 1.24×10^{-3}
S	Surface recombination velocity	m/s
V_{bi}	Built-in potential of a junction or Schottky contact	V; typically 0.5–1 V
W	Depletion-region width (in Week 13 W_B, W_E are base and emitter widths; in Week 14 W is channel width)	μm or m
C, C_j	Capacitance (junction or depletion capacitance)	F (pF = 10^{-12} F)
Φ_m	Work function of a metal	eV; Al 4.28, Au 5.1
Φ_s	Work function of a semiconductor, $\chi + (E_c - E_F)$	eV
χ	Electron affinity: energy from E_c up to the vacuum level	eV; Si 4.05
Φ_B	Schottky barrier height seen from the metal	eV (or V)

Abbreviations

Abbreviation	Meaning
Si	Silicon
Ge	Germanium
GaAs	Gallium arsenide
VPE	Vapour-phase epitaxy
LPE	Liquid-phase epitaxy
MBE	Molecular-beam epitaxy
n^+	Very heavily doped n-type (similarly p^+)

Practice Paper 1

Answer any FIVE questions. Each question carries 20 marks. Time: 3 hours.

1. (a) Define a semiconductor and explain how its conductivity differs from that of a conductor and an insulator. *(8 marks)* (b) Define Miller indices. *(4 marks)* (c) A plane makes intercepts of a, 2b and ∞ (parallel to the z-axis) on the axes of a cubic silicon crystal. Find its Miller indices and the spacing between neighbouring planes of this family. *(8 marks)*

2. (a) Describe the major stages in converting metallurgical-grade silicon into electronic-grade silicon. Give the chemical equations. *(10 marks)* (b) Explain the principle and operation of the Czochralski method. *(10 marks)*

3. (a) Explain the principle of ion implantation and its major advantages over thermal diffusion. *(10 marks)* (b) Compare diffusion and ion implantation as doping techniques. *(10 marks)*

4. (a) What is meant by an intrinsic semiconductor? *(3 marks)* (b) Derive the expression for the intrinsic carrier concentration. *(12 marks)* (c) For GaAs at 300 K, $N_c = 4.7\times10^{23}$ m^{-3}, $N_v = 7.0\times10^{24}$ m^{-3} and $E_g = 1.42$ eV. Calculate n_i. *(5 marks)*

5. (a) Explain how the Fermi level changes with doping in n-type and p-type semiconductors. Illustrate with band diagrams. *(10 marks)*
(b) Silicon is doped with $N_A = 2\times10^{21}$ m^{-3} acceptors. Assuming complete ionisation, find the hole and electron concentrations at 300 K and the position of the Fermi level relative to E_i. *(10 marks)*

6. (a) Derive the expression for the drift current density due to electrons and holes. *(12 marks)*
(b) n-type silicon has $n = 1\times10^{21}$ m^{-3}. Calculate the electron drift current density in a field of 5000 V/m. *(8 marks)*

7. (a) Explain the Shockley–Read–Hall recombination mechanism. *(12 marks)* (b) Define minority-carrier lifetime and explain its significance in semiconductor devices. *(8 marks)*

8. (a) Explain the formation of the depletion region and built-in potential in a p-n junction. *(10 marks)*
(b) A silicon p^+n junction has $N_A = 1\times10^{24}$ m^{-3} and $N_D = 1\times10^{21}$ m^{-3}. Calculate the built-in potential and the zero-bias depletion width at 300 K. *(10 marks)*

Marking guide, Paper 1

Question	Model answers to mark against	Answers to new numerical parts
1	Q1, Q6, Q7	Intercepts 1, 2, ∞ → reciprocals 1, ½, 0 → × 2 → **(2 1 0)**; $d = a/\sqrt{(2^2 + 1^2 + 0^2)} = 0.543\ \text{nm}/\sqrt{5} =$ **0.243 nm**
2	Q9, Q10	—
3	Q20, Q21	—
4	Q27	$n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT} = \sqrt{(4.7\times10^{23} \times 7.0\times10^{24})} \times e^{-1.42/0.0518} =$ **2.3×10^{12} m^{-3}**
5	Q33, Q34	$p = N_A =$ **2×10^{21} m^{-3}**; $n = n_i^2/p =$ **5×10^{10} m^{-3}**; $E_i - E_F = 0.0259 \ln(2\times10^{21}/10^{16}) =$ **0.32 eV below E_i**
6	Q37, Q38	$J_n = qn\mu_n E = 1.6\times10^{-19} \times 10^{21} \times 0.135 \times 5000 =$ **1.08×10^{5} A/m^2**
7	Q43, Q44	—
8	Q47	$V_{bi} = 0.0259 \ln(10^{24} \times 10^{21}/10^{32}) =$ **0.775 V**; $W \approx \sqrt{(2\varepsilon_s V_{bi}/qN_D)} =$ **1.0×10^{-6} m (1.0 µm)**, almost all on the n side

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Practice Paper 2

Answer any FIVE questions. Each question carries 20 marks. Time: 3 hours.

1. (a) Distinguish between elemental and compound semiconductors, giving two examples of each. *(8 marks)* (b) Compare the properties and applications of Si, Ge and GaAs. *(12 marks)*

2. (a) What is a crystal lattice? Explain the significance of crystal structure in semiconductor materials. *(10 marks)* (b) Explain the difference between covalent, ionic and metallic bonding, and identify the dominant bonding in silicon. *(10 marks)*

3. (a) State four advantages and two limitations of the Czochralski method. *(8 marks)* (b) Compare VPE, LPE and MBE in terms of growth mechanism, temperature, precision and applications. *(12 marks)*

4. (a) What is doping, and why is it necessary in device fabrication? *(6 marks)* (b) Differentiate between n-type and p-type semiconductors. *(6 marks)* (c) Explain the difference between donor and acceptor impurities, with examples. *(8 marks)*

5. (a) State the time-independent Schrödinger equation and explain its terms. *(6 marks)* (b) Explain the Kronig–Penney model and how it leads to energy bands. *(10 marks)* (c) Distinguish between the valence band, the conduction band and the forbidden gap. *(4 marks)*

6. (a) Explain the Fermi–Dirac distribution function and its significance. *(8 marks)* (b) Explain what happens to the distribution at absolute zero. *(4 marks)* (c) Find the probability that a state 0.1 eV above the Fermi level is occupied at 300 K, and the probability that a state 0.1 eV below it is empty. *(8 marks)*

7. (a) Explain diffusion current in a semiconductor and state Fick's law. *(10 marks)*
(b) State and explain the Einstein relation. Use it to find the electron diffusion coefficient in a sample where $\mu_n = 0.10$ m^2/V·s at 300 K. *(10 marks)*

8. (a) Differentiate between a Schottky barrier diode and an ohmic contact, and explain their energy-band characteristics. *(14 marks)*

(b) Nickel (Φ_m = 5.15 eV) is deposited on n-type silicon with $N_D = 1\times10^{22}$ m^{-3}. Using the ideal model, find the barrier height Φ_B and the built-in potential V_{bi}. *(6 marks)*

Marking guide, Paper 2

Question	Model answers to mark against	Answers to new numerical parts
1	Q2, Q4	—
2	Q5, Q8	—
3	Q11, Q13	—
4	Q14, Q15, Q16	—
5	Q23, Q24, Q25	—
6	Q30, Q31	$f = 1/(1 + e^{0.1/0.0259})$ = **0.021** (2 %); the state 0.1 eV below E_F is empty with probability $1 - f$ = **0.021** too, by symmetry
7	Q39, Q40	$D_n = (kT/q)\mu_n = 0.0259 \times 0.10$ = **2.59×10^{-3} m^2/s**
8	Q49	$\Phi_B = \Phi_m - \chi = 5.15 - 4.05$ = **1.10 eV**; $\Phi_s = 4.05 + 0.0259 \ln(2.8\times10^{25}/10^{22})$ = 4.26 eV; $V_{bi} = \Phi_m - \Phi_s$ = **0.89 V** (ideal; pinning lowers both in practice)

Practice Paper 3

Answer any FIVE questions. Each question carries 20 marks. Time: 3 hours.

1. (a) Explain why silicon is widely used in device fabrication. *(8 marks)* (b) Explain the principles of VPE, LPE and MBE. *(12 marks)*

2. (a) Define Miller indices and explain how they identify crystal planes. *(8 marks)* (b) Determine the Miller indices of a plane with intercepts 2a, 3b and 1c, and sketch the plane. *(12 marks)*

3. (a) Describe the mechanism of dopant diffusion in a semiconductor. *(10 marks)* (b) State and explain Fick's first and second laws of diffusion. *(10 marks)*

4. (a) Explain the basic assumptions of the quantum-mechanical model of electrons in a semiconductor. *(8 marks)* (b) Explain the difference between direct and indirect band-gap semiconductors. *(8 marks)* (c) Calculate the longest wavelength of light that GaAs (E_g = 1.42 eV) can absorb or emit. *(4 marks)*

5. (a) Explain the difference between intrinsic and extrinsic carrier concentrations. *(6 marks)* (b) Silicon has $n_i = 1.0\times10^{16}$ m^{-3}. If the electron and hole concentrations are equal, determine them. *(4 marks)* (c) Define the density of states and explain its importance in finding carrier concentrations. *(10 marks)*

6. (a) Define carrier mobility and explain the factors that affect it. *(8 marks)* (b) Explain the mechanisms of carrier scattering. *(6 marks)* (c) Explain velocity saturation and hot-carrier effects. *(6 marks)*

7. (a) What is meant by generation and recombination of carriers? *(5 marks)* (b) Explain the continuity equation for minority carriers and its application. *(9 marks)* (c) What is surface recombination velocity? Explain how surface states affect carrier lifetime. *(6 marks)*

8. (a) Derive the ideal Shockley diode equation. *(12 marks)*
(b) A Schottky diode of area 1×10^{-7} m^2 gives a straight line of $1/C^2$ against V with slope -1.2×10^{21} F^{-2}V^{-1}, crossing the

voltage axis at +0.80 V. Find the doping concentration of the semiconductor and the built-in potential. *(8 marks)*

Marking guide, Paper 3

Question	Model answers to mark against	Answers to new numerical parts
1	Q3, Q12	—
2	Q6, Q7	—
3	Q17, Q18	—
4	Q22, Q26	$\lambda = 1.24/E_g = 1.24/1.42 =$ **0.873 µm** (near infrared)
5	Q28, Q29, Q32	$\mathbf{n = p = 1.0\times10^{16}\ m^{-3}}$
6	Q35, Q36, Q41	—
7	Q42, Q45, Q46	—
8	Q48, Q50	$N = 2/(q\varepsilon_s A^2\lvert\text{slope}\rvert) = 2/(1.6\times10^{-19} \times 1.04\times10^{-10} \times 10^{-14} \times 1.2\times10^{21}) =$ **1.0×10^{22} m^{-3}**; V_{bi} = intercept = **0.80 V**

REVISION TOOLS

Last-Minute Flashcards

One line per question: the facts, equations and numbers that every full answer must contain. Use this page in the last days before the exam. Cover the right-hand column, say the answer aloud, then check. If you cannot, go back to the full model answer.

Symbols and abbreviations used on this page

Every symbol, constant and abbreviation on this page is listed here, even if another page already defined it, so you can study this page on its own.

Constants and units

Symbol	Meaning	Units / typical value
q	Magnitude of the electronic charge	1.6×10^{-19} C
k	Boltzmann constant	8.617×10^{-5} eV/K = 1.381×10^{-23} J/K
kT	Thermal energy	0.0259 eV (often 0.02585 eV) at 300 K
kT/q	Thermal voltage	0.0259 V (about 26 mV) at 300 K
T	Absolute temperature	kelvin (K); room temperature = 300 K
ħ	Reduced Planck constant, h/2π	1.055×10^{-34} J·s
eV	Electron-volt, a unit of energy	1 eV = 1.6×10^{-19} J
µm	Micrometre (micron)	1 µm = 10^{-6} m

Symbols

Symbol	Meaning	Units / typical value
E_c (E_C)	Energy of the bottom of the conduction band	eV
E_v (E_V)	Energy of the top of the valence band	eV
E_g	Band-gap energy, $E_c - E_v$	eV; Si 1.12, Ge 0.66, GaAs 1.42 at 300 K
E_F	Fermi level: the energy at which a state has a 50 % chance of being occupied	eV
E_i	Intrinsic Fermi level, close to mid-gap	eV
E_t	Energy level of a trap (recombination centre) in the gap	eV
E_a	Activation energy of dopant diffusion	eV; about 3–4 eV
f(E)	Fermi–Dirac function: probability that a state at energy E is occupied	dimensionless, 0 to 1
g(E), g_c, g_v	Density of states: allowed states per unit energy per unit volume	states/(eV·m^3)
N_c	Effective density of states in the conduction band	m^{-3}; Si 2.8×10^{25} at 300 K
N_v	Effective density of states in the valence band	m^{-3}; Si 1.04×10^{25} at 300 K
n	Electron concentration (free electrons per m^3)	m^{-3}
p	Hole concentration (holes per m^3)	m^{-3}
n_i	Intrinsic carrier concentration (n = p = n_i in pure material)	m^{-3}; Si ≈ 1×10^{16} at 300 K
N_D	Donor concentration	m^{-3}
N_A	Acceptor concentration	m^{-3}
Δn, Δp	Excess electron and hole concentrations above equilibrium	m^{-3}
m^*, m_n^*, m_p^*	Effective mass of electrons or holes	kg, or as a multiple of m_0
ψ	Wave function ($\lvert\psi\rvert^2$ is the probability density)	—
α (Kronig–Penney)	$\sqrt{(2mE)}/\hbar$, set by the electron energy	m^{-1}
P (Kronig–Penney)	Barrier strength $mV_0ba/\hbar^2$	dimensionless
λ	Wavelength	μm or nm; λ(μm) = 1.24/E(eV)
(h k l), {h k l}	Miller indices of a plane; family of equivalent planes	integers
C_s	Surface concentration of a diffusion (or solid concentration when freezing)	m^{-3}
C(x, t)	Dopant concentration at depth x and time t	m^{-3}
D	Diffusion coefficient (dopants in a furnace, or carriers)	m^2/s
D_0	Diffusion constant in $D = D_0e^{-E_a/kT}$	m^2/s
F	Particle flux in Fick's law	particles/(m^2·s)
Q	Dose: dopant atoms per unit area (in Weeks 13–14, Q also means charge)	atoms/m^2 (C or C/m^2 for charge)
erfc	Complementary error function, erfc(u) = 1 − erf(u); read from a table	dimensionless

Symbol	Meaning	Units / typical value
R_p	Projected range: average depth at which implanted ions stop	μm
R_s	Sheet resistance (or diode series resistance in Week 12)	Ω/□ (ohms per square); Ω
E (field)	Electric field (on band diagrams, E is energy instead)	V/m
μ, μ_n, μ_p	Mobility: drift velocity per unit field (electrons, holes)	$m^2/(V\cdot s)$; Si μ_n 0.135, μ_p 0.048
v_d	Drift velocity	m/s
v_{sat}	Saturation velocity at high field	$\approx 10^5$ m/s in Si
σ	Conductivity, $q(n\mu_n + p\mu_p)$ (in Week 8, σ is also a capture cross-section)	S/m (siemens per metre)
ρ	Resistivity, 1/σ	Ω·m
J, J_n, J_p	Current density (total, electrons, holes)	A/m^2
D_n, D_p	Diffusion coefficients of electrons and holes	m^2/s; Si 3.5×10^{-3} and 1.24×10^{-3}
T_e	Electron temperature of hot carriers	K
G, G_L	Generation rate of electron–hole pairs (G_L: by light)	$m^{-3}s^{-1}$
R	Recombination rate	$m^{-3}s^{-1}$
U	Net recombination rate, R − G	$m^{-3}s^{-1}$
τ, τ_n, τ_p	Minority-carrier lifetime (τ_{eff}: effective, including surfaces)	s; Si 1 ns – 1 ms
N_t	Density of traps (recombination centres)	m^{-3}
n_1, p_1	SRH parameters $n_{ie}^{(E_t - E_i)/kT}$ and $n_{ie}^{(E_i - E_t)/kT}$	m^{-3}
L, L_n, L_p	Diffusion length $\sqrt{(D\tau)}$: distance a minority carrier travels before recombining	μm or m
S	Surface recombination velocity	m/s
N_{st}	Density of surface traps	m^{-2}
U_s	Recombination rate per unit area at a surface	$m^{-2}s^{-1}$
V_{bi}	Built-in potential of a junction or Schottky contact	V; typically 0.5–1 V
W	Depletion-region width (in Week 13 W_B, W_E are base and emitter widths; in Week 14 W is channel width)	μm or m
I_s, J_s (I_S)	Saturation current (density) of a diode or transistor	A; A/m^2
I	Current	A
η	Ideality factor of a diode, 1 (ideal) to 2	dimensionless
C, C_j	Capacitance (junction or depletion capacitance)	F (pF = 10^{-12} F)
Φ_m	Work function of a metal	eV; Al 4.28, Au 5.1
Φ_s	Work function of a semiconductor, $\chi + (E_c - E_F)$	eV
χ	Electron affinity: energy from E_c up to the vacuum level	eV; Si 4.05
Φ_B	Schottky barrier height seen from the metal	eV (or V)
β	Common-emitter current gain, I_C/I_B (also h_{FE})	dimensionless; 50–500

Symbol	Meaning	Units / typical value
V_T (V_{th})	Threshold voltage of a MOSFET	V
v_{dn}, v_{dp}	Drift velocity of electrons and holes	m/s
μ_i, τ_i (μ_L, μ_I, N_I)	Mobility and scattering time for one scattering mechanism (L: lattice, I: ionised impurity); N_I = ionised-impurity density	$m^2/(V\cdot s)$; s; m^{-3}
G_n, G_p, R_n, R_p	Generation and recombination rates of electrons and holes	$m^{-3}s^{-1}$
τ_b (τ_{bulk}), $\tau_{surface}$	Lifetime from bulk recombination only; from surface recombination only	s
U_s, p_s, Δp_s	Surface recombination rate; hole density and excess at the surface	$m^{-2}s^{-1}$; m^{-3}
p_n, n_p	Minority-carrier density: holes on the n side, electrons on the p side	m^{-3}
erfc(u), $erfc^{-1}$	Complementary error function, 1 − erf(u), and its inverse (read from a table)	dimensionless
N(x), N(W)	Doping concentration at depth x (or at the depletion edge W), from C-V profiling	m^{-3}

Abbreviations

Abbreviation	Meaning
Si	Silicon
Ge	Germanium
GaAs	Gallium arsenide
InP	Indium phosphide
SiO_2	Silicon dioxide (silica), the oxide insulator
CB	Conduction band
VB	Valence band
III–V	Compound of Group III and Group V elements (e.g. GaAs)
MG-Si	Metallurgical-grade silicon (≈ 98 % pure)
EG-Si	Electronic-grade silicon (99.9999999 % or better)
CZ	Czochralski crystal growth
FZ	Float-zone crystal growth
CVD	Chemical vapour deposition
VPE	Vapour-phase epitaxy
LPE	Liquid-phase epitaxy
MBE	Molecular-beam epitaxy
MOCVD	Metal-organic chemical vapour deposition
UHV	Ultra-high vacuum (≈ 10^{-10} Torr)
SRH	Shockley–Read–Hall (trap-assisted) recombination
LDD	Lightly doped drain
I-V	Current–voltage (measurement or characteristic)
C-V	Capacitance–voltage (measurement or characteristic)
HEMT	High-electron-mobility transistor
BJT	Bipolar junction transistor
HBT	Heterojunction bipolar transistor
IC	Integrated circuit
VLSI	Very-large-scale integration
LED	Light-emitting diode
RF	Radio frequency
LHS	Left-hand side (of an equation)
IR	Infrared

Material physics and semiconductor growth (Q1–Q13)

Q	Topic	Must-know answer
1	Semiconductor vs conductor and insulator	Conductivity between metal and insulator, controllable by doping, T, light. Metal: no gap, σ falls with T. Insulator: gap > 4 eV. Semiconductor: gap ≈ 1 eV, two carriers, σ rises with T.
2	Elemental and compound semiconductors	Elemental: one element, Group IV, covalent, e.g. **Si, Ge**. Compound: two or more elements (III–V, II–VI), tunable and often direct gap, e.g. **GaAs, InP**.
3	Why silicon	Abundant and cheap; excellent native SiO_2; 1.12 eV gap gives low leakage to 150 °C; large perfect crystals; strong, good thermal conductor; well-understood doping.
4	Si, Ge and GaAs	Gaps 1.12 / 0.66 / 1.42 eV; indirect / indirect / direct; μ_n 0.135 / 0.39 / 0.85 $m^2/V{\cdot}s$. Si for ICs and power; Ge for IR and SiGe HBTs; GaAs for RF, LEDs, lasers.
5	Crystal lattice and structure	Lattice = periodic array of points; crystal = lattice + basis. Structure sets bands and gap, mobility, anisotropy, doping sites, epitaxy, cleavage and defects.
6	Miller indices	Smallest integers proportional to reciprocals of intercepts. Steps: intercepts → reciprocals → clear fractions → (h k l). Infinite intercept → 0; negative → bar.
7	Miller indices of intercepts 2a, 3b, 1c	2, 3, 1 → 1/2, 1/3, 1 → × 6 → **(3 2 6)**. In a cubic crystal d = a/7.
8	Covalent, ionic and metallic bonding	Covalent: shared pairs, directional. Ionic: electron transfer, Coulomb attraction. Metallic: electron sea. **Si is covalent**: four sp^3 bonds, tetrahedral, 109.5°.
9	MG-Si to EG-Si	$SiO_2 + 2C \rightarrow$ MG-Si (98 %); $Si + 3HCl \rightarrow SiHCl_3$ (300 °C); fractional distillation; Siemens CVD $SiHCl_3 + H_2 \rightarrow Si + 3HCl$ (1100 °C) → EG-Si (9N–11N) polysilicon; then CZ/FZ.
10	Czochralski growth	Seed dipped into melt at 1414 °C and slowly pulled (≈ 1 mm/min) with counter-rotation; melt freezes epitaxially on seed; Dash neck removes dislocations; shoulder, body, tail.
11	Advantages and limitations of CZ	Advantages: large diameter, dislocation-free, controlled doping, low cost, oxygen strengthens and getters. Limitations: O and C contamination from crucible (limits resistivity), axial doping variation from segregation.
12	VPE, LPE and MBE principles	VPE: gas reacts on hot substrate ($SiCl_4 + 2H_2 \rightarrow Si + 4HCl$). LPE: precipitation from cooling saturated melt. MBE: atomic beams in UHV with shutters, monolayer control.
13	VPE, LPE and MBE compared	Temperature VPE 1000–1250 °C > LPE 700–900 °C > MBE 500–650 °C. Precision MBE (atomic) > VPE > LPE. Uses: VPE production Si/MOCVD LEDs; LPE LEDs, thick layers; MBE quantum wells, HEMTs.

Impurity control and doping (Q14–Q21)

Q	Topic	Must-know answer
14	What doping is and why	Controlled addition of impurities (10^{20}–10^{26} m^{-3}). Needed because intrinsic σ is too low and T-sensitive; sets carrier type, makes junctions, sets V_{bi}, V_T, β, contacts.
15	n-type and p-type	n-type: Group V donors, electrons majority, E_F near E_c, $n \approx N_D$. p-type: Group III acceptors, holes majority, E_F near E_v, $p \approx N_A$.
16	Donors and acceptors	Donor: one extra valence electron, gives it to CB, becomes +, level near E_c (P, As, Sb). Acceptor: one fewer, takes an electron creating a hole, becomes −, level near E_v (B, Al, Ga, In).
17	Mechanism of dopant diffusion	Thermally activated jumps down a gradient. Vacancy (substitutional, slow: B, P, As), interstitial (fast: Au, Cu, Fe), interstitialcy. $D = D_0 e^{-E_a/kT}$. Predeposition + drive-in.
18	Fick's first and second laws	First law $F = -D\partial C/\partial x$ (flux ∝ gradient). Second law $\partial C/\partial t = D\partial^2 C/\partial x^2$ (from continuity), gives the profile's time evolution.
19	Constant-source diffusion profile	Constant source → $C = 10^{26}\,\mathrm{erfc}(x/2\sqrt{(Dt)})$ m^{-3}. Maximum (10^{26}) at the surface, falling monotonically with depth; penetrates as √t; dose $Q = (2/\sqrt{\pi})C_s\sqrt{(Dt)}$.
20	Ion implantation and its advantages	Ionised, mass-selected dopant ions accelerated (keV–MeV) into the wafer; Gaussian profile at R_p; anneal. Advantages: precise dose, independent depth, low temperature, resist masks, shallow and buried profiles.
21	Diffusion vs ion implantation	Diffusion: high T, surface-peaked erfc/Gaussian, coupled dose/depth, no damage, cheap batch. Implant: room T, peak at R_p, ±1 % dose, damage needs anneal, channelling, costly, standard for VLSI.

Energy bands and carrier statistics (Q22–Q34)

Q	Topic	Must-know answer
22	Assumptions of the quantum model	Wave nature; fixed nuclei (Born–Oppenheimer); perfect periodic potential; independent electrons in average potential; Bloch functions; Pauli principle; effective mass; Fermi–Dirac statistics.
23	Schrödinger equation	$-(\hbar^2/2m)d^2\psi/dx^2 + V\psi = E\psi$. Kinetic-energy term + potential-energy term = total energy E; $\|\psi\|^2$ is probability density; E eigenvalues give quantisation.
24	Kronig–Penney model	Periodic square wells and barriers; Bloch condition gives $P\,\sin(\alpha a)/\alpha a + \cos \alpha a = \cos ka$. Only energies with LHS between ±1 are allowed → bands and gaps; edges at $k = n\pi/a$.
25	Valence band, conduction band, gap	VB: highest full band at 0 K (bond electrons, holes). CB: lowest empty band (free electrons). Gap $E_g = E_c - E_v$: no states.
26	Direct and indirect band gaps	Direct: CB min and VB max at same k, photon only, efficient light (GaAs). Indirect: different k, phonon needed, poor emitter (Si, Ge). $\lambda = 1.24/E_g$ μm.
27	Intrinsic semiconductor; derive n_i	Pure, $n = p = n_i$. $n = N_{ce}^{-(E_c - E_F)/kT}$, $p = N_{ve}^{-(E_F - E_v)/kT}$; multiply: $\mathbf{n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT}} \propto T^{3/2} e^{-E_g/2kT}$.
28	Intrinsic and extrinsic concentrations	Intrinsic: thermal only, $n = p = n_i$, strongly T-dependent. Extrinsic: from dopants, $n \neq p$, $n \approx N_D$ or $p \approx N_A$, nearly T-independent. Linked by $np = n_i^2$.
29	n and p when $n = p = n_i$	$n = p$ and $np = n_i^2$ ⇒ $\mathbf{n = p = 1.0\times10^{16}\ m^{-3}}$ ($1.0\times10^{10}\ cm^{-3} = 1.0\times10^{16}\ m^{-3}$).
30	Fermi–Dirac function	$f(E) = 1/(1 + e^{(E-E_F)/kT})$: occupation probability of a state; ½ at E_F. Gives n and p, defines E_F, constant E_F in equilibrium sets junction physics.
31	Fermi–Dirac at absolute zero	At 0 K f is a step: 1 below E_F, 0 above. VB full, CB empty, pure semiconductor is an insulator.
32	Density of states	g(E) = states per unit energy per unit volume, $g_c \propto (m_n*)^{3/2}\sqrt{(E - E_c)}$. $n = \int g\, f\, dE$; gives N_c, N_v; zero in gap.
33	Fermi level and doping	n-type: E_F rises toward E_c, $E_F - E_i = kT \ln(N_D/n_i)$. p-type: falls toward E_v. 60 mV per decade; degenerate at very high doping; returns to E_i at high T.
34	Majority carriers for $N_D = 5\times10^{22}\ m^{-3}$ ($5\times10^{16}\ cm^{-3}$)	$n \approx N_D = \mathbf{5\times10^{22}\ m^{-3}}$ ($5\times10^{16}\ cm^{-3} = 5\times10^{22}\ m^{-3}$). ($p = n_i^2/N_D = 10^{32}/5\times10^{22} = 2\times10^9\ m^{-3}$; $E_F - E_i = 0.40$ eV in Si.)

Carrier transport and injection (Q35–Q46)

Q	Topic	Must-know answer
35	Mobility and what affects it	$\mu = v_d/E = q\tau/m*$. Depends on effective mass, temperature (lattice $T^{-3/2}$, impurity $T^{3/2}$), doping, field, defects, surfaces, orientation, strain.
36	Scattering mechanisms	Lattice (acoustic, optical, polar phonons), ionised impurity, neutral impurity, carrier–carrier, defects, surface roughness, alloy, intervalley. Matthiessen $1/\mu = \Sigma 1/\mu_i$.
37	Drift current density	$v_{dn} = -\mu_n E$, $v_{dp} = \mu_p E$; $J = \rho v$: $J_n = qn\mu_n E$, $J_p = qp\mu_p E$; $\mathbf{J = q(n\mu_n + p\mu_p)E}$, $\sigma = q(n\mu_n + p\mu_p)$.
38	Electron drift current density	Convert first: $n = 2\times10^{22}$ m^{-3}, $\mu_n = 0.135$ m^2/V·s, $E = 1000$ V/m. $J = qn\mu_n E = 1.6\times10^{-19} \times 2\times10^{22} \times 0.135 \times 1000 = \mathbf{4.32\times10^5\ A/m^2}$ (= 43.2 A/cm^2).
39	Diffusion current and Fick's law	Current from carriers moving down a concentration gradient by random motion. Fick: $F = -D\,dC/dx$. $J_n = qD_n\,dn/dx$, $J_p = -qD_p\,dp/dx$.
40	Einstein relation	$\mathbf{D/\mu = kT/q}$ (0.0259 V). From zero equilibrium current in a non-uniform sample. Si: $D_n = 3.5\times10^{-3}$, $D_p = 1.24\times10^{-3}$ m^2/s.
41	Velocity saturation and hot carriers	High field: optical-phonon emission caps v_d at $v_{sat} \approx 10^5$ m/s. Hot carriers ($T_e \gg T$): impact ionisation, avalanche, oxide injection and V_T drift, substrate current; LDD as cure.
42	Generation and recombination	Generation creates e–h pairs (thermal, optical, impact); recombination annihilates them (radiative, SRH, Auger, surface). Equal in equilibrium; net $U = \Delta p/\tau$.
43	SRH recombination	Recombination via mid-gap trap: capture/emission of electrons and holes. $U = (np - n_i^2)/[\tau_p(n + n_1) + \tau_n(p + p_1)]$; maximum for $E_t \approx E_i$; $\tau = 1/\sigma v_{th} N_t$.
44	Minority-carrier lifetime	Average time an excess minority carrier survives: $\Delta p(t) = \Delta p(0)e^{-t/\tau}$. Sets $L = \sqrt{(D\tau)}$, diode leakage, switching speed, BJT gain, solar-cell efficiency.
45	Continuity equation	$\partial p/\partial t = -(1/q)\partial J_p/\partial x + G - R$; minority form $\partial \Delta p/\partial t = D\partial^2\Delta p/\partial x^2 - \Delta p/\tau + G$. Used for diodes (exponential profile), BJTs, lifetime and Haynes–Shockley, solar cells.
46	Surface recombination velocity	S defined by $U_s = S\Delta p_s$ (m/s), $S = \sigma v_{th} N_{st}$. Dangling-bond surface states act as SRH centres and shorten the lifetime: $1/\tau_{eff} = 1/\tau_b + 2S/W$ when $SW/2D \ll 1$ (for very large S the loss is limited by diffusion to the surface). Passivate with SiO_2, SiN_x, Al_2O_3.

Junctions, interfaces and characterisation (Q47–Q50)

Q	Topic	Must-know answer
47	Depletion region and built-in potential	Diffusion of carriers uncovers fixed ions → depletion region and field; drift balances diffusion; flat E_F. $\mathbf{V_{bi} = (kT/q)\ ln(N_A N_D/n_i^2)}$; $W = \sqrt{(2\varepsilon_s V_{bi}(N_A + N_D)/qN_A N_D)}$.
48	Shockley diode equation	Assumptions; law of junction $p_n = p_{n0e}^{qV/kT}$; diffusion equation → exponential decay over L_p; currents at edges; $\mathbf{I = I_s(e^{qV/kT} - 1)}$, $J_s = qn_i^2(D_p/L_pN_D + D_n/L_nN_A)$. Forward: exponential rise; reverse: $-I_s$.
49	Schottky diode vs ohmic contact	Schottky: $\Phi_m > \Phi_s$ (n-type), upward bending, depletion, barrier $\Phi_B = \Phi_m - \chi$, rectifying, thermionic, fast. Ohmic: $\Phi_m < \Phi_s$ or heavy doping, accumulation or tunnelling, linear I-V.
50	I-V and C-V characterisation	I-V: ln I vs V gives I_s, η, R_s, breakdown, Φ_B. C-V: $C = \varepsilon_s A/W$; $\mathbf{1/C^2 = 2(V_{bi} - V)/(q\varepsilon_s NA^2)}$; slope gives $N = 2/(q\varepsilon_s A^2 \cdot \text{slope})$, intercept gives V_{bi}; local slope gives profile N(W).

If you remember only ten things

1. $np = n_i^2$, and $n_i(\text{Si}) = 1\times10^{16}\ \text{m}^{-3}$ ($1\times10^{10}\ \text{cm}^{-3}$) at 300 K.
2. $n \approx N_D$ in n-type, $p \approx N_A$ in p-type (complete ionisation).
3. $n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT}$: a wider gap means far fewer carriers.
4. $f(E) = 1/(1 + e^{(E - E_F)/kT})$; E_F moves 60 mV per decade of doping.
5. $J = q(n\mu_n + p\mu_p)E$ for drift; $J_n = qD_n\, dn/dx$ for diffusion; $D/\mu = kT/q$.
6. Fick: $F = -D\, \partial C/\partial x$ and $\partial C/\partial t = D\, \partial^2 C/\partial x^2$; constant source gives an erfc profile.
7. Lifetime τ and diffusion length $L = \sqrt{(D\tau)}$ control every injection device.
8. $V_{bi} = (kT/q)\ln(N_A N_D/n_i^2)$; $W \propto \sqrt{(V_{bi} - V)}$ and is wider on the lightly doped side.
9. $I = I_s(e^{qV/kT} - 1)$: forward current rises tenfold per 60 mV; reverse current saturates at $-I_s$.
10. Schottky: $\Phi_B = \Phi_m - \chi$, rectifying, thermionic, fast. Ohmic: heavy doping makes the barrier thin enough to tunnel through. C-V: the slope of $1/C^2$ against V gives the doping, the intercept gives V_{bi}.

REVISION TOOLS

Glossary — Key Terms, Symbols, Constants and Abbreviations

Every key term, symbol, constant and abbreviation used in the 50 model answers. Each question also defines its own key terms, and each page has its own symbols table.

Key terms A–Z

Term	Meaning	Questions
Absolute zero (0 K)	the lowest possible temperature, where thermal energy kT = 0.	Q31
Acceptor	a Group III atom (B, Al, Ga, In) that takes an electron from a bond, creating a hole and making silicon p-type.	Q14, Q16, Q33
Acceptor and donor ions (N_A^-, N_D^+)	dopant atoms that have taken or given up an electron; they are charged and fixed in the lattice, so they cannot move.	Q47
Activation energy (E_a)	the energy barrier an atom must overcome to make one jump, in eV.	Q17
Allowed band	a range of energies for which electron states exist.	Q24
Allowed state	an energy level that an electron may occupy in the crystal (in the conduction or valence band, not in the gap).	Q30, Q32
Annealing	heating after implantation to repair lattice damage and place dopants on lattice sites where they become electrically active.	Q20, Q21
Autodoping / out-diffusion	unwanted transfer of dopant from the substrate into the growing layer, by evaporation and re-deposition / by diffusion at high temperature.	Q13
Background doping (N_B)	the doping the wafer already has, of the opposite type to the diffused dopant.	Q19
Band gap (E_g)	the forbidden energy range between the top of the valence band (E_v) and the bottom of the conduction band (E_c), $E_g = E_c - E_v$, in electron-volts (eV).	Q1, Q2, Q3, Q4, Q25, Q27, Q32
Band-gap engineering	tuning E_g by alloying compounds, e.g. $Al_xGa_{1-x}As$.	Q2
Barrier strength (P)	a dimensionless number measuring how tightly each atom holds its electrons.	Q24
Basis	the atom or group of atoms attached to each lattice point; crystal = lattice + basis.	Q5
Bloch's theorem	in a periodic potential the wave function satisfies $\psi(x + L) = e^{ikL}\psi(x)$, where L is the period; it is a travelling wave times a periodic function.	Q24
Boltzmann approximation	replacing the Fermi–Dirac function by an exponential when E_F is more than about 3kT from the band edge.	Q27, Q30
Brillouin-zone boundary	the values $k = \pm n\pi/a$ where the electron wave is Bragg-reflected by the lattice and band edges occur.	Q24
Built-in potential (V_{bi})	the voltage across the depletion region in equilibrium; the barrier majority carriers must climb to cross the junction.	Q47, Q49
Bulk lifetime (τ_{bulk})	the lifetime set by recombination inside the volume alone (s).	Q46
C-V characterisation	measuring small-signal capacitance as the DC reverse bias is stepped, to find doping and V_{bi}.	Q50
Capture / emission	a trap taking a carrier from a band / giving a carrier back to a band.	Q43
Capture cross-section (σ_n, σ_p)	the effective target area a trap presents to an electron or a hole (m^2).	Q43
Carrier concentration (n, p)	the number of free electrons (n) or holes (p) per m^3.	Q30

Term	Meaning	Questions
Channelling	ions travelling unusually deep along open crystal directions; prevented by tilting the wafer about 7°.	Q20
Chemical vapour deposition (CVD)	growing a solid from gases that react on a hot surface.	Q9
Compensation	partial cancelling when both donors and acceptors are present.	Q33
Complementary error function (erfc)	erfc(u) = 1 − erf(u); a tabulated function that equals 1 at u = 0 and falls rapidly towards 0 as u grows.	Q19
Complete ionisation	every dopant atom has given up its electron (donor) or hole (acceptor); true for Si at room temperature.	Q28, Q34
Compound semiconductor	a semiconductor made of two or more different elements in a fixed ratio.	Q2
Concentration (C)	the number of dopant atoms per unit volume, m^{-3}.	Q18
Concentration gradient ($\partial C/\partial x$)	the rate at which concentration changes with depth, m^{-4}.	Q18, Q39
Conduction band	the band above the valence band; electrons in it are free to move and carry current.	Q1, Q31
Conduction band (E_c) and valence band (E_v)	the band of free-electron states and the band of bonding-electron states; E_c and E_v are their edges.	Q16
Conduction band / valence band	the band of energies where electrons move freely / the band of energies occupied by the bonding electrons; they are separated by the band gap E_g (1.12 eV in Si).	Q42
Conduction-band minimum / valence-band maximum	the lowest point of the conduction band and the highest point of the valence band on the E–k diagram.	Q26
Conductivity (σ)	how easily a material carries electric current, in S/m; its inverse is resistivity $\rho = 1/\sigma$ in Ω·m.	Q1, Q37
Constant-source diffusion (predeposition)	diffusion in which a dopant supply keeps the surface concentration fixed at C_s for the whole process.	Q19
Continuity (conservation of atoms)	atoms are neither created nor destroyed, so any change in the number stored in a slice equals inflow minus outflow.	Q18
Continuity equation	a book-keeping statement that carriers are conserved: change = net inflow + generation − recombination.	Q45
Covalent bond	a pair of electrons shared between neighbouring atoms (four per Si atom).	Q25
Crystal lattice	an infinite, periodic three-dimensional array of points in which every point has identical surroundings.	Q5
Crystal plane	a flat sheet passing through lattice points of a crystal.	Q6
Current density (J)	charge crossing unit area per unit time (A/m^2).	Q37, Q38, Q45
Czochralski (CZ) method	growth of a large single crystal by slowly pulling a seed crystal from molten silicon.	Q3, Q11
Dangling bond	an unpaired bond left on a surface atom because the crystal ends there.	Q46

Term	Meaning	Questions
Dash necking	pulling a thin neck (≈ 3 mm wide) quickly at the start so that dislocations grow out of the sides.	Q10
Degenerate semiconductor	one so heavily doped that E_F is within 3kT of a band edge or inside a band.	Q33
Delocalised electrons	electrons not tied to any one atom, free to move through the whole crystal.	Q8
Density of states (g(E))	the number of allowed states per unit energy per unit volume at energy E ($m^{-3}eV^{-1}$).	Q32
Depletion / accumulation layer	a surface layer with fewer / more majority carriers than the bulk; bands bend up / down in n-type.	Q49
Depletion capacitance (C)	capacitance of the carrier-free depletion layer, $C = \varepsilon_s A/W$ (F).	Q50
Depletion region (space-charge region), width W	the layer around the junction emptied of mobile carriers, containing only fixed ion charge.	Q47
Diamond cubic / zinc blende	the crystal structures of Si and of GaAs; both have each atom bonded to four neighbours, but in zinc blende the two kinds of atom alternate.	Q2, Q4
Diamond cubic structure	two interpenetrating face-centred cubic (FCC) lattices offset by a quarter of the body diagonal; each atom has four tetrahedral neighbours (the structure of Si and Ge).	Q5
Diffusion	the net movement of atoms from a region of high concentration to one of low concentration, caused by their random thermal motion.	Q17, Q39, Q47
Diffusion coefficient (D)	a measure of how fast a dopant spreads, in m^2/s.	Q17, Q18, Q39, Q40, Q44, Q46, Q48
Diffusion current density	the current carried by the diffusion flux of charged carriers (A/m^2).	Q39
Diffusion length (L)	average distance a minority carrier diffuses before recombining, $L = \sqrt{(D\tau)}$.	Q44, Q45, Q48
Diffusion length ($\sqrt{(Dt)}$)	the characteristic distance dopant spreads in time t; the profile's depth scale is $2\sqrt{(Dt)}$.	Q19
Direct / indirect band gap	direct means an electron can drop across the gap and emit light without changing its momentum; indirect means a lattice vibration (phonon) must also take part, so light emission is weak.	Q2, Q4
Directional bond	a bond that points along fixed directions, so it fixes the angles between neighbouring atoms.	Q8
Dislocation	a line defect, a row of atoms out of place; it causes leakage and must be removed.	Q10, Q11
Donor	a Group V atom (P, As, Sb) that gives a free electron to silicon, making it n-type.	Q14, Q16, Q33
Donor / acceptor	an impurity that gives an electron to the conduction band (e.g. P in Si; concentration N_D) / takes an electron from the valence band, creating a hole (e.g. B in Si; concentration N_A).	Q28

Term	Meaning	Questions
Donor concentration (N_D)	number of donor atoms per m^3; each ionised donor gives one free electron.	Q34
Dopant	an impurity added on purpose to control conductivity, e.g. boron (B) or phosphorus (P).	Q3, Q10, Q14
Doping	adding a small, controlled amount of a chosen impurity to change the number of carriers.	Q1
Doping profile N(x)	doping concentration as a function of depth below the surface.	Q50
Dose (Q)	the total number of dopant atoms introduced per m^2 of surface (the area under the profile).	Q19, Q20, Q21
Drift	carrier motion caused by an electric field.	Q37, Q38, Q47
Drift and diffusion	carrier motion driven by an electric field E, and by a concentration gradient.	Q45
Drift velocity (v_d)	the small average velocity carriers gain in the direction of an electric field (m/s).	Q35, Q37, Q38, Q41
E_c, E_v	the energy of the bottom of the conduction band and the top of the valence band (eV).	Q25
Effective density of states (N_c, N_v)	the number of states per m^3 that, placed at the band edge, give the same carrier count as the whole band; Si at 300 K: $N_c = 2.8\times10^{25}$, $N_v = 1.04\times10^{25}$ m^{-3}.	Q27, Q32, Q33
Effective lifetime (τ_{eff})	the lifetime actually measured, including both bulk and surface losses (s).	Q46
Effective mass ($m*$)	the mass a carrier appears to have when pushed by a force inside the crystal.	Q22, Q32, Q35, Q37
Effusion cell	a small heated oven that evaporates one element to make an atomic beam (MBE).	Q12
Eigenvalue	an allowed value of E for which the equation has an acceptable solution.	Q23
Electron affinity (χ)	energy from the conduction-band edge E_c up to the vacuum level; 4.05 eV for Si, independent of doping.	Q49
Electron concentration (n)	the number of free electrons in the conduction band per m^3 (m^{-3}).	Q29
Electron mobility (μ_n)	drift velocity per unit electric field, $m^2/V\cdot s$; higher mobility means faster devices.	Q4
Electron-volt (eV)	the energy an electron gains across 1 V; 1 eV = 1.6×10^{-19} J.	Q25
Electronegativity	how strongly an atom attracts electrons towards itself. A large difference between two atoms favours ionic bonding.	Q8
Electronic-grade silicon (EG-Si)	polycrystalline silicon of 99.9999999 % ("9N") purity or better, i.e. at most about one impurity atom per billion silicon atoms.	Q9
Electron–hole pair	a free electron plus the empty bond (hole) it leaves behind when a covalent bond breaks.	Q8, Q42
Elemental semiconductor	a semiconductor made of a single chemical element.	Q2

Term	Meaning	Questions
Energy band	a wide range of closely spaced allowed electron energies formed when many atoms join into a solid.	Q1, Q25
Epitaxy	growth of a thin single-crystal layer that copies the crystal arrangement of the substrate.	Q5, Q12, Q13
Excess carrier concentration (Δp, Δn)	the carrier density above its equilibrium value, $\Delta p = p - p_0$ (m^{-3}).	Q44, Q45
Excess carrier density (Δp, Δn)	the amount by which a carrier density exceeds its equilibrium value.	Q48
Extrinsic semiconductor	a doped semiconductor whose carrier density is set by the dopants, not by n_i.	Q14, Q28
E–k diagram (band structure)	a graph of allowed electron energy E against k.	Q26
Fermi level (E_F)	the energy at which a state has a 50 % chance of being occupied; its position shows whether electrons or holes dominate.	Q15, Q27, Q30, Q31, Q33, Q40, Q47
Fermi level (E_F) and intrinsic level (E_i)	the 50 %-occupancy energy, and its value in the pure material (near mid-gap).	Q28, Q34
Fermion	a particle, such as the electron, that obeys the Pauli exclusion principle.	Q30
Fermi–Dirac distribution f(E)	the probability that an electron state of energy E is occupied in thermal equilibrium.	Q22
Fermi–Dirac function (f(E))	the probability that an allowed electron state at energy E is occupied in thermal equilibrium.	Q31, Q32
Float-zone (FZ) method	the crucible-free alternative, in which a molten zone is moved along a polysilicon rod.	Q11
Flux (F)	the number of atoms crossing unit area per unit time, atoms/($m^2 \cdot s$).	Q18, Q39
Forbidden gap	a range of energies for which no electron states exist.	Q24
Fractional distillation	separating liquids by repeatedly boiling and condensing them, using their different boiling points.	Q9
Free carrier	an electron in the conduction band or a hole in the valence band, able to carry current.	Q31
Generation rate (G)	number of electron–hole pairs created per unit volume per unit time ($m^{-3}s^{-1}$).	Q42
Group IV, III–V, II–VI	columns of the periodic table; e.g. a III–V compound joins a group III element (Ga, In, Al) with a group V element (As, P, N).	Q2
Hamiltonian ($\hat{H}$)	the total-energy operator, kinetic plus potential.	Q23
HEMT / MESFET	high-speed field-effect transistors made in GaAs for microwave circuits.	Q4
Hole	the empty place left in a bond when an electron is freed; it moves through the crystal like a positive charge.	Q1, Q15, Q25
Hole concentration (p)	the number of holes (empty bond states in the valence band) per m^3 (m^{-3}).	Q29

Term	Meaning	Questions
Homoepitaxy / heteroepitaxy	growing the same material (Si on Si) / a different material (AlGaAs on GaAs).	Q12
Hot carrier	a carrier whose average energy is far above the thermal value (3/2)kT; described by an electron temperature T_e much greater than the lattice temperature T.	Q41
I-V characterisation	measuring current as the DC voltage is stepped, to extract saturation current, ideality factor, series resistance, leakage, breakdown and barrier height.	Q50
Ideality factor (η)	1 for pure diffusion (or thermionic emission) current, up to 2 when depletion-region recombination dominates.	Q50
Impact ionisation	a hot carrier breaks a covalent bond and creates a new electron–hole pair.	Q41
Impurity (dopant)	a foreign atom sitting on a lattice site of the semiconductor in place of a host atom (a substitutional impurity).	Q16
Ingot (boule)	the large cylindrical single crystal that is pulled; later sliced into wafers.	Q10
Intercept	the distance from the origin at which a plane cuts a crystal axis, measured in units of the lattice constants a, b, c.	Q6, Q7
Interface abruptness	how sharply the composition or doping changes from one layer to the next.	Q13
Interface trap density (D_{it})	the number of unwanted electron states at the Si/SiO_2 boundary per m^2 per eV.	Q3
Interplanar spacing (d_{hkl})	the perpendicular distance between neighbouring parallel planes of the same (h k l) set.	Q6, Q7
Interstitial site	a space between normal lattice sites where a small atom can sit.	Q17
Intrinsic carrier concentration (n_i)	the number of electrons (equal to holes) per m^3 created by heat in pure material; 1×10^{16} m^{-3} for Si at 300 K.	Q3, Q4, Q14, Q27, Q28, Q29, Q47
Intrinsic gettering	trapping of metal impurities by oxygen precipitates deep in the wafer, away from the devices at the surface.	Q11
Intrinsic level (E_i)	the Fermi level of undoped material, close to mid-gap (about 0.56 eV below E_c in Si).	Q33, Q40, Q43
Intrinsic semiconductor	a pure semiconductor whose free electrons and holes come only from breaking bonds by heat, so $n = p = n_i$.	Q14, Q27, Q29
Ion	an atom that has lost electrons (positive ion, e.g. Na^+) or gained them (negative ion, e.g. Cl^-).	Q8
Ion energy (E)	the kinetic energy gained by an ion of charge q through accelerating voltage V, E = qV (keV to MeV).	Q20
Ion implantation	doping by firing accelerated dopant ions into the wafer.	Q20, Q21
Ionisation energy	the small energy needed to free the donor's electron ($E_c - E_D$) or to fill the acceptor's empty bond ($E_A - E_v$); about 0.04–0.05 eV in Si.	Q16

Term	Meaning	Questions
Ionised impurity	a dopant that has given up or taken an electron and so carries a fixed charge (N_D^+ or N_A^-).	Q16, Q36
Junction depth (x_j)	the depth where the diffused concentration equals N_B, so a p-n junction forms.	Q19
kT	the typical thermal energy of a particle; 0.0259 eV at 300 K (k = Boltzmann's constant, T = absolute temperature).	Q1
Lateral diffusion	sideways spread of dopant under the edge of a mask window.	Q21
Lattice	the regular, repeating arrangement of atoms in a crystal; its repeat distance is the lattice period a.	Q22
Lattice constant (a)	the edge length of a cubic unit cell; Si 0.543 nm (1 nm = 10^{-9} m).	Q5, Q6
Lattice constants (a, b, c)	the edge lengths of the unit cell along x, y and z.	Q7
LDD (lightly doped drain)	a MOSFET drain extension with lower doping that reduces the peak field.	Q41
Leakage current	the small unwanted current through a reverse-biased junction, caused by thermally generated carriers.	Q3
Lifetime (τ_p)	average survival time of an excess minority carrier (s).	Q45, Q48
Low-level injection	excess carrier density much smaller than the majority density.	Q43, Q44, Q48
Lowest common multiple (LCM)	the smallest number that all the denominators divide into exactly.	Q6, Q7
LPE (liquid-phase epitaxy)	epitaxy by precipitation from a saturated solution in molten metal as it cools.	Q13
Majority / minority carrier	the carrier present in larger / smaller numbers (electrons / holes in n-type).	Q28
Majority carrier	the carrier present in larger number (electrons in n-type, holes in p-type).	Q15, Q34
Mass-action law	in equilibrium $np = n_i^2$, so raising one carrier lowers the other.	Q15, Q28, Q29, Q34
Matthiessen's rule	when several scattering mechanisms act, $1/\mu = 1/\mu_1 + 1/\mu_2 + \dots$	Q35, Q36
MBE (molecular-beam epitaxy)	epitaxy from beams of atoms in ultra-high vacuum.	Q13
Mean free time (τ_m)	the average time between two collisions of a carrier (about 0.1–1 ps).	Q35, Q36, Q37
Melt	the molten silicon in the crucible, held just above silicon's melting point of 1414 °C.	Q10
Metallurgical-grade silicon (MG-Si)	silicon about 98 % pure, made by reducing quartz with carbon in an arc furnace; too impure for devices.	Q9
Miller indices (h k l)	the smallest integers proportional to the reciprocals of a plane's intercepts on the crystal axes; they name the plane's orientation.	Q7
Minority carrier	the carrier present in smaller number (holes in n-type, electrons in p-type).	Q15, Q34, Q39, Q44, Q48
Minority-carrier lifetime (τ)	average time an excess minority carrier survives before it recombines.	Q42

Term	Meaning	Questions
Mobility (μ)	drift velocity per unit electric field ($m^2/V \cdot s$); it falls when carriers are scattered.	Q5, Q35, Q36, Q37, Q38, Q40
MOCVD	metal-organic chemical vapour deposition, the version of VPE used for III–V compounds.	Q12
Monolayer	one atomic layer of the crystal, about 0.28 nm thick for GaAs.	Q13
Mott–Schottky plot	$1/C^2$ against V; straight for uniform doping.	Q50
n-type semiconductor	a semiconductor doped mainly with donors, so free electrons outnumber holes.	Q15, Q34
Native oxide	an oxide that grows directly on the material by heating it in oxygen or steam; for silicon it is silicon dioxide (SiO_2).	Q3
Negative differential mobility	a field range where more field gives less drift velocity (GaAs electrons).	Q41
Non-degenerate semiconductor	doped lightly enough that Boltzmann statistics apply (E_F more than about 3kT from the band edges).	Q40
Nuclear and electronic stopping	energy loss by collisions with silicon nuclei (causes damage) and with electrons (little damage).	Q20
Octet (full outer shell)	eight electrons in the outer shell, the stable arrangement that atoms reach by bonding.	Q8
One-sided junction	p^+n, n^+p or Schottky diode; the depletion region lies almost entirely in the lightly doped side.	Q50
Optical phonon	a high-energy lattice vibration (about 0.063 eV in Si) that can absorb a large amount of carrier energy in one collision.	Q41
p-n junction	the boundary between a p-type and an n-type region in one continuous crystal.	Q47
p-type semiconductor	a semiconductor doped mainly with acceptors, so holes outnumber free electrons.	Q15
Partial derivative (∂)	a rate of change with respect to one variable while the other (x or t) is held fixed.	Q18
Passivation	treating a surface so that it has few active surface states.	Q46
Pauli exclusion principle	no two electrons can occupy the same quantum state (at most one electron per state, counting spin).	Q30
Periodic potential	a potential energy V(x) that repeats with the lattice period, $V(x + a) = V(x)$.	Q22, Q24, Q36
Phonon	a quantum (packet) of lattice vibration.	Q22, Q26, Q35, Q36, Q43
Photon	a quantum of light; it carries energy $h\nu$ but almost no momentum compared with an electron in a crystal.	Q26
Polycrystalline silicon (polysilicon)	solid silicon made of many small crystal grains pointing in different directions.	Q9
Predeposition / drive-in	the first diffusion step with a constant surface source, and the second step that spreads a fixed dose deeper.	Q17

Term	Meaning	Questions
Profile	the curve of dopant concentration C against depth x.	Q21
Projected range (R_p)	the average depth at which the ions stop.	Q20
Projected range (R_p) and straggle (ΔR_p)	mean stopping depth of implanted ions and its spread.	Q21
Quantum well / heterostructure	a very thin layer (a few nm) of one semiconductor sandwiched between another; it needs atomically sharp interfaces.	Q13
Quasi-equilibrium	carriers in the depletion region stay so close to equilibrium that Boltzmann statistics still link the densities on either side of it.	Q48
Quasi-neutral region	a region outside the depletion layer where the net charge, and so the field, is nearly zero.	Q45
Radiative recombination	an electron falls into a hole and the energy leaves as a photon.	Q26
Reciprocal	one divided by a number; the reciprocal of 2 is 1/2, and of ∞ is 0.	Q6, Q7
Recombination rate (R)	number of electron–hole pairs destroyed per unit volume per unit time ($m^{-3}s^{-1}$).	Q42, Q44
Reduced Planck constant ($\hbar$)	$\hbar = h/2\pi = 1.055\times10^{-34}$ J·s.	Q23
Resistivity (ρ)	how strongly the material opposes current (Ω·m); set mainly by the doping.	Q11, Q14
RHEED (reflection high-energy electron diffraction)	an electron beam bounced off the growing surface; its pattern shows the surface order and counts atomic layers.	Q12
Saturation current (I_s, J_s)	the small reverse current of the diode, and the scale factor of the exponential (A, or A/m^2 for J_s).	Q48
Saturation velocity (v_{sat})	the maximum drift velocity at high field, about 10^5 m/s in Si.	Q38, Q41
Scattering	any collision that randomises a carrier's direction and destroys the velocity it gained from the field.	Q35, Q36
Schottky barrier height (Φ_B)	barrier an electron in the metal must climb to enter the semiconductor (eV); fixed by the interface.	Q49
Schrödinger equation	the wave equation whose solutions give the allowed wave functions and energies of an electron.	Q22
Seed crystal	a small, perfect single crystal cut to the wanted orientation (e.g. ⟨100⟩) that starts the growth.	Q10
Segregation coefficient (k_0)	the ratio of dopant concentration in the freezing solid to that in the liquid, $k_0 = C_s/C_l$; for most dopants $k_0 < 1$ (B 0.8, P 0.35).	Q11
Series resistance (R_s)	resistance of the bulk, contacts and leads in series with the junction (Ω).	Q50
Siemens process	the CVD step in which purified $SiHCl_3$ and hydrogen deposit pure silicon on heated silicon rods.	Q9
SiGe HBT	a fast bipolar transistor with a silicon–germanium base.	Q4
Single crystal	a solid with the same lattice and orientation throughout; *polycrystalline* material is made of many small crystals (grains), *amorphous* has no long-range order.	Q5, Q10

Term	Meaning	Questions
Stationary state	a state whose energy and probability distribution do not change with time.	Q23
Stoichiometry	the exact ratio of the elements in a compound (1 : 1 for GaAs).	Q2
Straggle (ΔR_p)	the standard deviation of the stopping depths about R_p.	Q20
Substitutional impurity	a dopant that occupies a normal lattice site in place of a silicon atom.	Q17
Substrate	the wafer on which the layer is grown.	Q12
Supersaturated solution	a liquid holding more dissolved material than it can keep at that temperature, so the excess comes out as solid.	Q12
Surface concentration (C_s)	the dopant concentration at x = 0, here 1×10^{26} m^{-3} (close to the solid solubility of B or P in Si).	Q19
Surface recombination velocity (S)	the rate of recombination per unit area divided by the excess minority density at the surface (m/s).	Q46
Surface states	localised energy levels in the band gap at a surface, caused by broken (dangling) bonds, adsorbed atoms and oxide defects.	Q46
Thermal budget	the total amount of high-temperature processing (temperature and time) a wafer receives; more budget means existing profiles spread more.	Q21
Thermal diffusion	doping in which dopant atoms move into a heated wafer (900–1200 °C) from a surface source by random thermal jumps.	Q21
Thermal energy (kT)	Boltzmann constant times absolute temperature; 0.0259 eV at 300 K.	Q27, Q30
Thermal equilibrium	the steady state of a sample at fixed temperature with no applied voltage, light or current.	Q29, Q40, Q42
Thermal velocity (v_{th})	the average random speed of carriers, about 1×10^5 m/s at 300 K.	Q43
Thermal voltage (kT/q)	Boltzmann constant × temperature ÷ electronic charge; 0.0259 V at 300 K.	Q40
Thermionic emission	carriers crossing *over* a barrier using thermal energy.	Q49
Trap	a defect or impurity with an energy level inside the band gap that helps electrons and holes recombine.	Q42, Q43
Trap density (N_t)	concentration of recombination centres in the gap (m^{-3}).	Q44
Trichlorosilane ($SiHCl_3$)	a silicon compound that is liquid at room temperature (boils at 32 °C), used as the carrier that can be purified.	Q9
Tunnelling (field emission)	carriers passing *through* a barrier only a few nanometres thick.	Q49
Ultra-high vacuum (UHV)	pressure of about 10^{-10} Torr, so low that atoms travel in straight lines without colliding.	Q12
Unit cell	the smallest block which, repeated by translation in three directions, rebuilds the whole crystal.	Q5
Vacancy	an empty lattice site where an atom is missing.	Q17
Vacuum level (E_{vac})	energy of an electron at rest just outside the material; the common reference.	Q49

Term	Meaning	Questions
Valence band	the highest band that is full at 0 K; it holds the electrons that form the bonds.	Q1, Q31
Valence electrons	the electrons in an atom's outermost shell; they are the ones that form bonds. Silicon has four.	Q8, Q16
VPE (vapour-phase epitaxy)	epitaxy from gases that react on a hot wafer; MOCVD is its metal-organic form for III–V compounds.	Q13
Wafer orientation	the crystal plane parallel to the wafer surface, e.g. a (100) wafer.	Q6
Wave function (ψ)	the mathematical wave that describes an electron; $\|\psi\|^2$ gives the probability of finding it at a point.	Q22, Q23
Wave vector (k)	$k = 2\pi/\lambda$; it measures the electron's momentum, $p = \hbar k$.	Q24, Q26
Work function (Φ_m, Φ_s)	energy from the Fermi level up to the vacuum level, for the metal and the semiconductor (eV).	Q49

Constants and units

Symbol	Meaning	Units / typical value
q	Magnitude of the electronic charge	1.6×10^{-19} C
k	Boltzmann constant	8.617×10^{-5} eV/K = 1.381×10^{-23} J/K
kT	Thermal energy	0.0259 eV (often 0.02585 eV) at 300 K
kT/q	Thermal voltage	0.0259 V (about 26 mV) at 300 K
T	Absolute temperature	kelvin (K); room temperature = 300 K
h	Planck constant	6.626×10^{-34} J·s
ħ	Reduced Planck constant, h/2π	1.055×10^{-34} J·s
m_0	Free-electron mass	9.11×10^{-31} kg
ε_0	Permittivity of free space	8.854×10^{-12} F/m
ε_s (ε_{si})	Permittivity of silicon, $11.7\varepsilon_0$	1.04×10^{-10} F/m
ε_{ox}	Permittivity of SiO_2, $3.9\varepsilon_0$	3.45×10^{-11} F/m
ε_r	Relative permittivity (dielectric constant)	Si 11.7, SiO_2 3.9, GaAs 12.9
eV	Electron-volt, a unit of energy	1 eV = 1.6×10^{-19} J
Å	Ångström, a unit of length	1 Å = 10^{-10} m = 0.1 nm (not SI: this course uses nm)
µm	Micrometre (micron)	1 µm = 10^{-6} m

Symbols

Symbol	Meaning	Units / typical value
E_c (E_C)	Energy of the bottom of the conduction band	eV
E_v (E_V)	Energy of the top of the valence band	eV
E_g	Band-gap energy, $E_c - E_v$	eV; Si 1.12, Ge 0.66, GaAs 1.42 at 300 K
E_F	Fermi level: the energy at which a state has a 50 % chance of being occupied	eV
E_i	Intrinsic Fermi level, close to mid-gap	eV
E_Fn, E_Fp	Quasi-Fermi levels for electrons and holes, used out of equilibrium	eV
E_D	Donor energy level, just below E_c	eV; $E_c - E_D \approx 0.045$ eV for P in Si
E_A	Acceptor energy level, just above E_v	eV; $E_A - E_v \approx 0.045$ eV for B in Si
E_t	Energy level of a trap (recombination centre) in the gap	eV
E_a	Activation energy of dopant diffusion	eV; about 3–4 eV
E_n	Allowed energy of level n of a particle in a box	J or eV
f(E)	Fermi–Dirac function: probability that a state at energy E is occupied	dimensionless, 0 to 1
g(E), g_c, g_v	Density of states: allowed states per unit energy per unit volume	states/(eV·m^3)
N_c	Effective density of states in the conduction band	m^{-3}; Si 2.8×10^{25} at 300 K
N_v	Effective density of states in the valence band	m^{-3}; Si 1.04×10^{25} at 300 K
n	Electron concentration (free electrons per m^3)	m^{-3}
p	Hole concentration (holes per m^3)	m^{-3}
n_i	Intrinsic carrier concentration ($n = p = n_i$ in pure material)	m^{-3}; Si $\approx 1\times10^{16}$ at 300 K
n_0, p_0	Equilibrium electron and hole concentrations	m^{-3}
p_{n0}, n_{p0}	Equilibrium minority concentrations: holes on the n side, electrons on the p side	m^{-3}; $p_{n0} = n_i^2/N_D$
N_D	Donor concentration	m^{-3}
N_A	Acceptor concentration	m^{-3}
N_D^+, N_A^-	Concentrations of ionised donors and acceptors	m^{-3}
Δn, Δp	Excess electron and hole concentrations above equilibrium	m^{-3}
m∗, m_n∗, m_p∗	Effective mass of electrons or holes	kg, or as a multiple of m_0
ψ	Wave function ($\lvert\psi\rvert^2$ is the probability density)	—
V(x)	Potential energy in the Schrödinger equation	J or eV
k (wave vector)	Wave vector, 2π/λ; measures the electron's momentum, p = ħk	m^{-1}
α (Kronig–Penney)	√(2mE)/ħ, set by the electron energy	m^{-1}

Symbol	Meaning	Units / typical value
P (Kronig–Penney)	Barrier strength $mV_0ba/\hbar^2$	dimensionless
V_0	Height of a potential barrier	J or eV
λ	Wavelength	μm or nm; λ(μm) = 1.24/E(eV)
ν, hν	Frequency of light, and photon energy	Hz; eV
a	Lattice constant: edge length of the cubic unit cell	nm; Si 0.543, Ge and GaAs 0.565
(h k l), {h k l}	Miller indices of a plane; family of equivalent planes	integers
[u v w], ⟨u v w⟩	A crystal direction; family of equivalent directions	integers
d_{hkl}	Spacing between neighbouring (h k l) planes, $a/\sqrt{(h^2 + k^2 + l^2)}$	nm
r_{nn}	Nearest-neighbour distance	nm; Si 0.235
k_0	Segregation coefficient, C_s/C_l at the freezing interface	dimensionless; B 0.8, P 0.35
C_s	Surface concentration of a diffusion (or solid concentration when freezing)	m^{-3}
C_l, C_0	Concentration in the melt (liquid); C_0 at the start	m^{-3}
f (growth)	Fraction of the melt already solidified	0 to 1
C(x, t)	Dopant concentration at depth x and time t	m^{-3}
D	Diffusion coefficient (dopants in a furnace, or carriers)	m^2/s
D_0	Diffusion constant in $D = D_0e^{-E_a/kT}$	m^2/s
J (Fick)	Dopant flux: atoms crossing 1 m^2 per second	atoms/(m^2·s)
F	Particle flux in Fick's law	particles/(m^2·s)
Q	Dose: dopant atoms per unit area (in Weeks 13–14, Q also means charge)	atoms/m^2 (C or C/m^2 for charge)
Φ (dose)	Implanted dose (ions per m^2)	m^{-2}
erfc	Complementary error function, erfc(u) = 1 − erf(u); read from a table	dimensionless
x_j	Junction depth: where the dopant concentration equals the background	μm
N_B (C_B)	Background doping of the wafer (in Week 13, N_B is the base doping)	m^{-3}
R_p	Projected range: average depth at which implanted ions stop	μm
ΔR_p	Straggle: spread of the implanted ions' stopping depths	μm
N_p	Peak concentration of an implanted profile	m^{-3}
S_n, S_e	Nuclear and electronic stopping power of an implanted ion	eV·m^2
N_Si	Atom density of silicon	5×10^{28} m^{-3}
R_s	Sheet resistance (or diode series resistance in Week 12)	Ω/□ (ohms per square); Ω
x_{mask}	Thickness of an implant mask	μm
E (field)	Electric field (on band diagrams, E is energy instead)	V/m

Symbol	Meaning	Units / typical value
μ, μ_n, μ_p	Mobility: drift velocity per unit field (electrons, holes)	$m^2/(V\cdot s)$; Si μ_n 0.135, μ_p 0.048
μ_L, μ_I	Mobility limited by lattice scattering, by ionised-impurity scattering	$m^2/(V\cdot s)$
v_d	Drift velocity	m/s
v_{th}	Thermal velocity of carriers	$\approx 10^5$ m/s at 300 K
v_{sat}	Saturation velocity at high field	$\approx 10^5$ m/s in Si
τ_m	Mean free time between collisions	s; about 0.1–1 ps
σ	Conductivity, $q(n\mu_n + p\mu_p)$ (in Week 8, σ is also a capture cross-section)	S/m (siemens per metre)
ρ	Resistivity, $1/\sigma$	$\Omega\cdot$m
J, J_n, J_p	Current density (total, electrons, holes)	A/m^2
D_n, D_p	Diffusion coefficients of electrons and holes	m^2/s; Si 3.5×10^{-3} and 1.24×10^{-3}
R_H, V_H, B	Hall coefficient, Hall voltage, magnetic flux density	m^3/C; V; T (Wb/m^2)
T_e	Electron temperature of hot carriers	K
G, G_L	Generation rate of electron–hole pairs (G_L: by light)	$m^{-3}s^{-1}$
R	Recombination rate	$m^{-3}s^{-1}$
U	Net recombination rate, R − G	$m^{-3}s^{-1}$
τ, τ_n, τ_p	Minority-carrier lifetime (τ_{eff}: effective, including surfaces)	s; Si 1 ns – 1 ms
N_t	Density of traps (recombination centres)	m^{-3}
σ_n, σ_p	Capture cross-section of a trap for electrons, holes	m^2
n_1, p_1	SRH parameters $n_{ie}^{(E_t - E_i)/kT}$ and $n_{ie}^{(E_i - E_t)/kT}$	m^{-3}
L, L_n, L_p	Diffusion length $\sqrt{(D\tau)}$: distance a minority carrier travels before recombining	µm or m
S	Surface recombination velocity	m/s
N_{st}	Density of surface traps	m^{-2}
U_s	Recombination rate per unit area at a surface	$m^{-2}s^{-1}$
V_{bi}	Built-in potential of a junction or Schottky contact	V; typically 0.5–1 V
V, V_a	Applied voltage (positive for forward bias)	V
W	Depletion-region width (in Week 13 W_B, W_E are base and emitter widths; in Week 14 W is channel width)	µm or m
x_n, x_p	Parts of the depletion region on the n side and on the p side	µm
E_{max}	Peak electric field, at the junction	V/m
I_s, J_s (I_S)	Saturation current (density) of a diode or transistor	A; A/m^2
I	Current	A
A (area)	Device or contact area	m^2

Symbol	Meaning	Units / typical value
η	Ideality factor of a diode, 1 (ideal) to 2	dimensionless
C, C_j	Capacitance (junction or depletion capacitance)	F (pF = 10^{-12} F)
V_{BR}	Breakdown voltage	V
Φ_m	Work function of a metal	eV; Al 4.28, Au 5.1
Φ_s	Work function of a semiconductor, $\chi + (E_c - E_F)$	eV
χ	Electron affinity: energy from E_c up to the vacuum level	eV; Si 4.05
Φ_B	Schottky barrier height seen from the metal	eV (or V)
A^{**}	Effective Richardson constant	$A/(m^2 \cdot K^2)$; n-Si $\approx 1.1\times10^6$
ρ_c	Specific contact resistance	$\Omega \cdot m^2$
E_{vac}	Vacuum level: energy of an electron just free of the material	eV
I_E, I_B, I_C	Emitter, base and collector currents	A (mA)
V_{BE}, V_{CE}, V_{CB}	Base–emitter, collector–emitter and collector–base voltages	V
α, β, γ (cell)	Angles between the edges of a unit cell	degrees; 90° for cubic
α	Common-base current gain, I_C/I_E	dimensionless; 0.98–0.999
β	Common-emitter current gain, I_C/I_B (also h_{FE})	dimensionless; 50–500
γ (BJT)	Emitter injection efficiency	dimensionless, close to 1
α_T	Base transport factor	dimensionless, close to 1
N_E, W_E, D_E; N_B, W_B, D_B	Doping, width and minority diffusion coefficient of the emitter; of the base	m^{-3}; µm; m^2/s
V_A	Early voltage	V
r_o	Output resistance	Ω
τ_B	Base transit time, $W_B^2/2D_n$	s
f_T	Transition (cut-off) frequency	Hz
C_{ox}	Gate-oxide capacitance per unit area, ε_{ox}/t_{ox}	F/m^2
t_{ox}	Gate-oxide thickness	nm
φ_F, ψ_s (φ_s)	Bulk Fermi potential $(kT/q)\ln(N_A/n_i)$; surface potential (band bending)	V
W_d, W_{dmax}, Q_{dmax}	MOS depletion width, its maximum at inversion, and the depletion charge then	µm; C/m^2
V_T (V_{th})	Threshold voltage of a MOSFET	V
V_{FB}	Flat-band voltage	V
Φ_{ms}, Q_f	Gate–substrate work-function difference; fixed oxide charge	V; C/m^2
γ (MOSFET)	Body-effect coefficient $\sqrt{(2q\varepsilon_{si}N_A)}/C_{ox}$	$V^{1/2}$
V_{GS}, V_{DS}, V_{SB}, V_G	Gate–source, drain–source, source–body and gate voltages	V
I_D	Drain current	A (mA)

Symbol	Meaning	Units / typical value
W/L	Channel width-to-length ratio	dimensionless
g_m	Transconductance, $\partial I_D/\partial V_{GS}$	S (A/V)
k (MOSFET)	Device parameter $\mu_n C_{ox}(W/L)$	A/V^2 (mA/V^2)
λ (MOSFET)	Channel-length modulation parameter	V^{-1}
λ_{max}	Longest wavelength a material absorbs, $1.24/E_g$ (μm, E_g in eV)	μm
σ_i, ρ_i	Conductivity and resistivity of intrinsic (undoped) material; σ_{doped} after doping	S/m; Ω·m
a_{layer}, a_{sub}	Lattice constants of the grown layer and of the substrate	nm
N_{total}	Total number of dopant atoms in a region	atoms
$\psi_k(x)$, $u_k(x)$	Bloch wavefunction of an electron, and its part that repeats with the lattice	—
F_n, F_p	Particle flux of electrons and holes (carriers crossing 1 m^2 per second)	$m^{-2}s^{-1}$
B_z, I_x, V_H	Hall effect: magnetic field (z), current (x) and Hall voltage	T (tesla); A; V
R_H	Hall coefficient, 1/qp (p-type) or −1/qn (n-type)	m^3/C
v_{dn}, v_{dp}	Drift velocity of electrons and holes	m/s
μ_i, τ_i (μ_L, μ_I, N_I)	Mobility and scattering time for one scattering mechanism (L: lattice, I: ionised impurity); N_I = ionised-impurity density	$m^2/(V\cdot s)$; s; m^{-3}
J_{drift}, J_{diff}	Drift and diffusion parts of the current density	A/m^2
E_{sat}	Field at which the drift velocity saturates, v_{sat}/μ	V/m
G_{th}, R_{th}	Thermal generation and recombination rates (equal in equilibrium)	$m^{-3}s^{-1}$
G_n, G_p, R_n, R_p	Generation and recombination rates of electrons and holes	$m^{-3}s^{-1}$
f_t, e_n, e_p	Probability a trap is occupied by an electron; emission rates of electrons and holes from traps	dimensionless; s^{-1}
r_a, r_b, r_c, r_d	SRH rates: electron capture, electron emission, hole capture, hole emission	$m^{-3}s^{-1}$
τ_{SRH}, $\tau_{radiative}$, $\tau_A$uger	Lifetimes set by trap (SRH), radiative and Auger recombination	s
τ_b (τ_{bulk}), $\tau_{surface}$	Lifetime from bulk recombination only; from surface recombination only	s
U_s, p_s, Δp_s	Surface recombination rate; hole density and excess at the surface	$m^{-2}s^{-1}$; m^{-3}
n_{n0}, p_{p0}, p_{n0}, n_{p0}	Equilibrium densities: majority (n_{n0}, p_{p0}) and minority (p_{n0} holes on n side, n_{p0} electrons on p side)	m^{-3}
Δp_n, Δn_p	Excess minority carriers at the depletion edges (holes on n side, electrons on p side)	m^{-3}
R_c	Contact resistance	Ω (specific: $\Omega\cdot m^2$)
Φ_Bn, Φ_Bp	Schottky barrier height for electrons (n-type) and for holes (p-type)	eV
$V_{junction}$	Part of the applied voltage that appears across the junction itself	V
$I_{E,n}$, $I_{E,p}$, $I_{C,n}$	Electron and hole parts of the emitter and collector currents	A

Symbol	Meaning	Units / typical value
$n_B(x)$	Minority-electron density in the base of an NPN transistor	m^{-3}
I_{gen}, J_{gen}	Generation current (density) from carriers generated in the depletion region	A; A/m^2
p_n, n_p	Minority-carrier density: holes on the n side, electrons on the p side	m^{-3}
ΔI_D, ΔV_{GS}	Small changes in drain current and gate voltage (their ratio is g_m)	A; V
Q_{charge}	Total charge delivered by the ion beam, I × t	C
B (recombination)	Radiative recombination coefficient, R = B np	m^3/s; GaAs ≈ 10^{-16}, Si ≈ 10^{-20}
C_n, C_p (Auger)	Auger recombination coefficients	m^6/s; Si ≈ 10^{-43}
τ_0 (τ_g)	Effective generation lifetime in a depletion region	s
f_m	Lattice mismatch, $(a_{layer} - a_{sub})/a_{sub}$	dimensionless, usually in %
v_{pull}	Pull rate of a Czochralski crystal	mm/min
λ_c	Cut-off wavelength of a detector, $1.24/E_g$ (µm, E_g in eV)	µm
c	Speed of light in vacuum	3×10^8 m/s
erfc(u), $erfc^{-1}$	Complementary error function, 1 − erf(u), and its inverse (read from a table)	dimensionless
N(x), N(W)	Doping concentration at depth x (or at the depletion edge W), from C-V profiling	m^{-3}
V_{EB}, V_{BC} (V_{CB})	Emitter–base and base–collector voltages	V
ΔV, ΔV_{BE}	A small change in a voltage	V
I_{C1}, I_{C2}	Collector currents at two operating points	A
r (atom)	Atomic radius (hard-sphere model), half the nearest-neighbour distance	nm
t	Time (or a thickness, where the text says so)	s (or m)
ΔV_T	Shift in threshold voltage (e.g. from body bias)	V

Abbreviations

Abbreviation	Meaning
Si	Silicon
Ge	Germanium
GaAs	Gallium arsenide
GaN	Gallium nitride
SiC	Silicon carbide
InP	Indium phosphide
GaP	Gallium phosphide
SiO_2	Silicon dioxide (silica), the oxide insulator
CB	Conduction band
VB	Valence band
FCC	Face-centred cubic lattice
BCC	Body-centred cubic lattice
SC	Simple cubic lattice
APF	Atomic packing factor
III–V	Compound of Group III and Group V elements (e.g. GaAs)
MG-Si	Metallurgical-grade silicon (≈ 98 % pure)
EG-Si	Electronic-grade silicon (99.9999999 % or better)
CZ	Czochralski crystal growth
FZ	Float-zone crystal growth
CVD	Chemical vapour deposition
VPE	Vapour-phase epitaxy
LPE	Liquid-phase epitaxy
MBE	Molecular-beam epitaxy
MOCVD	Metal-organic chemical vapour deposition
UHV	Ultra-high vacuum (≈ 10^{-10} Torr)
RHEED	Reflection high-energy electron diffraction (counts atomic layers during MBE)
RTA	Rapid thermal annealing
SIMS	Secondary-ion mass spectrometry
TLM	Transmission-line method (contact resistance)
SRH	Shockley–Read–Hall (trap-assisted) recombination
LDD	Lightly doped drain
I-V	Current–voltage (measurement or characteristic)

Abbreviation	Meaning
C-V	Capacitance–voltage (measurement or characteristic)
LCR	Inductance–capacitance–resistance meter
SMU	Source-measure unit
MOS	Metal–oxide–semiconductor
MOSFET	Metal–oxide–semiconductor field-effect transistor
CMOS	Complementary MOS (n- and p-channel MOSFETs together)
FET	Field-effect transistor
JFET	Junction field-effect transistor
MESFET	Metal–semiconductor field-effect transistor
HEMT	High-electron-mobility transistor
BJT	Bipolar junction transistor
HBT	Heterojunction bipolar transistor
NPN	BJT with n-type emitter, p-type base, n-type collector
PNP	BJT with p-type emitter, n-type base, p-type collector
E–B	Emitter–base (junction)
C–B	Collector–base (junction)
IC	Integrated circuit
VLSI	Very-large-scale integration
LED	Light-emitting diode
RF	Radio frequency
IGBT	Insulated-gate bipolar transistor
MEMS	Micro-electro-mechanical systems
CAT	Continuous Assessment Test
PGD	Postgraduate Diploma
DC	Direct current (steady)
AC	Alternating current (small signal)
n^+	Very heavily doped n-type (similarly p^+)
LCM	Lowest common multiple
LHS	Left-hand side (of an equation)
MATLAB	A numerical computing program (alternative to Python)
KOH	Potassium hydroxide (an anisotropic silicon etchant)
UV	Ultraviolet
IR	Infrared
VCSEL	Vertical-cavity surface-emitting laser

Abbreviation	Meaning
PCD	Photoconductive decay (lifetime measurement)
TCAD	Technology computer-aided design (device simulation software)
FD	Fermi–Dirac (distribution)